Prevention of Inland Vessel Accidents: Laws and Rules

English translation — Part 0: Front matter and table of contents

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Translator’s note: This is an English translation of the Bengali compilation অভ্যন্তরীণ নৌযানের দুর্ঘটনা প্রতিরোধ: আইন ও বিধি *(H A*

Publishers, Dhaka, 2005), prepared from a scanned copy. Terms given in English in the original are kept as printed. Small grey markers such as p.49 show where each printed page of the book begins. Bengali numerals and lettering have been converted to Western numerals and (a), (b), (c)… Apparent misprints in the original are reproduced as printed and flagged [sic] or [as printed]; a consolidated list of doubtful readings is given in the Appendix at the end. Note that in this copy of the book the English text of the Inland Shipping Ordinance, 1976 (pp. 357–429) is not present; in its place the book contains a Bengali rendering of the Ordinance (pp. 431–483) followed by some supplementary material (duties of supervisors, storm-warning signals). That Bengali text has been translated back into English here; for the authoritative wording, the original English Ordinance should be consulted.

Title page

Complimentary copy

Prevention of Inland Vessel Accidents: Laws and Rules

Md. Moinuddin Zulfiquar

H A Publishers

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Dedication: To those who lost their lives in river accidents over the years.

(Handwritten note on the dedication page: “VIP cabin = 3 m², Family cabin = 3 m² — from Kamrul via Sil”.)

Compiler’s note

Although the book entitled *Prevention of Inland Vessel Accidents: Laws and Rules* is a compilation published in Bengali on the subject of ensuring the safety of vessels, the Inland Shipping Ordinance, 1976 and all of its amendments have been included in English so that there is no departure from the original law. Only the updated principal law and rules relating to this subject have been included in this compilation. The rules and the law have been compiled in such an order that they run continuously from the construction of a vessel through to its operation; as a result, the English portion of the principal law has been placed at the end.

Permission to publish the book is still being processed, but given the demands of the present situation, only the compiled portion is being published, in the hope that the laws and rules for the prevention of river accidents reach everyone concerned. No elaboration or commentary has been included. It is hoped, however, that publication of this compilation will bring forward appropriate proposals on the rules that need amending in light of the amendments to the principal law, and proposals for correcting the errors that exist in the rules; the discussion will gain reasoning and breadth. After obtaining permission to publish, I hope to begin the modest effort of producing a book on this subject — drawing on long working experience — that can be kept to hand and used in the day-to-day work of this sector.

If this compilation is of even the slightest use to those concerned, my small effort will have been worthwhile. It is not unusual for a compiled book to contain errors. If any error, including printing errors, comes to the attention of a kind reader, or if there is any advice on improving the book, the compiler humbly requests that it be communicated to him.

Md. Moinuddin Zulfiquar

Foreword

I am pleased to learn that the book compiled by Mr. Md. Moinuddin Zulfiquar, Engineer and Ship Surveyor of the Department of Shipping, entitled *Prevention of Inland Vessel Accidents: Laws and Rules*, is being published. In this book the principal law, ordinance, amendments, rules and regulations on the prevention of vessel accidents have been compiled in sequence. I believe the book will be of benefit to everyone connected with vessels — in particular those involved in ensuring navigational safety, vessel owners, employees, government officials, journalists, persons engaged in the shipbuilding business, and anyone interested in knowing about these matters. I wish the compiler's effort every success.

(signed) 26/7/05

Lt. Col. Akbar Hossain (Retd.), Bir Protik
Minister
Ministry of Shipping
Government of the People's Republic of Bangladesh
Bangladesh Secretariat, Dhaka

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p.13 ## Chapter 1 — Preliminary

1. Short title and application

1. These rules shall be called the Inland Steel-Built Ships Construction Rules, 2001.

2. These rules shall apply to the construction of all vessels whose construction in steel commences from the date on which these rules come into force, and to all steel-built vessels plying on inland waterways which have a propulsion unit of more than 12 kilowatts and all non-propelled steel-built vessels of more than 20 metres overall length.

2. Definitions

Unless there is anything repugnant in the subject or context, in these rules—

a. “Department” means the Department of Shipping;

b. “Ordinance” means The Inland Shipping Ordinance, 1976 (LXXII of 1976);

c. “Weathertight” means all those devices which prevent the entry of rain and water falling from above into the ship or floating equipment;

d. “Coastal crossing endorsement / partial bay-crossing vessel” shall mean those inland vessels which, having fulfilled the conditions mentioned in these rules, ply between the inland water limit of the Meghna river estuary and the water limit of the Karnaphuli river estuary, or within 20 kilometres of the coast; such vessels shall have a coastal crossing endorsement;

e. “Weather deck” means the uppermost water- or weathertight deck;

f. “the Authority” means the Inland Shipping Safety Administration (ISA) or any authority designated for this purpose by the Government;

g. “Depth” means the vertical distance in metres, at the mid-length (L) of the vessel, from the upper edge of the keel to the upper edge of the main deck beam, or in the case of an open vessel, to the upper edge of the vessel;

p.14 (h) “Emergency switch board” means a switch board by which, in the event of failure of the main electrical power supply system, electrical power can be supplied directly from the emergency source of electrical power or the temporary source of emergency power, and which is designated for the distribution of electrical power to the emergency services;

i. “Tanker” means any cargo ship constructed for the carriage of liquid materials in bulk;

j. “Deadweight” means the difference, in tonnes, between the displacement (Displacement) of the vessel in water at full-load draft and the lightweight of the vessel; in this case the ship’s crew, passengers, passengers’ luggage, stores, fuel, water and cargo shall be deemed to be included;

k. “Draft” means the vertical distance, in metres, at the mid-length of the vessel, from the upper edge of the keel to the load water line or loaded waterline;

l. “Schedule” means a Schedule annexed to these rules;

m. “Length” means the distance along the waterline from the forward edge of the main post of the fore part of the vessel (Stem line) to the after edge of the rudder post, or, if there is no rudder post, to the centre of the rudder stock;

n. “Vessel” means any ship, whether power-driven or not;

o. “Watertight” means such deck and devices as prevent the entry of water into the ship or floating equipment under all conditions;

p. “Breadth” means the maximum moulded breadth of the vessel measured over the outer edges of the frames;

q. “Freeboard” means the distance between the depth and the draft of a vessel or floating equipment having a watertight freeboard deck;

r. “Freeboard deck” means the uppermost watertight deck;

s. “Main source of electrical power” means the electrical source installed for the purpose of supplying all services necessary for keeping the ship in normal operational and habitable condition;

t. “Main steering gear” means the machinery, rudder, actuator, steering gear, power unit, if any, and auxiliary equipment,

p.15 and the necessary means for applying torque to the rudder stock in order to move and operate the rudder for the purpose of steering the ship under normal service conditions;

u. “Main switch board” means that switch board to which electrical power is supplied from the main source of electrical power and which is installed for the distribution of electrical power to the ship’s services;

v. “Passenger” means any person on board a vessel other than the master, the crew, any other person employed or engaged in any capacity on board the vessel or floating equipment in the business of the vessel or floating equipment, and a child under one year of age;

w. “Engineer and Ship Surveyor” means the engineer and ship surveyor of the “Inland Shipping Safety Administration” located nearest to the place where the construction work of the vessel is carried out;

x. “Maximum speed” means the speed for which the ship is designed in order to maintain the maximum speed of the ship when in service at the design draft;

y. “Table” means a table described in the Schedule.

3. Form, proportion and speed (Form, Proportion & Speed)

The form, proportion and speed of the ship shall be normal, and these rules shall not apply to ships of unusual form, proportion or speed, or to ships constructed for the carriage of special cargoes or for special service.

4. Materials

1. A ship built of fully welded steel for the purpose of navigation on inland waterways shall also be suitable for navigation on large lakes or canals, and the Authority may, if it so wishes, also grant approval for navigation on inland waterways to vessels built of any other material.

2. If any previously used material is re-used in a ship, the average thickness of the plate shall be increased by 25% over the dimension mentioned in these rules, and in the case of other materials such as round steel, profiles and pipes, the dimension of the material concerned shall likewise be increased by 20%.

p.16 (3) The thickness of the plates of ships constructed for a partial bay-crossing / coastal crossing endorsement shall be 20% greater than the thickness mentioned in these directions. For those structural components (Structural component) which are determined by their sectional modulus, the sectional modulus shall be increased by 25%. In the case of previously used materials the sectional modulus shall be increased by 40%.

5. Stress-bearing capacity

When constructing a ship, it shall be so constructed that, if excessive load-induced stress falls on the ship owing to special features of the plan or design and to an unusual loaded condition or ballast condition, it possesses that additional stress-bearing capacity.

6. Survey during construction

1. The materials used for the construction or repair of the hull of a ship or for the installation or repair of machinery shall be of a standard approved by the Authority, and the Engineer and Ship Surveyor shall ensure this.

2. Those parts of the ship for which approval is required shall be approved by the Department, and the Engineer and Ship Surveyor shall inspect them in order to ensure conformity with the approved plans.

3. The Engineer and Ship Surveyors shall inspect the joining work of the ship and the fitting of machinery and electrical installations, and together with this shall carry out the quality-of-workmanship and operational tests.

4. After completion of construction, during the trial voyage, the Engineer and Ship Surveyor shall test and examine all the equipment, machinery and electrical installations of the ship in working condition.

5. In order that the Engineer and Ship Surveyor may perform his duties, he shall have right of access to the ship and to the workshops in which the parts to be examined are to be manufactured and assembled.

6. The shipyard or construction unit shall provide the Engineer and Ship Surveyor free of charge with all material and manpower assistance so that he may carry out all tests. The vessel owner shall bear all expenses.

p.17 ### 7. Construction plans for approval (Plans)

1. For the purpose of ensuring conformity with these rules, six copies of whichever of the following construction plans are applicable, clearly showing the arrangement, form and dimensions, shall be submitted for approval to the concerned office of the Inland Shipping Safety Administration, namely:—

a. In the case of dry cargo vessels, tankers, passenger vessels and special vessels of more than 24 metres length, the following required drawings, wherever applicable, shall be sent for approval to the Design and Development Section of the Administration, namely:—

1. General arrangement (General arrangement);
2. Safety plan (Safety plan);
3. Lines plan (Lines plan);
4. Hydrostatic curves (Hydrostatic Curves);
5. Cross curves of stability (Cross curve of stability);
6. Stability booklet (Stability booklet);
7. Capacity plan (Capacity plan);
8. Midships section and shell expansion plan (Midships section and sheel expansion plan);
9. Drawing of longitudinal sections and deck plans including seating and openings (Drawing of longitudinal sections and deck plans including seating and openings);
10. Shell expansion including welds, openings, stringers, frames etc. (Shell expansion including welds, openings, stringers, frames etc.);
11. Hatch coamings and covers (Hatch comings and covers);

12. Bulkheads including tank bulkheads (Bulkheads including tank bulkheads);
13. Tank top and separate tanks (Tank top and separate tanks);
14. Bottom structure in engine room (Bottom structure in engine room);

p.18 (15) Main engine, auxiliary machinery and equipment, deck equipment, davit etc. foundations (Main engine, auxiliary machinery and equipment, deck equipment, davit etc. foundations); (16) Rudder and steering arrangement (Rudder and steering arrangement); (17) Calculation of equipment number (Calculation of equipment number); (18) Anchoring and mooring arrangement (Anchoring and mooring arrangement); (19) Masts including navigation lights, and the ship's sails and the ropes required for rigging the masts etc. (rigging); (20) Superstructure and deck house (Superstucture and deck house); (21) General arrangement of machinery spaces (General arrangement of machinery spaces); (22) Shafting and propeller (Shafting and propeller); (23) Piping diagrams for bilge sounding (Sounding), fire line, air escape, fuel oil, lubrication oil, deck washing pipes and all pipes under pressure (Piping diagrams for bilge, fire, sounding, air, fuel oil, lubrication oil, deck washing and all pipes under pressure); (24) Cargo piping and pump room including ventilation for tankers (Cargo piping and pump room including ventilation for tankers); (25) Electrical installations (Electrical installations); (26) Placing and ventilation of accumulators (Placing and ventilation of accumulators); (27) General characteristics for main and auxiliary engines and equipment (General characteristics for main and auxiliary engine and equipments);

p.19 (28) Accommodation plan of officers, crews and passengers (Accommodation Plan of Officers, Crews and Passangers).

b. In the case of fishing vessels, the following required drawings, wherever applicable, shall be sent for approval to the Design Section of the Administration, namely:—

1. General arrangement including safety plan (General arrangement including safety plan);
2. Midships section and shell expansion (Midships section and shell expansion);
3. Rudder and steering arrangement (Rudder and steering arrangement);
4. Masts including navigation lights, and the ship's sails and the ropes required for rigging the masts etc. (Masts and rigging including navigation lights);
5. General arrangement of machinery space or spaces (General arrangement of machinery spaces);
6. Shafting and propeller (Shafting and propeller);
7. Piping diagrams for bilge, fire line, sewage, water, oily water, fire and fuel oil pipes (Piping diagrams for bilge, fire and fuel oil-pipes);
8. Electrical installations (Electrical installation);
9. General characteristics for main and auxiliary engines and equipment (General characteristics for main and auxiliary engines and equipment);
10. Lines plan (Lines plan);
11. Hydrostatic curves (Hydrostatic curves);
12. Cross curves of stability (Cross curve of stability);
13. Stability booklet (Stability booklet);
14. Accommodation plan (Accommodation plan).

p.20 (c) In the case of dumb barges, the following required drawings shall be sent for approval to the concerned office of the Administration, namely:—

1. General arrangement including safety plan and capacity plan (General arrangement including safety plan and capacity plan);
2. Lines and stability booklet (Lines and stability booklet);
3. Midships section and shell expansion (Midships section and shell expansion);
4. Navigation lights or masts, and the ship's sails and the ropes required for rigging the masts etc. (Masts and rigging including navigation lights);
5. Piping diagrams including pipes for bilge, oily water, ballast and sounding arrangements (Piping diagrams for bilge, ballast and sounding pipes);
6. Accommodation plan (Accommodation plan).

d. In the case of dry cargo vessels, tankers, passenger vessels and special vessels of up to 24 metres length, the following required drawings shall be sent for approval to the concerned office of the Administration:—

1. General arrangement (General arrangement);
2. Safety plan (Safety plan);
3. Midships section (Midships section);
4. Drawing of longitudinal sections and deck plans including seating and openings (Drawing of longitudinal sections and deck plans including seating and openings);
5. Shell expansion including welds, openings, stringers, frames etc. (Shell expansion including welds, openings stringers, frames etc);

p.21 (6) Bottom structure in engine room (Bottom structure in engine room); (7) Rudder and steering arrangement (Rudder and steering arragement); (8) Calculation of equipment number (Calculation of equipement number); (9) Shafting and propeller (Shafting and propeller); (10) Cargo piping and pump room including ventilation for tankers (Cargo piping and pump room including ventilation for tankers); (11) Placing and ventilation of accumulators (Placing and ventilation of accumulators); (12) General characteristics of main and auxiliary engines and

equipment (General characteristics of main and auxiliary engines and equipment); (13) Accommodation plan (Accommodation plan).

e. In the case of fishing vessels, the following required drawings shall be sent for approval to the concerned office of the Administration:—

1. General arrangement including safety plan and capacity plans (General arrangement including safety plan and capacity plan);
2. Midships section and shell expansion (Midships section and shell expansion);
3. Accommodation plan (Accommodation plan).

f. In the case of flat dumb barges, the following required drawings shall be sent for approval to the concerned section of the Administration:

1. General arrangement including safety plan and capacity plans (General arrangement including safety plan and capacity plan);
2. Midships section and shell expansion (Midships section and shell expansion);
3. Accommodation plan (Accommodation Plan).

p.22 (2) Various required information may be shown on one or more plan drawings, and in the case of vessels of less than forty metres length built by the novel or primitive method, the requirement of plan approval for registration may be waived/relaxed.

3. If any vessel has been built in accordance with the current rules of a recognised international classification society, then the plan drawings under these rules shall be sent to and approved by the said international classification society.

Chapter 2

8. Material dimensions (Material dimensions)

1. In the case of new plates and previously used plates, plates of minimum thickness 4 millimetres and 6 millimetres respectively shall be used, and in the case of previously used plates the thickness shall always be increased by 25% over the thickness mentioned in the tables of this Chapter.

2. If previously used material is used, the sectional modulus of all scantlings (Scantlings) shall always be increased by 20%.

3. In these rules, previously used plate or section shall mean those plates or sections which have been collected from a vessel abandoned by the Government or by an organisation registered or approved by the Government, which are in good condition, and in which the extent of wastage shall not exceed 10% and, by area, shall not exceed 20% of the area.

9. Keel plating

1. For those vessels whose bottom does not rise gradually from the centre line at the two sides (Without rise of floor), a keel plate (Keel Plate) shall not be required, and the thickness of the bottom plate including the central strake shall be 20% greater than that of the bottom plates.

2. For those vessels whose bottom rises gradually and slightly from the centre line at the two sides of the floor, a keel plate (Keel Plate) of breadth and thickness not less than the value obtained as prescribed in Table ‘1’ of the Schedule shall be fitted on the centre line of the vessel.

p.23 (3) The thickness of the flat keel plate (Keel plate) connected to the sole piece shall be increased by at least 20% over the values given in these rules for a length of at least 4 metres.

10. Bottom plate

The thickness of the bottom plating shall not be less than the value prescribed in Table ‘2’ of the Schedule, and if the frame spacing is greater than the value prescribed in Table ‘2’ of the Schedule, the thickness of the bottom plate shall be increased at the rate of 0.20 millimetre for every 25 millimetres of increase in the spacing.

11. Plating of the widest part of the curve at the two edges of the bottom (Bilge plating)

1. The thickness of the plating of the curved part at the two edges of the bottom of the vessel shall not be less than the value prescribed in Table ‘3’ of the Schedule, and where the bilge plating is rounded, the radius of curvature in that case shall not be less than five times the thickness of the plating.

2. Where the bottom edge of the vessel is rounded (Rounded bilge), the bilge plating shall be extended up to the upper edge of the curve, and where the radius of the widest part of the bottom (Bilge) exceeds 300 millimetres, the thickness of the plating shall be at least 1.5 millimetres greater than the tabular thickness of the side plating.

3. Where the radius of curvature is actually equal to the depth of the floor, the thickness of the bilge plate (Bilge plate) shall be 1.15 × t millimetres. In that case, t = the thickness of the bottom plating at the midship section.

4. In the case of a smaller radius of curvature, the thickness of the bottom bilge plate shall be (1.15 × t + 1) millimetres, where t = the thickness of the bottom plating at the midships section.

5. The plate of both parts of the bottom bilge shall extend 100 millimetres on the side of the rounded part and shall reach up to 150 millimetres above the upper edge of the floor.

6. In the case of a square chine, the thickness of the bilge angle bar shall not be less than the following dimensions, and if it is inside the structure of the angle bar—

p.24 t + 2 millimetres, and t + 2 millimetres in other cases, t = the thickness, in millimetres, of the bottom plating in the midships region shall be taken.

12. Side shell plate

The thickness of the side shell plating shall not be less than the value prescribed in Table ‘4’ of the Schedule.

13. Deck plate

1. The thickness of the deck plating at the side of the hatchway shall not be less than the value prescribed in Table ‘5’ of the Schedule.

2. The thickness of the whole deck plating shall not be less than the value prescribed in Table ‘6’ of the Schedule.

3. If the vessel has a double bottom, the thickness of the inner bottom plate shall be in accordance with Table ‘7’ of the Schedule and shall not be less than that of the bottom plate.

14. Main post of the fore part (Stem)

1. The thickness of steel-built plate stems shall not be less than the value prescribed in Table ‘7’ of the Schedule; however, above and below the loaded waterline the thickness may be gradually reduced towards the stem head and the bottom, and at the upper edge of the stem it may be equal to the thickness of the side shell plating, and at the lower edge of the stem it may be made equal to the thickness of the bottom plate.

2. Horizontal ribs (horizontal ribs) shall be fitted on the stem plates at a spacing not exceeding one metre, and where the radius of curvature at the forward edge of the stem is large, proper stiffening shall be provided in that case by fitting a centre line stiffener or by other means.

3. The breadth of the stem plate shall be sufficient for it to extend throughout the fore region of small radius, and the plate stem shall be stiffened by vertical web plates together with the stem, and the mutual spacing of the web plates may be at most 550 millimetres.

4. A stiffener made from flat bar of the same thickness as the stem shall be connected on the centre line of the stem.

5. Where the radius of the stem is greater than that of bar stems, a centre web shall be required in that case.

p.25 ### 15. Bar stems

1. Bar stems may be of round or rectangular section.

2. The height and thickness of bar stems shall not be less than the value prescribed in Table ‘8’ of the Schedule, or a circular bar may be used for the same section modulus.

3. Bar stems shall extend about 1 metre beyond the forefoot (Forefoot), and in the case of vessels not propelled by their own power the height and thickness of bar stems may be reduced by 10%.

16. Stern frame

1. The scantlings of rectangular stern frames must correspond to the scantlings of bar stems.

2. The stern frame must be connected to the shell plating. Within the structure of the ship, the stern frame shall be joined to the plate, the thickness of which shall be equal to at least 20% of the thickness of the stern frame.

3. The extent of the horizontal flange at the upper edge of the stern frame must be equal to at least 10 times the plate thickness, and in the case of a flat plate keel the lower end of the stern frame shall be extended into the structure by at least more than two frame spacings by means of a horizontal component-part (Component-part) having the scantlings of the stern frame.

4. The breadth and thickness of bar stern frames shall not be less than the value prescribed in Table-‘9’ of the Schedule.

17. Propeller post

1. The propeller post of a plate stern frame shall be of a form suitable for the streamline (Streamline) in the after part of the ship’s hull, and the scantlings shall not be less than the value prescribed in Table-‘10’ of the Schedule.
2. Below the propeller boss, the breadth and thickness of the propeller post shall be gradually increased in such a way that the sole pieces may also be proportionally strengthened and stiffened, and the propeller post shall be strengthened by webs and ribs of suitable thickness and number.
3. From the bottom up to the boss, the girders of the propeller post must be kept at least in accordance with the value prescribed in Table-‘10’ of the Schedule, and

p.26 the scantlings (Scantlings) of the upper part connected to the rudder post may be gradually reduced where the propeller post is connected to the rudder post.

18. Rudder post

1. Whether solid or hollow, and whether streamlined or not, that part which completely closes off the aperture of the after part is the rudder post of the vessel, and it may be fastened in its position by welding or by bolts.
2. The rudder posts shall be made sufficiently strong so that when the ship touches the ground while under way or while in dry dock, there shall be no danger of their bending owing to the weight of the after part.
3. When fastened by bolts, at least 4 bolts shall be used at the upper end of the rudder post.

19. Sole piece

1. In no case shall the cross-sectional area of a solid sole piece be less than 2700 square millimetres.
2. The thickness and breadth of any solid sole piece shall not be less than the value prescribed in Table-‘11’ of the Schedule.
3. The thickness of the steel plates forming the main part of the sole piece of the stern frame shall not be less than the thickness of the steel plates forming the main part of the propeller post.
4. The ribs (Ribs) in the sole piece shall be arranged under the propeller post, braces etc. and in other suitable positions.
5. In the construction of the connection of the heel with the keel, the sole piece shall always be an integral part.
6. As a general rule, depending on the arrangement of the aperture, the after part of the sole piece shall be of rounded form as far as possible from the propeller post.

20. Heel piece

The length of the heel piece of the stern frame shall be at least three times the frame spacing and shall be strongly connected with the keel (bottom).

p.27 ### 21. Propeller boss

After machining, the thickness of the propeller boss shall not be less than the value prescribed in Table-‘12’ of the Schedule.

22. Bottom construction (Bottom construction)

1. Floor plates shall be connected to every frame, and the floors shall be continuous with the keelson webs between the floors.
2. Face plates shall be connected to the floors, or they shall be provided with flanges.
3. Floors shall not be connected to the bottom shell by means of flanges.
4. The necessary limber holes shall be kept in the floors.
5. In self-propelled vessels, the floors in the after part of the stuffing box bulkhead shall be raised above the stern tube.
6. The section modulus for floors shall be in accordance with Table-‘13’ of the Schedule.
7. In the case of considerable rise of floor at the two ends, the section modulus of the floor at one-quarter of the breadth of the vessel shall not be less than half of the section modulus at the centre line of the ship.

8. The depth of the floor shall not be less than 1.5 times the frame depth measured at the end of the overlap of the frame with the floor.

9. The scantlings of floor plates shall not be less than the values prescribed in Table-‘13’ of the Schedule.

10. The ratio of the height of the floor to the thickness of the web shall not exceed 60.

11. In ships having a longitudinally framed bottom, the floors shall be so arranged that the spacing between them does not exceed 3.5 metres.

23. Keelsons

1. All ships shall have girders or keelsons composed (Composed) of web plate and face plate, which shall be extended forward and aft as far as practicable.

p.28 (2) The number of keelsons to be fitted shall depend on the breadth of the vessels in accordance with the direction given in Table-‘14’ of the Schedule.

3. The thickness of the webs and the sectional area of the face plate of side girders or keelsons shall not be less than the values prescribed in Table-‘15’ of the Schedule.

4. The keelson must be provided with a web plate whose depth shall be equal to the depth of the floor and whose thickness shall be equal to the thickness of the floors as per the rules.

5. The thickness of the web plates and face plate of the centre keelson or girder, and the sectional area and breadth of the face plates, shall be in accordance with Table-‘16’ of the Schedule.

6. The thickness and sectional area of the webs and face plates of centre and side girders and keelsons may be gradually reduced to 0.85 times the values prescribed in Table-‘16’ of the Schedule.

24. Shell and deck scantlings

The section modulus for various rolled bars and built-up constructions shall be in accordance with the values prescribed in (Table-13).

25. Side frame

1. The standard frame spacing shall be 500 mm, and the main frames shall be arranged throughout the length of the ship, and the section modulus of the frames shall not be less than the values prescribed in Table-‘17’ of the Schedule.

2. For transverse framing the maximum frame spacing shall be 450 + 2L millimetres and in no case shall it exceed 500 millimetres at the ends, and the maximum longitudinal frame spacing shall be 550 + 2L (millimetres).

3. If the frame spacing differs from the standard frame spacing, the standard section modulus shall be reduced or increased at a linear (Linearly) rate.

4. The section modulus of the frames of tanks shall be increased by 10%.

p.29 ### 26. Web frames

1. For transversely framed vessels, the maximum spacing of web frames shall not exceed the value prescribed in Table-‘18’ of the Schedule.

2. The section modulus of web frames shall be at least 150% in the case of vessels built with combined framing and 200% in the case of vessels built with transverse framing.

27. Side stringer

1. If the depth of the ship exceeds 2.5 metres, a side stringer shall be fitted at a depth of about 0.6 metre above the base (base), and the horizontal extent of the said stringer shall be from the fore end up to the point where the loaded waterline reaches its maximum breadth.

2. The connections shall be extended to the full depth of the web plate, and where the stringers are of the same depth as the web frames, the vertical flanges shall be connected over the face of the webs by means of a suitable strip (efficient strip) or gusset (gusset).

3. The section modulus of web frames and transverse frame hold stringers shall not be less than the values prescribed in Table-‘19’ of the Schedule.

4. Where no effective compensation is made for the cut-away portions of the frame for any purpose, the depth of the stringer shall not be less than 2.5 times the depth of the cut-away portions.

28. Deck beam

1. In the case of transversely built vessels, a deck beam shall be connected at every frame.
2. The beams under winches, windlass, steering gear and other concentrated weights shall be properly strengthened, and in vessels having continuous hatch coamings, adequate flanged brackets shall be fitted at every frame in place of deck beams.
3. The depth of the side brackets below the deck shall be half the deck breadth and the thickness equal to that of the side frame; the brackets shall be extended below the deck and firmly connected to the continuous hatch coaming.

p.30 (4) The section modulus of deck beams shall not be less than the value prescribed in Table-‘20’ of the Schedule.

5. The length-to-depth ratio of transverse beams on the strength deck shall be 30 or less, and for effective decks (decks below the strength deck which are considered to contribute strength to the longitudinal strength of the hull) and superstructures, as far as practicable, the said ratio shall be 40 or less.

29. Deck web beams

Deck web beams shall be connected to the web frames. The section modulus of a deck web beam shall not be less than twice that of the deck beam.

30. Deck girder

1. Where transverse deck beams have an unsupported span of more than 0.5B, deck girders shall be provided, and the deck girders shall be made of plate pieces which shall be welded to the deck plate.
2. The height of the deck girder shall not be less than 1/25 of the unsupported span, and for continuous deck beams the depth of the scalloped girders shall be at least 1.5 times the depth of the beams.
3. The deck girders shall be connected to the supports.
4. The purpose of the connection to the deck girder is correct stress transmission, and for this purpose the connection of the girder to the bulkhead shall be made with a bracket of twice the depth of the girder, or an equivalent arrangement shall be adopted.
5. At concentrated weights, in the way of stanchions and at places of heavy tension of beam scallops, tripping stiffeners or brackets shall be provided on the deck girder at every fourth beam.
6. The section modulus of the deck girder shall not be less than the value prescribed in Table-‘21’ of the Schedule, and if more than one girder is fitted, the modulus given in Table ‘21’ may be reduced at the rate of 100 (N−1) (N+1) per cent, where N shall be taken as the number of effective longitudinal girders.

p.31 ### 31. Stanchion or pillar

1. Where it is not possible for the watertight bulkheads or deck girders to bear the loads applied on them by themselves, arrangements shall be made for fitting stanchions or pillars there.
2. The rows of stanchions or pillars of the superstructure shall be fitted in vertical line with the rows of stanchions or pillars of the hull, and as far as practicable the stanchions or pillars shall be located on the vertical plane of the keelsons or intercostal supports, and they shall have adequate scantlings (scantlings) at the base plates, and the position of the plates shall be such that proper distribution of the load over the various floors is ensured.
3. Stanchions or pillars shall be specially provided under capstans, winches, windlass, machinery spaces, and the breadth and corners of deckhouses.
4. In the engine room, strong stanchions or pillars shall be fitted in conjunction with the bottom girders and web beams, and special care shall be taken in the arrangement of girders, stanchions or pillars and steel bulkheads in order to reduce vibration of the deck house, and in cargo holds stanchions shall be avoided as far as possible.
5. The load which the stanchions shall bear shall be calculated at the rate of 10 KN/m^2 for the weather deck and 3.0 KN/m^2 for the accommodation deck, and for circular stanchions or pillars, whether solid or hollow, the cross-sectional dimensions shall be calculated in accordance with Table-‘22’ and Table-‘23’ of the Schedule in relation to their length and the capacity P of the load to be borne.
6. In a row of stanchions, the load P borne by any stanchion shall be increased by the sum of the loads acting on the stanchions or pillars located above it.

32. Bulkhead (Bulk head)

1. All vessels of length (L) exceeding 20 metres shall have a collision bulkhead (Collission Bhead) and an aft bulkhead.

2. Vessels of length exceeding 30 metres shall have a bulkhead at each end of the engine room, and all watertight bulkheads shall extend up to the uppermost continuous deck or to the top of the hatch side coaming, where applicable.

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3. Transverse watertight bulkheads shall be of the flat type with vertical stiffeners (Vertical stiffeners), and watertight bulkheads shall be fitted opposite the collision bulkhead and the forward bulkhead/stiffener-fitted cargo-hold compartments of the engine room.

33. Collision and after peak bulkheads

1. The distance of the collision bulkhead from the forward end of the load waterline shall not be less than the value given in column 2 of Table 24 of the Schedule, nor more than the value given in column 3 of the said Table.

2. The thickness of the plating and the section modulus of the stiffeners of the collision bulkhead and the after bulkhead shall not be less than the values specified in Table 25 of the Schedule.

3. In self-propelled vessels the after bulkhead shall normally be located at a distance of at least three frame spacings from the forward end of the propeller boss, and all such bulkheads shall be determined in the same manner as stuffing box bulkheads; however, in vessels with the engine in the after part, this after bulkhead may form the after boundary of the machinery space.

34. Other bulkheads (Bulk head)

1. There shall be complete bulkheads within the holds, and the mutual spacing between such bulkheads shall not exceed 0.15L + 6.5 metres, where L is the waterline length of the vessel in metres.

2. The thickness of the watertight bulkhead plating and the section modulus of the stiffeners shall be as specified in Table 26 of the Schedule below.

35. Bulkhead penetration

Watertight doors shall be fitted in all watertight bulkheads except collision bulkheads, and the doors shall be strong enough to withstand the water pressure which may come upon them; however, where bulkhead stiffeners are cut in the way of a watertight door, the cut portions shall be strengthened by frames in order to maintain the full strength of the bulkheads.

36. Watertight bulkhead penetration

1. The number of watertight bulkhead (Water tight door) penetrations shall be kept as small as possible, and all penetrations shall be made watertight.

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2. Penetrations shall, as far as practicable, be kept at the height of the bulkhead, and no penetration of the collision watertight bulkhead shall be permitted for anything other than drainage or filling pipes.

37. Hull tanks (Hull tanks)

1. Where any deep tank is used for carrying water, fuel oil and other liquid substances, it shall be considered as a part of the structure, and for this purpose, where such tanks extend from one side to the other (tanks for consumable stores), such tanks shall be fitted with a longitudinal bulkhead, which may be constructed as a wash bulkhead.

2. Each tank shall be fitted with a screened air pipe, overflow pipe and sounding pipe, and the air pipes shall be led above the weather deck.

3. The air pipes shall be so arranged that, when the tanks are completely filled, the height of the pipes above the surface of the open decks is at least 450 millimetres.

4. Sounding pipes shall be led down to the bottom of the open tank, and a striking plate of adequate size and thickness shall be fitted to the bottom plating directly below their open end; but the forward or after peak tanks (forepeak or after peak tank) shall not be used for carrying oil.

5. The thickness of the plating shall not be less than the value specified in Table 27 of the Schedule, and the section modulus of the stiffeners connected with brackets shall not be less than the value specified in Table 28 of the Schedule.

38. Independent tanks

1. In order that the independent tanks located in the ship’s structure remain stationary in the various conditions created by the motion of the ship, they shall be adequately secured, and no separate fuel oil tanks shall be kept in cargo holds.

2. Drip trays shall be fitted on the outer sides of the tank to collect oil leaking through any hole, and the sight glass shall be of approved oil-resistant material, and a self-closing (Self Closing) valve shall be fitted at the lower end of the sight glass.

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3. The thickness of the plate shall not be less than 4 millimetres and the section modulus of the stiffeners not less than 2 cubic centimetres, and only new plate shall be used in fuel tanks.

39. Drinking water tanks

1. Drinking water tanks shall be kept separated by cofferdams from those tanks which do not contain drinking, ballast or distilled water.

2. In no case shall sanitary arrangements or the associated piping arrangements be placed directly above drinking water tanks.

3. The manhole of a drinking water tank shall be placed on the upper side of the tank, and a sill (Sill) of modest height shall be fitted around that manhole.

4. Pipes carrying liquids other than water shall not pass through drinking water tanks.

5. The air and sounding (Sounding) pipes connected to drinking water tanks shall be kept separate from the pipes connected to other tanks.

40. Wash bulkheads

The plate thickness of wash bulkheads shall be at least 5 millimetres and the section moduli of the stiffeners at least 5 cubic centimetres, and the stiffeners shall be fitted at least at every second frame, and the lower free edge of the wash bulkhead shall be adequately strongly secured.

41. Watertight test

1. All tanks shall be tested for watertightness by a head of water of 1.0 metre above the top of the tank, or by the pressure of the highest point of the overflow or air pipe, whichever is greater, and the said test shall be completed before the vessel is floated and before any kind of cementing or piping.

2. Oil-tight or oil-resisting decks and bulkheads shall be inspected before the testing work is carried out; however, if after the test they are pierced for the fitting of pipes or for any other purpose, then a second

p.35

test shall be conducted, and the said test may be carried out while the vessel is afloat.

3. In the case of tanks located in the ship’s structure, the watertight test shall be carried out by a head of water of 3 metres above the top of the tank, or by the pressure of the highest point of the overflow and air pipe, whichever is greater.

42. Machinery spaces

1. Machinery spaces shall be suitably strengthened by means of web frames, strong beams and stanchions or pillars.

2. Machinery, shafting etc. shall be effectively supported, and the adjacent structures shall be adequately strengthened.

3. In every machinery space, besides the normal means of exit, an emergency escape route shall be provided.

4. In ships with a single bottom, the top girders of the main engines shall be seated on thick rider plates placed transversely, and the holding-down bolts of the main engines shall pass through the rider plates, and the suitably extended thrust bearings shall be bolted to effective seatings.

43. Floor plates

Floor plates shall be uniform, and no flanges shall be acceptable in them, and the floors shall be made in T-section.

44. Keelsons

1. The keelsons beneath the engine shall extend over the whole length of the engine room and shall be carried, by means of thick brackets, at least one frame spacing beyond the bulkheads on both sides; and where, owing to the position of the vessel, the required arrangement is not possible in the after part, the keelsons shall terminate at a web frame of correspondingly effective strengthened deep floor characteristics.

2. The keelsons beneath the engine shall normally be continuous and the floors shall be intercostal.

3. Over more than half of the length of the centre, the keelson section modulus shall not be less than the value specified in Table 15 of the Schedule.

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45. Web frames in the engine room

1. The spacing between the web frames of the engine room shall be at least equal to the value specified in Table 18 of the Schedule, and shall at no time exceed 4 times the frame spacing or 2.4 metres.

2. The section modulus of the web shall not be less than the value specified in Table 29 of the Schedule.

46. Side stringers in the engine room

1. Where the depth of the ship at half the length of the engine room exceeds 2.0 metres, a side stringer shall be fitted at half the depth of the ship, and its section modulus shall not be less than the section modulus of the web frames of the engine room.

2. The side stringers shall be carried in the form of brackets up to two frames beyond the bulkhead forward of the fore part and forward of the after part of the engine room.

3. Where floors and keelsons have to be cut to make room for the engine sump, they shall be adequately strengthened there to compensate for the loss of strength.

47. Cargo hatchways (Longitudinal hatch Coamings)

1. Where the cargo hatchway is located above the cargo hold from the fore end to the after end of the ship, both longitudinal coamings shall be considered as an essential part of the longitudinal structure.

2. To resist buckling of the fore and after hatch coamings, a longitudinal coaming with a horizontal stiffener shall be fitted at the least possible distance from the upper edge of the coaming, and the upper edge of the coaming shall not be less than 450 millimetres above the deck.

3. Longitudinal hatchway coaming stiffeners shall be provided to resist tripping by means of a vertical stay (mast-stay rope) attached above every web frame or transversely placed watertight bulkhead, and these may be of the same section as the horizontal stiffeners of the upper edge of the coaming, or brackets with flanged edges whose thickness shall be equal to the thickness of the coaming and whose height shall be equal to 10 times the thickness.

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4. In the case mentioned in sub-rule (3), where the upper portion of the coaming has to be partially cut away for seating the hatchway beams, the edges of the cut portion shall be rounded, and a doubling plate or a plate of increased thickness shall be fitted to ensure the adequate load-bearing capacity of the hatchway beams.

5. The extended depth of the hatch coaming below the deck shall not be less than the depth of the deck web beams.

6. The thickness of the hatch coamings of 450 millimetre high coamings shall not be less than the dimensions specified in Table 30 of the Schedule.

48. Transverse hatch coamings (Transverse Hatch Coamings)

1. Transverse hatch coamings shall be fitted with horizontal stiffeners at their upper edge, kept continuous with the longitudinal stiffeners.

2. Transverse coamings shall be strengthened, like the longitudinal stiffeners, by means of vertical stay plates (stay), and these shall be connected with the horizontal stiffeners.

3. The thickness of the hatchway coaming shall not be less than the value specified in Table 31 of the Schedule.

4. The extended depth of the hatch coaming below the deck shall not be less than the depth of the deck web beams.

49. Supporting items for hatch covers

1. Where the hatch covers are not of a self-supporting nature, transverse hatchway beams shall be provided for

their support, and the width of the supporting surface shall be at least 65 millimetres.

2. The section modulus of the transverse steel hatchway beams (channel bars) shall not be less than the value specified in Table 32 of the Schedule.

50. Fore and after beams of hatchways

Where, for supporting the transverse hatchway beams, a hatchway beam secured by bolts only is provided on the centre line of the ship, the section modulus shall not be less than the value specified in Table 33 of the Schedule.

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51. Side fore and after hatchway beams

The section modulus of the side fore and after hatchway beams arranged between the centre line and the fore and after hatch coamings shall not be less than the values specified in Table 33 of the Schedule.

52. Hatch covers or hatch lids

1. The thickness and section modulus specified in Table 34 are to be followed for vessels plying on inland waterways, and if the vessel is built with the intention of partial bay-crossing trade, all dimensional measurements shall be increased by 20%.

2. The thickness of hatch covers shall be at least 4 millimetres, and they shall be so designed that the hatch covers can be preserved.

3. The sides of the hatch covers shall be of such shape that they can prevent the entry of water into the cargo hold.

4. If the vessel is built with the intention of partial bay-crossing trade, gaskets shall be fitted on the hatch covers, and there shall be arrangements for locking the hatch covers to ensure that the lids remain in place.

5. The section modulus of the stiffeners or strengthening members of the hatch covers shall not be less than the value specified in Table 34 of the Schedule.

53. Wooden hatch covers or wooden hatch lids

1. The thickness of wooden hatch covers shall not be less than 30 millimetres for spans up to 1.5 metres and 50 millimetres for spans up to 2.5 metres, and if the vessel is built for the purpose of bay-crossing trade, the thickness shall not be less than 50 millimetres.

2. In the case of partial bay-crossing service, the wooden hatch covers shall be made up of three broad planks placed side by side, and they shall be connected to one another by transverse wooden boards at least 40 millimetres thick and about 125 millimetres wide on their under side, and the maximum spacing shall be 1.0 metre.

3. For fitting the tapering ends of the wedges of vessels built for service in rough waters or for partial bay-crossing trade,

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cleats shall be fitted, and the cleats shall be at least 50 millimetres wide, and the spacing between them shall not exceed 1 millimetre [sic], and the distance of the cleats from the hatch corners shall not exceed 150 millimetres.

4. The batten wedges used for holding the tarpaulins shall be in effective and good condition, and the wedges shall be of wood and shall taper at a rate of not more than 1:6, and the thickness at the head shall not be less than 12 millimetres.

5. When any vessel plies in rough waters it shall be kept covered with tarpaulin, and during bay-crossing two tarpaulins shall be used, and the tarpaulins shall be kept firmly secured by battens and wedges as mentioned in sub-rules (2), (3) and (4).

6. The tarpaulins shall be secured by transversely fitted steel bars or steel wire, and there shall be at least one securing arrangement for each tarpaulin cover, and in rough waters the securing arrangements shall be not more than 3 metres apart, and during bay-crossing they shall be 1.5 metres apart.

54. Openings in decks, shell plating and weather deck

1. The corners of openings for doors, ports, hawse pipes, side scuttles, water inlets (waterinlet) etc. must be rounded corners.

2. The openings shall be properly compensated and strengthened.

55. Access ways and doors

1. Stairways or structure doors leading to the main deck shall be enclosed by structure on all sides, or there shall be an equivalent strong or rigid structural access arrangement.

2. The doors of vessels plying on inland waterways shall be weathertight, and in the case of partial bay-crossing vessels there shall be watertight doors, and the height of the door sills (sills) shall be at least 150 millimetres.

3. Weathertight (wathertight) doors shall be made of steel, and the edges of the doors shall be of such shape as to resist the entry of water, and each door shall have at least two hinges.

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4. Watertight doors shall be made of steel, and the edges of the doors shall be of such shape as to resist the entry of water, and gaskets shall be provided on the door frames, and each door shall have at least two hinges.

56. Openings of machinery spaces

1. The openings of machinery spaces shall be well protected from the accommodation spaces by strongly constructed steel casings, and in those places where the casing is open towards the outside, watertight bulkheads of at least 1 metre height shall be fitted. Machinery spaces shall be mechanically ventilated.

57. Openings of cargo holds

1. The access hatches of cargo holds shall have 450 millimetre high coamings, and the lids shall be made of steel and shall be of such shape that they can prevent the entry of water into the cargo hold (cargohold), and they shall have at least 2 toggles or bolts.

2. The ventilation air inlets of cargo holds shall be so designed that they can resist spray and rain water, and there shall be arrangements for closing the air inlets by means operable from outside, and they shall have arrangements for opening and closing.

58. Windows and side scuttles

1. No windows and side scuttles shall be permitted in the shell plating or below the freeboard deck within a distance of less than 25% of the vessel’s length from the fore end.

2. Those windows and side scuttles which are fitted in the after part of the ship, the superstructure and deckhouse bulkheads shall be very well constructed, and when closed they shall be weathertight (Weather-tight).

3. If the vessel is built for the purpose of partial bay-crossing trade, the windows and side scuttles above the freeboard deck shall be fitted with permanently attached deadlights, and when closed they shall be watertight (Weather-tight).

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59. Manholes and flush deck plugs

Manholes and flush deck plugs located under the open sky or on top of tanks shall be fitted with steel covers, or they shall be strongly built, and they shall be fitted with gaskets (gaskets) having arrangements to ensure the prevention of the entry of water.

60. Freeing ports for drainage of water

1. Arrangements shall be ensured for the rapid drainage of water from the deck, and where the bulwark (Bulwark) of the upper part of the weather deck may trap water, freeing ports of adequate sectional area shall be provided for draining the water.

2. The cross-section of the freeing ports for draining water on both sides of the freeboard deck shall be at least $0.1 \times l\ m^2$, where l is the length in metres of the continuous bulwark (Bulwark) of the upper part.

3. The freeing ports for draining water on the superstructure decks shall be made at the rate of 50 per cent of those of the freeboard deck. If the height of the freeing port for draining water exceeds 250 millimetres, horizontal bars shall be fitted at not more than 250 millimetres apart.

61. Drainage pipe (Drainage pipe)

1. Water drainage pipes of sufficient number and adequate size shall be provided so that water can certainly flow out from every weather deck (Weather deck).

2. There shall be arrangements for the water of all other decks inside the hull or superstructure to flow into the bilge, and non-return valves (Non-Return Valve) shall be connected to the water drainage pipes.

3. The drainage pipes from the upper part of the weather deck extending to the outside of the hull shall be made of such pipe whose thickness shall, as a rule, not be less than the thickness of the shell plate (Shell Plate).

62. Superstructure and deckhouse

1. The side bulkhead (Bulk-head) decks shall be in accordance with the dimensions of the ship’s structure.

2. Superstructures and deckhouses, bulkheads, partition bulkheads and side bulkheads made with vertical stiffeners and ceiling beams

shall be skilfully secured by the practicable maximum number of continuous and complete gantries.

a. “Superstructure” means a decked enclosed structure above the freeboard deck which extends from one side of the ship to the other, or whose side plating is connected to the shell plating at not more than 4 per cent of the ship’s breadth (B) inboard, and this expression shall include a raised quarter deck and superstructure;

b. “Deck house” (Deck House) means a decked structure above the freeboard deck other than a superstructure.

3. Superstructures and deckhouses shall be well supported (Supported), and, as far as practicable, the side bulkhead stiffeners shall be located on the same plane as the frames of the main hull, and the stiffeners shall extend to the deck, although it is not necessary that they be welded to the deck.

4. Deckhouse side bulkheads shall be very skilfully attached to the deck, and the beams to which they are connected shall be supported by under-deck girders of at least the same depth and thickness as those beams.

5. Deckhouse stiffeners shall normally be placed in the same plane (Plane) as the beams and frames, and where necessary additional stiffeners shall be used in way of the bulkheads and between the strong beams.

63. Bulkheads (Bulk-head)

1. The plate thickness of the watertight bulkheads on the outer sides of the ship and of the outside deckhouse bulkheads shall not be less than 3 millimetres, and the section of the outside stiffeners (Outiside Stiffeners) of the deckhouse shall not be less than the values specified in Table 35 of the Schedule.

2. For forward bulkheads the section modulus shall be increased by 20%, and the bulkhead shall be strengthened below or above the windows or at other openings (Opening) by means of horizontal stiffeners of the same size as the vertical stiffeners (Stiffeners).

3. The plate thickness of the effective bulkheads inside the ship shall not be less than 3 millimetres, and the section moduli for the stiffeners shall be at least 85% of those of the outside bulkheads.

64. Engine protection structure

The plate thickness of engine casings shall not be less than 4 millimetres, and the section moduli for the stiffeners shall be at least 10% of those of the outside bulkheads.

65. Decks

1. The thickness of the deck plating of internal deckhouse decks shall not be less than 4.0 millimetres for a deck beam spacing of up to 500 millimetres, and it shall be at least 5.0 millimetres for a greater deck beam spacing.

2. For open decks the deck plate thickness shall be increased by 1.0 millimetre.

3. The section moduli of beams and girders shall not be less than the values specified in Table 36 and Table 37 of the Schedule respectively.

66. Rudder and steering gear

1. To ensure adequate steering qualities, hydraulic/electro-hydraulic/mechanical/electric steering system equipment shall be installed in every ship, and all equipment necessary for steering, including the rudder and steering stand, shall be arranged as an emergency steering system from the steering gear.

2. In all existing passenger ships of length (L) exceeding 40 metres, power steering shall be fitted within 5 years of these rules coming into force, and in all new ships of the same length.

3. The steering gear compartment shall be kept free of all such devices (devices) which may obstruct the operation of the steering gear.

4. If the rudder stock of any ship engaged in pushing or towing work (Pusher/Towing/Tug) is fitted, its diameter shall be increased by 10% from what has been required in this Chapter.

67. Rudder stock

1. The diameter of the rudder stock shall not be less than the values specified in Table 38 and Table 39 of the Schedule respectively.

2. In the case of rudders without bearings below the neck bearing, the diameter of the stock shall be at least 10% greater than the values specified in Table 38 and 39 of the Schedule respectively.

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3. The connection of the lower part of the stock with the upper part shall be made above the lower bearing by a gradual reduction of diameter.

68. Pintles

1. The diameter of the pintles shall not be less than the value specified in Table 40 of the Schedule.

2. The length of the pintle shall be 20% greater than the diameter of the pintle.

3. The bearing length shall not be less than the diameter of the pintle, and the taper (Taper) of the pintle shall be at the rate of 1:6 of the diameter.

69. Coupling flange (Coupling flange)

1. Where the rudder stock and the main piece are coupled (Coupled) by flanges, the thickness of the flanges shall not be less than the diameter of the flange bolts.

2. The distance between the bolts and the edge of the flange shall not be less than 2/3 of the diameter of the bolts, and suitable arrangements shall be made for securing the nuts.

70. Coupling bolts

1. The coupling shall be effected in such a way that it can transmit the full torque (Full torque) of the rudder stock.

2. Each bolt shall not be less than the value specified in Table 41 of the Schedule.

71. Rudder blade

1. For a single plate rudder the thickness of the rudder plate shall not be less than 8 millimetres, and the rudder plating shall be skilfully strengthened by means of horizontal bars.

2. In the case of a double-plated rudder the thickness of the plate shall not be less than 6 millimetres. The inside of the rudder plating shall be skilfully strengthened by means of horizontal webs and vertical frames.

3. The rudder shall be watertight.

72. Main steering gear

1. Power-operated main steering gears shall be capable of turning the rudder from 30 degrees on one side to 30 degrees on the other side within 30 seconds or less while the ship is proceeding ahead.

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2. Hand-operated steering gears shall be acceptable for rudder stock diameters up to 100 millimetres, and if the number of rudders is two, the maximum permitted shaft diameter shall be 80 millimetres, and if the number is three it shall be 60 millimetres.

3. Not more than 30 turns of the hand wheel shall be required to turn the rudder from one hard-over position to the other hard-over position, and the force required to turn the hand wheel shall not exceed 200 Newtons (N).

73. Emergency steering gear

1. In passenger ships of length exceeding 40 metres an emergency steering gear shall be provided, and it shall be capable of turning the rudder from 15 degrees on one side to 15 degrees on the other side within 60 seconds while proceeding ahead at half the ship’s design speed or 5 knots, whichever is greater.

2. If the steering gear is hydraulic, the emergency steering gear may be hand-operated hydraulic, and the emergency steering shall be kept ready for emergency use at all times.

3. In all vessels other than those mentioned in sub-rule (1), such arrangements shall be provided that the vessel may receive directional control even if the ordinary steering system becomes inoperative.

4. For power-operated steering gears, rudder angle indicators shall be fitted at the main and emergency steering positions and on the steering gear.

74. Tillers and quadrants

1. The tillers and quadrants of power-operated rudders shall, in addition to being secured by means of a key of suitable size, be shrunk onto the rudder stock or secured by bolts.

2. The tillers of hand-operated rudders shall be skilfully connected with the rudder stock.

3. The radius of the quadrant shall not be less than 10 times the diameter of the rudder stock, and the thickness of the quadrant plate shall not be less than 1% of its diameter or less than 5 millimetres, whichever value is greater.

4. In comparison with the required rudder stock diameter d, the diameter of the hub (hub) shall not be less than the following values, namely:

For the main tiller or quadrant— Outside diameter = 1.8d Height = 1.0d, and

For the emergency tiller or quadrant— Outside diameter = 1.5d Height = 0.8d, and

when fitting a split hub (Split hub) on both sides of the rudder stock, it must be fitted by at least two bolts.

5. The diameter of the bolts shall not be less than the value specified in Table 42 of the Schedule.

6. The value of the rectangular cross-section of the arm of the tiller or quadrant shall not be less than the value specified in Table 43 of the Schedule.

75. Stoppers or locking arrangement

The rotation of the quadrant or tiller shall be limited from both sides by stoppers, and in every steering gear such arrangements shall be provided for fixing the rudder in any position.

76. Piping (of hydraulic steering)

When a hydraulic steering system is to be installed, the following rules shall be followed, namely:

a. the pipes of the hydraulic steering gear system shall be installed so as to ensure maximum safety while remaining easily accessible;

b. the pipes shall be installed as close to the centreline as possible;

c. as far as practicable it shall be seen that the pipes do not pass through the cargo spaces, and if the steering gear pipes have to be led through the cargo holds, arrangements shall be made to protect the pipes by steel sheets;

d. the pipes shall be installed in such a way that they remain free from stress and vibration;

e. the pipes of the main and emergency systems shall be separate;

f. the steering gear pipe system shall not be connected with any other system;

g. protected oil level indicators shall be fitted to the tanks used in the hydraulic steering system so that the minimum permitted oil level can be observed; and

h. a piping system filter shall be installed for cleaning the operating oil.

77. Electrical installation

1. If the steering gear is of the type electrically operated from the steering position, the main and emergency electrical supplies shall be separate.

2. The wiring arrangement systems of the main and emergency steering shall be on separate sides of the vessel.

78. Steering chains and rods

1. The sizes and dimensions of chains and rods shall not be less than the values specified in Table 44 of the Schedule.

2. The connection of the tiller and quadrant with the chain shall be in accordance with the strength of the chain.

3. Stud-less short link chain shall be used, and the shackles, shackle pins, chain couplings and connecting pieces and other connections shall be suitable for the size of the chain and shall conform to recognised standards.

4. The steering rods and chains shall be installed in such a way that they can be easily reached and cannot be damaged.

5. The leads of the steering chains shall be as direct or straight as possible, and sharp bends shall be avoided, and unless placed inside tubes, the steering rods and chains shall not be allowed to be led through passenger spaces or cargo spaces.

79. Equipment

1. The required equipment for anchors, chain cables and ropes shall be determined in accordance with Table 45 of the Schedule as a function of the equipment number Z.

2. For all normal inland cargo vessels the equipment number Z shall be determined from the following formula:

$$Z = Cb \times L \times B \times T$$

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where—

Cb = block coefficient of the vessel at the maximum permitted draught. L = length of the vessel, in metres. B = breadth of the vessel, in metres. T = maximum permitted draught, in metres.

3. For passenger vessels and vessels which receive a considerable amount of wind resistance, the equipment number according to the formula mentioned in sub-rule (2) shall be in accordance with the value of A.

$$Zl = Z + A$$

A = the sum of the longitudinal area of the vessel above the draught T and the longitudinal area of superstructures and deckhouses of breadth less than B/4 (Windage area).

4. When calculating the equipment number for tugs used in pushing and towing alongside, the windage area (Windage area) shall be taken into account, and the Z number shall be increased by 300%.

5. When Z falls between two numbers in the table, the value of the nearest number shall be considered as the equipment number.

6. The requirements of tow lines and moorings in Table 45 of the Schedule shall be considered as guidance, and the total weight of those main anchors of which two are used at the same time shall be in accordance with Table 45 of the Schedule; however, in vessels of equipment number up to 100, one anchor of the weight given in the said table may be used.

7. In the after part of all vessels a stern anchor of at least 25% of the total weight of the main anchors shall be kept; however, in the case of vessels of equipment number less than 400 it shall be considered optional.

8. For anchors of weight up to 750 kilograms, grapnels of equal weight and the associated chains may be substituted by hemp rope or steel wire rope.

80. Anchors

1. Anchors must be of an approved type.

2. The individual weight of the main anchors may vary up to 7% from the required weight on the condition that the total weight of the anchors shall not be less than the total rule weight.

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3. The weight of the head of stockless anchors, including the pin and accessories, shall not be less than 60% of the total weight of the anchor.

4. Where stock anchors are provided, the weight of the anchor (excluding the stock) shall not be less than 80% of the weight shown in the table given for stockless anchors, and the weight of the stock shall be at least equal to 25% of the weight of the anchor excluding the stock but including the connecting shackle.

5. Where special anchors approved by the Authority are used as “high-holding power anchors”, the weight of the anchor may be reduced by 25%.

81. Requirements for chain cables

1. Chains of proper gauge size shall be used as anchor chain cable.
2. Short-link or stud-link chain cables may be used as anchor chain cable.
3. The following grades may be used for chain material, namely:

a. Grade K-1 (ordinary quality);

b. Grade K-2 (higher tensile, High tension quality).

4. The breaking load (Breaking load) of any anchor chain cable shall not be less than 35 times the weight of the attached anchor, and where a “high-holding power anchor” is used, the breaking load of the anchor chain cable shall not be less than 47 times the actual weight of the anchor.
5. The chain cable diameter of short-link and stud-link chains may be taken from the values specified in Table 46 of the Schedule as a function of the breaking load.
6. The length of anchor chain in metres shall be as follows, namely:

For the main anchor: L + 10 metres. For the stream anchor: 2/3 L + 10 metres. where L is the length of the vessel.

The length of anchor chain shall not be less than 40 metres; but it need not exceed 60 metres either.

82. Requirements for ropes (Tow & moving lines)

1. Table 45 shall be followed as guidance for tow-lines and moorings, and steel cables shall be made of galvanised steel of the following types, namely:

a. 1370 N/mm² steel, 114 wires (6 × 19) 6 strands with intermediate fibre core, for breaking loads (Breaking weight) of less than 147 KN;

b. 1370 N/mm² steel, 144 wires (6 × 24) 6 strands with 7 fibre cores, for breaking loads from 147 KN to 490; and

c. 1570 N/mm² steel, 222 wires (6 × 37) 6 strands with central fibre core, for breaking loads of less than 490 KN.

2. Synthetic mooring lines may be used on the condition that their breaking load shall not be less than F = KFo, where—

a. Fo = the breaking load of the mooring lines shown in the table;

b. K = 1.30 for polypropylene mooring lines; and

c. K = 1.20 for mooring lines made of other synthetic materials.

3. The length of a tow-line in metres shall not be less than (L + 50) nor more than 200.
4. Each mooring length in metres shall not be less than (L + 50) nor more than 100.
5. In no case shall the diameter of fibre rope be less than 20mm.
6. Steel wire shall not be used in vessels carrying flammable liquids.

83. Bulwark and railing (Bulwark and Railing)

1. On all weather decks forward and aft, bulwarks (Bulwark) of a minimum height of 600 millimetres shall be provided, and for passenger vessels the height of the bulwark or guard rail shall not be less than 900 millimetres.
2. The thickness of the bulwark shall not be less than the value specified in Table 47.

3. At the upper edge of the bulwark (Bulwark) a bulb section of not less than 100 × 8 millimetre size or an angle bar of not less than 60 × 30 × 5 mm shall be fitted, and the bulwark shall be supported by means of strong stays (Stay) made of flat bar of not less than 60 × 6 millimetre size.
4. The spacing between the stays shall not exceed 1200 millimetres, and the stays (Stay) shall be arranged over deck beams or brackets, and every fourth stay shall be of a shape corresponding to the top rail.
5. On all open decks without bulwarks there shall be railings of a height equal to the height required for bulwarks, and the diameter of the top rail shall be at least 50 millimetres, and the diameter of the rail or rails below it shall be at least 25 millimetres.

6. The spacing between the rails shall not exceed 300 millimetres, and the open railings of passenger vessels shall be fitted on vertical stays (Stay) placed not more than 200 millimetres apart.

84. Chain lockers

1. Chain lockers shall be of such capacity and depth that the chain can enter the chain pipes directly and freely, even if it is tied together.

2. The chain pipes shall, as far as possible, be kept at the centre of the locker, and there shall be separate lockers for the port and starboard side chains.

3. The minimum thickness of the walls of the forward chain lockers shall be 6 millimetres, and the minimum chain locker wall thickness of vessels of length 50 metres or more shall not be less than the values specified in Table 48 of the Schedule.

4. The wall thickness of the after chain lockers (Stem Chain Lockers) may be reduced by 1 millimetre.

5. The vessel's end of the chain shall be reliably secured in the chain locker.

6. The design of the chain locker pipes must be such that water cannot enter the locker during heavy rainfall.

7. There shall be arrangements for easily pumping out the water of the chain lockers from the narrow part of the ship's fore end, or arrangements for it to drain out to the bilge (bilge) of the ship's bottom.

85. Propeller brackets

1. Where the strut (Strut) of the propeller shaft is fitted, it may be of the V (V) or I (I) type, and the length of the strut barrel or boss shall be at least three times the diameter of the propeller shaft, and if possible a square angle (square angle) shall be formed by the struts or propeller brackets.

2. The width and thickness of each V-strut arm shall not be less than the values specified in Table 49 of the Schedule.

3. The width and thickness of the I-strut arm (I-Strut arm) construction shall not be less than the values specified in Table 50 of the Schedule.

4. The length of the longer leg of the V-strut or the leg of the I-strut, measured from the outer perimeter of the strut barrel or boss to the outer side of the shell plating, shall not exceed 20 times the diameter of the tail shaft, and where this length is greater, the width and thickness of the strut shall be increased.

5. The struts shall pass through the shell plating and they shall be connected to the floor plates or web frames, and in way of the propeller brackets a plate shall be fitted on the shell, whose thickness shall be increased by 50% or by a doubling plate.

86. Anchor windlass (Anchor Windlass)

1. In every vessel, a windlass or capstan shall be fitted on the fore and after decks in accordance with the anchors corresponding to the requirements described in rules 77 and 78 of these rules.

2. For anchors up to 500 kilograms a hand-operated anchor windlass or capstan may be approved.

3. Windlasses shall be fitted with a coupling suitable for disconnection between the cable lifter and the drive shaft, and hydraulic or electric couplings shall be suitable for being opened by hand.

4. Cable lifter windlasses capable of holding a load equal to 0.8 times the minimum breaking load of the chain shall be fitted, and for all chains there shall be a mechanical chain stopper carrying a load equal to that of the windlass brake, and the deck structures beneath the chain stopper shall be strengthened.

5. Cable lifters must have at least five snugs.

6. Mechanical, hydraulic and electric windlasses shall be capable of producing, as a guide, a pull (Nominal pull) of $Z = Xd^2$ without obstruction, where—

a. a minimum average speed of 0.15 M/S for at least 30 minutes without any break;

b. Z = pull in kilonewtons (KN);

c. d = diameter of the chain cable in millimetres.

7. In the case of windlasses driven by physical power, the average hoisting speed, together with the pull indicated in sub-rule (6), shall be at least 0.3 M/S.

87. Mooring equipment

1. Anchor windlasses and capstans may also be built to serve as mooring winches.
2. In every vessel or floating equipment, at least one double bollard shall be fitted on the port and starboard sides at the fore and after ends, and beneath the bollards (iron posts for tying the ropes that hold the ship fast to the jetty) there shall be supporting plates and deck reinforcement.

88. Bottom ceiling

1. Solid wooden ceiling shall be fitted extending to the upper part of the bilge of the bottom of the cargo holds, and the thickness of the said wood shall not be less than 50 millimetres.
2. The ceiling on the horizontal flat of the floor shall be fitted as a portable section, or arranged so as to be easily removable for cleaning, painting or inspection of the inner bottom as required.
3. The bottom ceiling shall rest upon and be bounded at the sides by the bulkheads and steel angles or flanged plates.

4. The bottom ceiling shall be flush (flush), and there shall be no overlapping or lashing (lash) there.

89. Side sparring

In all those cargo spaces which are built for the purpose of general transport, sparring of at least 25 millimetres thickness and 75 millimetres width shall be fitted at 250 millimetre intervals above the bilge ceiling, and the side sparring shall be well secured to every fourth frame, and side ceiling may be omitted according to the wish of the shipowner.

90. Cementing

1. The bottom of single-bottom ships and the bilges of all ships shall be carefully cleaned, and a coating of good mortar made of first-class cement shall be applied therein, including the plates and frames, up to the upper turn of the bilge.
2. With the approval of the Authority any other equivalent mixture such as coal tar may also be used. With regard to water ballast, cement grout may be used in place of paint.

91. Painting

1. Good quality paint shall be applied to all steel work other than oil tanks, and a coating of epoxy paint/wash cement may be given to the steel work of water tanks.
2. In those parts of the ship’s structure where cement is not to be applied, at least two coats of proper paint shall be applied.
3. Before applying any paint, the metal parts must be cleaned, and mill scale and accumulated rust shall be completely removed.
4. If a considerable time elapses between the floating of the vessel in water and its delivery, the underwater parts of the vessel shall be painted again before being put into service.

92. General welding requirements

1. The requirements of these rules shall be complied with in the welding work of ship hull construction, and suitable permanent marks shall be drawn on the sides of the bulkheads for guidance.

2. The adopted welding method and the filler metals applied shall be such that the resulting welding is capable of having strength and rigidity comparatively equal to those of the parent metal.
3. Surface welds of underwater joints shall be welded with corrosion-resistant filler material.
4. The welding method of all connections shall be determined, to the satisfaction of the Administration, for the welding process, electrode type, edge preparation, welding procedure and the proposed work.

93. Arrangements for ensuring continuity of strength of connections

1. When passing through bulkheads (i.e. in tanks), except in special cases where the continuity of girder webs is required, in other cases the connection work may be completed by means of brackets having a welding sectional area at the girder equal to 1.1 times the sectional area (Sectional area) of the girder and 1.36 times the sectional area of the girder, and these brackets shall generally be continuous when crossing the

bulkhead.

2. Where the bracket edges are obstructed because of the bulkhead, two half-brackets shall be fitted in the same plane, and after chamfering (Champering) the bracket edges of the girder cross-section (at least 1.35 times the sectional area of the bracket) they shall be welded fast to the bulkhead, and the throat (throat) thickness of the welding shall not be above 0.7 times the thickness of the bulkhead.

94. Connections

1. Those parts which are to be welded shall be connected one to the other by means of bolts or welded assemblers or any other method ensuring adequate tightness and perfect alignment, while at the same time being as free as possible for shrinkage (shrinkage), and where lugs (lug), steer-ups or spangles (Spanges) are used, those attachments shall be removed after welding.

2. During the connection work, the alignment of the various parts shall be achieved without the use of excessive force likely to create high residual stresses.

3. The arrangements adopted for the lay-out of the connections and welding design shall be made in such a way that no kind of internal or external stress can occur in the construction.

4. Sharp variation (Sharp variation) in the sectional area of the stiffening part of the supported plating or at the welding location of the stiffener shall be avoided as far as possible.

5. Two parallel welds shall not be closer to each other than 300 millimetres, and the scallops described as secondary stiffeners, the ends of brackets and, with them, their intersection paths with the primary stiffeners shall be removed.

6. Where the parts to be connected are made up of a combination of both welds and rivets, the rivets around the welds of specified length shall not be placed until the welding is completed.

7. In the case of butt welding or fillet welding of thick plates, it shall be specially examined whether gaps have been left at the root of their edges in the preparation of the joints of all materials to be welded in order to make welding possible, and the angles and spacing of the edges shall be adequate so that they give the opportunity to properly complete the first welding work at the root of the weld.

8. When connecting plates of equal thickness, it shall be checked whether they have been correctly aligned in height.

9. If reduction by normal means is not possible, misalignment (Misalignment) shall be acceptable on the condition that it shall not exceed 1 millimetre.

10. In the case of the connection of two plates of different thicknesses t1 and t2, welding shall be done after back-welding, and where the difference in the thickness of the plates is 2.5 millimetres or more, the thicker plate shall be reduced over a minimum width of 3 × (t1 — t2).

95. Fillet welds of lap joints

1. Fillet joints in lap joints shall be used only in those cases where the connection is to be made through the application of moderate stress,

and this weld may be used for the horizontal joints of shell plating.

2. Fillet welds may be used for the bottom and side butts of some types of heavily built vessels including double bottoms, which shall be butt welded in accordance with the requirements applicable in the case of these vessels.

3. The width of overlap of these welds shall not be less than the value specified in Table 51 of the Schedule.

4. Butt welding on permanent backing, i.e. butt welding of the connections of plates stiffened (Backed) by flanges or of the frame plates of any stiffener, shall be permissible there where, owing to the design of the structure (Structure) (rudder, peak, mast etc.), a backing run is not possible.

96. Plug welding

1. Plug welding shall be used only in those cases where fillet welding is not possible.

2. The spacing between plug welds shall be at least equal to their length, and the width of the open part shall not be less than the thickness of the plate; at the same time the edges of the vessel shall also be welded.

97. Tack welding

1. Principally where the thickness of the backing plate is small, tack welding may be used in place of plug welding, and all such welding shall be equivalent to scalloped welds having the same pitch and throat depth.
2. The spacing between tack welds shall be at least twice their length, and the width of the open part shall be such that it can be easily filleted along its perimeter.
3. Unless special approval is obtained from the Authority, vertical down welding (Vertical down yard welding) shall not be applied for the welding of main structural parts subject to high and dynamic stresses.
4. All materials used for welding must be approved by the Authority.

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5. Where it is intended to weld steels other than the ordinary steel used for ship hull construction, welding materials of the respective grade shall be used in accordance with the approval of the Authority.
6. For structural items with thick walls, forgings and castings, specially basic welding materials shall be used.
7. Where, before welding, the plates, sections and structural parts require a coating of corrosion-resistant primary shop primer, the Engineer and Ship Surveyor shall be satisfied that the said primer does not impair the quality of the weld and does not damage the shop primer, and where necessary a test of its quality shall be carried out by the manufacturer.

98. Preparation of welding work and workmanship of welding

1. The structural parts of the part where welding is to be done must be clean and dry, and if there is flame-cutting grease, paint and mud, scale, rust or slag (slag), it shall be carefully removed before welding.
2. During the preparation of weld connections, attention shall be paid to the requirements of edge preparation and weld gaps at the time of alignment.
3. Where the width of the gap is slightly more than the acceptable distance, it may be reduced by means of deposition welding at the connection edges with the permission of the Surveyor.
4. With the consent of the Surveyor, excessive gaps may be closed by means of a strip of at least 50 millimetres, 4 times the plate thickness, but it shall be welded with care.
5. Plates and sections shall be in perfect alignment, especially where the structural parts are obstructed by bulkheads, partition walls or girder webs, and misalignment of more than 15 per cent of the plate thickness or 2 millimetres shall not be acceptable.
6. Tack welding shall be carried out by trained workers, and where the quality of the stitch webs is not equivalent to the finished weld, they shall be carefully cut away before completing the weld.

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7. Temporary fitting plates, strutting pieces, fairing bolts etc. shall be made of steel of good weldability, and the fittings for completing the welding of structural parts shall be carefully removed in such a way that the surface of the structural part is not damaged, and thereafter the weld shall be ground smooth.
8. Above all, in the case of automatic welding, temporary small run-on and run-off plates shall be fitted for clearing the end craters and welding defects at the start of welding.
9. When welding in open positions, measures shall be taken for protection from storms and rain, and where, owing to the thickness of certain specified materials or special types of welding methods etc., pre-heating of the structural parts is required in accordance with the instructions of the material manufacturers, that shall be done.
10. All welding work shall be carried out in the most favourable welding position, and for vertical downward welding, the opportunity of unrestricted shrinkage of the structural parts shall be provided through the adoption of a suitable welding sequence.
11. Care shall be taken to ensure that all welded connections obtain uniform fusion through penetration (Penetration) with smooth weld surfaces without excessive reinforcement, and in the case of multi-run welding the slag shall be carefully removed, and major pores and slag-inclusion locations shall not be welded over but rather cut out and welded again.
12. Defects created by poor workmanship or defective materials shall be repaired only with the consent of the Surveyor.
13. Minor surface defects shall, as far as possible, be ground smooth, and major surface defects shall be carefully cut out and welded again with adequate heat, and where cracks (Cracks) are repaired with the consent of the Surveyor, the location and length of the crack shall be identified by a suitable method, and a portion longer than the length of the crack must be cut out and thereafter welded again.

99. Testing of welded connections

The skilled and accurate work of the welds, and with it the completeness of the work, shall be ensured through careful inspection by the workers concerned with the work, and when the work is finished, after inspection by the person concerned with the work, the welds shall be presented at the appropriate stages of construction for inspection by the Engineer and Ship Surveyor, and for this purpose the welded connections shall, in accordance with the rules, remain in an unpainted and accessible condition, and the Engineer and Ship Surveyor may reject inadequately pre-inspected structural parts and, where considered necessary, direct that they be presented again for inspection after successful inspection and repair by the workers concerned.

100. Testing of welds

1. The shell plate welds of all passenger vessels, tankers and cargo vessels of length exceeding 50 metres shall be tested by X-ray or ultrasonic methods.
2. Where radiographs (i.e. X-rays) are taken, their number (relating to film of 580 millimetre length) shall be at least 1/3 of the length of the vessel.
3. The radiographs shall principally be taken from the crossing portions of longitudinal and transverse connections, pre-fabricated (Pre-fabricated) butt connections and also from butt connections welded in difficult or constrained positions, and the butts between girders and stiffeners shall also be tested.
4. Radiographic testing, rating of image quality and with it the evaluation and rating of the radiographs shall be carried out in accordance with the prescribed rules of welding, and defects shall be repaired in accordance with the directions of the Surveyor.
5. In place of the required radiographs, subject to the consent of the Authority, ultrasonic tests may be carried out, and they shall be carried out to the same extent as the X-ray tests, and weld sections of one metre length shall be tested.
6. The Engineer and Ship Surveyor shall indicate the locations of the tests.
7. All tanks shall be tested hydraulically.

8. The underwater structure shall be tested for water penetration (Penetration), and the other parts of the structure shall be tested for water penetration or by hose pipe.
9. Where there is doubt about the standard of workmanship, or where the types of defects and their location, or the need for testing, are considered to exist, the Engineer and Ship Surveyor shall direct that such testing is required, or may direct an extension of the scope of the testing or testing by one or more other methods, namely: ultrasonic and surface crack detection testing etc.

Chapter 3

101. General instructions

1. The instructions which have been set out in this Chapter for the construction and installation of main propulsion machinery (principal propelling machinery), auxiliary machinery and equipment, including the arrangement of pumps and pipe installations, shall be considered mandatory in the case of steel-built vessels and floating equipment on inland waterways.
2. Besides the requirements included in the instructions mentioned in this Chapter, the Authority shall reserve the right to impose additional requirements for all kinds of machinery where it considers it indispensable on the basis of new information or operational experience, or in appropriate cases the Authority may grant permission for deviation from the rules.

102. Dimensions of machinery parts

1. All machinery parts must be capable of withstanding the special service conditions of the ship, especially the ship’s motion, vibration, wear etc., and their dimensions must be in accordance with recognised marine engineering practice.
2. The machinery installations and pipe arrangement systems shall be of design and construction adequately suitable for the work for which they are intended, and the moving parts, hot surfaces and other hazards shall be so installed and protected, taking suitable measures, that any risk to the persons on board the ship is reduced to a minimum.

103. Temperature:

The design and construction of the main and auxiliary machinery shall be suitable for tropical conditions, including a minimum temperature of 50 degrees Celsius and high humidity, and a minimum water temperature of 35 degrees Celsius.

104. Effect of inclination (Inclination):

During the vessel's normal operation, the machinery installations shall operate without fault notwithstanding all its motions and inclinations, and to meet this requirement trouble-free operation shall be ensured at the following permitted angles of inclination (Angle), namely:

a. a static list (List) of 15 degrees to port or starboard, and

b. a static trim (trim) angle of 5 degrees.

105. Fuel:

The flash point of oil or fuel used for running the engines shall not be less than 55 degrees Celsius, unless it is specially approved.

106. Astern power:

Sufficient astern power (Astern Powers) shall be provided to maintain control of the ship in all normal circumstances.

107. Control and regulation:

1. Machinery shall be arranged and fitted in such a way that it can be controlled in accordance with the requirements of operation.

2. All essential automatic or remote-controlled systems shall be suitable for manual (mannually) operation, and there shall be an arrangement for stopping the main engine from outside the engine room.

3. Adequate means of communication shall be provided between the navigation bridge and the control stations for the main propulsion machinery, and between the navigation bridge and the steering gear compartment.

108. Gear transmissions and couplings:

Where the main propulsion plant (main propulsion plant) is operated by means of change-over gears or couplings, the control system (Control System) shall be designed in such a way that

the installation cannot be overloaded or damaged, and the propulsion engine cannot be over-driven, stopped or run at excessive speed, through operator error.

109. Turning gear (Turning gear):

The turning gear (Turning gear) necessary for turning the main engine shall be fitted.

110. Trials and tests:

1. During dock trials and trial voyages, the performance of the machinery installations and their components shall be tested to the satisfaction of the Surveyor.

2. The pressure-withstanding capacity of pipe installations and equipment under pressure shall be tested by means of 1.5 times the working pressure.

3. All safety devices shall be tested and checked.

111. Safety equipment and their protection arrangements:

1. The layout and safety provisions of the entire machinery installation shall be made so comprehensively that the risk of accident is kept as low as practicable.

2. Flywheels, chain and belt drives, couplings and other moving parts which may cause an accident to any person shall be separately guarded.

3. When the engine is running, an arrangement shall be provided whereby the engine-starting cranks may be automatically disengaged.

4. Vessels under pressure shall be fitted with pressure gauges having a scale of at least 150 percent of the maximum pressure, and the maximum working pressure shall be indicated in red on the scale.

5. Level glass gauges shall be of a material approved by the Authority and shall be protected against mechanical damage, and a shut-off device shall be fitted at the lower end.

6. Safety and shut-off valves shall be suitable for safe operation.

7. Safety valves shall be spring loaded and shall be set in such a way that the maximum permissible working pressure can never be exceeded, and they shall be protected against unauthorised interference.

112. Machinery spaces:

There shall be adequate and easily accessible space in machinery spaces for the servicing and maintenance of machinery.

113. Ventilation:

Adequate ventilation shall be provided in machinery spaces to prevent any accumulation of flammable or toxic gases, and mechanical ventilation shall be provided according to the type of vessel to the satisfaction of the Engineer and Ship Surveyor.

114. Machinery seatings:

The seatings of engines and machinery shall, to the satisfaction of the Authority, be strongly built, adequately connected to the structure, and in accordance with the recommendations of the machinery manufacturer.

115. Bilges:

1. All bilges shall be easily accessible and suitable for easy cleaning, and filters shall be fitted to the bilge suction lines.

2. Precautions and arrangements shall be taken so that bilge water cannot enter any electrical machinery as a result of motions of any kind and heeling during operation.

3. Nothing which causes damage to the environment shall be discharged directly into the water.

116. Fire prevention:

1. Fuel oil, lubricating oil, cargo oil and other flammable oil systems shall not be installed directly above exhaust pipe lines, silencers or other very hot surfaces, and shall in practice be arranged as far as possible away from all such places, unless proper arrangements are made for leaking oils.

2. Machinery installations and pipe systems shall, as far as practicable, be kept free from leakage of fuel oil, lubricating oil and other flammable oils.

3. Drip pans shall be installed beneath equipment carrying a risk of oil leakage.

4. Proper arrangements shall be provided for conveying leaked oils to some other safe place.

5. The driving units of ventilating fans located in machinery spaces, cargo spaces and control stations, and of boiler fans, shall be provided with arrangements suitable for stopping them, in the event of fire in the spaces where they are located or in their vicinity, from an easily accessible position outside the space concerned.

6. Where machinery installations and pipe systems and the structures surrounding them are built of wood and similar flammable materials, protection shall be provided against fire and the escape of harmful gases.

7. Where the driving units of fuel oil burning pumps, fuel oil transfer pumps, other similar fuel oil pumps, fuel oil purifiers and cargo oil pumps are located, arrangements shall be provided suitable for stopping them, in the event of fire in those spaces or in their vicinity, from an easily accessible position outside the space concerned.

8. The use of any plastic and non-reinforced rubber pipe in fuel and lubricating oil piping systems shall not be permitted.

117. Internal combustion engines:

1. The directions set out in this Chapter shall apply to internal combustion engines used as propulsion engines and to the engines of auxiliary machinery.

2. The specified power output of the engines shall be in accordance with the maximum continuous rating at approximately rated speed, and at the time of trials and tests the rate of power output of the engine shall be calculated on the basis of proof of the engine running at 10% overload for 1/2 hour.

118. Design:

The design of the engine shall be acceptable for operation in a marine environment with tropical conditions.

119. Cylinder relief valves:

1. Engines with a bore diameter of more than 230 millimetres shall be fitted with one cylinder relief valve on each cylinder, and the valves shall be loaded to not more than 40% above the maximum continuous combustion pressure developed.

2. In the case of auxiliary engines, consideration shall be given to replacing the relief valve by an efficient warning device for over-pressure.

120. Protection against overspeed:

1. Each engine shall be fitted with an efficient governor adjusted in such a way that its speed cannot exceed the maximum rated speed by more than 15%.

2. The auxiliary engines of electric generators, and main engines which may be fitted with declutched (declutched) or controllable (controllable) pitch propellers, shall be fitted with governors adjusted in such a way that, on sudden loss of full load, the speed variation does not exceed the following standards, namely:

a. momentary variation: 10% of the maximum rated speed;

b. permanent variation: 5% of the maximum rated speed.

121. Alarm devices:

Diesel engines with a continuous output of more than 40 kilowatts shall have, with their lubricating oil systems, alarm devices which give visible and audible alarms if the lubricating oil supply fails or if there is a significant drop in the lubricating oil pressure.

122. Crankcase:

1. In the case of engines with an output exceeding 180 kilowatts, lightweight spring-loaded relief valves or quick-acting and self-closing devices shall be provided to relieve crankcase pressure in the event of an internal explosion in the crankcase and to prevent the subsequent inrush of air.

2. If the cylinder bore exceeds 200 millimetres, there shall be at least two crankcase safety valves.

3. Each safety valve shall have a free cross-sectional area of at least 45 square centimetres, and the total free cross-sectional area shall not be less than 115 square centimetres per cubic metre of crankcase volume.

4. If crankcase ventilation pipes are fitted, they shall be kept as small as practicable in order to minimise the rapid inrush of air after any explosion.

5. Ventilation pipes leading from the engine crankcase shall be led to a safe position on deck or in another approved place.

123. Fuel oil system:

Only metallic joining pipes shall be permitted for the joining of fuel injection piping; for engines with a cylinder bore exceeding 250 millimetres the high-pressure fuel oil injection pipe system shall be effectively protected and shielded.

124. Exhaust system:

1. The exhaust pipes of separate engines shall normally be led separately to the atmosphere through silencers, and where the exhaust pipes from two or more engines lead to a common silencer, each exhaust pipe shall have an isolating device.

2. Where exhaust lines are led overboard near the waterline, arrangements shall be made so that water cannot enter the engine.

3. To keep heat transfer to a minimum, exhaust pipes shall be efficiently lagged with non-combustible insulating material, and the surface shall not become hotter than 60 degrees Celsius, and special care shall be taken so that exhaust gases cannot enter the accommodation areas.

4. Efficient silencers shall be fitted to the exhaust lines.

125. Lubricating oil system:

1. Arrangements shall be made for installing suitable equipment at appropriate places on the lubricating oil systems to indicate the flow condition or pressure of the lubricating oil.
2. Arrangements shall be provided for lubricating oil sampling connections at appropriate places on the lubricating oil systems.
3. Lubricating oil drain pipes from the engine crankcase sump to the sludge tank shall be submerged at their outlet ends, and the lubricating oil drain pipes of two or more engine units shall not be interconnected.
4. Oil filters shall be fitted to the lubricating oil lines, and the filters shall not be placed over excessively rotating parts or near hot parts, and drip pans shall be placed beneath the filters.

126. Cooling water or oil system:

1. Temperature-measuring equipment shall be fitted to the outlet pipe lines of the cooling water and cooling oil.
2. Drain cocks shall be fitted at the lowest positions of the water jackets and water pipe lines.

127. Engine starting arrangements—

a. Air compressor.—(1) For engines with a compressed air starting system, the following arrangements shall be provided, namely:

b. where there is one main-engine-driven compressor, an additional hand compressor or a compressor driven by a hand-started auxiliary engine shall be fitted;

ii. where there is no main-engine-driven compressor, a compressor driven by hand, or electrically, or by an auxiliary engine shall be fitted.

2. The air compressor shall have sufficient capacity to fill the starting air receiver within one hour.
3. The capacity of the air receiver shall, without recharging, be sufficient to start each main engine at least 6 consecutive times in the case of non-reversible engines and at least 12 times in the case of reversible engines.
4. The number of air receivers shall be at least two.

b. Batteries.—

1. Where main engines are started electrically, a separate starter battery arrangement shall be provided for each engine, and this battery shall be capable of starting the main engine from the cold condition.
2. There shall be a separate battery for electrically started auxiliary engines.
3. Starter batteries shall be used only for the purpose of starting the engine.

4. In the case of electrically started main engines, arrangements shall be provided for charging the engine-starting batteries.
5. If an auxiliary-engine-driven generator is fitted, arrangements shall be made for charging the batteries by the generator, and for such a plant the minimum number of batteries shall be two.
6. Starting batteries shall be capable of starting the main engines, and the starter batteries of auxiliary engines shall have the capacity to make at least 4 starts within 30 minutes.

Gearing and Couplings

128. Gearing and couplings:

1. The requirements relating to gearing and couplings set out in this Chapter shall, subject to the other provisions of these Rules, apply to any type of coupling installed, including the spur and bevel gearing of the main propulsion engine.
2. The requirements set out in this Chapter shall also apply to important auxiliary machinery such as electric generator sets, couplings, winches, windlasses, lubrication, cooling and bilge pumps.

129. Materials (Materials):

All gearing and coupling components transmitting torque from the ship’s propulsion plant, such as the participating shafts, intermediate shafts, propeller shafts, wheel rims and pinions, shall be made of forged or cast steel; however, rolled bars may be used for plain, keyless shafts.

130. Permissible tooth load:

The tangential loads (Tangential loads) corresponding to the maximum torque transmitted by the reduction gear in continuous running shall not exceed the maximum permissible load in respect of tooth surface pressure and tooth bending stress.

131. Tests:

1. The gear teeth of the main propulsion plant shall, before the trial voyage, be coated with a suitable paint in order to establish the contact pattern, and

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all ahead and astern speeds shall be tested during the trial voyage.

2. The couplings of the propulsion plant and of the electric generator sets shall be verified in all the different kinds of running.

132. Materials of main propulsion shafting:

1. Propeller, intermediate and thrust shafts, together with flange couplings, shall be of forged steel, and couplings may be made of cast steel or ductile cast iron.
2. Rolled round steel may be used for plain flangeless shafts.
3. The tensile strength (Tensile strenght) of the steel used for shafting shall be between 450 and 750 N/mm^2.
4. Propeller shafts and stern tube shafts shall be made of corrosion-resistant material, or shall be effectively protected against corrosion as far as practicable, namely:
5. The minimum shaft diameter shall be determined by the following formula:

$$d = 120 \times \sqrt[3]{(P / n)}$$

where,

d = required minimum outside diameter of the shaft in millimetres.

p = power transmitted by the shaft in kilowatts.

n = shaft speed in rpm or revolutions per minute.

6. A test certificate or verification certificate shall be issued for all materials used in the propulsion plants.

133. Torsional vibrations (Torsional Vibrations):

If the power per shaft exceeds 400 kilowatts, a report on torsional vibrations shall be submitted to the Authority.

134. Couplings:

1. For normal shaft designs, the thickness of the forged steel coupling flange shall be at least 25% of the shaft diameter.
2. The fillet radius at the base of the coupling flange shall not be less than 0.8 of the shaft diameter.

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135. Stern bush:

The length of stern bushes for the different types of cooling systems shall be as follows, namely:

a. the length of water-lubricated bearings lined with lignum vitae (Lignum Vitae) or with pieces of rubber composition or plastic material shall not be less than 3–4 times the required diameter of the screw shaft;

b. for oil-lubricated bearings with white metal liners and oil sealing glands, the length of the liner shall not be less than 2 times the screw shaft diameter;

c. for oil-lubricated bearings with grey iron casting or bronze liners with oil sealing glands, the length of the liner shall not be less than 4 times the screw shaft diameter.

136. Piping installation:

1. The necessary pumps and piping arrangements shall be provided for bilge suction, ballast water, fuel oil, fresh water, drainage, etc.
2. Piping installations shall be sufficiently strong and provided with suitable supporting arrangements against vibration.
3. Piping shall be installed with the minimum number of bends and joints and shall be as straight as possible.
4. Piping systems shall be properly designed allowing for expansion due to thermal stress and allowing for deflection of the ship’s structure by means of expansion joints or bends.
5. Piping in cargo spaces shall be installed with good clearance between the stiffeners and shall be protected from possible damage.
6. Pipes shall be joined by sockets, flanges and butt welds, and gaskets shall be provided in the flanges.
7. Flange connections shall be fitted in suitable places, and for the convenience of repair work, the fresh water and sanitary water systems in accommodation spaces shall be provided with a suitable number of shut-off valves.
8. Drain cocks shall be fitted at the lowest point of all pipes for drainage. Pipes shall be led without passing through the crew’s cabins.

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9. Pipes running (Running) fore and aft shall, as far as practicable, be led without passing through the crew’s cabins.
10. For concealed piping arrangements, access to the pipes beneath curtains or furniture, etc. shall be ensured.
11. The materials used for pipes, valves, pipe fittings and auxiliary equipment, and their construction, shall be adequate for their locations and in conformity with recognised standards.
12. Special materials such as rubber hose, plastic pipe, single pipe, aluminium alloy, etc. may be used with the prior approval of the Authority, having regard to safety against fire and flooding as well as the working conditions.
13. All power-operated valves shall also be suitable for hand operation.

137. Pressure test:

1. Before being put into service, and at specified intervals, the pressure of the piping systems shall be tested at 1.5 times the working pressure.
2. All valves and fittings shall be tested at not less than 1.5 times the maximum working pressure.
3. Low-pressure pipes such as bilge or ballast pipes shall, after fitting in the vessel, be tested at a pressure at least equal to the maximum pressure at which they will work in service.

138. Wall thickness:

1. The wall thickness of pipes shall be adequate on the basis of internal pressure, pipe service, location, etc., and the recommended minimum thickness of pipes shall, where applicable, be equal to the standards specified in Table 54, Table 55 and Table 56 of the Schedule.
2. The thickness of pipes under high internal pressure, such as high-pressure hydraulic lines, etc., shall be determined by calculation.
3. All pipes passing through the shell plating shall have a thickness at least corresponding to that of the shell plating.

139. Joints:

1. Flanged couplings shall normally be used for detachable pipe joints, and for less important low-pressure pipes, namely:

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sanitary systems, hot water systems, etc., threaded sockets may be used.

2. Flanges may be cut from steel plate or made by forging or casting, and the use of pipes of the said material shall be suitable for the design temperature.
3. Welded flanges shall not be fitted tightly on the pipe, and at no place shall the gap between the bore of the flange and the outside diameter of the pipe exceed 1.5 millimetres, and the sum of the gaps on diametrically

opposite sides shall not exceed 2.5 millimetres.

140. Fitting:

1. Provision for expansion shall be made in pipe line systems as far as possible, by means of pipe bends, expansion compensators, gland-type joints and similar means, on account of rise in temperature and deformation of the ship’s structure.

2. Pipes shall, as far as possible, be kept away from the vicinity of electrical equipment such as generator switchboards, control gear, etc., and where this is not possible, unless arrangements are made for stopping any kind of leakage from openings in the equipment, arrangements shall be made so that no flange or joint falls over or near the electrical equipment.

3. All pipes, valve cocks, pipe fittings, valve operating rods, handles, etc. located in cargo spaces or in positions on the weather deck, wherever there is a possibility of damage, shall be given adequate protection, and the covering provided for protection shall be constructed in such a way that it can be easily removed for inspection.

4. Proper measures shall be taken for the prevention of corrosion of pipes installed in places inaccessible for maintenance and inspection.

5. Pipe lines liable to be subjected to an internal pressure exceeding the design pressure shall be protected by means of relief valves or alternative over-pressure prevention devices, and the discharge ends of the relief valves or over-pressure prevention devices shall be led to safe places.

6. Where considered necessary, pressure and temperature measuring devices shall be provided.

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7. Unless measures are adopted for the prevention of accidental contamination during operation, fuel oil pipes, lubricating oil pipes, sea water pipes, fire pipes, fresh water pipes and bilge pipes shall be kept separate according to the purpose of the pipe.

8. When led through bulkheads or tank walls, pipes shall be led in a watertight or oil-tight manner, and no bolts shall be used inside them, and no holes shall be drilled in the tank walls for the insertion of screws or bolts.

141. Ship’s shore connections:

1. Valves suitable for connecting the ship’s shore connections shall be fitted, and connection to the ship’s side shall be provided by means of thick flanges.

2. The ship’s side valves shall be easily accessible, and sea water inlet and discharge valves shall be capable of being operated from above the floor plates, and the ship’s side cocks shall be fitted in such a way that the handle can be removed only when the cock is closed.

3. The positions of discharge outlets on the outside of the vessel shall be such that, when lowering into the water, water will not be discharged onto the lifeboats and liferafts in the positions included in the launching arrangements, unless special measures are adopted at all those places to prevent the discharge of water.

4. Sea cooling water shall be provided from suction valves fitted to two or more sea chests or sea suctions.

142. Bilge and ballast systems:

a. Layout: (1) The layout of the bilge lines and bilge suctions shall be made in such a way that the vessel’s bilges can be completely pumped out even when the vessel is in a trimmed condition (Trim).

2. Bilge suctions shall be arranged on both sides of the ship, and one bilge suction shall be fitted for the spaces at the ends of the ship.

3. Spaces located forward of the collision bulkhead (Collission Bhd) shall not be connected to the general bilge system, unless specially approved by the Authority.

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4. Each space separated by a bulkhead shall have its own bilge suction branch line.

5. Bilge pipes shall not be led through lubricating oil tanks or tanks of water used for daily purposes.

6. Ballast lines may be connected to the bilge system on the condition that a non-return valve is fitted between the main bilge line and the pump suction pipe.

b. Connection with the bilge system:

1. In order to prevent the flow of water from one space to another through the bilge system, each bilge branch

line connected with the different spaces shall be connected to the main bilge line by means of a screw-down non-return valve.

2. In the case of small ships with cargo holds, the branch bilge lines connected with the different spaces may be connected to the bilge pump through change-over cocks.

3. Mud boxes and rose boxes shall be fitted at suitable positions on all bilge suction pipes except the emergency bilge suction pipes of the hold spaces, engine room and shaft tunnels.

4. In the case of bilge pumps installed for ballasting, the requirements set out in rule 140 (a) shall be complied with.

5. In the case of ships 50 metres or more in length, in addition to the requirements set out in rule 140 (a), arrangements shall be made, in accordance with the provisions set out in sub-rules (6), (7), (8) and (9) of this rule, for one direct bilge suction pipe and one emergency bilge suction pipe for pumping from the engine room.

6. The direct bilge suction pipe shall be directly connected to an independent power-driven pump as set out in rule (d) and rule 141.

7. The emergency bilge suction pipe, fitted with a screw-down non-return valve with a hand wheel easily operable from above the engine room platform, shall be fitted to the main cooling water pump or to the largest power-driven pump kept in the engine room, in addition to the bilge pumps set out in rules (d) and (e).

8. The internal diameter of the direct bilge suction pipe shall not be less than the internal diameter of the bilge main line.

9. The internal diameter of the emergency bilge suction pipe shall be equal to the internal diameter of the pump suction.

c. Size of bilge lines:

1. The diameter of the main bilge line shall not be less than the size given in Table 57 of the Schedule.

2. In ships with a propulsion power of 50 kilowatts to 250 kilowatts, one engine-driven bilge pump shall be fitted, which may be driven by the main engine.

3. In the case of ships with a propulsion power exceeding 250 kilowatts, a second independently driven bilge pump shall be provided.

4. In passenger launches, the second bilge pump shall be installed separately from the main bilge pump.

5. All power-driven pumps essential for bilge service shall be of the self-priming (Self-priming) type.

e. Capacity of pumps: (1) The capacity of each bilge pump shall not be less than the value given by the following formula, namely:

$Q = 4.0/10000\ d^2$ (for vessels up to 30 metres in length)

$Q = 5.75/10000\ d^2$ (for vessels [up to] 30 metres in length) where

Q = capacity in m^3/hour

d = internal diameter of the main bilge line in millimetres.

143. General service pumps:

The essential bilge pumps mentioned in rule (e) may be used for fire-fighting or for general service work of an intermittent nature.

144. Air pipes:

1. Air pipes shall be fitted to all tanks and void spaces at their highest points, the ends of which shall normally terminate above the open deck.

2. Air pipes shall be fitted at the end of the tank opposite to that at which the filling pipes are, or at the highest part of the tank, and air pipes shall be fitted so that under normal and well-arranged conditions they are self-draining.

3. In the case of all tanks which may be pumped either by the vessel’s pumps or by pumps on shore through the filling main line, the total cross-sectional area (Total cross sectional area) of the air pipes connected to each

tank shall not be less than 25% more than the effective area of the filling pipes concerned.

4. Such arrangements shall be adopted that tanks and spaces do not become completely devoid of air when being emptied by pump.

5. Air pipe heads shall be of the automatically opening type which can prevent excessive pressure in the tanks.

6. Goosenecks or non-return valves shall be fitted to the air pipes.

7. Cofferdams or void spaces provided with bilge suctions shall be provided with air pipes with their ends terminating in the open air.

145. Sounding pipes (Sounding pipe):

1. All tanks and bilges in all compartments which are not at all times readily accessible shall be provided with sounding arrangements, and the sounding arrangements shall, as far as possible, be led close to the suction pipes.

2. Sounding pipes shall be as straight as practicable.

3. All sounding pipes shall be led to accessible positions above the bulkhead deck, and striking plates of adequate thickness and size shall be fitted beneath them.

4. In no case shall the diameter of a sounding pipe be less than 32 mm.

146. Gauge glass (Gauge glass):

The glass gauges used for tanks used for carrying fuel oil, lubricating oil and other flammable oils shall be in accordance with the following conditions, namely:

a. the glasses used for oil level indicators shall be of flat-plate type and of heat-resistant quality, and adequate measures shall be taken to protect them from mechanical damage;

b. for tanks with a capacity not exceeding 1000 litres, however, round glass gauges may be used subject to the approval of the Authority; and

c. means shall be provided for closing the valves with the cocks at the lower end of the glass gauge.

147. Overflow pipe: (Overflow pipe):

1. Overflow pipes shall be in accordance with the following conditions, namely:

a. overflow pipes shall be provided for tanks which can be filled by pump;

b. the total sectional area (sectional area) of the air pipes shall be not less than 1.25 times the mean sectional area of the filling pipes;

c. the open ends of the air pipes connected to the tanks shall be below any open part.

2. The arrangement of the overflow pipes of fuel oil settling (Settling) tanks and fuel oil service tanks shall be as follows, namely:

a. the overflow pipes connected to fuel oil, lubricating oil and other flammable oil tanks shall be connected to other tanks with sufficient capacity reserved for the overflow; and

b. suitable means such as return valves for preventing the entry of water from the river shall be connected to the overflow pipes.

148. Fuel oil systems:

1. The compartments in which fuel oil burning systems, fuel oil settling and service tanks, fuel oil purifiers, etc. are located shall be easily accessible and provided with good ventilation.

2. Fuel oil systems in the engine compartment shall be arranged and fitted so that maintenance and inspection are easy.

3. Proper precautions shall be taken, with such arrangements that oil leakage cannot cause an accident such as fire and that oil leakage can be easily detected.

4. All valves and cocks shall be capable of being operated from above the platform.

5. Valves, cocks and other fittings connected to fuel oil tanks shall be fitted in safe positions so that they cannot be damaged from outside.

6. Stop valves or cocks shall be fitted on both the suction and delivery sides of fuel oil pumps.

7. The pressure relief valves installed on the delivery side of fuel oil pumps shall be arranged so that the discharged oil can be taken to the suction side of the pumps.

8. Metal drip trays with sufficiently deep edges shall be placed beneath burners, fuel oil pumps, fuel oil tanks such as fuel oil settling and service tanks, and other equipment used for fuel oil which has frequently to be opened for cleaning or maintenance.

9. Fuel oil filling pipes from outside shall be solely for entirely separate use, and the open ends of all such pipes shall be led over the weather deck and shall be provided with strong covers.

10. Self-closing type drain valves or cocks shall be fitted to the fuel oil settling tanks and service tanks.

11. Stop valves or cocks directly attached to the tank wall shall be fitted on the fuel oil suction pipes from fuel oil tanks located at a position higher than the double bottom.

12. Except for tanks with a capacity not exceeding 1000 litres, in other cases, in addition to the local closing arrangement, the valves and cocks shall be capable of being closed from a safe position outside the space in which they are located.

13. Fuel oil filters shall be provided in the fuel oil supply pipe lines, and the filters required for the main propulsion machinery shall be capable of being cleaned without stopping the supply of filtered oil.

149. Fire prevention:

1. The exhaust gas lines from the main and auxiliary engines shall be adequately insulated.

2. Each exhaust pipe of the engine shall be insulated with non-combustible material in such a way that its surface cannot become hotter than 60 degrees Celsius.

3. The insulation shall be protected by galvanised steel plate.

4. When exhaust gas lines and pipes pass through bulkheads, they require thermal insulation.

5. Fuel pumps, exhaust and inlet fans, motors and cargo pumps (for oil tanks) shall be equipped with arrangements for stopping them in an emergency, and there shall be an arrangement for quick-closing devices on the delivery valves supplying fuel oil from the service tanks.

6. All stopping and shut-off devices shall be capable of being operated from outside the space in which they are located.

150. Spare parts and tools:

1. Spare parts, devices and tools for maintenance or repair work, according to the purpose and number of the machinery and piping installations, shall be kept in stock in the engine compartment, boiler room or other convenient place.

2. The spare parts specified in rule 142 shall be kept on board the ship in accordance with the recommendations of the manufacturer.

151. Additional spare parts to be kept on board the vessel:

1. For the main engine—

a. 1 set of complete main thrust block or bearing;

b. 1 set of complete exhaust valves;

c. 1 set of complete air inlet valves;

d. if the engine is air-started, 1 starting air valve;

e. 1/4 set of fuel oil valves of each size and type;

f. 1 set of special gaskets and packing for the cylinder cover and liner of each type, for one cylinder.

2. For gears and couplings—

a. 1 set of thrust pads for the thrust bearing; and

b. 1 set of elastic elements for the coupling.

3. For the air compressor—

a. 1 set of piston rings;

b. 1/2 set of suction and delivery valves.

4. For pumps—

a. 1 bearing of each type and size;

b. 1 rotor seal of each type and size;

c. in the case of piston pumps, 1 set of piston rings of each type and size.

5. For other purposes—

a. 1 safety valve for each vessel under pressure;

b. 10% of the hoses and additional parts fitted in the machinery and piping systems.

152. Tools:

A sufficient number of suitable tools and special tools, according to the size of the machinery installation, shall be kept in stock on board the ship for carrying out maintenance and repair work.

153. Fittings:

A supply of screws, bolts, nuts, gaskets and filters in very common use shall be kept on board, and at least one spring of each type used in the machinery shall be kept on board.

Chapter 4

Electrical Installations

154. Applicability of Chapter 4:

The provisions of Chapter 4 shall apply to self-propelled or non-propelled ships navigating by night on inland waterways, or providing service for more than 6 hours, or having a total engine power exceeding 12 kilowatts.

155. Passenger vessels:

1. The services essential for the safety of passenger vessels shall be maintained in an emergency, and the safety of the ship and persons from electrical accidents shall be ensured.

2. The essential services mentioned in sub-rule (1) shall mean the services necessary for the propulsion and safety of the ship; however, if the installation is not performed physically, the following items shall be included among the essential services, namely:

a. auxiliary machinery connected with the propulsion of the vessel;

b. engine starting devices;

c. steering gear;

d. windlasses (Windlasses);

e. mooring;

f. side thrusters, if any;

g. winches (Winches) on emergency boats, if any;

h. fire, bilge and ballast pumps;

i. watertight doors, if any;

j. automatic sprinkler pumps, if any;

k. ventilating (Ventilating) fans for the engine room;

l. navigation lights;

m. the main lighting system for generally accessible (accessible) parts of the vessel and parts used by passengers and persons;

n. radio wireless equipment;

o. main engine remote control system;

p. fire-fighting, extinguishing and warning systems.

156. Plans and data:

The plans and data to be submitted shall include a complete feeder list plan and data showing, for each size of cable, the voltage drop for the longest length of cable, the type of cable, the rating or setting of circuit breakers, the rating of fuses and switches, and the breaking capacity of circuit breakers, to be submitted to the Authority.

157. Oil tankers:

1. The following arrangements shall be provided for oil tankers, namely:

a. a general arrangement of the ship showing the location of hazardous electrical equipment;

b. a schedule of electrical equipment of safe type located in hazardous zones or spaces, with detailed particulars of the type of equipment fitted;

c. all spaces of the tanker in which explosive gases may normally accumulate, the compartments designated for battery accumulators, paint lockers, acetylene stores, etc., to be identified as hazardous zones or spaces and protective arrangements made.

2. Equipment connected with direct current (D-C current) shall operate satisfactorily with a voltage variation of 6 percent increase and 10 percent decrease.

3. A-C current equipment shall operate satisfactorily with a voltage variation of 6 percent increase and 10 percent decrease at the rated frequency, and with a frequency fluctuation (Frequency fluctuation) of 5 percent increase or decrease at the rated voltage.

4. Contactors (Contractors) and similar equipment shall not drop out at an overload of less than 85 percent.

158. Temperature rating:

The cooling air and cooling water temperatures applicable to ships designated for operation in Bangladesh shall be as follows, namely:

a. primary cooling water supply 35 degrees Celsius;

b. cooling air temperature 50 degrees Celsius.

159. Installation of electrical equipment:

1. Electrical equipment shall be fitted or protected in such a way as to keep to a minimum the possibility of damage from mechanical damage or from dust, oil vapour or liquid leakage.

2. Skylights and airtight barriers shall be installed in order to avoid the possibility of machinery or equipment being submerged in water.

3. Arrangements for enclosure shall be made where necessary for the protection of moving parts of machinery.

4. All nuts and screws used for making connections with current-carrying parts and with working parts shall be effectively locked.

160. Inclination (Inclination):

Such arrangements shall be provided that all machines and apparatus can work satisfactorily even when the vessel heels to 22.5 degrees, or when the vessel’s trim (Trim) is 10 degrees, or when both conditions occur together.

161. Protection from bilge water:

All generators or electric couplings shall be so fitted that they cannot be damaged by bilge water, and if necessary, watertight coamings shall be provided around the base of such equipment to form a well, with arrangements for removing water from the well.

162. Accessibility:

1. The design and installation layout of electrical machinery shall be such that those parts which require inspection or adjustment are accessible.

2. Armatures, field coil motors and rotating fields shall be removable, and where air ducts are used, means of access shall be provided there.

163. Watertight equipment:

All unenclosed electrical equipment, or all electrical equipment located near the sea so as to be splashed by water or in any other place of intense humidity, shall be of watertight type or shall be protected by being kept in a watertight enclosure.

164. Corrosion-resistant parts:

Bushings of enclosed spaces and other parts of electrical equipment which may be destroyed or rendered ineffective by corrosion shall be made of corrosion-resistant material or of material rendered corrosion-resistant.

165. Explosion-proof:

1. Electrical installations in paint, oil and battery lockers or stores, and in the cargo spaces of vessels approved for the carriage of dangerous goods, shall be explosion-proof.

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2. Electrical installations on all open decks and in the pump rooms of oil tankers shall be explosion-proof.

166. Earthing (Earthing):

1. The frames and bases of permanently installed generators, motors, controllers, apparatus and similar equipment, in the case of which the layout and method of installation do not give the assurance of certain earthing, shall be earthed through separate conductors and provided with protection against damage.

2. The metal frames of all portable electric lamps, apparatus and similar tools supplied as ship’s equipment and rated at a voltage exceeding 55 volts shall be conveniently earthed through a conductor, unless ensured by double insulation or an insulating transformer.

3. Depending on the insulating material and the nature of the over-voltage created by switching and fault conditions, the distances between parts, and between live parts and earthed metal, whether through surface or through air space, shall be adequate for the working voltage.

4. Rubber carpet shall be provided on the floor or deck around the main switchboard.

167. Distribution systems:

1. The following distribution systems may be used, namely:

parallel systems with constant voltage;

a. direct current: single wire.

b. alternating current: single-phase two-wire, and three-phase three-wire or four-wire with neutral earthed.

2. Except for 2-pole insulated final sub-circuits, in other cases systems using hull return may be adopted, and hull return systems shall not be approved for oil tankers, gas-carrying vessels and vessels carrying dangerous chemical substances.

3. In the case of parallel systems with constant voltage, for generation, power, coupling and cooling equipment permanently connected with fixed wiring, neither A-C nor D-C current shall exceed 500 volts, and

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shall not exceed 220 volts for the lighting arrangements of cabins and public rooms, for the apparatus of cooling equipment, and for all equipment not described above.

168. Earthing indication:

In every insulated distribution system, arrangements shall be made by means of lamps or other means for indicating the condition of the insulation to earth.

169. Vessel services:

All vessels using electricity for auxiliary power or for lighting shall have at least two sources of electrical power,

and the total capacity of the sources of power shall be sufficient to carry the load required in normal operation, with one source of power kept in reserve.

170. Oil tankers:

Where electrical power is required for essential equipment, the generating plant and converting plant shall be of such capacity that all such equipment can be operated satisfactorily even with one generating set or converting set out of service.

171. Reserve capacity:

In A-C (Alternating Current) current systems, where additional equipment is required for use in emergency work, if one generating set breaks down the remaining sets shall have sufficient reserve capacity to start the ship's largest motor without causing the stopping of any motor or the failure of any other mechanical device due to excessive voltage drop in the system.

172. Emergency source of electrical power in passenger vessels:

1. All passenger vessels carrying more than 200 passengers shall be provided with an emergency source of electrical power having the characteristics described in sub-rules (2) and (3).

2. Where emergency generating sets are fitted, they shall be capable of being started immediately even when they are cold.

3. If hand starting is shown to be practicable, alternative means of starting shall be required, and where hand starting is not possible, other means shall be provided

which shall be capable of starting at least 6 times in thirty minutes without the assistance of the sources within the machinery space.

4. The available electrical power shall be sufficient to supply all services necessary for the safety of passengers and crew in an emergency, and at the same time due regard shall be given to similar services as may have to be operated simultaneously.

5. Special consideration shall be given to all alleyways, stairways and machinery spaces and control stations (that is, all places where the radio, main navigating or central fire recording equipment or the emergency generator is located), fire detection and alarm systems, electrically driven fire pumps, if any, automatic sprinkler systems, if any, where electrically driven steering gear is fitted, radio equipment and navigation lights.

6. The electrical power shall be sufficient for a period of at least 3 hours.

7. Where a non-automatic starting system is installed for the emergency generating set, in the event of failure of the main source of electrical power, the emergency lights and emergency alarm, together with the communication systems, shall be capable of being brought into automatic operation by an emergency battery system.

8. All passenger vessels carrying 200 passengers shall be provided with an emergency set of batteries capable of supplying electrical power to all alleyways, stairways and machinery spaces and control stations (that is, all places where the radio, main navigating or central fire recording equipment or the emergency generator is located), fire detection and alarm systems, radio equipment and navigation lights, and the electrical power shall be sufficient for a period of at least 1 hour.

173. Emergency electrical system:

1. The emergency source of electrical power may be a generator driven by a suitable prime mover with an independent fuel supply and satisfactory starting arrangements, and in that case the flash point of the fuel used shall not be less than 43 degrees Celsius.

2. As an alternative to the source mentioned in sub-rule (1), the emergency source of electrical power may be an accumulator (storage) battery capable of carrying the emergency load without recharging or excessive voltage drop, and in that case the arrangement of the accumulator batteries shall be such that the emergency lighting is brought into operation automatically immediately upon failure of the main supply to the lights, and when the accumulator battery is discharging,

an indicator installed in the machinery space or on the wheelhouse shall indicate this.

3. There shall be a separate charger set for the emergency batteries.

4. As far as practicable, during emergency operation electrical power may be supplied from the main switchboard to the emergency switchboard.

5. The emergency source of electrical power, the emergency switchboard and the associated transformer shall be installed above the uppermost continuous deck, shall be directly accessible from the open deck, and shall not be installed forward of the collision bulkhead (Collision Bhd).

174. Electrical circuits:

1. Circuits from which two or more final sub-circuits are supplied with electricity shall be rated (Rated) according to the total connected load, subject to the application of a diversity factor (diversity factor).

2. Where spare connections are kept on any section or distribution board, allowance for future increase of load shall be made with the total connected load before applying any diversity factor, and the diversity factor may be used in determining the size of cables, the rating of switchgear and the rating of fuse gear.

175. Motor circuits:

A separate feeder sub-circuit shall be provided for each fixed motor required for essential services and having a full load, and the carrying capacity of the conductors shall not be less than 125% of the motor’s full-load current rating, and no branch circuit shall have conductors smaller than No. 14 AWG wire.

176. Branch circuits:

All branch circuits shall be provided with protection only at the panel board.

177. Lighting circuits:

1. Lighting circuits shall be supplied with electricity by final sub-circuits separate from the circuits used for heating (heading), cooling (cooling) and power, and the power feeder supplying the apparatus necessary for the navigation of the vessel shall not be used for the essential supply.

2. No final sub-circuit with a rating exceeding 15 amperes shall supply electricity to more than one point, and the number of lighting points supplied by a final sub-circuit with a rating of 15 amperes or less shall not exceed the number described below, namely:

a. for 24 to 55 volt circuits ————–– 10

b. for 110 to 127 volt circuits ————- 14

c. for 220 to 250 volt circuits ———— 18

3. Where lamp holders are closely spaced (intrinsically safe), the number of points shall be unregulated except for the final sub-circuits of that cornice lighting, panel lighting and electrical lines:

Provided that the sub-circuit maximum operating current shall not exceed 10 amperes.

4. The lighting of cargo spaces shall be controlled by multipole-connected switches located outside those spaces, and arrangements shall be made for keeping all such circuits completely isolated, and they shall be kept locked in the off position.

5. The arrangement of lights in machinery spaces shall be such that such spaces cannot be left in darkness even if any one branch circuit fails.

178. Navigation light circuits:

1. All lights located in or controlled from the wheelhouse shall be supplied with electricity through a separate feeder or feeders from the emergency switchboard, if provided.

2. Direct feeders with indicator panels shall be without switches, or shall be without overcurrent protection from the feeder supplying electricity to the wheelhouse.

179. Running lights:

1. Duplicate lamps or other dual light sources, which may be oil or electric, shall be fitted to the Engineer and Ship Surveyor’s masthead, port, starboard, range, UNC and stern lights; however, if electric, they shall be controlled by an indicator panel.

2. The indicator panel shall be provided with a fused feeder disconnect switch, and the fuses located at the supply end shall have at least twice the capacity of the line fuses of the indicator panel.

3. For smaller vessels (power-driven boats) the requirements described in sub-rules (1) and (2) may be modified, and the emergency navigation lights (navigation lights) shall also receive electrical supply from the emergency source of electrical power.

4. Each insulated pole shall be controlled and protected by a switch and fuse breaker fitted on each navigation light distribution board, and if the lights are not visible from the bridge, an automatic failure indicator shall be provided.

180. Arc searchlight circuits:

1. Where arc searchlights are used, the circuits shall have circuit breakers.

2. All vessels navigating between 6 p.m. and 6 a.m. shall be provided with navigation lights and arc searchlights.

181. Motor control:

Good arrangements shall be provided for starting and stopping each electric motor, and they shall be fitted in such a position that the person controlling the motor can easily operate them, and control apparatus shall be provided for each motor above 1.0 kilowatt.

182. Heating or cooling and cooking equipment:

Each item of heating or cooling and cooking equipment shall be controlled as a complete unit by a multipole-connected switch fitted near the equipment, and in the case of cabin coolers a single-pole switch shall also be acceptable.

183. Remote control:

1. There shall be arrangements for stopping the exhaust ventilation fans used in machinery and cargo spaces, and they shall be suitable for operating all these arrangements from outside in the event of fire.

2. Remote controls located outside the engine or boiler space shall be fitted for the machinery driving boiler fans, independently driven pumps supplying oil to the main propulsion machinery for bearing lubrication and piston cooling, fuel oil transfer pumps and other similar fuel pumps, and for other fuel pumps also remote controls shall be fitted outside the space concerned

so that the said pumps can be shut down in the event of fire in the spaces where they are located.

3. In passenger ships, master controls shall be fitted to all power-driven exhaust ventilation systems so that all fans can be stopped from any one of the positions located as far apart as practicable.

184. Steering gear:

Where it is necessary to install an electrical control system for the steering system, an alternative control system shall be installed, which may be a duplicate electrical control system or a control system by other means.

185. Application of cables:

1. The types of the various cables of the vessel shall, unless otherwise specifically approved, be in accordance with the general requirements, and for single-conductor A-C cables they shall be non-magnetic.

2. The voltage rating of any cable shall not be less than the nominal voltage of the circuit in which it is used.

3. For propulsion power cables, cables insulated with ethylene rubber, cross-linked polyethylene or silicone rubber shall be used; however, where the normal ambient temperature will not exceed 50 degrees Celsius, polyvinyl chloride insulated cables may be used.

4. All cables shall have a suitable moisture-resistant jacket, and except in the case of single-conductor A-C cables, the use of an impervious metallic sheath may be considered in other cases.

186. Distribution cables:

1. All electrical cables of power, lighting, communication control and electronic circuits shall have insulation suitable for a conductor temperature of not less than 75 degrees Celsius.

2. The rated comparative temperature of the insulating material shall be at least 10 degrees Celsius higher than the maximum ambient temperature likely to exist or be produced in the place where the cable is installed.

 (3) At those positions where the electrical cables installed are liable to be damaged during the normal operation of the vessel, braided metal armour shall be provided, or other suitable measures shall be taken to

protect them from mechanical damage.

187. Portable and flexible electrical cables:

1. Electrical cables and cables likely to be used in flexible service shall be armoured.
2. Cables fitted on open decks and in bathrooms, cargo storage spaces, machinery spaces or any such place where condensation of water may occur or harmful vapour (i.e. oil vapour) may be present shall have an impervious sheath, and in permanently [wet] spaces cables of the submersible type shall be used.
3. Except in final circuits, i.e. where the cables are installed inside metal conduit sheathing pipes of not more than 25 millimetres internal diameter and the conduits remain mechanically and electrically continuous, all other cables shall be of the flame-retardant or fire-resistant type.

188. Current rating:

1. The maximum uninterrupted load carried by any cable shall not exceed its current rating, and in the plans to be submitted for approval the diversity factor of the individual loads and the duration of the maximum demand shall be shown at the time of estimating that maximum uninterrupted load.
2. In normal service conditions, when the cables carry the maximum current, the voltage drop from the bus bars (Bas bar) of the main switchboard to any point of the installation shall not exceed 6% of the nominal voltage.
3. When determining the current rating of the circuits of the lighting system, it shall be determined for each lamp holder on the basis of the maximum load which may be connected to it, taking a minimum of 60 watts, unless the fitting is so connected that only a lamp of a rating less than 60 watts can be connected to it.

p.93 (4) For the supply of cargo-handling gear, the cables supplying electricity to winches, cranes, windlasses and capstans shall be rated suitably for their duty.

5. If the duty does not require a longer-time rating, the cables of winch or crane motors may be rated on a half-hour basis on the basis of the half-hour kilowatt rating of the motors.
6. The motors of windlass and capstan motors shall not be rated for less than one hour on the basis of the one-hour kilowatt rating of the motor, and in all cases the rating shall be subject to the voltage drop within the specified limits.
7. The current rating of cables for the maximum conductor temperatures shall be of the values specified in the tables from Table 58 to 63 of the Schedule.
8. For higher temperatures silicone rubber and mineral insulation may be used (0.15 degrees Celsius for silicone rubber [as printed], unlimited for mineral insulation) when these need not be touched by the persons on board the ship, and all such proposals for higher-temperature applications shall be specially considered.
9. The temperature shall be taken as the combined temperature of the ambient temperature and the temperature rise due to the load.
10. For intermittent loads and different temperature conditions, the correction factors of Tables 64 and 65 of the Schedule may be applied for the half-hour and one-hour ratings, and in no case shall a rating less than the half-hour rating be used.

189. Protection:

The installation, including short circuits, shall be protected against accidental over-current flow, and the protective arrangements shall be suitable for full and co-ordinated provision with a view to ensuring the following matters, namely:

a. continuity of service even in a faulty condition through the discriminative action of the protective devices; and

b. elimination of the fault so as to reduce damage to the system and the risk of fire.

p.94 ### 190. Overload protection:

Circuit breakers and automatic switches used for overload protection shall have tripping characteristics suitable for the system, and fuses above 100 amperes shall not be used for overload protection, but they may be used for short-circuit protection.

191. Short-circuit protection:

1. Protective arrangements against short-circuit current shall be provided.
2. The breaking capacity of every protective device shall not be less than the maximum value of the short-circuit current which may flow at the point of installation.

3. The making capacity of every switch-operating device used which is capable of being closed when required during a short circuit shall not be less than the maximum value of the short-circuit current at the point of installation.

4. In A.C. current this maximum shall correspond to the maximum permissible value for maximum asymmetry (maximum asymmetry).

5. In the absence of proper data, the following short-circuit currents at the machine terminals shall be assumed, namely:

a. D.C. system: ten times the full-load current for the generators (including spares) normally connected;

b. six times the full-load current for the motors operating at the same time; and

c. A.C. system: ten times the full-load current for the generators.

192. Protection of circuits:

1. Short-circuit protection shall be provided on every live pole of a D.C. system and on every phase (Phase) of an A.C. system.

2. Overload protection shall be provided in the following cases, namely:

a. in a two-wire D.C. or single-phase A.C. current system, at least one line or phase (Phase);

b. in a three-wire D.C. system, both outer lines;

p.95 (c) in an insulated three-phase A.C. system, at least two phases (phase);

d. in an earthed (Earthed conductor) three-phase A.C. system, all three phases.

3. A non-linked switch (non-linked switch) or non-linked circuit breaker shall be installed in an earthed conductor (Earthed conductor), and the switch or circuit breaker which is fitted shall operate simultaneously on the earthed conductor and the insulated conductors; and this shall not preclude the need to provide an isolating link (for the purpose of testing) usable only when the other conductors are disconnected.

193. Protection of generators:

1. Generators not arranged to be run in parallel shall have a circuit breaker or conductor arranged to keep all insulated poles (Pole) simultaneously within the specified position limit, or, in the case of generators rated at 50 kilowatts, a multipole linked switch with a fuse on each insulated pole.

2. For generators arranged to be run in parallel, there shall be a circuit breaker or conductor arranged to open all insulated poles simultaneously and the following arrangements, namely:

a. for D.C. generators, instantaneous reverse-current protection operating at not more than 15 percent of the rated current;

b. for A.C. generators, reverse-power protection with a time delay (time delay) selected and set within the specified limits of 2 percent to 15 percent of full load according to the characteristics of the prime mover.

3. The reverse-current protection shall be sufficient to deal with reverse-current conditions arising from the ship’s network, i.e. from the winches.

194. Feeder circuits:

1. Isolation and protection of every main distribution circuit shall be ensured by multipole circuit breakers or switches and fuses.

p.96 (2) The protective devices shall be suitable for allowing the excessive current to flow during the normal acceleration of the motors.

3. The cables of the circuits of motors supplied with electricity fitted with overload protection may be provided with short-circuit protection.

4. Every motor of rating exceeding 0.5 kilowatts shall be individually protected against overload and short circuit, and the circuit may be protected by the same protective device adopted for the motor and its supply cable.

5. If the owner so desires, the overload protection may be replaced by an overload alarm.

6. The primary circuits of power transformers shall be protected against short circuit by circuit breakers and fuses.

7. When transformers are arranged to be run in parallel, means of isolation shall be provided on the secondary

windings.

8. The switches and circuit breakers shall be capable of withstanding excess current.

195. Lighting circuits:

1. Lighting shall have overload and short-circuit protection.

2. Voltmeters, voltage coils of measuring instruments, earth-indicating devices and pilot lamps shall be protected together with their connecting loads, and if a pilot lamp is fitted in the same casing as an integral part of any other equipment, the pilot lamp need not be individually protected.

3. Batteries other than starter batteries shall be protected against short circuit by a fuse or multipole circuit breaker for each insulated conductor at a position adjacent to their compartment.

196. Installation or laying of cables:

1. Cable runs shall be as straight as practicable, and cables having insulating materials of different maximum rated conductor temperatures shall not be bunched together, and where this is impracticable the cables

p.97 shall be so operated that no cable reaches a temperature higher than the lowest temperature in the bunch, and cables having protective coverings which may damage the coverings of other cables shall not be bunched together with other cables.

2. The minimum internal radius of bend of installed cables shall generally be in accordance with the following formula:

4d – for cables without braiding, foiled armouring or other metal sheet and of overall diameter not exceeding 25 millimetres.

6d – for other cables.

d = overall diameter of the cable measured in millimetres.

3. So that the cables are not damaged, they shall be effectively supported and secured without damage to their outer coverings, and when passing through watertight bulkheads or decks, individual watertight glands or wrapped watertight boxes carrying several cables or wrapped watertight boxes carrying several cables shall be used.

4. Cables passing through decks shall be protected by deck tubes or pipes, and where cables pass through the frames or non-watertight steel bulkheads the holes shall be lined with lead or other approved material of a non-corrosive nature, and the cut-outs or rectangular holes for the passage of cables shall be radiused.

5. Cables shall, as far as practicable, be installed on trays or in pipes.

6. The radius of metal conduits carrying electrical cables shall not be less than that of the conduit or pipes, and in the case of pipes of 64 millimetres diameter it shall not be less than twice the diameter of the pipe, and cables belonging to the same circuit shall be installed in the same conduit.

7. Non-metallic conduits or pipes shall be of flame-retardant material, and PVC conduits shall not be used in refrigerated spaces or on open decks.

p.98 (8) A.C. cables shall be non-armoured or armoured with non-magnetic material.

9. When installing two, three or four single-core cables or cables forming respectively single-phase circuits or three-phase and neutral circuits, the cables shall as far as possible be in contact with one another, and in no case shall the distance between adjacent cables exceed one diameter.

10. If any joint (Joint) is required, all conductors should be adequately protected, insulated and protected from atmospheric action.

197. Power and lighting batteries:

1. Power and lighting batteries shall be of the acid or alkaline type, and the power cells shall be so constructed that the electrolyte cannot spill out while the ships are moving (up to 40 degrees from the vertical).

2. The cells of the battery or accumulator shall be assembled in crates with handles or on platforms for convenience of handling, and the total mass of one set shall not exceed 100 kilograms.

3. The batteries shall be so arranged that the trays are accessible and there is at least 300 millimetres of head room over them.

4. To every crate or platform a description plate shall be fitted containing information regarding its type name, the name of the manufacturer, the nominal capacity at a specified discharge rate and the density of the

electrolyte at the time when the battery is fully charged.

5. The cell containers shall not be constructed of materials sensitive to electrical shock, like glass, or of readily combustible materials.

198. Battery location and installation:

1. The place for keeping accumulator batteries shall be a house, and this shall not be installed in cargo storage spaces.

2. Batteries shall not be kept in any places where excessive heat, excessive cold, splashing water or the possibility of a flow of water exists, or where the operation of the battery may be impaired or their deterioration accelerated.

p.99 (3) Engine starter batteries shall be kept as close as practicable to the engines concerned and shall be so kept that there is convenient access for replacement, testing, checking, topping up and cleaning, and they shall be so installed that no damage to the surrounding equipment occurs as a result of the emission of battery vapour.

4. Lead batteries and alkaline batteries shall not be installed in the same room.

5. Those batteries which require electrical power exceeding 2 kilowatts for charging (calculated from the maximum charging current of the charging equipment and the nominal voltage of the battery) shall be installed in a protected place, and if they are kept on deck they shall be kept in a cupboard or box.

6. Those batteries which require electrical power equal to or less than 2 kilowatts for charging shall be installed in a place specified for keeping them or in a box or cupboard, and they may also be installed below deck in a place with free air circulation, in the machine room or in a place with the convenience of a source of air circulation, but the falling of any object and the falling of water on them shall be prevented, and the container for keeping them shall be designed to be suitable for the free escape of gas.

7. The arrangement giving opportunity for the escape of gas shall provide protection and at the same time prevent sparks or flames likely to initiate a fire from easily reaching the gas vent holes of the cells.

8. The cells or supports shall be carefully wrapped and secured so that they cannot in any way be displaced by reason of the motion of the vessel or its heeling over.

9. In lockers or boxes, the interior of the spaces specially set apart for batteries, including the box shelves, and the other structural components shall be protected from deterioration by a coat of paint or a covering of electrolyte-resisting material.

10. If the ambient temperature inside the rooms specially marked for keeping batteries falls below 5 degrees Celsius, arrangements for heating them shall be provided.

p.100 ### 199. Instruction plates:

1. Instruction plates bearing the words that entry with smoking and naked flame is prohibited shall be affixed to the doors of the compartments for keeping batteries.

2. In cupboards and lockers protecting accumulators too, instruction plates shall be affixed written to the effect that there are dangerous accumulators and a risk of explosion.

200. Ventilation:

1. To avoid the accumulation of flammable gas, there shall be ventilation arrangements in battery rooms, and if the pipes are led directly from the battery ceiling upwards to the open air, natural ventilation arrangements shall be provided, and if the provision of natural ventilation arrangements is not possible, mechanical exhaust ventilation arrangements shall be made by installing an intake fan at the top of the room.

2. The fans shall be capable of changing the air of the room completely within two minutes, and the fans shall be fire-resistant, and an open space shall be kept near the floor for the entry of air.

3. In the same manner as for the battery room, battery lockers shall, if possible, be provided with ventilation arrangements through pipes to the open air from the top of the locker or through exhaust ventilation pipes, but the pipes shall reach at least 915 millimetres above the locker of machinery spaces provided with ventilation arrangements and similar compartments, and water-holding holes [louvres] or similar arrangements shall be provided near the bottom for the entry of air.

4. For the prevention of water entry, there shall be a gooseneck, mushroom head or its connection with a deck box together with an arrangement for the pipe to reach at least 1220 millimetres above the top of the box, and holes for the entry of air shall be kept on at least two opposite sides of the box, and the whole deck box including the open parts of the ventilation shall be watertight so as to prevent the entry of splashing water or rain water.

5. In a small battery box no ventilation arrangement shall be required other than an open part near the top for

the escape of gas.

p.101 ### 201. Prevention of corrosion:

1. The interior of battery rooms, including the structural members and shelves and the places of air entry and exit, shall be painted with corrosion-resisting paint.

2. The shelves of the battery room or the lockers for acid batteries shall have a watertight lining of lead sheet of minimum thickness 1.6 millimetres up to a height of 76 millimetres on all sides.

3. For alkaline batteries, the shelves shall likewise be given a steel lining of at least 0.8 millimetres thickness.

4. As an alternative, a watertight lead pan may be fitted in the battery room for alkaline batteries [as printed], and it shall be carried over the whole deck and at least 152 millimetres up on all sides.

5. Deck boxes shall be lined in accordance with the alternative method mentioned in sub-rule (4), and small battery boxes shall be lined to a depth of 76 millimetres in conformity with the method described in this rule.

6. Where conductors enter the battery room, the holes shall be covered so as to make them watertight.

7. All electrical connections of the battery rooms shall be resistant to splashed or leaking electrolyte, and the current-carrying capacity of the conductor shall not be exceeded.

8. When determining the cable, the charge-starting rate or the maximum discharge rate, whichever is greater, shall be taken into consideration.

202. Overload and reverse-current protection:

1. Every battery conductor shall be provided with an overload protection device, but in engine-starting batteries or in any battery of nominal voltage 6 volts or less overload protection shall not be required.

2. The required overload protection device shall be provided in every battery conductor outside and adjacent to the room, and in place of a circuit [breaker] fuses may be used for the protection of emergency lighting storage batteries, and in the charging equipment of all batteries of line voltage exceeding 20%, other than rectifiers, automatic protective devices against reversal current shall be provided.

p.102 ### 203. Electrical equipment:

If it is so certified that the keeping of electricity-consuming devices or data-recording equipment in a hazardous atmosphere is safe, then resistance thermometers, thermo-couples, liquid level measuring devices or alarms and gas-detecting equipment may be installed in tanks or in enclosed spaces adjacent to cargo oil tanks.

Chapter 5

Passenger vessels

204. Applicability of Chapter 5:

1. The provisions mentioned in this Chapter shall, in addition to the rules described in Chapters 3 and 4, apply in the case of propeller-driven passenger ships with or without sleeping arrangements, with machinery fitted forward or amidships.

2. The scantlings (Scantling) and arrangements shall, as far as practicable, be in accordance with Chapter 3 and in accordance with the requirements mentioned in this Chapter.

3. The structural requirements mentioned in this Chapter have been framed for the purpose of vessels of length less than 70 metres, whose length-to-depth ratio does not exceed 65 [as printed] and generally whose breadth-to-depth ratio does not exceed 5.

205. Watertight bulkheads:

1. In addition to the watertight bulkheads described in Chapter 2, additional watertight bulkheads shall be fitted in order to satisfy the floodability requirements of the “Inland Ships Stability Rules, 2001”, and they shall extend up to the bulkhead deck.

2. The spaces designated for passengers shall be separated from the machinery spaces, boiler room and cargo storage spaces by watertight bulkheads.

3. In the collision bulkhead and the bulkhead separating the passenger spaces from the machinery space, no openings or watertight doors shall be kept below the bulkhead deck.

p.103 (4) Where openings are required to be kept in other bulkheads, their number shall be limited to the minimum number required for the normal operation of the ship, and watertight doors shall be fitted in all openings.

5. Those watertight doors which cannot be operated by remote control may be permitted only where passengers do not enter, but the doors shall be kept closed, and those doors may be opened for passengers but shall again be closed immediately.

6. Pipe systems with open ends and ventilation pipes shall be so arranged that if the compartment under consideration is flooded, no other space or tank can be flooded.

7. Where a number of compartments have open connection through pipe lines or ventilation pipes, in all such cases the pipes and pipe lines shall pass through the watertight bulkheads above the waterline in the most unfavourable flooded condition, and where this is not possible with the pipes, valves remotely controlled from above the bulkhead deck shall be fitted.

8. When the pipe system has no open end in any compartment, that pipe line shall not be considered as damaged if that compartment is flooded, provided that its distance from the bottom of the ship is not less than 0.5 metres.

9. Cables passing through watertight bulkheads shall be so arranged that the watertightness of the bulkheads is not impaired.

206. Arrangement of embarkation and disembarkation, access ways, stairs, etc.:

1. Around all decks which are not enclosed by bulkheads and are built for the purpose of use by passengers, bulwarks or railings of height not less than 0.9 metres shall be so fitted that children cannot fall down through them.

2. The access arrangements for embarkation and disembarkation as mentioned in sub-rule (1) shall be protected, and the gangways also shall be at least 0.6 metres wide and provided with hand-railings on both sides.

p.104 (3) Passageways, stairs, doors and exit routes for the use of passengers shall be at least 0.8 metres wide, and the width of the doors of cabins and other small spaces may be reduced somewhat; and if only one passageway or stair leads to any place designated as a passenger space, the minimum width shall be at least 1.0 metre, which may be reduced to 0.8 metres in small ships.

4. In the case of spaces or groups of spaces designated for 80 passengers, the total clear width of the exit routes or the exit routes for their use in an emergency shall be on the basis of at least 15 millimetres for each passenger.

5. Escape routes (Escape route) and for leaving the space in an emergency there shall be a clear open space of at least 0.6 by 0.6 metres, and the escape routes (Escape route) and the routes for leaving the space in an emergency should lead to a suitable open deck located on the lifeboat deck where the lifeboats are secured to the vessel.

6. Doors of passenger accommodation, other than doors leading to passageways, shall open outwards or be of the sliding type, and shall be suitable for not being capable of being locked by unauthorised persons during the voyage.

207. Structural configuration (Structural Configuration):

1. Unless otherwise mentioned, the structural configuration of the vessel shall be built in accordance with Chapter 2.

2. Transverse or longitudinal framing may be adopted, and in large ships of length 40 metres or more longitudinal stiffening may be applied to the bottom and the strength deck, and as an alternative transverse framing may be followed there in the strength deck too.

3. Passenger vessels of length exceeding 50 metres shall have a double bottom and a freeboard or strength deck.

4. Passenger vessels of length exceeding 36 metres shall be of the single-compartment type in accordance with the “Stability Rules” for inland ships, and shall have the necessary watertight bulkheads.

p.105 ### 208. Single-bottom vessels:

1. Where parts of the spaces below the upper deck of a single-bottom decked vessel are designated for use as cargo storage or as a godown, a well-constructed ceiling shall be fitted on the basis of the actual loading.

2. If a double bottom is not fitted with an open lounge, its roof shall be well constructed.

209. Longitudinal strength (Longitudinal Strength):

1. The design bending moments shall be calculated for determining the longitudinal strength (Longitudinal Strength) to be maintained amidships of the ship.

2. The bending moment shall be determined by direct calculation.

3. The design bending moments shall take into consideration the following loading conditions:

a. the actual condition for the voyage, fully equipped, with fresh water, fuel and lubricating oil tanks completely full, the maximum permitted number of passengers and the weight of cargo, the crew and stores present on board, and the ballast tanks partially or completely full according to the actual condition;

b. the condition of the ship fully equipped, with fresh water, fuel and lubricating oil tanks 95% empty and the ballast tanks partially or completely full according to the actual condition;

c. the condition representing the higher values of bending moments during the voyage in any condition of the ship; and

d. where an effective superstructure is attached, it shall be taken into account when calculating the section modulus of the structure, and the deckhouse shall be included when calculating the section modulus of the structure on condition that the longitudinal and transverse bulkheads are at least 0.75 B, well constructed, well supported and of length at least 0.2 L or 10 metres, whichever is greater, and located amidships of the ship.

p.106 ### 210. Keel, side plating, openings and structural plating:

1. The thickness of the open [sheer-strake/plating] shall not be less than the requirement of the relevant table in Chapter 2, but in the case of vessels of length exceeding 40 metres the thickness of the bottom plating shall need to be increased in order to satisfy the requirements described in rule 212.

2. The breadth and thickness of the keel plate of the ship shall be in accordance with the requirements of the tables mentioned in Chapter 2, and it shall be maintained over the full length of the ship.

3. The thickness of the bottom plating shall be in accordance with the requirements of the relevant tables in Chapter 2.

211. Bilge plating:

1. The thickness of the bilge plating shall be maintained, including the midship portion of the ship, from amidships to well forward and aft of the bilge.

2. The bilge radius shall be at least 10 times the thickness of the bilge plating, and the bilge strake shall extend at least 200 millimetres on both sides of the radius.

3. In solid round, square or square bilges built with an angle-bar connection on the outer covering, the bottom plating and side opening plating adjacent to the round, square or angle bars shall not be increased over the bottom plating in the pathway.

212. Side plating:

The thickness of the side plating shall be in accordance with the requirements of the tables mentioned in Chapter 2, and it shall be maintained over 0.5 [L] amidships of the ship, and it may begin to be reduced up to the thickness obtained in the after region.

213. Hull openings:

1. The corners of the openings in the hull plating shall be well rounded, and compensating arrangements shall be provided for openings of breadth exceeding 250 millimetres.

2. For side scuttles and windows, the edges of the hull openings shall be stiffened from the inside by an angle bar of at least 75 millimetres height, and the side door, porthole and the openings connected

p.107 with it shall be protected, and only in the case of large portholes shall compensating arrangements be provided for the cutting of the shell plating.

3. The openings below the board, or below the depth from which the compensating safety distance is calculated, shall be circular.

214. Deck plating:

1. Where openings are made in the deck plating for access or a deck house is built let into the deck, the thickness of the deck plating in the way shall be increased by compensation for the cut material, and when the loss of cross-sectional area has been taken into account at the time of calculating the actual midship hull section modulus, no compensating arrangement shall be required.

2. Compensating arrangements shall be required for circular openings in the deck plating of diameter less than 150 millimetres if they are located well away from other openings and the portion cut transversely on the deck does not exceed 3 percent of the total cross-sectional area.

3. Where plate-panel openings are cut, the edges of their openings shall be adequately stiffened.

4. For vessels of length 40 metres and above, the deck plating thickness shall not be less than the thickness in accordance with the requirements of Chapter 2, and the deck plating shall be increased for satisfying the requirements mentioned in rule 212.

5. Below superstructures or deck houses of length at least 0.2 L or 10 metres, whichever is greater, and located in the midship portion of the ship, the thickness of the deck plate shall be in accordance with the tables mentioned in Chapter 2.

6. The exposed deck plating of the sides of the deck house shall have the deck house let in to a width equal to at least one frame spacing in the way thereof, and the exposed deck plating shall extend over two frame spacings at all ends of the superstructure or deck house.

215. Single-bottom structure:

1. There shall be a centre and keelson or girder in accordance with the provisions mentioned in Chapter 2.

p.108 (2) Plate floors shall be fitted to every frame.

3. Transverses shall generally be fitted at a spacing not exceeding 3.5 metres.

4. If the ship has rise of floor (Rise of floor), then the height of the floors and the height of the transverses shall be nearly parallel to the rise of floor.

5. Except for the seating of the engine and the requirements of the propulsion system, the floors shall not be cut at the centre line for any other reason, and in the case of cutting at the centreline the centreline girder web plate shall be continuous subject to maintaining the transverse strength of the floors.

6. The floors shall be provided with drain holes of adequate number and shape.

7. The depth of the transverses shall generally be not less than twice the depth of the slots of the longitudinals.

8. The longitudinals shall generally pass through watertight floors and transverse walls, and if they stop without passing through the watertight floors and transverse walls, brackets shall be fitted in line with the longitudinals.

9. The scallops between the longitudinals may be fitted without end connections, overlaps with the transverses or tripping brackets in the path.

216. Double bottom structure (Double bottom structure):

1. The requirements specified in this rule and other rules shall apply to transversely framed or longitudinally framed double bottoms.

2. The depth of the double bottom at the centreline shall not be less than 800 millimetres.

3. The double bottom shall be accessible for cleaning, inspection and survey, and there shall be at least one watertight manhole connected with every compartment in the double bottom.

4. Arrangements for free air and water flow from all parts of the double bottom compartments to the air pipes and suctions shall be made, and the calculation of the pumping rate required for this shall be taken into consideration.

5. The scantlings (Scantlings) used in the construction of the transversely or longitudinally framed double bottom shall be as follows, namely:

p.109 (a) the thickness of the inner bottom plating equal to the keel plate;

b. the dimensions of the scantlings of the standard described in Chapter 2;

c. web frames and keelsons extending from the bottom of the ship to the inner bottom;

d. where access openings are cut in the floors and girders, the height of the openings generally within 50 percent of the depth of the double bottom;

e. no access opening kept at the end path of the floors and girders; and

f. the openings at the centre of the tank depth.

6. Where a margin plate is proposed for the bilge, the thickness required for the margin plating shall be more than 1 millimetre.

7. Where the double bottom (Double bottom) is not continuous over the full breadth of the ship, the transverse strength of the bottom structure shall be maintained, and that shall be in accordance with the minimum

requirements of this rule.

217. Watertight bulkheads in double bottom (Double bottom) tanks:

1. If the breadth of the vessel is 8 metres or more, there shall be a watertight bulkhead at its centreline.
2. Below every watertight bulkhead there shall be a transverse watertight bulkhead in the double bottom tank.
3. [as printed] The thickness of the plate of the watertight bulkheads shall be at least equal to the side plating.

218. Shell framing:

1. The requirements specified in this rule shall apply to both longitudinal and transverse frames, and where longitudinals are not adopted over the full length of the ship, they shall be efficiently integrated with the transverse frames.
2. The structure of the side frames of the midship portion of the ship, dependent on the end connections, shall be in accordance with the requirements mentioned in Chapter 2, and where brackets differing in arm dimensions from the standard brackets are fitted, the frame modulus shall be corrected in accordance with Chapter 2.

p.110 ### 219. Transverse shell framing (Transverse Shell Framing):

1. Where a combined frame system is adopted (hull transverse and bottom and deck longitudinal frames), a transverse ring system shall be provided, and in such a case the section modulus of the side frames in the way of the bottom and deck transverses shall be increased by 100 percent, and the end connections shall be in accordance with the requirements described in Chapter 2.
2. The brackets at the top and bottom of the side frames shall be connected with the relevant adjacent deck and bottom longitudinals.
3. When the frames are carried through the bilge radius, they shall be connected directly with the bottom transverses.

220. Longitudinal shell framing (Longitudinal Shell Framing):

1. The longitudinal stiffening members shall be supported by hull transverses satisfying the requirements of the relevant tables mentioned in Chapter 2, and if the hull transverses at the intersection of the bottom transverses are at least equal to the bottom transverses and at the intersection of the deck transverses at least equal to the deck transverses, then the brackets may be omitted.
2. The face bar of the side transverse shall be welded to the face bars of the bottom and deck transverses, and the web plates of the bottom and deck transverses shall be stiffened rigidly in line with the face bar of the side transverse.
3. The shell transverses shall form a ring system together with the bottom and deck transverses.

221. Deck structures:

1. The transverse and longitudinal deck supporting structure shall be in accordance with the standards specified in the tables mentioned in Chapter 2.
2. Deck girders and transverses may be fitted in connection with load-bearing bulkheads or pillars for the support of deck beams or deck longitudinals.
3. Effective arrangements shall be made for the distribution of load at the head and heel of all pillars, and tripping brackets or equivalent brackets shall be fitted to the web plates of the transverses or girders in line with the pillar.
4. Where concentrated loads are supported, the pillars shall be strengthened for the additional bending moment imposed on them.

p.111 ### 222. Erections (Erections):

1. Erections shall be built in accordance with the requirements described in Chapter 2.
2. Where any erection is to be fitted in accordance with the provisions mentioned in Chapter 2, the thickness of that deck plating shall be in accordance with the standards specified in the tables mentioned in Chapter 2.
3. In vessels of length 40 metres or more, the thickness shall need to be increased to satisfy the requirements of rules 212, 213 and 214.
4. The side plating of effective erections shall be well stiffened and secured at the ends, and arrangements for a strong connection with the main structure shall be made.

223. Guidance on methods of direct calculation:

1. Where direct calculation is adopted as an alternative to the structures obtained by the rule formulae, or where all supporting data, i.e. supporting conditions, for the scantling calculation of structural members not included in the guidance are to be determined, they shall be submitted for approval together with the calculations.

2. In addition to the permissible stresses, the stress values mentioned in sub-rules (3) and (4) shall be applied.

3. For structural members included in the hull section modulus, the maximum bending stress, shear stress and equivalent stresses shall not exceed the values mentioned in Table 66 of the Schedule.

4. For structural members not included in the hull section modulus, the maximum bending stresses, shear stresses and equivalent stresses may exceed [as printed] the values given in Table 67 of the Schedule.

Chapter 6

Tankers

224. Application of Chapter 6:

The rules mentioned in this Chapter shall apply in the case of tankers carrying flammable, toxic and dangerous liquids as additional to the directions given in Chapters 2, 3 and 4.

p.112 ### 225. Tanker:

The following shall be deemed to be the intended tankers, namely:

a. ships carrying flammable liquids in tanks built as part of the hull of the ship; and

b. ships with independent tanks carrying flammable liquids, separate from the hull of the ship, in which the tanks are enclosed.

226. Liquid cargo storage spaces:

The cargo storage spaces specified within the scope of this rule include cargo tanks, spaces containing cargo tanks, cofferdams, cargo pump rooms and, along with them, the bilges belonging to them.

227. Permissible cargoes:

1. Flammable liquid materials shall be deemed permissible for carriage in tanks.

2. In addition to the plans mentioned in these Rules to be sent for plan approval, the following plans also shall be submitted to the Authority for approval, namely:

a. plans for the cargo tanks;

b. plans for the oil measuring arrangement in the cargo tanks, the extended [venting] material and the level measuring equipment; and

c. plans for the bilge and ballast water pipes within the cargo tank area.

228. Construction materials:

1. The hull, superstructure and tanks of the ship shall be built of steel or iron used for structural construction, satisfying the requirements of the rules relating to structural construction.

2. Wood shall not be used in the following cases, namely:

a. fenders and masts;

b. carpentry work and panelling and linings of the accommodation, galley provision rooms;

c. furniture and furniture of the engine room and work tables, benches and places for keeping tools;

p.113 (d) window and lamp frames, removable parts of the wheel house, sheathing of the after deck, [unclear] plating, the lists of equipment and the boat; and

e. the protective locker for the storage of batteries.

3. At both ends of the liquid cargo storage space there shall be a cofferdam of 800 millimetres width.

229. Cargo pumps:

1. Cargo pumps above the deck shall be installed within the forward and after cofferdams.
2. Cargo pumps below the deck shall be installed in special pump rooms separated by gastight and fire-resisting partition walls.
3. The pump rooms and their access ways must be located in the cargo area, and they shall have no connection with the machinery spaces or other spaces containing materials liable to catch fire.

230. Equipment and operation:

1. Pressure relief devices shall be fitted to the positive displacement pumps among the cargo pumps, and from there the discharge shall be recirculated through the suction side of the cargo discharge pumps.
2. The cargo pumps shall be provided with arrangements for stopping from outside the pump room.
3. The cargo pump output shall be capable of being controlled from outside the pump room.
4. For monitoring the pump pressure, there shall be a pressure gauge arrangement at the pump control station.

231. Drive:

1. The driving machinery shall be installed outside the cargo area, and exceptional arrangements may be approved for hydraulic or intrinsically safe electrical drives.
2. The arrangements for passing the pump drive shafts into the pump room bulkheads shall be approved, and they shall be of gastight construction, and in this case only type-tested seals shall be accepted.

p.114 (3) The bulkhead penetrations above the deck for the pump drive shafts shall be so constructed that gas cannot flow through them.

232. Laying of pipes:

1. The cargo pipe system shall be permanently laid and shall be completely separate from other pipe systems, and they shall not be extended aft of the cargo tank area.
2. The cargo pipe lines must be so laid that the liquid remaining in the pipes can be drained into the cargo tanks, and the cargo pumps and filters of pump rooms below the deck shall be so arranged that they can be safely drained at all times.
3. The cargo filling lines shall be laid as close to the tank bottom as practicable.
4. As required, expansion bends, corrugated expansion pipes or other recognised means of expansion compensation shall be provided.
5. Cargo pipe lines may be installed in cargo tanks below the deck if they are located in a stop valve operable from the deck of the tanks with which they are connected, and in addition a stop valve shall be provided in the pump room for each pipe line leading to the tanks.

233. Pipeline design:

1. For the design of the cargo lines the relevant rules of Chapter 3 shall be followed.
2. The cargo lines shall be joined by welding.

234. Valves, fittings and equipment:

1. Shut-off devices made of cast steel, steel or other approved materials shall be fitted at the hose connections, and they shall be protected from unauthorised opening.
2. There shall be an indicator showing whether the stop valves are open and how far they are open, and the operating rods of the shut-off devices inside the cargo tanks shall pass through the tank top in an oil-tight manner.

p.115 ### 235. Bilge lines:

1. The bilge pumping arrangements of the machinery spaces must not be used for pumping the bilges of the spaces of the cargo area, and the bilge pumping arrangements of the machinery spaces shall be determined in accordance with the rules given in the machinery part of these Rules.
2. In the spaces of the cargo tanks and in the void spaces of the cargo area there shall be special bilge pumping arrangements, i.e. explosion-proof electrically or hydraulically operated pumps located inside the cargo area.
3. The main bilge shall be in accordance with the provisions of Chapter 3, but the internal diameter shall not be less than 62 millimetres.

4. The internal diameter of every branch bilge line shall be 75% of the main bilge line, but it shall not be less than 42 millimetres.

236. Cargo pump room bilge:

1. For pumping the cargo pump room bilges there shall be a special bilge pumping arrangement within the cargo area.

2. In special circumstances, bilge operation shall be possible also from a place from which removal of the cargo pump room bilges is not easy, and in that case the necessary equipment must be capable of being operated from outside the pump room.

237. Filling and draining of cofferdams:

1. A bottom valve operable from the deck shall be fitted in the cofferdams, by means of which they can be filled with water, and they shall be capable of being filled by means of a hose with suitable coupling from the fire extinguishing line through a pipe system permanently installed in the cofferdam.

2. Liquid shall be drained from the cofferdams only by using a pump located within the cargo area.

238. Ballasting facilities within the cargo and tank area:

1. The ballast systems for the cargo tanks within the cargo tank area or for special ballast water shall be independent of the pipe systems forward or aft of the cofferdams.

p.116 (2) Ballast water may be brought from outboard by a pipe line through the cofferdam, and this line shall be fitted with the help of a screw-down non-return valve and may be connected with the cargo pumps, and its function shall be to prevent the discharge of water or flammable liquids over the ship.

3. The spaces forward of the cargo area and the ballast tanks may be filled or emptied using the fire-extinguishing line.

239. Ventilation of pump rooms, cofferdams, cargo tanks and void spaces:

1. Effective means of ventilation of pump rooms and cofferdams shall be provided, and such arrangements shall not be connected with the other ventilation systems of the ship.

2. Pump rooms shall be ventilated by means of power-driven fans of the extraction type, and fresh air may be supplied by natural draught through the entrance to the pump room from above.

3. Cargo pump rooms shall have suitably applied ventilation arrangements (exhaust fans) ensuring at least 20 air changes per hour.

4. The suction pipe shall terminate close to the bottom of the pump room.

5. The air intake and exhaust arrangements shall be located in the cargo area, and they shall be as high above the deck as practicable and as far as practicable from the vent openings of the cargo tanks.

6. Arrangements shall be provided so that the air supply to the pump room can be effectively stopped from the fire deck.

7. Approved flame arresters shall be fitted in the air inlets from cofferdams, cargo tank spaces and void spaces, and the air inlets must be located within the cargo area.

240. Ventilation of cargo tanks:

The inlets for ventilation within the tanks shall be at least 500 millimetres above the liquid level.

241. Flame screens (Flame Screen):

1. The material of the flame screens shall be made of conductive metallic material, and they must be approved by the Authority.

p.117 (2) High-velocity vents with an exit velocity of at least 30 metres/second for the removal of gas or gas mixtures from the ship may be used as flame arresters if they are approved by the Authority for such a purpose.

242. Level-indicating equipment and sampling equipment:

1. The type and design of the equipment shall depend on the type of the ship and of the cargo carried, and their nature shall be as follows, namely:

a. access equipment (sighting pipe, observation access);

b. equipment with limited access angle (failing cover or flaps);

c. closed equipment (observation glass, floats, pneumatic system, etc.); and

d. sounding pipes, observation access ways.

2. Sounding pipes shall be provided in the pump room, cofferdams and void spaces, and the sounding pipes shall be led through the deck in an oil-tight manner, and they shall terminate well above the deck, at a height of at least 450 millimetres from the deck.

3. Observation plates with sight glasses shall be permanently fitted on the deck covers, and they must be capable of being closed by watertight covers or flaps.

4. The type testing of closed liquid-level measuring equipment shall be subject to approval by the Authority.

243. Sampling:

The design of the equipment for taking samples from the tanks shall ensure that gas mixtures cannot escape during sampling.

244. Prevention of sparks emitted from engine and boiler exhausts (Spark Arrester):

1. The exhaust lines of diesel engines shall be connected with spark arresters.

2. Spark-emission-preventing means (Spark Arrester) shall be fitted to the funnels of boilers and (kitchen) galley stoves.

p.118 ### 245. Engine installation:

1. The main propelling engines shall be installed in a specially enclosed room outside the cargo space limits.

2. Engine spaces above the deck for keeping propulsion engines shall be considered as special spaces for the purposes of such directions.

3. The engine room and casing shall be completely fire-insulated from the surroundings by means of steel bulkheads and doors.

4. The air inlets shall be fitted at the highest position practicable and fitted with fire-resisting closures (closur).

5. In the case of propulsion installations installed on deck, i.e. rudder propellers, special engine chamber arrangements may be adopted if the driving engines remain stopped during loading and unloading of cargo.

Chapter 7

246. Vessels which shall be surveyed and inspected:

Subject to the provisions of the Ordinance, all registered vessels, boats and floating equipment shall be surveyed and inspected, and non-registered vessels, boats and floating equipment shall be surveyed and inspected if there is any such reason to suspect, regarding their size and type, that they are fit for registration.

Provided that vessels, boats and floating equipment kept laid off (Lay off) shall not need to be surveyed if the relevant Engineer and Ship Surveyor has been informed thereof in writing.

247. Responsibility for having survey done:

The owner or master shall, paying the prescribed fee for the survey, make a written request 1 month before the expiry of the period, and they shall ensure that the prescribed date for carrying out the various kinds of survey has not been exceeded. A separate fee shall be paid for each survey.

248. Kinds of survey:

Surveys shall be of the following kinds, namely:

a. annual survey;

b. dry-docking survey;

p.119 (c) special survey;

d. new construction survey;

e. initial survey;

f. conversion survey;

g. damage and repair survey;

h. laying-up and recommissioning survey; and

i. unscheduled survey.

249. Scope of inspection:

1. During inspection, the certificate documents of the vessels, safety equipment, navigation lights, day signals and the certificates of the crew shall be examined.

2. During the survey the following matters also shall be examined, namely:

a. the location of the life-saving appliances and equipment and the fire safety equipment, including their condition of ready availability for easy access and use;

b. marking of the life-saving appliances and equipment;

c. the absence of obstructions in the exit routes and the open or structural access ways which may be used as emergency escape routes; and

d. testing the navigation lights and searchlights and verifying the day signals.

3. If any valid notable statement or recommendation is found on the vessel, it shall be recorded in the remarks section of the “Survey Report” form of the report.

250. Annual surveys:

1. The internationally specified periodic “class”, load line and safety equipment [surveys] shall be co-ordinated at the time of the annual survey.

2. The annual survey shall be conducted in the manner described below, namely:

a. examining whether the number of crew has been fixed in accordance with the “Minimum Safe Manning Certificate”;

b. conducting the annual survey up to the report form on the basis of the “Record of Equipment”;

c. keeping the vessel as far as practicable in an empty condition during the survey so that the Engineer and Ship Surveyor can easily enter to examine the internal scantlings (Scantling) and the “tank tops”;

p.120 (d) paying special attention at the time of carrying out the survey to the location of the life-saving appliances and equipment and the fire safety equipment, including their condition of ready availability for easy access and use;

e. marking of the life-saving appliances and equipment;

f. the absence of obstructions in the exit routes and open or structural access ways which may be used as emergency escape routes;

g. air pipes and intake passages and the means of closing them;

h. weathertight doors and hatch covers firmly secured;

i. the fire screens located in the ventilation or air pipes of oil tanks;

j. fire and bilge pumps;

k. navigation lights, searchlight, whistle or horn;

l. communication between the wheel house engine and the wheel house emergency steering position;

m. steering gear;

n. emergency source of electricity;

o. testing the quick-closing valves of fuel oil by operating them;

p. starting the main engines and auxiliary engines and running them on trial; and

q. examining the anchoring and [mooring] equipment as far as practicable, without ranging the anchors or chains.

3. The survey certificate valid for one year shall be endorsed (endorsed), and the annual survey report shall be submitted at the survey office of the Inland Water Administration (ISA).

251. Dry-docking survey:

1. When the ship, boat or floating equipment is brought into dock or hauled up on a slipway for examining the underwater parts from outside and the survey is carried out, at that time the underwater hull, the forward stem post, the sea-water intake components in the after part,

p.121 the discharge openings, rudders, propellers and propeller shafts shall be examined from outside.

2. The stern bushes and rudder bearings shall be measured.

3. The “Dry Dock Survey Report” form shall be submitted at the office of the Inland Water Administration (ISA) and the survey certificate shall be got endorsed (endorsed).

4. For the operational tests of the ship, the driving survey with the ship in the afloat condition and the dock survey shall be conducted together with the annual survey.

252. Special survey:

1. Special Survey-1 shall be conducted by adding the following matters as an extension of the annual and dry-dock survey, namely:

a. before painting, the underwater [hull], the open cargo holds, the cargo tanks, the fore and aft peak parts shall be cleaned and examined, and if the Surveyor has reason to suspect the presence of wastage, he may request thickness measurement;

b. pressure testing the tanks or verifying them by filling;

c. partially removing the covering of the tank top and the coverings above the floors of the cargo holds so that the Engineer and Ship Surveyor can examine the upper part of the tank and the bottom scantlings;

d. ranging and cleaning and examining the anchor chains and anchors, and also cleaning and examining the chain locker;

e. testing the derricks of the cargo gear by means of a minimum 25% overload and the SWL being supported;

f. examining the main engine, auxiliary engines and the holes [bearings] of the deck equipment;

g. opening up the main engine including valves and injectors, the cylinder covers and at least 50% of the pistons together with their corresponding bearings;

p.122 (h) open at least 50% of the main bearings of the main engine, dismantle them piece by piece, clean and examine them, and examine as far as possible the camshaft drive, gear couplings and thrust bearings;

i. run all pumps and check them;

j. open and examine the sea-chests and the ship’s-side valves, and where necessary make the sea-chests fully or properly conform with the river bed;

k. examine the pipelines and their fittings and, as far as possible, carry out an operational test;

l. overhaul and test the installation of the enclosures of all electrical equipment, the electrical control panels and switch-boards, as necessary for repair or testing according to the specifications;

m. clean, paint and varnish the generators and electrical motors and test them;

n. examine the cable wires and the bulkhead and deck penetrations, as far as practicable;

o. examine the installation of the accumulators and all accommodation arrangements; approve and endorse the pump rooms, the paint, the accumulator and pump valves as being explosion-proof.

2. The measures described in sub-rule (1) are the minimum required for Special Survey No. 1, and if the Engineer and Ship Surveyor finds defects or has doubts about any of them, he may recommend that the survey be extended.

3. All damage and defects found during the survey, and all items of unfinished work noted, shall be repaired or rectified and the vessel shall be kept clean.

4. An interim survey certificate valid for a period not exceeding 2 months may be issued, and a special survey report shall be submitted to the survey office.

p.123 (5) The principal repairs and alterations carried out during the special survey shall be reported separately.

253. Special Survey:

Special Survey No. 2 shall be carried out to the extent of Special Survey No. 1 with the following modifications, namely:—

a. all water tanks within the hull shall be cleaned and examined internally;

b. the propeller shaft shall be withdrawn for examination of the stern bush, bearing glands, keyway and taper, and shall be checked by NDT (Non-destructive) means to see whether it has cracked;

c. all pistons, connecting rods and main bearing parts of the main engine or engines shall be taken out or dismantled, and at least 50% of the liners shall be withdrawn for examination and measurement, and the crankshafts shall be [checked for] alignment; and

d. pressure vessels pressurised at 4 bar or more shall be pressure tested by applying + 50% over-pressure, and the pressure of the CO2 bottles shall also be checked.

254. Special Survey No. 3:

Special Survey No. 3 shall be carried out to the extent of Special Survey No. 2 together with the following additions, namely:

a. all tanks shall be cleaned, examined internally and pressure tested, and if the pressure tests are carried out by means of air the pressure shall be 0.2 bar;

b. if the vessel is of single-bottom construction, the top covers of all tanks or the floor coverings above them shall be removed 100%;

c. the linings of portholes or windows shall be removed for inspection of the condition of the hidden structures;

p.124 (d) the rudder and rudder stock shall be dismantled and opened for examination;

e. the keyways and trepans shall be checked by NDT (Non-Destructive Test) to see whether they have cracked;

f. all pipes shall be pressure tested;

g. the structural members and the anchor chains shall be measured, and if it is suspected that the anchors have become worn and corroded they shall be weighed.

255. New vessel / construction survey:

1. For approval in accordance with the Construction Rules for inland steel-built ships and the Stability Rules for inland ships, these shall be sent to the office of the Inland Shipping Administration (ISA).

2. When carrying out the construction survey of a new vessel, the Engineer and Ship Surveyor shall, by inspection and testing, ensure that the vessel is built in accordance with the approved plans and design with good workmanship and quality.

3. All installations and equipment shall be examined and verified.

4. Special attention shall be paid to the following matters, namely:

a. the welds of the underwater hull, plates and tanks;

b. checking the welds by pressure test or hose test;

c. requesting X-ray, ultrasonic or penetration tests before use, by the Engineer and Ship Surveyor;

d. checking and verifying the tightness of the watertight or weathertight doors and hatches and the arrangements for closing them;

e. fitting flame-screens on the openings of air pipes and oil tanks;

f. also verifying on deck the operation of fire flaps and quick-closing valves;

g. testing the ventilation of the accumulator lockers and pump rooms;

p.125 (h) checking that the escape routes and emergency escape routes are free of all kinds of obstruction;

i. examining the installation of the life-saving appliances and fire-fighting equipment and their easy accessibility and readiness for use; and

j. testing all pipe systems under pressure in operation, and testing the pressure vessels at 50% over-pressure.

5. A sea trial covering the following matters:

a. speed test at 50% and 100% engine power;

b. test of astern running capability;

c. test by operating the anchor;

d. turning test or ship-turning test to starboard and to port, including measurement of the turning circle;

e. test of the ability to stop instantly to avoid collision, and measurement of the distance; and

f. engine running test at 25% power for 1 hour, at 50% power for 1 hour, at 75% power for 1 hour, at 100% power for 2 hours and at 110% power for 5 hours.

6. When required, an inclining test shall be carried out and the freeboard mark shall be confirmed.

7. Floating equipment shall be tested by operation, and if there is any equipment on the box it shall be examined and verified.

8. The report and protocol of the necessary verifications and operational tests shall be prepared and sent to the office of the Inland Shipping Administration (ISA).

256. Initial survey:

1. An initial survey to the extent of an annual survey shall be carried out on newly built ships and on previously used ships brought from abroad, and in the case of old ships the Engineer and Ship Surveyor shall complete and hand over to the ship the following interim certificates and documents, namely:

a. record of equipment;

b. freeboard certificate;

p.126 (c) safe minimum manning certificate.

2. The Engineer and Ship Surveyor shall supervise the work of permanently engraving the register number.

3. The Engineer and Ship Surveyor shall send two sets of copies of the documents to the office of the Inland Shipping Administration (ISA).

4. The due dates of all surveys shall be reckoned from the date of the initial survey.

257. Conversion survey:

1. If any vessel is converted or rebuilt to such an extent that the principal particulars of the vessel, such as length, breadth, draught, depth, carrying capacity or engine power, are altered, the said ship, boat or floating equipment shall be surveyed as a newly built vessel.

2. After completion of the rebuilding or conversion work, the due date of the survey shall be fixed from a new date of putting the vessel concerned into service.

258. Damage and repair survey:

1. When any registered ship, boat or floating equipment is damaged, the master or the owner or their representative shall request an Engineer and Ship Surveyor to attend, and in that case the Engineer and Ship Surveyor shall examine the damage and defects and shall state how much repair is to be carried out, and at the same time the Engineer and Ship Surveyor shall also approve the method of repair.

2. The Engineer and Ship Surveyor shall supervise the repair work and carry out the necessary tests and the verification and comparative tests.

3. If the damage or defects are of a minor nature and do not affect the safety, seaworthiness or operation of the ship, boat or floating equipment, the said repair work may be deferred to a later date prescribed by the Engineer and Ship Surveyor.

4. The Engineer and Ship Surveyor may also mention the imposition of restrictions on the navigation, cargo or number of passengers, and in that case the Engineer and Ship Surveyor shall state the final date relating to the repair and to the navigation, cargo carriage and number of passengers, and that letter of recommendation shall be kept on board the vessel and shown to the inspectors when required.

p.127 (5) The Engineer and Ship Surveyor shall be given a written report comprising the damage or defects found, the repairs carried out and the recommendations, if any.

259. Lay-up and recommissioning survey:

1. If any registered ship, boat or floating equipment is permanently laid up for a period of at least 6 months, the owner shall inform the Registrar of the Inland Shipping Administration (ISA) of this in writing.

2. If any survey has become overdue, the said overdue survey shall be completed before lay-up.

3. When any ship, boat or floating equipment is to be brought back into service, its survey shall be carried out in the following manner, namely:

a. lay-up period less than 1 year: to the extent of the last expired survey, but at least to the extent of an annual survey;

b. lay-up period 1 to 2 years: to the extent of the last expired survey, but at least to the extent of an annual and dry dock survey;

c. lay-up period more than 2 years: to the extent of the next special survey reckoned from the date of construction of the ship, boat or floating equipment, and the new due survey date shall be reckoned from the survey date.

260. Unscheduled surveys:

1. When such a suspicion arises regarding any registered vessel that the ship's manning is not proper, that safety equipment is missing, that damage has occurred, or that the ship is not fit for river or sea voyage, unscheduled surveys may be carried out on that ship.

2. When such a suspicion arises regarding unregistered vessels that, on the basis of size or engine power, they ought to be registered, unscheduled surveys may be carried out on them.

261. Due dates of survey inspections:

1. The date on which the due survey is to be carried out, which shall be reckoned as the due date from the date of the initial survey.

p.128 (2) If any major conversion or rebuilding is carried out, the new due date shall be reckoned from the date of completion of the said work.

3. If the ship, boat or floating equipment is kept laid up for more than 2 years, the new due date shall be reckoned from the date of recommissioning.

262. Inspection:

1. Inspections shall be unscheduled and may be carried out at any time.

2. In the case of dumb barges, only a dry dock and special survey every three years; and it shall always be carried out whenever defects or neglect are suspected.

263. Annual survey:

1. An annual survey shall be carried out every year within a period from 2 months before to 2 months after the due date.

2. In the case of dumb barges, only a dry dock and special survey shall be carried out every three years.

264. Dry dock survey:

A dry dock survey shall be carried out in the case of passenger launches and tankers every second year, and in the case of dry-cargo ships, dumb barges and other floating equipment every fourth year, within a period of 2 months before and after the due date.

265. Special survey:

1. A special survey shall be carried out once every fourth year, and that survey may be carried out 3 months in advance and shall be completed by the due date.

2. When the ship, boat or floating equipment is 4 years old, the said survey shall be at least to the extent of Special Survey No. 1; when 8 years old, to the extent of No. 2; and when 12, 16, 20 etc. years old, to the extent of Special Survey No. 3.

266. Damage and repair survey:

When any damage occurs, or repair work on the hull or machinery items or on structures employed for the safety of the crew, passengers or cargo p.129 is carried out, it shall always be surveyed and repaired to the satisfaction of the Surveyor.

Jurisdiction and powers of the Engineer and Ship Surveyor, and responsibilities and duties of the Inspector

267. Inspections:

1. Unscheduled inspections shall be carried out by the Inspectors, and during inspection the Inspectors shall check whether the route permit, freight rates and time-table are recorded, and in addition they shall also supervise that overloading does not occur.

2. If the Inspector finds any gross negligence, he shall immediately submit a report to the Field Station Engineer and Ship Surveyor and, through the head office of the “Inland Shipping Safety Administration” (ISA), for the purpose of forwarding it to the Marine Court.

268. Surveys:

1. All surveys shall be carried out by the Engineer and Ship Surveyors.

2. If a vessel is classed with any recognised classification society, the surveyor of the class concerned shall carry out the various surveys concerned and submit a report to the office of the “Inland Shipping Safety Administration” (ISA).

269. Responsibilities of the Engineer and Ship Surveyors:

1. The Engineer and Ship Surveyors shall be responsible for surveying all existing machinery, equipment, manning determination and documents, at least to the extent of the actual form or forms and in accordance with the requirements of these Rules.

2. If damaged parts, defects, absences or neglect are found, steps shall be taken to have the owner carry out the repair work, install the missing equipment, make corrections, etc., and the repair and correction work etc. shall be approved to the satisfaction of the Surveyor.

270. Inspector:

1. When the Inspector suspects any kind of neglect he shall carry out an unscheduled inspection, and shall pay special attention to the supervision of overloading before and after the main defects.

p.130 (2) Where necessary the Inspectors shall inspect jointly with the Inspectors of the BIWTA, so that cargo is not stowed in or on the spaces for passengers.

271. Documentation:

1. The Inspectors shall prepare documents of all inspections carried out and shall send the report for filing to the survey office of the “Inland Shipping Safety Administration” (ISA), and shall keep one copy of it on file at the survey station.

2. The Inspectors shall keep a note diary for all inspections carried out in the course of inspection work. For record-keeping the said diary shall be sent to the survey office of the “Inland Shipping Safety Administration” (ISA).

272. Powers of the Engineer and Ship Surveyors:

1. The Engineer and Ship Surveyors shall have the power to request the repair, completion and correction of damage found, defective equipment, missing equipment, safety equipment not kept in a proper manner, obstructions located in escape routes and emergency escape routes, and other neglect.

2. If the damage and defects are of a minor nature, and if they do not affect the operation of the ship, boat or floating equipment in respect of the safety of cargo and crew, seaworthiness or operation, the work of repair and of rectifying the defects in equipment may be deferred until a subsequent date stated by the Engineer and Ship Surveyor, but it may not be later than the next dry dock survey.

3. The Engineer and Ship Surveyor may also impose restrictions on the navigation of vessels, such as — if there are no navigation lights, and in accordance with the Rules, navigation only during daytime, etc.

4. If the damage or defects are of such extent that they affect the operation of the ship, boat and floating equipment in respect of the safety of passengers, cargo and crew, seaworthiness or operation, their navigation shall be stopped, and they shall not be permitted to navigate until the repair and defect-rectification work has been carried out to the satisfaction of the Surveyor.

p.131 (5) If it is necessary for the repair work to take the vessel to a workshop or dock, the Engineer and Ship Surveyor may grant a special permit for the ship, boat or floating equipment to proceed, or to be towed, to the nearest workshop or repair yard:

Provided that he shall be satisfied regarding the safety arrangements necessary for such a voyage.

6. When the navigation of any vessel is to be stopped, the Engineer and Ship Surveyor may inform the port or gate authority and the river police of this, and in such case may bring the “Certificate of Registration” and the survey certificate ashore. Where necessary he may hand over the said vessel to the law-enforcing agency.

7. The defects on account of which the navigation of a ship, boat or floating equipment shall be stopped until they are rectified are as follows, namely:

a. lack of life-saving appliances and equipment and their unsatisfactory stowage;

b. escape routes and emergency escape routes, including the access routes of the hull;

c. navigation at night without proper navigation lights;

d. the structure or the watertight bulkheads are leaking; and

e. if the propulsion or steering arrangements are damaged.

273. Inspectors:

1. The Inspectors shall have the power to request the completion or rectification of damage found, absences, improper assembly of equipment, obstruction of escape and emergency escape routes, and other neglect.

2. If the defects are of a minor nature and do not affect the safety of passengers, cargo and crew or the seaworthiness of the ship, boat or floating equipment, they may be deferred until a date stated by the Inspector, but the period of deferment may not exceed one month.

3. The Inspector may impose a restriction to the effect that, if the navigation lights are not working and are not in accordance with the rules, the vessel may navigate only during daytime.

p.132 (4) If the defects are such that they would affect the safety of passengers, cargo or crew, or the seaworthiness or operation of the ship, boat or floating equipment, the Inspector may stop the navigation of the vessels and may refrain from granting permission to navigate until the defects are removed.

5. When the navigation of any vessel is stopped by the Inspector, it shall be reported without delay to the “Inland Shipping Safety Administration” (ISA) Field Station Engineer and Ship Surveyor, the port authorities and the river police, and in such case the certificate of registration and the survey certificate shall be brought ashore.

6. The defects on account of which the navigation of a ship, boat or floating equipment shall be stopped until the defects are removed are as follows, namely:

a. lack of life-saving appliances and fire-fighting equipment and their unsatisfactory maintenance;

b. obstruction of escape routes and escape routes, including the access routes of the hull;

c. overloading;

d. keeping cargo in or on the passenger spaces; and

e. navigation at night without proper navigation lights.

274. Drawings and documents to be kept on board the vessel:

The following drawings and documents shall be kept on board the vessel, namely:—

a. certificate of registration;

b. survey certificate;

c. freeboard declaration;

d. record of equipment;

e. general arrangement (GA plan), including the safety plan;

f. stability, including various loading conditions;

g. safe minimum manning certificate;

h. master’s licence;

i. engine driver’s licence;

j. certificate of the sukani (helmsman) and, if a tanker, of the pumpman; and

k. check or test certificate of the fire-fighting appliances.

p.133 **Some guiding principles for carrying out surveys**

275. Thickness measurement of structural members:

1. Thickness shall be measured by an approved and sworn person and by means of calibrated equipment.

2. Measurements shall be taken from parts free of rust and contamination, after removing rust and contamination.

3. A maximum reduction of 20% of the thickness of the hull, deck, tank top and bilge plating, and of up to 25% in the case of bulkheads, deck longitudinals, hatch coamings, stringers, keelsons and equivalent structures, shall be acceptable.

4. When the Engineer and Ship Surveyor suspects that the amount of wastage of the structural members may be below the permitted standard, he may at any time request the thickness measurement of the structure concerned.

5. At Special Survey No. 3, in the case of vessels 12 years old, at least all structural plates including the deck, tank top and hatch coamings shall be measured in 2 girth belts around the vessel, and in that case the measuring location shall be at 25% of the length from the midship of the vessel, and there shall be at least 1 measuring point in each of the plates mentioned.

6. At Special Survey No. 4, in the case of vessels 16 years old, in addition to the above measurements, at least all hull plates between the low and deep water lines, keelsons, stringers, deck girders, and the weather deck plating and access openings around the hatches shall be measured.

7. At Special Survey No. 5 and subsequent surveys, in the case of vessels 20 years old or older, a third girth belt around the vessel and the bulkheads of the ballast tanks shall be measured.

276. Anchor chain cables:

Beginning from Special Survey No. 3 and at all subsequent special surveys, in the case of vessels 12 years old or older, the anchor chain cables shall be measured with 2 cross-measurements over the length of each link, and the maximum permissible reduction of the cross-sectional area shall be 20%, and if at the time of measurement the reduction at any point is more than that, the chain shall be replaced entirely.

p.134 ### 277. Anchors:

If the Engineer and Ship Surveyor suspects that the weight of the anchor or anchors has been reduced, he may request that the anchor be weighed, and if the loss of weight is more than 10% of the required weight, the anchor or anchors shall be replaced.

278. Propeller shaft bearing clearances:

As guidance for the renewal of stern tube bearings, the following maximum permissible clearances may be kept, namely:

a. if the clearance is more than 0.8 millimetre plus 0.6% of the shaft diameter, the bearing shall be renewed, or remetalled in the case of white metal lubricated by grease or water, and cast iron;

b. if the bearing is oil lubricated, the seal manufacturer’s recommendations shall be followed; and

c. if rubber bearings are used, the permissible clearance may be increased by 30% calculated from the basic formula.

279. Rudder bearings:

When the clearance for the bearing exceeds 0.6% of the shaft diameter plus 2 millimetres, and for synthetic or fibre bearings 0.5% of the shaft diameter plus 3.0 millimetres, the bearing shall be renewed or remetalled.

280. Pistons and liners:

Pistons or liners shall be renewed in accordance with the manufacturers’ recommendations.

281. Crankshaft deflection:

The engine or engines shall be realigned in accordance with the manufacturer’s recommendations, but the difference shall in no case be more than 0.05 millimetre.

282. Indents:

1. If smooth indents occur in the side and bottom plating of the ship, and the dented parts are not deeper than 5% of the breadth of the indents, they may be disregarded.

p.135 (2) If the said depth is between 5 and 10%, a statement shall be prepared about it, and if the depth is more than 10% it shall be repaired.

3. If the indents are sharp or acute and produce a bulge or hollow in it, they shall be repaired.

283. Measurement of electrical insulation resistance:

The minimum insulation resistance shall be at least 0.5 megohm, and the said measurement shall be taken at 500

volts DC.

284. Fire-fighting equipment:

Every two years the fire-fighting appliances shall be inspected by a company recognised by the Authority, and a check label shall be affixed on the fire-fighting appliances, and the inspecting company shall leave on board the vessel a dated inspection certificate regarding the inspection carried out.

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p.137 # SCHEDULE

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p.139 ### Table 1

Thickness of keel and inner bottom plate:

Frame spacing (millimetres) → / Length of vessel L (metres) ↓	450	500	550	Breadth of keel plate (millimetres)
	Thickness (mm)	Thickness (mm)	Thickness (mm)	
L = 20	4.5	5	5.5	850
L = 30	5	5.5	6	900
L = 40	5.5	6	6.5	950
L = 50	6	6.5	7	1000
L = 60	6.5	7	7.5	1050
L = 70	7	7.5	8	1100

For intermediate values, the next higher value shall be selected for the keel and plate thickness.

Table 2

Thickness of bottom plate:

Frame spacing (millimetres) → / Length of vessel L (metres) ↓	450	500	550
	Thickness (mm)	Thickness (mm)	Thickness (mm)
20	4.5	5.0	5.5
30	5.0	5.0	5.5
40	5.0	5.5	6.0
50	5.5	6.0	6.5
60	6.0	6.5	7.0
70	7.0	7.0	7.5

Table 3

Thickness of bilge plate (millimetres):

Frame spacing (millimetres) → / Length of vessel L (metres) ↓	450	500	550
	Thickness (mm)	Thickness (mm)	Thickness (mm)
20	6	6.5	7
30	6.5	7	7.5
40	7	7.5	8
50	7.5	8	8.5
60	8	8.5	9
70	8.5	9	9.5

p.140 ### Table 4

Thickness of side shell plate (millimetres):

Frame spacing (millimetres) → / Length of vessel L	450	500	550

(metres) ↓			
	Thickness (mm)	Thickness (mm)	Thickness (mm)
20	4.5	5	5.5
30	5	5	5.5
40	5	5.5	6
50	5.5	6	6.5
60	6	6.5	7
70	6	7	7.5

Table 5

Thickness of deck plating in way of hatchways:

Length of vessel (L) up to (metres)	Thickness for transverse framing (millimetres) (Transversal framing thickness)	Thickness for longitudinal framing (millimetres) (Longitudinal framing thickness)
20	5.0	8.5
30	6.0	5.5
40	7.0	5.5
50	7.5	7.0
60	8.5	7.5
70	9.0	8.0

Table 6

Thickness of deck plating overall:

Length of vessel (L) up to (metres)	Thickness for transverse framing (millimetres) (Transversal framing thickness)	Thickness for longitudinal framing (millimetres) (Longitudinal framing thickness)
30	8.0	8.0
40	8.5	8.0
50	5.0	8.5
60	5.5	5.0
70	6.0	5.5

p.141 ### Table 7

Thickness of steel plate stem:

Length of vessel (L) (metres)	Thickness (millimetres)
20	6.0
30	7.0
40	8.0
50	9.0
60	10.0
70	11.0

In the case of non-self-propelled vessels the above values may be reduced by up to 20%.

Table 8

Stem bar, rectangular

Length of vessel (L) (metres)	Breadth (millimetres)	Thickness (millimetres)
20	100	20
30	110	22
40	125	25
50	135	30
60	150	35
70	165	40

Table 9

Stern bar, rectangular

Length of vessel (L) (metres)	Breadth (millimetres)	Thickness (millimetres)
20	100	20
30	110	25
40	125	30
50	135	35
60	150	40
70	165	45

p.142 ### Table 10

Propeller post

Length of vessel (L) (metres)	Breadth (millimetres)	Thickness (millimetres)	Radius (millimetres)
20	150	28	24
30	172	37	28
40	200	45	32
50	225	55	36
60	250	64	40
70	275	75	45

Table 11

Sole piece

Length of vessel (L) (metres)	Breadth (millimetres)	Thickness (millimetres)
20	30	90
30	40	120
40	50	150
50	60	170
60	70	190
70	75	210

(Note: the printed column headings read “Breadth” then “Thickness” as above; a handwritten annotation in the book marks the first value column as “thickness” and the second as “breadth”.)

Table 12

Propeller boss

d	t
30	11
40	14
50	18
60	21
75	27
90	32
105	38
120	42
140	50
160	58

p.143 Provided that, in the said Table 12, d = diameter of the propeller shaft t = thickness of the propeller boss shall be taken.

Table 13

Section modulus (cm³) against breadth of vessel (metres), draught (metres) and frame spacing (millimetres):

Draught (metres)	Frame spacing (mm)	B = 2	B = 3	B = 4	B = 6	B = 8	B = 10	B = 12
1	450	8	18	31	70	123	192	280
1	500	9	20	25 [unclear; possibly 35]	77	138	215	310

1	550	10	22	38	85	150	235	340
1.5	450	12	26	46	105	185	288	415
1.5	500	13	30	52	115	205	320	465
1.5	550	15	32	58	128	225	355	510
2.5 [as printed]	450	16	35	62.5	140	246	385	550
2.5 [as printed]	500	22	48	86	194	342	535	770
2.5 [as printed]	550	24	53	94	212	375	590	845
3	450	23	52	94	208	370	578	830
3	500	26	58	104	232	410	640	925
3	550	28	65	114	255	450	705	1014
3.5	450	27	62	108	242	430	675	968
3.5	500	30	68	120	270	480	750	1076
3.5	550	33	75	115 [unclear; possibly 135]	300	526	825	1183
4	450	30	70	123	278	492	765	1106
4	500	34	78	138	310	548	854	1129 [unclear; possibly 1229]
4	550	38	85	150	340	600	940	1352

p.144 ### Table 14

Breadth of vessel (metres)	Number of keelsons — centre	Number of keelsons — side
up to 5	1	—
5 to 10	1	2
10 to 15	1	4

Table 15

Thickness of web plate and sectional area

Length of vessel (L) (metres)	Thickness of web plate (millimetres)	Sectional area of face plate (cm²)
20	6.5	18
30	7.0	23
40	7.5	28
50	8.0	32
60	8.5	36
70	9.0	42

Table 16

Centre keelson

Length of vessel (L) (metres)	Thickness of web plate (millimetres)	Sectional area of face plate (cm²)	Breadth of face plate (millimetres)
20	7.0	21	200
30	7.5	27	230
40	8.0	33	250
50	8.5	39	275
60	9.0	45	300
70	9.5	51	320

p.145 ### Table 17

Section modulus for side frames

Depth of vessel (metres)	Section modulus (cm³)
1.5	15
2.0	15

2.0	15
2.5	27
3.0	46
3.5	73
4.0	110
4.5	155
5.0	215
5.5	285
6.0	365
6.5	470
7.0	590

Table 18

Length of vessel L (metres)	20	25	30	35	40	45	50	55	60
Maximum spacing of web framing (millimetres) [as printed; values appear to be in metres]	3.0	3.25	3.5	3.75	4.0	4.25	4.5	4.75	5.0

Table 19

Depth of vessel (metres)	Modulus at distance L (cm³): L = 1.5	L = 2	L = 3	L = 4	L = 5
2.5	21	36	82	145	230
3	30	53	118	210	330
3.5	40	71	160	285	450
4	52	93	209	371	580

Here L = the distance (in metres) between the web frames, or between a web frame and a watertight bulkhead.

p.146 ### Table 20

Section modulus for deck beams

S = (metres)	Modulus at distance L (cm³): frame spacing 450 mm	Frame spacing 500 mm	Frame spacing 550 mm
2	6	7	8
3	13	15	16
4	23	26	28
5	36	40	44
6	52	57	63
7	70	78	85
8	91	100	110
10	142	158	174
12	205	228	250

In this Table, S = the distance of the unsupported span (unsupported spacing), whichever is greater, between deck girders or between a girder and the side shell.

Table 21

Section modulus for deck girders

Mean breadth of deck (metres)	Unsupported span (metres) (Unsupported Span M): 1	2	3	4	5	6	7	8
	Section modulus (cm³)							
3	40	110	240	400	615	880	1200	1550

4	47	145	305	530	815	1170	1585	2065
5	55	175	375	655	1015	1455	1975	2575
6	65	210	450	785	1215	1745	2370	2760
7	75	240	520	910	1415	2030	2760	3600
8	85	270	890	1040	1615	2320	3150	4110
9	95	305	665	1170	1815	2610	3545	4625
10	105	335	735	1295	2015	2895	3935	5135

p.147 It is to be noted that Table 21 has in fact been prepared on the basis of a uniform load of 20 kN/m², and if the intended load is greater, the modulus shall be increased in direct proportion.

Table 22

Solid pillars or columns, rule diameter (millimetres)

H, m = height; P, kN = load. Diameter (millimetres):

H (m) ↓ / P (kN) →	10	20	30	40	60	80	10 [as printed; presumably 100]	12 [120]	15 [150]	18 [180]	21 [210]	14 [as printed; presumably 240]	28 [280]	32 [320]
1	21	25	28	30	34	37	40	42	45	47	50	52	55	58
1.2	23	28	31	33	37	40	43	45	48	51	54	55	59	61
1.4	25	30	33	36	40	43	43	46	48	51	56	59	62	64
1.6	27	32	35	38	43	46	49	51	54	58	60	62	65	67
1.8	29	34	37	40	45	48	51	54	57	61	63	65	69	71
2	30	35	39	42	47	51	53	56	59	63	65	68	71	74
2.2	32	37	41	44	49	53	56	59	62	66	68	71	75	77
2.4	33	39	43	46	51	55	58	61	64	68	71	74	77	79
2.6	34	41	44	48	53	57	60	64	67	71	73	76	80	82
2.8	35	42	46	50	55	59	62	66	69	74	76	79	82	85
3	36	44	48	52	57	61	64	68	72	76	79	81	85	88
3.2	38	45	49	53	58	63	66	70	74	78	81	84	87	90
3.4	39	46	50	54	60	65	69	72	76	80	83	86	90	93
3.6	40	47	51	56	61	66	70	73	78	82	85	87	92	95
3.8	41	47	51	56	61	66	70	73	78	82	85	87	92	95
4	42	50	54	59	64	69	73	77	82	86	90	93	97	100

In this Table H shall be taken as the height of the pillar (metres).

p.148 ### Table 23

Tubular (hollow) pillars and columns — rule diameter and thickness, CC

Pillar / column: outside diameter (millimetres) and wall thickness (millimetres), against length of pillar / column (metres):

Outside diameter (mm)	Wall thickness (mm)	< 1	1.5	2	2.5	3	3.5	4	4.5
38.0	2.0	28	16	8	5	4	3	2	2
38.0	4.0	40	20	11	7	5	4	3	2
38.0	5.0	48	23	13	8	6	4	3	3
51.0	2.6	46	35	21	14	9	7	5	4
51.0	4.0	68	51	30	19	13	10	7	6
51.0	5.0	83	60	35	23	16	12	9	7
63.5	2.9	69	59	45	30	21	15	12	9
63.5	4.0	93	79	60	39	27	20	15	12
63.5	5.6	126	106	78	50	35	26	20	16
76.1	2.9	37 [as printed; possibly 97]	78	67	52	36	27	20	16
76.1	4.5	131	118	100	76	53	39	30	23

76.1	5.6	160	144	120	91	63	46	35	28
88.9	3.2	144	106	95	81	64	47	36	29
88.9	5.0	174	162	144	122	94	69	53	42
88.9	5.6	194	180	160	135	104	76	58	46
101.6	3.6	149	141	130	117	100	79	61	48
101.6	5.0	204	193	178	158	134	106	81	64
101.6	6.3	253	239	220	195	164	128	98	78
127.0	4.0	211	204	195	182	167	149	129	105
127.0	5.6	291	282	268	251	229	204	175	142
127.0	6.3	326	315	299	280	256	227	194	157
139.7	4.0	234	228	219	208	194	178	159	138
139.7	5.6	323	314	302	286	267	244	218	188
139.7	7.1	405	394	378	358	333	304	270	232

p.149 Table 23 (continued)

Outside diameter (mm)	Wall thickness (mm)	< 1	1.5	2	2.5	3	3.5	4	4.5
152.4	4.5	288	281	272	261	274	230	211	189
152.4	6.3	398	389	376	360	340	316	289	258
152.4	7.1	446	435	421	402	380	353	322	288
165.1	4.5	313	307	299	288	275	260	242	222
165.1	7.1	486	476	463	446	425	401	373	340
165.1	8.0	544	534	519	499	476	448	416	380

Table 24

Position of the collision watertight bulkhead

Length of vessel (L) (metres)	Distance from the fore side of the stem at the load line, metres (minimum)	Distance from the fore side of the stem at the load line, metres (maximum)
(1)	(2)	(3)
20	1	1.6
30	1.5	2.4
40	2	3.2
50	2.5	4
60	3	4.8
70	3.5	5.6

(Note: the two value columns are printed with identical headings “distance from the fore side of the stem at the load line, metres”; they evidently give the lower and upper limits.)

Table 25

Thickness of collision and after peak watertight bulkheads

h	Plate thickness (mm) at frame spacing 450 mm	500 mm	550 mm	Section modulus of stiffeners (cm³) at frame spacing 450 mm	500 mm	550 mm
1.5	4	4	4	11	12	13
2	4	4	4.5	23	23	28
2.5	4	4.5	4.5	38	42	47

p.150 Table 25 (continued)

h	Plate thickness (mm) at frame spacing 450 mm	500 mm	550 mm	Section modulus of stiffeners (cm³) at frame spacing 450 mm	500 mm	550 mm
3	4.5	4.5	5	64	72	80

3	4.5	4.5	5	64	72	80
3.5	4.5	5	5.5	100	110	120
4	5	5	5.5	150	165	180
4.5	5.5	5.5	6	210	230	255
5	5.5	5.5	6	285	315	345

In this Table h shall be taken as the height of the watertight bulkhead at the centre line (metres).

Table 26

Thickness and section modulus of watertight bulkheads

h	Plate thickness (mm) at frame spacing 450 mm	500 mm	550 mm	Section modulus of stiffeners (cm^3) at frame spacing 450 mm	500 mm	550 mm
1.5	4	4	4	9	10	11
2	4	4	4	18	19	21
2.5	4	4	4	32	35	38
3	4	4	4	52	57	64
3.5	4	4.5	5	85	90	95
4	4.5	4.5	5	120	130	140
4.5	4.5	5	5.5	170	185	200
5	5	5	5.5	230	255	275

In this Table h shall be taken as the height of the watertight bulkhead at the centre line (metres).

Table 27

Thickness of plating, deep tanks

h (metres)	Plating thickness (mm) at frame spacing 450 mm	500 mm	550 mm
1.5	4	4	4
2	4	4	4.5

p.151 Table 27 (continued)

h (metres)	Plating thickness (mm) at frame spacing 450 mm	500 mm	550 mm
2.5	4	4.5	4.5
3	4.5	4.5	5
3.5	4.5	5	5.5
4	5	5	5.5
4.5	5	5.5	6
5	5.5	5.5	6

In this Table h = 0.7 times the vertical distance measured from the lower edge of the plate to a point 2.0 metres above the top of the overflow.

Table 28

Tank stiffeners

h (metres)	Section modulus of stiffeners (cm^3) at frame spacing 450 mm	500 mm	550 mm
1.5	11	12	13
2	23	26	28
2.5	38	42	47
3	64	72	80
3.5	100	110	120
4	150	165	180
4.5	210	230	255
5	285	315	345

In this Table L = the span (metres) between the adjacent supports of the stiffeners, including the length of the connection (metres), and where girders are fitted, L shall be the distance from the heel of the end connection to the first girder, or the distance between the girders.

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Table 29

Section modulus of web frames in the engine room

Depth of vessel (metres)	Frame spacing (metres): 1.8	2	2.2	2.4
(Section modulus in centimetres³)				
1.5	45	47	52	57
2	101	112	123	134
2.5	197	219	241	262
3	341	378	415	453
4	705	785	862	941
4.5	821	910	1000	1090

Table 30

L metres	t millimetres
20	5.0
30	6.0
40	7.0
50	7.5
60	8.0
70	8.5

Where L shall be the length of the vessel and t the thickness of the coaming. The thickness of the hatch coamings of vessels crossing partial bays (estuaries) shall be increased by 10% and rounded to the nearest 0.5 millimetre.

Table 31

B metres	t millimetres
up to 5	6.0
6	6.5
7	7.0
8	7.5
9	8.0

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B metres	t millimetres
10	8.5
11	9.0
12	9.5

In this table B is the breadth or beam of the vessel and t the thickness of the hatch coaming. The thickness of the hatch coamings of vessels crossing partial bays (estuaries) shall be increased by 10% and rounded to the nearest 0.5 millimetre.

Table 32

Section modulus of portable beams (in centimetres³)

bp b	1.0	2.0	3.0	4.0	5.0
(Section modulus in centimetres³)					
1.0	5	7	15	26	40
1.5	5	10	22	38	60
2.0	5	13	30	52	80
2.5	5	16	36	64	100

In this table bp = spacing between the beams (in metres). b = half-breadth of the hatch opening (in metres).

Table 33

Section modulus of hatchway beams forward and aft of the centre line (in centimetres³)

b I	1.5	3	4	5	6	7	8
(Section modulus in centimetres³)							
1	5	16	30	45	65	88	114
2	8	32	58	90	130	175	230
3	12	48	86	135	195	265	345
4	16	65	115	180	260	250	460
5	20	82	145	225	325	475	575
6	24	98	175	270	390	530	690
7	30	115	200	315	450	615	800

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In this table b = half-breadth of the hatch and opening, in metres. I = the spacing between the hatch and the coaming and the cross beams, or between two adjacent cross beams, and, if no cross beam is fitted, between the coamings.

Table 33 (continued)

Section modulus of the adjoining forward and aft hatchway beams

b I	1.5	3	4	5	6	7	8
(Section modulus in centimetres³)							
1	5	8	14	21	32	42	54
2	5	15	27	42	62	84	108
3	6	24	42	64	92	124	162
4	8	32	54	84	122	165	215
5	10	38	68	105	152	206	267
6	12	46	82	126	182	248	325

In this table b = half-breadth of the hatch opening, in metres. I = the spacing between the hatch and the coaming and the cross beams, or between two adjacent cross beams, and, if no cross beam is fitted, between the coamings.

Table 34

Section modulus of hatch cover stiffeners

I S	450	500	550
(Section modulus in centimetres³)			
2	5	5	5
3	5	5	5
4	7	8	9
5	12	13	14
6	17	18	20
7	22	25	27
8	30	34	38

In this table I = unsupported span of the stiffeners, metres. S = unsupported span [sic — spacing] of the stiffeners, metres.

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Table 35

Section modulus of deckhouse outboard stiffeners, centimetres³

L S (centimetres)	450	500	550
2.0	10	11	12
2.2	12	12	14
2.4	14	15	16
2.6	16	18	20
2.8	18	20	22

3.0	20	23	26

In this table L = length of the unsupported span, in metres. S = spacing between the stiffeners, in millimetres.

Table 36

Section modulus of beams and longitudinals for decks of all classes

L S (centimetres)	450	500	550
1.5	6.0	6	7
2	7	8	8
3	12	12	13
4	17	19	20
5	25	27	30
6	33	38	40
7	44	48	54
8	56	62	68
9	70	77	84
10	85	94	104
11	102	113	124
12	121	135	150

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In this table L = unsupported span of the deck beams and longitudinals, measured in metres. S = spacing between the beams and longitudinals, measured in millimetres.

Table 37

Section modulus of deck girders

L Spacing of deck girders, millimetres	2.0	2.4	2.8	3.2
2	16	19	24	26
3	38	46	54	62
4	68	80	91	104
5	105	125	145	165
6	146	175	205	235
8	260	315	370	420
10	405	490	570	650
12	590	700	820	950
14	800	960	1120	1270

In this table L = unsupported span (Span) of the deck girder longitudinals, measured in metres.

Table 38 — Supported rudder | Table 39 — Unsupported rudder

Table 38: A	d	Table 39: A	d
0.25	29	0.25	38
0.50	48	0.5	58
0.75	61	0.75	74
1.0	72	1	87
1.25	82	1.25	100
1.50	92	1.5	110
1.75	101	1.75	120
2.0	109	2	128
2.25	116	2.25	138
2.50	123	2.5	147

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In these tables A = area of the rudder, metres² d = diameter of the rudder stock, in millimetres.

Table 40

A	d

0.25	22
0.5	32
0.75	42
1	50
1.25	55
1.5	60
1.76	64
2	67
2.25	69
2.5	71

Pintle diameter: in this table A shall be the area of the rudder blade in square metres (m^2) and d shall be the diameter of the pintle in millimetres.

Table 41

Diameter of coupling bolts in millimetres

D	Number of bolts: 3	4	5
	(Diameter of coupling bolts, in millimetres)		
30	12	10	9
40	16	14	12
50	20	16	15
60	24	20	18
70	27	24	20
80	30	27	24
90	35	30	26

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Table 41 (continued)

D	3	4	5
100	39	33	30
110	42	36	32
120	48	40	35
130	50	45	38
140	55	48	42

In this table D shall be the diameter of the rudder stock in millimetres.

Table 42

d	Diameter of bolt, millimetres
30	8
40	11
50	14
60	16
70	18
80	22
90	24
100	26
110	30
120	32
130	35
140	38

In this table d = stock diameter.

Table 43

1	1.25 d	0.50 d	0.70 d	0.30 d
2	1.00 d	0.40 d	0.55 d	0.25 d
3	0.90 d	0.30 d	0.50 d	0.20 d

If the main and emergency steering gear operate on the same tiller or quadrant, then in that case the cross-section of the arm shall be increased by 10%.

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Table 44

Sizes of chain, rod, shackles and straining screws

d	I	d1	d2	SWL
30	400	5.0	6.0	250
40	500	5.0	7.0	250
50	600	6.5	10.0	400
60	700	6.5	10.0	400
70	800	8.0	10.0	630
80	900	8.0	12.0	630
90	1000	9.5	13.0	1000
100	1100	11.0	14.0	1250
110	1200	13.0	16.0	1600
120	1300	16.0	18.0	2000
130	1350	16.0	20.0	2000
140	1400	16.0	22.0	2500

In this table d = diameter of the rudder stock in millimetres. I = length of the tiller in millimetres. d1 = nominal diameter of the chain in millimetres. d2 = diameter of the rod in millimetres. SWL = Safe working load of the shackles and straining screws, kg.

Table 45

Anchors and ropes

Equipment Z	Main anchor: total weight, Kg	Tow-line (wire or fibre): breaking load, KN	Mooring lines: number of lines	Breaking load, KN
1	2	3	4	5
30	75	27	2	12
50	90	35	2	15

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Table 45 (continued)

1	2	3	4	5
100	180	60	2	30
150	270	80	2	40
200	360	100	2	50
250	450	115	2	60
300	540	125	2	70
400	670	150	2	85
500	790	170	2	100
600	920	185	2	110
700	1040			
800	1170	215	2	130
900	1280			
1000	1390	235	3	150
1100	1490			
1200	1580			
1300	1670			
1400	1750			
1500	1820	275	4	170
1600	1870			
1700	1930			
1800	1990			

1900	2050			
2000	2110	305	4	185
2200	2200			
2400	2290			
2600	2380			
2800	2470			
3000	2560	330	4	200
3200	2640			
3400	2740			

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1	2	3	4	5
3600	2830			
3800	2920			
4000	3010	355	4	210

Table 46

Anchor chain cable

Stud-link: Chain diameter	Grade (a1): Proof load	Grade (a2): Nominal breaking load	Short-link chain: Chain diameter	Grade (a1): Proof load	Grade (a2): Nominal breaking load
12.5	6700	9400	8	2500	3500
14	8400	11800	9	3000	4200
16	10900	15300	10	4000	5600
17.5	13000	18300	13	6400	9000
19	15300	21500	16	10000	14000
20.5	17800	24900	18	12600	17600
22	20400	28600	20	16000	22400
24	24200	33900	23	20000	28000
26	28300	39700	26	25200	35300
28	32700	45800	28	30000	42000
30	37500	52400	30	34000	47600
32	42500	59800	33	40000	56000
34	47700	66800	36	50000	70000
36	53300	74600	39	56000	78400
38	59200	82800	42	68000	95200
40	65300	91400	45	76000	106000
42	71700	100000	48	84000	118000
44	78400	110000			
46	85300	119900			
48	92600	130000			

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Table 47

Thickness of bulwark (Bulwark) plating

Length of vessel L (metres)	Thickness (millimetres)
20	3
25	3.5
30	4
35	4
40	4.5
45	4.5
50	5
55	5
60	5.5
70	6

Table 48

Thickness of the walls of forward (Bow) chain lockers

Length of vessel (metres)	Thickness (millimetres)
50	5.5
55	6
60	6
65	6.5
70	7

Table 49

Dimensions of V-Strut

D	W	t
25	56	9
40	86	15
60	132	22
75	168	28
90	203	33

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D	W	t
105	234	38
125	278	45
150	334	55

In this table— D = diameter of the propeller shaft, in millimetres. W = breadth of each strut (Strut), in millimetres. t = thickness of each strut, in millimetres.

Table 50

Dimensions of I-Strut

D	W	t
25	80	12
40	130	21
60	190	30
75	240	40
90	290	48
105	340	54
125	400	68
150	480	76

In this table— D = diameter of the propeller shaft, in millimetres. W = breadth of each strut (Strut), in millimetres. t = thickness of each strut, in millimetres.

Table 51

(t1 + t2) / 2	Breadth (b) mm
5.5	20
6 to 7.5	25
8 to 12.0	30

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Table 52

Section modulus together with plating

Modulus	Angle	Bulb	Flat bar	Size of bracket	
1	2	3	4	5	6
			50×5		
5			50×6		

6			50×7		
7			65×6		
9			65×6		
10			60×8	100×6.5	
11		60×4	65×7		
12		60×5	75×6		
13			65×8		
14		60×6	75×7		
16	60×40×5		75×8		
19	60×40×6		75×9		
20		80×5			
21	60×50×5		75×10		
22	60×40×7				
23		80×6	90×8		
25	75×50×5	80×7			
26			90×9		
27	75×55×5		100×8		
29	80×40×6; 65×50×60		90×10		
31			100×9		
32			110×8		
33			90×10		
34		100×6			
35	75×50×7		100×10		

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1	2	3	4	5	6
36			90×12		
37	75×55×7; 80×40×8		110×9	120×6.5	
38		100×7	120×8		
39	80×65×6		100×11		
42		100×8	110×10	130×6.5	
43	90×60×6				
44	75×50×9		120×9		
45	100×50×6				
47	75×55×9		110×11		
49		120×6	120×10	140×6.5	
52	80×65×8		110×12		
54		120×7			
55			120×7		
57	90×60×8		130×10	150×6.5	
58			150×8		
59	90×75×7; 100×50×8				
60		120×80	120×12		
61	100×65×7				
64	800×65×10		140×10	160×6.5	
66	100×65×7	140×68	120×13		
72			140×11		
74	100×50×10	140×7	150×10	170×6.5	
76			120×14		
78	100×65×9				
80			140×12		
82		140×8			
84			160×10		

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1	2	3	4	5	6

88	100×75×9		140×13	180×7	180×50×6.5
92		140×9	150×12		
96	100×65×11		140×14		
130 [sic]	130×65×8			190×7	190×50×6.5
105	100×75×11; 120×80×8	160×7	160×12; 180×10		
110	130×75×8		160×8		
115			160×13	200×7.5	200×50×6.5
120		160×9	140×16		
125	130×65×10		160×14		
130	120×80×10		180×12	21×7.5	210×50×6.5
135	130×75×10		160×15		
140			180×13		
145		180×8	160×16		
150	130×65×12; 150×75×9			220×8	22×50×6.5
155	120×80×12; 139×90×10		180×14		
160		180×9	160×17		
165	130×75×22		180×15	230×18	230×50×7
170			160×18		
175	120×80×14	180×10			
180	150×75×11; 130×90×12		180×16		
185		180×11	200×14	240×8.5	240×50×7
190	150×90×10; 160×80×10		180×17		
200	150×100×10		200×1.5	245×8.5	245×50×7
210		200×9	180×118	250×8.5	250×50×7

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1	2	3	4	5	6
220		210×10	210×16; 210×14		
230	150×90×12; 160×80×12		200×17	160×6	260×50×7.5
240	150×100×12	200×11	220×15		
250	180×90×10	200×11.5	200×18		
260	160×80×14		220×16; 240×14	270×9	270×50×7.5
270			200×19		
280	150×100×14	220×10	200×20; 220×17; 240×15	280×9.5	280×50×8
290	180×90×12				
300	200×100×10		220×18		
310		220×11.5	240×16; 260×14; 260×18	290×9.5	290×55×8
320			220×9		
330	180×90×14		220×20; 240×17; 260×15		
340		240×10			
350			240×18; 260×16	300×14	300×55×8
360	200×100×12				
370		240×11	240×11		
380	250×90×10		260×17	310×10.5	310×55×8.5
390		240×12	240×20		
400			260×18	315×10.5	315×55×8.5
410	200×100×14		280×16		

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1	2	3	4	5	6
420		260×10		320×10.5	320×55×85
430			260×19; 280×17		
450		260×11			
460	250×90×12		260×20; 280×18	330×11	330×60×9

470	200×100×16	260×12			
490			280×19; 300×17	340×11	340×60×9
500		260×13			
520		280×20 [sic]; 300×18			
530	250×90×14	280×11		350×11.5	350×60×9
550			280×21		
560			300×19		
570		280×12			
580			280×22	360×11.5	36×60×9.5
590			300×20; 320×18		
600	250×90×16	280×13		370×12	370×65×9.5
640		280×14; 300×11	300×21; 320×19		

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Modulus	Angle	Bulb	Flat bar	Size of bracket: without flange	With flange
660			300×21; 320×19		
680		300×12	320×20	380×12	380×65×10
700		300×13	300×23		
720			300×21; 320×19		
740		300×14	300×24; 320×22	390×12.5	390×65×10
760			340×20		
780		300×15; 320×12	320×23	400×13	40×70×10.5
800			340×21	400×13	400×70×10.5
820		320×13	320×24		
840			340×22; 360×20		
860		320×14		411×13	410×70×10.5
880			320×25; 340×12		
900		320×15; 340×12			
920			320×26; 340×24	420×13.5	420×75×10.5
940		320×16			
960		340×13	340×25	430×13.5	430×75×11
1000		340×14			
1020			340×26; 360×24		
1040		340×15	380×22		
1080		340×15		440×14	440×808×11
1100			340×28		
1120			360×26	450×14	450×808×11.5
1140			380×24		

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Table 53

Section modulus of T and angle girders built up together with plating

T girder: Section modulus (centimetres³)	Depth of web (millimetres)	Thickness of web (millimetres)	Area of face plate (centimetres²)	Angle girder: Depth (millimetres)	Width of flange (millimetres)	Thickness (millimetres)
1	2	3	4	5	6	7
100	128	5	6.4	179	69	5.5
120	143	5	6.8	195	72	5.5
140	155	5	7.2	208	75	5.5
160	167	5	7.5	221	75	6
180	178	5	7.1	233	78	6
200	189	5	8.3	245	81	6
220	199	5.5	8.3	256	81	6.5
240	209	5.5	8.6	267	84	6.5
260	219	5.5	8.9	277	86	6.5

280	228	5.5	9.2	286	88	6.5
300	238	5.5	9.5	295	88	7
320	247	5.5	9.8	304	89	7
340	255	5.5	9.8	304	89	7
360	263	6	10.1	321	93	7
380	271	6	10.3	329	95	7
400	279	6	10.6	377	95	7.5
420	287	6	10.8	345	96	7.5
440	294	6	11.0	352	97	7.5
460	301	6	11.2	360	98	7.5
480	308	6	11.4	367	99	7.5
500	315	6	11.7	373	99	8
550	333	6.5	11.7	389	101	8
600	350	6.5	11.1	404	104	8
650	366	6.5	12.5	419	106	8
700	381	6.5	12.9	425	106	8.5

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1	2	3	4	5	6	7
750	395	6.5	13.3	447	108	8.5
800	409	6.5	13.8	462	111	8.5
850	423	7	13.8	476	113	8.5
900	436	7	14.1	489	113	9
950	449	7	14.4	502	115	9
1000	462	7	14.7	513	117	9
1050	475	7	15.0	525	119	9
1100	487	7	15.3	537	120	9
1150	499	7	15.6	548	120	9.5
1200	511	7	16	559	122	9.5
1250	522	7.5	16	570	123	9.5
1300	533	7.5	16.3	580	124	9.5
1350	544	7.5	16.6	590	125	9.5
1400	555	7.5	16.9	600	125	10
1450	465 [sic]	7.5	17.2	610	126	10
1500	575	7.5	17.5	619	128	10
1600	575	7.5	17.8	638	130	10
1700	615	7.5	18.2	673	132	10.5
1800	635	8	18.2	673	132	10.5
1900	654	8	18.6	690	135	10.5
200 [sic — 2000]	673	8	19.0	708	138	10.5
2100	691	8	19.4	724	141	10.5
2200	709	8	19.7	740	141	11
2300	727	8	20.1	755	142	11
2400	744	8	20.5	770	143	11
2500	761	8.5	20.5	785	145	11
2600	777	8.5	20.8	799	147	11
2700	793	8.5	21.1	813	147	11.5
2800	809	8.5	21.1	813	148	11.5
2900	825	8.5	21.7	839	149	11.5

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1	2	3	4	5	6	7
3000	841	8.5	22.0	852	151	11.5
3100	856	8.5	22.3	864	153	11.5
3200	871	8.5	22.6	877	153	12
3300	886	8.5	22.9	889	154	12
3400	900	8.5	23.1	901	155	12

3500	914	9	23.2	913	156	12
3600	928	9	23.5	925	157	12
3700	942	9	23.8	937	159	12
3800	956	9	24.1	948	161	12
3900	969	9	24.4	956	163	12

Table 54

Steel pipe

Outside diameter of pipe (millimetres)	Wall thickness (millimetres)
up to 17.2	1.8
from 21.3	2
from 26.9	2.3
from 33.7	2.6
from 60.3	2.9
from 88.9	3.2
from 114.3	3.6
from 139.7	4
from 168.3	4.5
from 193.7	5.4
from 219.1	5.9

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Table 55

Copper pipe

Outside diameter of pipe (millimetres)	Wall thickness (millimetres)
up to 15	1.5
from 28	2
from 35	2.5
from 133	3

Table 56

Copper alloy pipe

Outside diameter of pipe (millimetres)	Wall thickness (millimetres)
up to 18	1
from 20	1.2
from 57	1.5
from 90	2
from 133	2.5
from 194	3
from 324	3.5

Table 57

Size of main bilge pipe

L	d
30	50
40	55
50	62
60	70
70	75

In this table: L = length of the vessel measured in metres. d = internal diameter of the main bilge line measured in millimetres.

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Table 58

Insulating materials with maximum rated conductor temperature

Material	Temperature (°C)
Elastomeric compounds	
Natural or synthetic rubber (for general service)	60
Butyl rubber	80
Ethylene propylene rubber	85
Cross-linked polyethylene	85
Silicone rubber	95
Thermoplastic compounds	
Polyvinyl chloride (for general service)	60
Polyvinyl chloride (heat-resistant quality)	75
Other materials	
Minerals	95

Table 59

Current rating for nominal cross-sections (N.C.) of general-service rubber and PVC (based on an ambient temperature of 40° degrees Celsius)

(N.C.) millimetres²	Single core, amperes	2 cores, amperes	3 or 4 cores, amperes
1	9	8	7
1.5	14	12	9
2.5	20	16	14
4	25	22	17
6	33	29	23
10	45	39	32
16	62	53	43
25	82	69	56
35	100	85	70
50	121	102	84
60	138	115	97

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(N.C.) millimetres²	Single core, amperes	2 cores, amperes	3 or 4 cores, amperes
70	155	132	108
95	190	161	132
120	219	185	153
150	253	215	177

Table 60

Heat-resistant PVC

Current rating for nominal cross-sections (N.C.) (based on an ambient temperature of 40° degrees Celsius).

(N.C.) millimetres²	Single core, amperes	2 cores, amperes	3 or 4 cores, amperes
1	14	12	10
1.5	18	15	13
2.5	26	22	18
4	35	29	24
6	44	38	31
10	62	53	43
16	82	69	57
25	108	93	77
35	135	113	95
50	162	137	113
60	189	162	135
70	205	174	144
95	248	211	174
120	292	248	205

120	292	248	205
150	335	286	232

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Table 61

Butyl

Current rating for nominal cross-sections (N.C.) (based on an ambient temperature of 40° degrees Celsius).

(N.C.) millimetres²	Single core, amperes	2 cores, amperes	3 or 4 cores, amperes
1	16	13	11
1.5	20	17	14
2.5	28	24	29 [sic]
4	37	32	27
6	48	41	34
10	67	57	47
16	90	76	63
25	118	96	82
35	150	127	105
50	177	150	123
60	188	171	139
70	230	166	161
95	278	235	195
120	321	273	225
150	364	310	255

Table 62

Silicone rubber, mineral

Current rating for nominal cross-sections (N.C.) (based on an ambient temperature of 40° degrees Celsius).

(N.C.) millimetres²	Single core, amperes	2 cores, amperes	3 or 4 cores, amperes
1	21	18	15
1.5	25	21	18
2.5	34	28	23

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(N.C.) millimetres²	Single core, amperes	2 cores, amperes	3 or 4 cores, amperes
4	44	38	30
6	58	49	36
10	79	67	55
16	105	89	74
25	142	121	99
35	173	147	121
50	210	185	152
60	242	205	168
70	268	228	188
95	325	278	226
120	378	320	263
150	431	368	301

Table 63

Ethylene propylene rubber, cross-linked polyethylene.

Current rating for nominal cross-sections (N.C.) (based on an ambient temperature of 40° degrees Celsius).

(N.C.) millimetres²	Single core, amperes	2 cores, amperes	3 or 4 cores, amperes
1	17	14	12
1.5	21	18	15
2.5	30	24	20

2.5	30	24	20
4	40	34	28
6	51	42	35
10	71	60	50
16	95	91	67
25	127	108	89
35	164	127	106
50	191	164	133

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(N.C.) millimetres²	Single core, amperes	2 cores, amperes	3 or 4 cores, amperes
60	212	180	148
70	239	202	166
95	292	248	205
120	339	294	237
150	387	329	270

Table 64

Insulation	Correction factor for ambient temperature: 40° C	45° C	50° C	55° C
Rubber or PVC (general purposes)	1.00	0.87	0.71	–
PVC (heat-resistant quality)	1.00	0.93	0.84	0.76
Butyl rubber	1.00	0.93	0.84	0.80
Ethylene propylene rubber, cross-linked polyethylene	1.00	0.94	0.89	0.82
Mineral, silicone rubber	1.00	0.95	0.90	0.85

Table 65

Correction factor	Half-hour rating: with metal sheath, mm²	Without metal sheath, mm²	One-hour rating: with metal sheath, mm²	Without metal sheath, mm²
1	up to 20	up to 75	up to 67	up to 230
1.1	21—40	76—125	68—170	231—400
1.15	41—65	126—180	171—290	401—600
1.2	66—95	181—250	291—430	

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Correction factor	Half-hour rating: with metal sheath, mm²	Without metal sheath, mm²	One-hour rating: with metal sheath, mm²	Without metal sheath, mm²
1.25	96—130	251—320	431—600	
1.3	131—170	321—400		
1.35	171—220	401—500		

Table 66

Maximum permissible stresses of longitudinal continuous members, KN/mm²

Item	Local bending stress	Combined (see Note 1)	Shear stress	Equivalent stress
Bottom girders	108 (11)	177 (18)	83 (8.5)	188 (19.2)
Bottom longitudinals, inner bottom longitudinals, side open longitudinals	137 (14)	177 (18)	83 (8.5)	188 (19.2)

Notes:

1. Combined stress shall mean the sum of the bending stress and the stresses due to local loads.
2. The figures in brackets are expressed in KN/mm² units.

Table 67

Maximum permissible stresses of local members, KN/mm²

Item	Bending stress	Shear stress	Equivalent stress
Floors, intercostal girders, bottom and deck transverses	124 (12.6)	83 (8.5)	177 (18)
Frames	133 (11.5)	83 (8.5)	172 (17.5)
Deck beams	137 (14)	83 (8.5)	186 (19)

The figures in brackets are expressed in KN/mm² units.

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p.181 # Inland Ships (Passenger) Rules, 2001

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p.185 ### 1. Short title

These Rules shall be called the Inland Ships (Passenger) Rules, 2001.

2. Definitions

1. Unless there is anything repugnant in the subject or context, in these Rules—

a. “Ordinance” means The Inland Shipping Ordinance, 1976 (LXXII of 1976);

b. “inland ship” and “ship” mean an “inland ship” as defined in Section 2(e) of the Ordinance;

c. “inland water” means ‘inland water’ as defined in Section 2(f) of the Ordinance;

d. “new ship” means any ship registered after these Rules come into force;

e. “BIWTA” means the Bangladesh Inland Water Transport Authority;

f. “BIWTC” means the Bangladesh Inland Water Transport Corporation;

g. “passenger” means any person on board an inland ship other than the master, officers and crew of the ship; provided that a child under one year of age shall not be included among the passengers.

3. Application

Unless otherwise provided, these Rules shall apply to all inland ships.

4. Passenger space

All spaces designated for passengers, that is to say, the passenger cabins, toilets, saloons, corridors, stairways and reserved open deck of a ship, shall be deemed to be passenger space.

5. Restrictions relating to passenger space

While the ship is under way, passengers shall not be allowed to enter the following places, namely:

p.186 (a) all spaces below the bulkhead deck;

b. the forward part of the deck house, or the structure above the weather deck in the mooring and anchor space;

c. the mooring deck at the after end of the ship, if any;

d. all decks which do not have an enclosure or railing of at least 0.9 metre height;

e. the top deck above the wheelhouse;

f. the wheelhouse, engine room and all similar spaces designated for the navigation of the ship;

g. spaces designated for the crew;

h. spaces designated for the provision of services;

i. other prohibited spaces.

6. Calculation of open deck space for passengers

1. In calculating open deck space for passengers, only covered deck areas shall be taken into account.

2. The following passenger areas shall not be included in the calculation of open space, namely:

a. passenger cabins and corridors;

b. areas containing furniture fixed permanently or otherwise to anything;

c. passengers’ sanitary spaces;

d. spaces designated for vendors of necessary goods;

e. stairways and the spaces beneath and around stairways;

f. hatch opening areas;

g. spaces adjacent to the engine room;

h. any other space which, in the opinion of the Surveyor, is not suitable for passengers.

3. 15% of the breadth of the open deck space shall be kept reserved as a passageway.

4. 5% of the open spaces shall be kept reserved for luggage.

7. Determination of the passenger-carrying capacity of an inland ship

The passenger-carrying capacity of an inland ship shall be determined in the following manner, namely:

p.187 (a) in all classes, one passenger for each chair and bunk;

b. in the case of benches:

c. for each third-class or inter-class passenger, a seat of 0.5 metre length;

ii. for each first- and second-class passenger, a seat of 0.6 metre length;

iii. where there is no designated space for sitting, the passenger-carrying capacity for navigation in calm waters during daytime shall be calculated from the open deck space of the ship as follows, namely:

iv. for each third-class passenger, a seat of 1.25 square metres;

v. for each second-class passenger, a seat of 1.75 square metres;

vi. for each first-class passenger, a seat of 2.75 square metres;

vii. two children under 12 years of age shall be counted as one person;

e. when navigating in calm waters at night, the number of passengers calculated on the basis of the open passenger area shall be reduced to two-thirds;

f. in the case of navigation in rough waters during daytime, the number of passengers calculated on the basis of the unrestricted passenger area shall be reduced to two-thirds;

g. in the case of navigation in rough waters at night, the number of passengers calculated on the basis of the unrestricted passenger area shall be reduced by half;

h. if, in calm waters, all the passengers gather on one half of the ship, either on the port or the starboard side, the number of passengers of any ship shall be limited such that the ship heels by not more than 10 degrees; and

i. for the purpose of any calculation of the heeling of the ship to port or starboard, the weight of each passenger shall be taken as equal to 75 kg.

p.188 ### 8. Installation of furniture

1. In the case of fixed seating, a clear space of 0.5 metre for third class and 0.6 metre for second and first class shall be kept in front of each row of benches or chairs.

2. Furniture which is not fixed to anything shall be tied down with ropes etc. in such a manner that it cannot shift from its position owing to the rolling or heeling of the ship.

3. When bad weather is anticipated, furniture which is not fixed to anything shall be tied down with ropes etc.

4. Seating arrangements and furniture fixed to anything shall be installed in such a manner that they cannot shift from their position owing to the rolling or heeling of the ship.

9. Toilet space

1. The following sanitary arrangements shall be provided on all passenger ships, namely:

a. for third- and second-class passengers, at least one toilet, and one additional toilet for every 50 passengers of those two classes;

b. for first-class passengers, at least one toilet, and one additional toilet for every 20 passengers of that class.

2. Wash basins shall be provided in all toilets.

3. Where the number of passengers exceeds 100, an additional toilet shall be provided for women.

4. There shall be at least one toilet and wash basin on each deck, and the toilets shall have flushing arrangements (connected to a low-down cistern).

5. This rule shall apply only to new ships, and ships existing on the date these Rules come into force shall comply with the provisions of this rule as far as practicable.

10. Space for women and children

1. On inland passenger ships carrying more than 100 passengers, a separate, marked space shall be provided for women and children, the area of which shall be at least equal to 10% of the total passenger area.

p.189 (2) The space referred to in sub-rule (1) shall have adequate light and ventilation and shall be weatherproof.

11. Space for preservation of privacy

To ensure privacy, on all passenger ships navigating at night with more than 200 passengers, a changing room shall be provided for preserving the privacy of open-deck passengers.

12. Safety enclosure and railing

1. The height of a safety enclosure or railing shall not be less than 0.9 metre, and its arrangement shall be such that no child can fall through the enclosure or railing.

2. The vertical bars of railings shall be spaced not more than 200 millimetres apart.

3. Safety enclosures and railings shall be fitted around all decks designated for the use of passengers, and they shall not be connected to the vertical partitions within the ship.

13. Exit routes

1. The width of passenger walkways, stairways, doors and exit routes constructed for the use of passengers must be at least 0.8 metre.

2. The width of doors to cabins and other small spaces shall not be less than the width referred to in sub-rule (1).

3. There shall be at least two exit routes from the spaces designated for passengers to the open deck, the width of one of which shall be at least 800 millimetres and that of the other at least 600 millimetres.

4. If the space referred to in sub-rule (3) is designated for 50 or more passengers, the width of both exit routes shall be at least 800 millimetres.

5. If there is only one route to a space designated for passengers, the minimum specified width shall be at least 1000 millimetres, which may be reduced to 800 millimetres in the case of open passenger launches.

p.190 (6) In the case of spaces designated for 80 or more passengers, all exit routes designated for passengers and for their use in an emergency shall be at least 15 millimetres per passenger.

14. Emergency exit routes

1. In all passenger spaces having only one ordinary exit route, there shall also be an emergency exit route capable of being opened outwards, with a clear opening of at least 600 × 600 mm.

2. If the ship has arrangements for carrying lifeboats, the safe escape routes and emergency routes shall be capable of reaching the open deck area.

15. Embarkation and disembarkation

1. Open spaces and embarkation and disembarkation arrangements shall be kept safe in the manner prescribed in rule 12.

2. Gangways shall be at least 0.6 metre wide and shall have hand rails or rope safety arrangements on both sides.

3. If the ship can generate electricity, the gangways shall be illuminated by floodlights, and if the ship has no arrangement for generating electricity, arrangements shall be made to illuminate the gangways by some other means.

4. Passengers must not embark on or disembark from the ship while cargo is being loaded or unloaded.

16. Doors

Except for doors connected to the passageways from passenger cabins, doors of passenger accommodation shall open outwards or be arranged to open in a sliding manner, and arrangements shall be made so that no unauthorised person can lock them while the ship is under way.

17. Obstructions

1. There shall be no obstruction of any kind in exit routes, emergency exit routes and stairways.

2. Any covering or canvas on the shell openings of passenger launches shall be capable of being easily opened or removed in an emergency.

p.191 (3) Openings which can be opened shall not be kept covered by ropes or nets which could obstruct the passengers’ way out in an emergency.

18. Dangerous goods

Dangerous goods specified in the Inland Ships (Dangerous Goods) Carriage Rules, 2001 shall not be carried on passenger ships or launches.

19. Conditions for allowing passengers to board or disembark from an inland passenger ship

1. The master of a ship, or any employee empowered in this behalf by the owner of an inland ship or its master, may refuse to accept any person as a passenger of an inland passenger ship if he—

a. has not paid his fare;

b. is an unaccompanied person of unsound mind;

c. is suffering from any infectious or contagious disease;

d. is intoxicated or incapable of taking care of himself;

e. is a person of disorderly disposition, or is otherwise in such a condition or behaves in such a manner as to cause, or be likely to cause, annoyance to other passengers;

f. the approved carrying capacity of the inland passenger ship, or of the class in which he wishes to travel, has already been filled.

2. No passenger shall—

a. travel in any class other than the class for which he has paid the fare;

b. travel beyond the destination for which he has paid the fare without having paid the fare for the additional distance in advance;

c. use or attempt to use his ticket on any date other than the date for which the ticket is valid;

d. make any alteration to or deface his ticket in such a manner as to render it illegible;

p.192 (e) take or attempt to take with him any luggage other than the luggage permitted to be taken with him free of charge, without having paid in advance the charges payable for such luggage.

3. When so directed by the master of the ship or by any person empowered in this behalf by the master or the owner, every passenger shall—

a. pay any unpaid fare;

b. produce his ticket and the receipt for his luggage, if any;

c. produce his ticket after arriving at the destination.

4. If any person is not permitted to board the ship, that person shall be entitled to a refund of the fare paid by him.

20. Conduct of passengers on an inland ship

1. No passenger of an inland ship shall—

a. take on board the ship or keep with him any kind of rotten meat, fish or vegetables, or any other kind of harmful foodstuff;

b. obstruct or create any hindrance to the master or any other officer or crew member of the ship in the performance of his duties;

c. in any way obstruct or interfere with the embarkation or disembarkation of any other passenger, or with the loading or unloading of cargo on the ship;

d. damage or attempt to damage any property of the ship;

e. board or leave the ship, or attempt to board or leave it, while the ship is in motion, to or from another ship;

f. enter, for any reason other than the performance of a legal duty, any room or space reserved for the use of other passengers and crew, or any reserved area; or refuse to leave such places when directed to do so by the master, an officer or a crew member of the ship;

p.193 (g) smoke, or keep fire or a naked light about him, in any part of the ship where smoking or the keeping of naked lights is prohibited;

h. consume intoxicants or behave in a disorderly manner;

i. commit any kind of nuisance or indecent act, or use indecent language;

j. disturb the comfort of other passengers for any reason other than the exercise of a legal right.

2. No male passenger shall, for any reason other than the performance of a legal duty, enter any room or space reserved exclusively for women, and no male person shall remain in such a place after having been requested by the master, an officer or any crew member to leave it.

3. No passenger shall enter or remain in any part of an inland passenger ship other than the spaces designated for passengers, and if any passenger is found in any of those places, he shall be bound to leave the place when told to do so by any crew member.

21. Preventive measures against the spread of epidemic disease

1. Whenever a case of cholera or any other dangerous epidemic disease occurs on an inland passenger ship, the master shall forthwith—

a. remove the patient, together with his bedding, utensils, water vessels and food, to a part of the after deck at the extreme end of the ship, where he shall be kept separate from other companions or passengers, as the case may be;

b. have all excreta, vomit, urine etc. of the patient cleaned with a disinfectant for infectious diseases;

c. arrange to obtain medical assistance at the nearest ghat and take steps to have the patient admitted to the nearest hospital;

d. after the patient has been taken off the ship and ashore, have the deck of the inland passenger ship properly washed, and have the accommodation and cabins disinfected against infectious disease;

p.194 (e) after all passengers have disembarked from the ship at the final destination, have all accommodation spaces of passengers and crew and all working spaces washed and disinfected.

2. If cholera or any other dangerous epidemic disease appears on the ship while the ship is at the ghat of departure, the master shall not sail the ship until the deck, all accommodation and working spaces have been washed and disinfected against infectious disease, and the patient has been put ashore.

3. If a patient dies on the ship, the dead body shall be wrapped in a cloth soaked in disinfectant medicine and handed over to his relatives or friends, and where no relative of the deceased can be found, it shall be handed over to the police for the funeral.

4. At least 5 litres of disinfectant shall be kept at all times on every inland passenger ship.

22. Restrictions on navigation

1. A ship or boat which is not registered under the Ordinance shall not embark or disembark any passenger within a radius of 1 kilometre of a BIWTA or BIWTC landing station or ghat.

2. No open passenger launch of less than 20 metres in length shall make any voyage of more than 2 hours’ total duration in rough waters.

3. No inland passenger ship of any kind shall navigate outside the inland water limits.

p.195 # Inland Ships (Free-board) Rules, 2001

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p.199 ## Chapter 1 — Preliminary

1. Short title

These Rules shall be called the Inland Ships (Free-board) Rules, 2001.

2. Definitions

Unless there is anything repugnant in the subject or context, in these Rules—

a. “Ordinance” means The Inland Shipping Ordinance, 1976 (LXXII of 1976);

b. “inland waterway” means an “inland ship” as defined in section 2(e) of the Ordinance; [sic — printed thus in the original]

c. “open ship” means a ship without a weather deck;

d. “open passenger launch” means an open ship used for carrying more than 12 passengers;

e. “ship” or “inland waterway” means “inland water” as defined in section 2(e) of the Ordinance; [sic — printed thus in the original]

f. “tug” means a vessel built for the purpose of towing or pushing ships or floating equipment;

g. “tanker” means an inland ship used for carrying large quantities of liquid in tanks or containers;

h. “draft” means the vertical distance, at the mid-point of the ship’s length (L), between the bottom edge of the ship’s bottom and the Load Water Line;

i. “length (L)” means the distance, on the load water line, from the forward edge of the stem at the fore end of the ship to the after edge of the rudder post, or, if the ship has no rudder post, to the centre of the rudder stock;

j. “new ship” means any ship registered after these Rules come into force;

k. “safety distance” means the watertight free-board, i.e. the depth of a deckless ship or floating equipment;

p.200 (l) “free-board deck” means the uppermost weathertight or watertight deck;

m. “year-round rough water limits” means that part of the inland waterways which the Authority, by notification in the official Gazette, declares to be year-round rough water limits;

n. “year-round calm water limits” means that part of the inland waterways which the Authority, by notification in the official Gazette, declares to be year-round calm water limits;

o. “floating equipment” means various non-self-propelled equipment, such as dredgers, floating cranes, etc., which are used for performing various kinds of work while afloat;

p. “fishing vessel” means a ship used for, or built for the purpose of, catching fish;

q. “cargo ship” means a self-propelled ship used for carrying cargo and not more than 12 passengers;

r. “seasonal rough water limits” means that part of the inland waterways which the Authority, by notification in the official Gazette, declares to be partially rough water limits;

s. “passenger” means any person on board an inland ship other than the master, officers and crew of the ship; provided that a child under one year of age shall not be included among the passengers;

t. “passenger ship” means a ship with a weather deck used for carrying more than 12 passengers, and includes ships with more than one deck;

u. “road ferry” means an open flush-deck ship engaged in ferrying 12 or more passengers or one or more vehicles; and

v. “service vessel” means a self-propelled ship, not being a passenger ship, cargo ship, tanker or fishing vessel, which is built for performing special duties.

p.201 ### 3. Application

Unless otherwise provided, these Rules shall apply to all ships and floating equipment irrespective of size and type.

Chapter 2 — Marking

4. Marking

1. The free-board and load line marks shall be marked and drawn on both sides of the ship or floating equipment by welding, engraving or cut-out designs, and shall be placed amidships.

2. These marks shall be white on a dark background and black on a light background.

3. The following marks shall be applied, namely:

a. the free-board line shall be a horizontal line 450 millimetres long and 25 millimetres wide, the upper edge of which shall be at the maximum permitted draft in calm water, and it shall be located directly below the deck line;

b. the deck line disc shall be a circle of 300 millimetres outside diameter and 25 millimetres width. The upper edge of the free-board line shall pass through the centre of the load line disc;

c. if the ship has several free-boards, those free-boards shall be marked by lines 150 millimetres long and 25 millimetres wide, the upper edges of which shall be at the actual maximum permitted draft, and shall be forward of the calm water limits free-board. The marks of the different free-boards shall be connected by a vertical line 25 millimetres wide;

d. the different free-boards shall be marked by a letter at the forward end of the free-board line; the said letter shall be of a size of at least 50 × 30 millimetres with a line thickness of 5 millimetres, and the pattern of marking shall be as follows, namely:

e. for calm waters, the letter S

ii. for rough waters, the letter R

p.202 (e) draft marks shall be permanently marked vertically on both sides of the ship, at the stem at the fore end of the ship and at the after end of the ship. In the metric system, each metre of draft shall be marked by English numerals 15 centimetres high and 15 millimetres thick, with the lower edge at the actual draft, and each 20 centimetres of draft shall be marked by English numerals 10 centimetres high and 10 millimetres thick, with the lower edge at the actual draft. If the draft is marked in feet, each foot shall be marked by numerals 6 inches high and 1 inch thick, with the lower edge at the actual draft.

Chapter 3 — Structural fitness

5. Application

This Chapter shall apply only to new ships and floating equipment.

6. Doors and sills

Those access openings in walls or superstructures through which the spaces below the free-board deck can be entered shall be watertight or weathertight, and the height of their sills shall be at least 150 millimetres. On tugs, the sill height shall be at least 300 millimetres.

Provided that, in the case of vessels built for the Chittagong route, the sill height shall be at least 450 millimetres.

7. Access hatches

All access hatches below the free-board deck shall be watertight and shall have coamings of at least 450 millimetres height.

8. Port holes and skylights

All port holes and skylights giving access to the spaces below the free-board deck shall be weathertight. There shall be no windows below the free-board deck.

p.203 ### 9. Cargo hatch coamings

The height of cargo hatch coamings shall be at least 450 millimetres. However, in the case of vessels built for the Chittagong route, the height of cargo hatch coamings shall be at least 750 millimetres.

10. Hatch covers

Cargo hatch covers shall be strong, sturdy and weathertight so that they will not be damaged by sun, wind, storm or disaster.

11. Air pipes

Air pipes entering tanks shall extend at least 450 millimetres above the free-board, and the said air pipes shall be fitted with goose necks. However, in the case of vessels built for Chittagong, the air pipes shall be at least 750 millimetres.

12. Scuppers and freeing ports

The scuppers and freeing ports of a ship for draining water from the deck, and the bulwarks, shall be of adequate size in accordance with the Inland Steel-built Vessel Construction Guidelines.

Chapter 4 — Free-board and safety distance

13. Passenger ships and open passenger launches

1. For passenger ships with a weather deck navigating in calm waters throughout the twelve months of the year and in seasonal rough waters during the calm season, the free-board shall be at least 300 millimetres. For open passenger launches (without a weather deck) navigating in calm waters throughout the twelve months of the year and in seasonal rough waters during the calm season, the safety distance shall be at least 500 millimetres.

2. The free-board of a ship shall be calculated in accordance with the requirements of the International Load Line Convention, 1996 [sic], or in the following manner, namely:

Determination of free-board (F) in millimetres—

a. ship’s length (L) up to 24 metres: F = 300 millimetres

b. ship’s length (L) 24 ≤ L ≤ 45 metres: F = 8 × (L − 24) + 300 millimetres

p.204 (c) ship’s length (L) 45 ≤ L ≤ 55 metres: F = 9 × (L − 24) + 300 millimetres

d. ship’s length (L) 55 ≤ L ≤ 60 metres: F = 8 × (L − 24) + 300 millimetres [printed thus in the original]

14. Passenger/motor-vehicle ferries

Passenger/motor-vehicle ferries shall cross only within year-round calm water limits. The free-board shall not be less than 5% of the breadth (B) or 500 millimetres, whichever is greater.

15. Road ferries

Road ferries shall cross only within year-round calm water limits. The free-board shall not be less than 5% of the breadth (B) or 500 millimetres, whichever is greater.

16. Cargo ships

1. In year-round calm water limits, and in seasonal rough water limits during the calm season, the free-board of cargo ships shall be at least 150 millimetres. If the hatch coamings of ships currently in use are less than 450 millimetres, the free-board shall be increased by exactly the amount by which the hatch coamings fall short of 450 millimetres:

Provided that, in the case of vessels built for the Chittagong route, the height of such hatch coamings shall not be less than 600 millimetres.

2. In year-round rough water limits, and in seasonal rough water limits during the rough season, the free-board specified in sub-rule (1) shall be increased by 100 millimetres.

17. Tankers

1. In year-round calm water limits, and in seasonal rough water limits during the calm season, the free-board of cargo ships [sic — printed thus; i.e. tankers] shall be at least 50 millimetres. If the access hatch coamings of existing tankers are less than 450 millimetres, the free-board shall be increased by 50% of the difference.

2. In year-round rough water limits and in seasonal rough water limits during the rough season, the free-board specified in sub-rule (1) shall be increased by 50 millimetres.

p.205 ### 18. Tugs

1. In year-round rough water limits [sic — printed thus] and in seasonal calm water limits during the calm season, the free-board of a tug shall be at least 5% of the breadth (B), but shall not be less than 250 millimetres.

2. In year-round rough water limits and in seasonal rough water limits during the rough season, the free-board

specified in sub-rule (1) shall be increased by 150 millimetres.

19. Service vessels

The provisions of rule 13 shall apply to service vessels:

Provided that, with the ship heeled at 12 degrees, there shall be a free-board of 100 millimetres.

20. Dumb barges

Dumb barges shall have a free-board of 5% of the breadth, but it shall not be less than 200 millimetres. If the height of the hatch coaming is less than 450 millimetres, the free-board shall be increased by 50% of the difference. If the barge is operated in rough water limits, the free-board shall be increased by a further 50% over the applicable requirement.

21. Pontoons

The minimum free-board shall be 5% of the breadth of the pontoon.

22. Floating equipment

Depending on the size and type of the floating equipment, and taking into consideration the safe operation of the equipment, the Inland Safety Administration (ISSA) shall determine the required free-board for each type of equipment.

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p.207 # Inland Ships (Stability) Rules, 2001

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p.210

p.211 ## Chapter 1 — Preliminary

1. Short title

These Rules shall be called the Inland Ships (Stability) Rules, 2001.

2. Definitions

1. Unless there is anything repugnant in the subject or context, in these Rules—

a. “Ordinance” means The Inland Shipping Ordinance, 1976 (LXXII of 1976);

b. “open passenger launch” means an open ship used for carrying more than 12 passengers;

c. “ship” or “inland ship” means an “inland ship” as defined in section 2(e) of the Ordinance;

d. “tanker” means an inland ship used for carrying liquid substances in bulk;

e. “dumb barge” means a ship, vessel or floating equipment used for carrying cargo or liquid substances, which is not self-propelled and does not carry any passenger;

f. “length (L)” means the distance on the load water line (loaded waterline) from the forward end of the stem at the fore part of the ship to the after end of the rudder post, or, if the ship has no rudder post, to the centre of the rudder stock;

g. “new ship” means any ship registered after these Rules come into force;

h. “pontoon” means a dumb barge having, on the weather deck, no openings other than manholes and air pipes, with covered exits and no gaps;

i. “free-board” means the distance between the depth of the ship or floating equipment, including the watertight free-board deck, and the draught;

p.212 (j) “free-board deck” means the uppermost weathertight or watertight deck of the vessel;

k. “existing ship” means a ship which is not a new ship;

l. “floating equipment” means various equipment, not self-propelled, such as dredgers, floating cranes, etc., which are used for carrying out various works in a floating condition;

m. “fishing vessel” means a ship used for, or built for the purpose of, fishing;

n. “passenger ship” means a ship with a weather deck used for carrying more than 12 passengers, and shall also include a ship having more than one deck;

o. “road ferry” means an open flush-deck ship engaged in ferrying 12 or more passengers or one or more vehicles;

p. “service vessel” means a self-propelled ship, which is not a passenger ship, cargo ship, tanker or fishing vessel, built for performing special duties.

2. In the case of ship design and stability, the following notations shall apply, namely:—

a. G — centre of gravity;

b. M — initial metacentre;

c. K — bottom of the ship’s bottom (keel);

d. S — point of projection of M;

e. GM — initial metacentric height;

f. KG — height of the centre of gravity above the ship’s bottom;

g. KM — height of the initial metacentre above the ship’s bottom;

h. GZ — righting lever (lever for returning to the correct position).

p.213 [Figure: diagram of a heeled ship section showing waterline W–L, points K (keel), G (centre of gravity), Z, M (metacentre) and S, with the righting lever GZ.]

3. Application

Unless otherwise provided, these Rules shall apply to all new ships irrespective of size and type.

4. Design and construction

The design, drawing, preparation of plans, construction and outfitting of all ships and floating equipment shall, subject to the satisfaction of the Surveyor, be in accordance with good naval-architectural standards, suitability for navigation on waterways, seaworthiness, safety and strength, to the satisfaction of the Surveyor.

5. Free-board

All ships and floating equipment shall fulfil the requirements of the provisions of the Inland Ships (Free-board) Rules.

Chapter 2 — Minimum requirements of stability

6. General

All ships of more than 24 metres in length, other than barges, pontoons and floating equipment, shall, in every loading condition, satisfy any one of the two criteria described below, namely:

p.214 **(1) Criterion A**

a. The area under the GZ curve (Curve) shall be not less than 0.055 metre-radians up to an angle of heel of 30° and not less than 0.09 metre-radians up to 40° or the angle of flooding, whichever is the lesser. In addition, the area under the GZ curve between the heel angles of 30° and 40°, or between 30° and the angle of flooding if this is less than 40°, shall be not less than 0.03 metre-radians;

b. The righting lever at an angle of heel equal to 30° shall be at least 0.20 metre;

c. The angle of heel at which the maximum righting lever occurs shall not be less than 25° and should preferably exceed 30°;

d. The initial metacentric height GM shall not be less than 0.15 metre for L ≥ 70 metres and 0.35 metre for L < 70 metres.

(2) Criterion B

a. The area under the righting-arm curve shall be not less than 0.070 metre-radians up to an angle of 15° when the maximum GZ occurs at 15°, and 0.055 metre-radians up to 30° when the maximum GZ occurs at 30° or above. Where the maximum GZ occurs between 15° and 30°, the corresponding area under the righting lever curve shall be `0.055 + 0.001 (30 − θmax)` metre-radians, where θmax is the angle of maximum lever;

b. The area under the righting lever curve between 30 degrees and the angle of flooding (flooding angle) or 40 degrees, whichever is less, shall not be less than 0.03 metre-radians;

c. At angles equal to or greater than 30 degrees and at the flooding angle, the righting lever (lever for returning to the correct position) shall be at least 0.02 metre;

d. The initial metacentric height GM shall not be less than 0.15 metre for L ≥ 70 metres and 0.35 metre for L < 70 metres;

p.215 (e) The maximum righting lever should occur at an angle of heel of more than 15 degrees.

Figure

Figure II/I

[Figure: graph of righting lever (GZ, metres, 0.05–0.30 on the vertical axis) against angle of heel (degrees, 10–70 on the horizontal axis), showing the righting lever curve with the hatched areas under the curve up to 30° and between 30° and 40°, and the GM line drawn to 1 radian (57.3°).]

7. Free Liquid Surface Effect

For all conditions, the initial metacentric height and the GZ curves shall be corrected for the effect of the free liquid surface of tanks in accordance with the following assumptions, namely:

a. To determine the effect of the surfaces on stability at all angles of inclination, the free surface of every kind of liquid (including water ballast tanks) contained in the tank or tanks that are to be considered shall be taken into account;

b. The tanks shall be either empty or as full as possible;

p.216 (c) The tanks assumed for the purpose of determining the free surface correction shall be those tanks which, when 50 per cent full, produce the maximum “free surface moment” Mfs at 30° inclination;

d. The value of Mfs for each tank may be calculated with the aid of the following formula, namely:—

`Mfs = v b ρ l k √δ`

Mfs = free surface moment (metre-tonnes);

v = total capacity of the tank (cubic metres);

b = maximum breadth of the tank (metres);

ρ = specific weight of the liquid in the tank, t/m³;

δl = block coefficient of the tank = `V / (b l h);`

h = maximum height of the tank in metres;

l = maximum length of the tank in metres;

k = dimensionless coefficient, derived or constructed from the attached table according to the ratio b/h. Intermediate values may be determined by interpolation. The value of 'K' may alternatively be calculated from the formula stated below the table.

Table of values of coefficient "k"

b/h	5°	10°	15°	20°	30°	40°	50°	60°	70°	b/h
10	0.01	0.11	0.12	0.12	0.11	0.10	0.09	0.07	0.05	10
5	0.04	0.07	0.10	0.11	0.11	0.11	0.10	0.08	0.07	5
3	0.02	0.04	0.07	0.09	0.11	0.11	0.10	0.09	0.08	3
2	0.01	0.03	0.04	0.06	0.09	0.11	0.11	0.10	0.09	2
1.5	0.01	0.02	0.03	0.05	0.07	0.10	0.11	0.11	0.11	1.5
1	0.01	0.01	0.02	0.03	0.05	0.07	0.10	0.12	0.13	1
0.75	0.01	0.01	0.02	0.02	0.04	0.05	0.08	0.12	0.15	0.75
0.5	0.00	0.01	0.01	0.02	0.02	0.04	0.05	0.09	0.16	0.5

`k = (sinθ/12)(1 + tan²θ/2)(b/h)`, where cotθ ≥ b/h

`k = (cosθ/8){1 + tanθ/(b/h)} − (cosθ/{12(b/h)²})(1 + cot²θ/2)`, where cotθ < b/h

p.217 ### 8. Passenger ships

1. This rule shall apply to all passenger ships and launches, new and old.
2. The stability shall be such that the free-board requirements are always fulfilled.
3. When all passengers are on one side of the ship, the ship shall not heel more than 10 degrees, and there shall then be sufficient free-board to withstand the rolling wave, wind and the centrifugal forces (CENTRIFUGAL FORCES) arising when turning, while the ship is in an agitated condition.
4. The metacentric height required on account of the crowding of passengers to one side shall be as follows, namely:—

`GM > (0.85 × B × WP) / D`

where,

B : moulded breadth, in metres

WP : total weight of passengers, in metric tonnes

D : displacement (quantity of water displaced), in metric tonnes

GM : metacentric height, in metres.

(In calculating the weight of each passenger including luggage, 75 kilograms has been taken.)

5. Passenger ships and launches shall not navigate in wind speeds exceeding 10 metres/second (36 kilometres/hour).
6. If, while under way, the wind speed exceeds 10 metres/second (36 kilometres/hour), passenger ships and launches shall immediately proceed towards the nearest bank, canal or sheltered waters.
7. In the case of any ship, only if it is found that its amount of heel is not more than 3 degrees may that ship navigate in winds of higher speed.
8. It must be ensured that the total combined temporary heel of the ship due to wind, turning and the crowding of passengers on one side shall never exceed 12 degrees.

p.218 (9) For the purpose of calculating the intact stability of the ship, the assumed matters shall be as follows, namely:—

a. The total weight of each passenger and his luggage shall be 75 kilograms;

b. The centre of the total weight of each passenger and his luggage shall be considered on average at 0.8 metre above the relevant deck;

c. The ability of the ship to withstand the combined effect of beam wind and rolling (weather criterion) shall, for each standard loading condition, be in accordance with the design determined by the Authority for this purpose.

Figure II/2

[Figure: graph of lever against angle of heel showing the GZ curve; the steady wind heeling lever lw1 and gust heeling lever lw2 drawn as horizontal lines; the angle of equilibrium θ0; the angle of roll θ1 to windward; the angles θ2 and θc (angle of flooding); and the hatched areas “a” (to windward of θ0, between the curve and lw2 down to θ1) and “b” (between lw2 and the GZ curve from θ0 to θ2).]

p.219 θ0 = the angle of heel of the vessel caused by a wind blowing at a constant speed (steady wind), which shall be the lesser of generally 16 degrees or 80% of the angle at which the deck edge is immersed.

θ1 = the angle of roll of the vessel to windward caused by waves, to be calculated according to the rule shown below.

θ2 = the leeward angle in the direction towards which the wind blows, being the least of the down-flooding angle, or 50 degrees, or the angle θc at which GZ and lw2 intersect.

d. For the combined effect of wind and rolling, the steady wind heeling lever (lw1) and the gust wind lever shall be calculated from the formula given below.

$$(lw1) = (P \times A \times Z) / D$$

$$(lw2) = 1.5\ lw1$$

where,

P = 0.0322 (tonne/metre);

A = lateral area exposed to wind;

Z = the vertical distance between the centre of A and the mid-point of the draught (metres);

D = displacement (water displaced) (metric tonnes).

For a ship to be considered stable in the above-mentioned condition, the area ‘b’ shall be equal to or greater than the area ‘a’.

e. The heeling moment for the angle of heel caused by turning may be calculated as follows:

$$MR = 0.02 \times (Vs^2 / L) \times \Delta \times (KG - T/2)$$

p.220 where,

MR = heeling moment, m;

Vs = designed speed, m/s;

Δ = displacement of the ship; t

KG = height of the centre of gravity (Gravity) above the keel (Keel) of the ship;

T = vertical distance from the top of the ship’s keel (Keel) to the Load Water Line;

L = length of the ship.

Figure 2 — Lateral area of superstructures

Figure II/3

[Figure: “LATERAL AREA OF SUPERSTRUCTURES” — side elevation of a ship showing the hatched lateral area A of the superstructures above the waterline, its centre of area CA, the vertical distance Z between CA and the centre of the underwater lateral area (VCB).]

10. The required GM for open passenger launches shall be not less than 0.8 metre and for passenger ships not less than 0.6 metre.

11. The following conditions shall apply only to new ships, namely:

a. The watertight subdivision of passenger ships of more than 36 metres in length shall be such that they satisfy the one-compartment (One Compartment) ship requirements of SOLAS 1974/78 and have a margin line of at least 75 millimetres. The subdivision shall be calculated by the method laid down in “SOLAS 1974/78, CHAPTER II-1 Part B”;

p.221 (b) This rule shall also apply to passenger, motor-car ferries and road ferries.

9. Cargo ships and tankers

For ships which do not fulfil the requirements of rule 6, the GM shall be not less than 0.6 metre and the minimum range of stability shall be not less than 40°.

10. Tugs and service vessels

The GM of tugs or service ships of length up to 15 metres or of power up to 250 BHP shall be not less than 0.6 metre, and the GM of tugs or service ships of length more than 15 metres and of power more than 250 BHP shall be not less than 0.75 metre.

11. Barges and pontoons

The GM of barges and pontoons shall be not less than 0.3 metre and the minimum range of stability shall be 20°.

12. Floating equipment

The minimum stability shall be determined differently in different cases.

Chapter 3 — Information to be supplied

13. Loading

1. Stability calculations shall be carried out for the principal loading conditions likely to occur which may be relevant to the operation of the ship.

2. If the ship owner does not supply sufficient detailed information relating to the likely standard loading conditions indicated under sub-rule (1), the stability calculations in the loaded condition shall be carried out following the conditions described below:

a. **Cargo ships and tankers:**

b. Departure condition, with the spaces designated for the carriage of full cargo evenly distributed, and full stores and fuel;

ii. Arrival condition corresponding to item (i), but with 10% stores and fuel;

p.222 (iii) Ballast and departure condition, being full stores and fuel;

iv. Ballast and arrival condition, being 10% stores and fuel.

b. **For any ship which may additionally carry deck cargo:**

c. Departure condition, with full cargo in the ship’s holds and on deck and full stores and fuel; the weight and height of the deck and cargo shall be stated;

ii. Arrival condition corresponding to item (i), but with 10% stores and fuel. The weight of water likely to be absorbed by the deck cargo during the voyage shall be added to the weight.

iii. **Passenger ships:**

iv. Departure condition, with full cargo, full stores and fuel and the full number of passengers with their luggage;

v. Arrival condition corresponding to item (i), but with 10% stores and fuel;

vi. Departure condition, without cargo but with full stores and fuel and the full number of passengers with their luggage;

vii. Arrival condition corresponding to item (iii), but with 10% stores and fuel.

14. Inland ship safety and administration and information to be supplied

1. In the case of new ships of 24 metres or more in length, the following information shall be supplied in triplicate to the inland ship safety administration, namely:

a. line plan;

b. general arrangement plan;

c. capacity plan, or weights and positions of centres of gravity, and information on volumes and centres of gravity for holds and tanks;

p.223 (d) Hydrostatic curves and tables;

e. Cross curves and corresponding tables;

f. Flooding angle as a function of draught;

g. Corrections to GM and KN for the free liquid surface effect in tanks;

h. KM as a function of draught and trim, in the form of a diagram (Diagram) and tables, if the figures change significantly with trim;

i. Inclining test report;

j. GZ curves (curve) and calculations for the loading conditions provided for in rule 14(1) mentioned above;

k. Minimum GZ curves or tables as a function of draught and trim, if the figures change significantly with trim.

2. For vessels of less than 24 metres in length or built by traditional methods, the general arrangement plan, plans and inclining test report need not be submitted.

3. When it appears that, in view of the size, construction and intended service of the ship, certain information mentioned in this rule is apparently unnecessary, the inland ship safety administration may exempt the ship from the requirement of supplying the corresponding information.

Chapter 4 — Tests

15. Inclining Test

1. The inclining test shall be carried out very accurately and a report on the said test shall be prepared describing all measured data and the method of carrying out the test.

2. The inland ship safety administration shall be informed in advance, by application, of the time and place of the inclining test. When carrying out the test the ship shall be moored slack and kept away from the jetty.

p.224 The number of persons present on the ship shall be kept as low as possible. This test shall be carried out in calm weather and still water. The ship shall not be in any noticeably heeled condition.

3. The readings (reading) of the angles of inclination shall be taken with the aid of inclination measuring equipment or, if possible, two pendulums. Draught readings shall be taken with the aid of all reliable draught marks. The distance through which the heeling weights are to be moved shall be so fixed that the angle of inclination on either side cannot be less than 1 degree.

4. When calculating the hydrostatic data corresponding to the inclining condition (hydrostatic data corresponding to the inclining letter [sic]), the draught readings (reading) shall be used taking into account the applicable corrections due to the rake of the stern (stem) and the stem (stem) at the fore part of the ship and the position of the draught marks. If the trim in the inclining condition is such that it necessarily affects the result of the test, the hydrostatic data shall be calculated taking the trim into account. The metacentric height shall be corrected for the free liquid surface effect.

5. The heeling weights shall be so placed that two heeling moments can be created on either side. Among the recorded angles of inclination at least four shall be suitable for use in the calculation. Sufficiently accurate calculation of the weights shall be provided for adding or subtracting.

16. Passenger ships of less than 24 metres in length

When the inclining test is carried out on a passenger ship of less than 24 metres in length, with all the passengers of the ship allowed to crowd and gather on one side, the angle of heel of the ship shall not exceed 10 degrees and likewise the remaining free-board shall not be less than 0.2 metre.

17. Qualifications for existing passenger ships

For the purpose of determining the maximum number of passengers permitted on the ship in accordance with these Rules and the Free-board Rules, all passenger ships and launches registered before the date fixed by the Government—

p.225 (a) shall carry out an inclining test;

b. a weight equal to the number of passengers, at an average of 75 kilograms per passenger, shall be placed on one side of the ship at half the distance between the centre line and the side;

c. it shall be ascertained that the amount of heel cannot exceed 10 degrees;

d. if the amount of heel for the permitted number of passengers, taking the seating arrangement or the required space, exceeds 10 degrees, arrangements shall be made to reduce the number of passengers until the amount of heel is reduced to a maximum of 10 degrees.

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p.227 # Inland Ships (Life Saving) Rules, 2001

p.228

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p.230

p.231 ## Chapter 1 — Preliminary

1. Short title

These Rules shall be called the Inland Ships (Life Saving) Rules, 2001.

2. Definitions

Unless there is anything repugnant in the subject or context, in these Rules:

a. “Department” means the Department of Shipping;

b. “Ordinance” means The Inland Shipping Ordinance, 1976 (LXXII of 1976);

c. “approved” means approved by the Department;

d. “open passenger launch” means an open ship used for carrying more than 12 passengers;

e. “ship” or “inland ship” means an “Inland Ship” as defined in section 2(e) of the Ordinance;

f. “tanker” means a ship used for carrying liquid substances in bulk in tanks;

g. “dumb barge” means a ship, vessel or floating equipment used for carrying cargo or liquid substances in bulk, which is not self-propelled and does not carry passengers;

h. “ferry” means a ship with a structure engaged in ferrying 12 or more passengers and vehicles;

i. “floating equipment” means various equipment, not self-propelled, such as dredgers, floating cranes, etc., which are used for carrying out various works in a floating condition;

j. “passenger” means any person among those on board a ship or inland ship other than the master, officers and crew of the ship, but a child passenger below one year of age shall not be included therein;

k. “passenger ship” means a ship with a watertight weather deck used for carrying more than 12 passengers, and shall also include a ship having more than one deck; and

p.232 (l) “road ferry” means an open flush-deck ship engaged in ferrying 12 or more passengers and one or more vehicles.

3. Application

Unless otherwise provided, these Rules shall apply to all inland ships.

4. Marking, etc.

1. All life-saving appliances and equipment shall be marked with the name of the ship and the name of the port of registry.

2. If the ship operates between 6 o’clock in the evening and 6 o’clock in the morning, retro-reflective (Retro-Reflective Tape) material shall be fitted on each side of the life-saving appliances and equipment.

Chapter 2

5. General requirements for life-saving appliances and equipment

1. All life-saving appliances and equipment:—

a. shall be constructed properly and of correct materials;

b. shall not be damaged at temperatures up to 65° degrees centigrade when stowed in the vessel, and at temperatures up to 38° degrees centigrade when immersed for use in sea or river water;

c. shall be rot-proof and corrosion-resistant and shall not be unduly damaged by sea or river water, oil or fungal attack;

d. shall be resistant to deterioration where exposed to sunlight;

e. all parts shall be of a highly visible (bright orange) colour so that they attract attention very easily; and

f. shall be approved by the Department.

2. During the conduct of a survey the Engineer and Ship Surveyor shall examine the condition of the life-saving appliances or equipment and, if at the end of the examination it appears that the said appliances or equipment have deteriorated to such an extent that,

p.233 their use would not be appropriate, then they shall be replaced by new appliances or, as the case may be, equipment.

3. The self-igniting lights of lifebuoys shall be renewed before the expiry of a period of five years from the date of manufacture.

4. Any buoyant (buoyant) apparatus shall fulfil the following conditions, namely:—

a. it shall be so strong that it can be thrown into the water from its stowage position without being damaged in any way;

b. its weight shall not exceed 120 kilograms;

c. it shall be made of approved materials and by an approved process of construction, and the use of air-filled buoyancy chambers shall be prohibited;

d. it shall have a painter for securing a boat;

e. whichever way up it floats, whether upstream or downstream, it shall remain effective and stable;

f. it shall be capable of supporting in fresh water up to 6 kilograms of iron attached to line beckets along the length of any of its sides up to 30 centimetres, so that no part of the upper surface of the buoyant apparatus is submerged below the water;

g. there shall be a life line firmly attached all around the buoyant apparatus, so that there is one becket for each person for whom the buoyant apparatus is intended; and

h. the buoyant apparatus shall be certified for a number of persons equal to the smaller of the numbers obtained by dividing by ten the weight of iron which the buoyant apparatus will keep afloat in fresh water, or by dividing by 30 the number of centimetres of the perimeter of the buoyant apparatus.

6. Properties of life buoys

All life buoys—

a. shall be capable of supporting a mass of 10 kilograms in fresh water;

p.234 (b) shall be made of approved materials and shall be capable of withstanding the effects of oil and oil products;

c. shall be of a bright orange colour;

d. shall have a mass of at least 2.5 kilograms;

e. shall have an inner diameter of 450 millimetres +10 per cent and an outer diameter of not more than 800 millimetres;

f. shall be fitted with a grab line around the circumference so that it can be grasped by hand;

g. shall be capable of withstanding the impact of being dropped into the water from a height of 20 metres;

h. the life line shall be of a non-kinking nature, buoyant, of not less than 8 millimetres diameter and of not less than 5 kilo-newtons (KN) [sic] strength; and

i. the self-igniting light of the life buoy shall be such that it is not extinguished by water and is capable of withstanding the impact of being dropped into the water from a height of 20 metres.

7. Conditions for life jackets

Life jackets—

a. shall be of an approved design and made of approved materials;

b. shall be capable of supporting a mass of 7.5 kilograms in fresh water for 24 hours;

c. shall be suitable for keeping the head of an exhausted or unconscious person above the water;

d. shall be so designed that there is no possibility of risk even if worn wrongly, and shall be suitable for wearing inside out;

e. shall be capable, after entry into the water, of turning the body of the wearer from a face-down vertical position to a slightly backward-inclined safe floating position;

f. shall be capable of withstanding the effects of oil and oil products;

g. shall be of a bright orange colour;

p.235 (h) shall be suitable for being put on easily and quickly and for remaining firmly fastened to the body; and

i. shall bear the following particulars, namely:

1. name of the manufacturer,

2. type,

3. year of manufacture, and

4. stamp of approval.

Chapter 3

8. General

The life-saving equipment applicable to all ships and floating equipment under these Rules, and their stowage, shall be complied with.

9. Life buoys

1. Two life buoys fitted with 30-metre-long buoyant life lines shall be kept within or at the wheelhouse, ready for use at any moment.
2. If the ship or floating equipment operates at night, that is, during the period from sunset to sunrise, self-igniting lights shall be attached to the life buoys.
3. If the ship is more than 20 metres but less than 40 metres in length, a total of two additional life buoys, one on each side of the ship, shall be kept on it ready for use at any moment.
4. If the ship or floating equipment is more than 40 metres in length, a further two life buoys, one forward and one aft, shall be kept on it ready for use at any moment.
5. During the operation of the ship or floating equipment, the life buoys shall be kept outside enclosed spaces and in a position from which they can be released without obstruction.

10. Life jackets

One life jacket for each crew member shall be kept in the crew cabins or in a clearly marked locker outside the walled spaces.

p.236 ### 11. Pyrotechnic distress signals

1. Every ship and floating equipment shall carry 3 approved parachute [rockets] and 2 approved hand flares.
2. Pyrotechnic distress signals shall be renewed before the expiry of three years from the date of manufacture.

12. Additional requirements for passenger launches or ferries and motor-car ferries

In addition to the life buoys and life jackets mentioned in rules 9 and 10, every passenger launch or ferry and motor-car ferry shall carry the following equipment, namely:

a. it shall carry a number of life buoys not less than 10% of the certified number of passengers, and all those life buoys shall be kept ready on the ship for use at any moment;

b. the buoyant apparatus which the ship is certified to carry for not less than 10% of the number of passengers shall be kept on the uppermost deck of the ship in a free-floating position; and

c. ships certified to carry 250 or more passengers at night shall carry two additional parachute rockets and two hand flares.

13. Additional requirements for road ferries

In addition to the life buoys and life jackets mentioned in rules 9 and 10, every road ferry shall carry the following equipment, namely:

a. in the case of a ferry of length up to 15 metres, four additional life buoys,

b. in the case of a ferry of length from 15 to 25 metres, eight additional life buoys and buoyant apparatus for 20 persons,

c. in the case of length more than 25 metres, twelve additional life buoys and buoyant apparatus for 40 persons, and

d. all additional life-saving appliances and equipment shall be readily available for use at any moment, and the buoyant apparatus shall be kept in a free-floating position.

p.237 ### 14. Additional requirements for tankers

In addition to the life buoys and life jackets mentioned in rules 9 and 10, every tanker shall be equipped with the following items, namely:—

a. two life buoys fitted with self-igniting lights, and

b. two parachute rockets and two hand flares.

15. Dumb barges

Every dumb barge shall carry not less than two life buoys.

Chapter 4

16. Ordinary exits

1. From every crew accommodation space and working space there shall be at least one exit of minimum 600 millimetres width leading to the open deck.
2. From every passenger space there shall be at least two exits to the open deck, one of which shall be at least 800 millimetres and the other 600 millimetres in width.
3. If the space is designed for 50 or more passengers, both exits shall be at least 800 millimetres wide.
4. If there is only one passageway or stairway for passengers to reach the space designated for them, its clear width shall be at least 1000 millimetres, which in the case of open passenger ships shall be 800 millimetres wide.
5. Exits shall be arranged to open outwards.

17. Emergency exits

From every space which has fewer than two exits, there shall be one emergency exit for escape of a size of at least 600 × 600 millimetres and arranged to open outwards.

18. Stairways

1. Stairways leading to connecting doors shall have a clear width of at least 600 millimetres.
2. The inclination (Inclination) of the angle of stairways from the horizontal shall be as follows, namely:

p.238 (a) in the case of passenger stairways, a maximum of 40 degrees,

b. in the case of crew stairways, a maximum of 50 degrees, and

c. in the case of engine rooms, pump rooms and similar spaces, a maximum of 55 degrees.

3. Emergency exits, ship’s holds and tanks may have vertical ladders.

19. Hand railings

1. All stairways having three or more steps shall have at least one hand railing.
2. If stairways are more than 800 millimetres wide, there shall be hand railings on both sides.

20. Obstructions

1. There shall be no obstruction in exits, emergency exits and stairways.
2. The covering or canvas awning of the open spaces of the structure of passenger ships shall be suitable for being easily opened or removed in an emergency.
3. Open spaces shall not be covered with ropes or nets.

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Chapter 1 — Preliminary

1. Short title

These Rules shall be called the Inland Ships (Fire Safety) Rules, 2001.

2. Definitions

Unless there is anything repugnant in the subject or context, in these Rules—

a. “Ordinance” means The Inland Shipping Ordinance, 1976 (LXXII of 1976);

b. “inland water” means “Inland water” as defined in the Ordinance;

c. “accommodation space” means spaces used by the general public, cabins, corridors, sanitary spaces and other similar spaces;

d. “open passenger launch” means an open ship without a weather deck used for carrying more than 12 passengers;

e. “weather deck” means that deck below which all open parts are provided with such arrangements for closing as will, in all conditions, prevent the entry of water into the ship or floating equipment;

f. “ship” or “inland ship” means an “inland ship” as defined in section 2(c) of the Ordinance;

g. “tanker” means an inland ship used for carrying large quantities of liquid substances in tanks or containers (in bulk);

h. “dumb barge” means a ship, vessel or floating equipment used for carrying cargo or liquid substances which is not self-propelled and does not carry any passengers;

i. “length (L)” means the distance on the load water line from the fore end of the stem at the forward part of the ship to the after end of the rudder post, or, where the ship has no rudder post, to the centre of the rudder stock;

j. “new ship” means any ship registered after the date these Rules come into force;

k. “existing ship” means a ship which is not a new ship;

l. “floating equipment” means various non-self-propelled equipment such as dredgers, floating cranes etc., which are used for performing various work in a floating condition;

m. “fishing vessel” means a ship used for, or built for the purpose of, catching fish;

n. “cargo ship” means a self-propelled ship used for carrying cargo and not more than 12 passengers;

o. “passenger” means any person among those on board a ship or inland ship other than the master, officers and crew of the ship, but shall not include a child passenger under one year of age;

p. “passenger ferry” means a ship having a structure engaged in ferrying 12 or more passengers and one or more vehicles;

q. “passenger ship” means a ship with a weather deck used for carrying more than 12 passengers, and shall also include an inland ship having more than one deck;

r. “road ferry” means an open flush-deck ship engaged in ferrying 12 or more passengers and one or more vehicles.

3. Marking

1. The name of the ship shall be clearly written on all fire-fighting equipment.

2. On equipment requiring testing or re-filling, the date of testing or, as the case may be, of re-filling shall be

marked.

3. The place of storage of fire-fighting equipment shall be clearly marked.
4. If any fire-fighting equipment is kept in a locker, the door of the locker concerned shall be marked.

4. Application

Unless otherwise provided, these Rules shall apply to all inland ships.

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Chapter 2 — Fire-fighting equipment

5. Fire pump

Where one or more fire pumps are required, the fire pump shall be as follows, namely:—

a. a minimum pressure of at least 2 bar shall be maintained at all hydrants;
b. the minimum capacity of the pump or pumps shall be at least 10 cubic metres per hour;
c. the pump or pumps may be combined bilge, ballast and general service pumps;
d. if the pump is capable of developing a pressure in excess of the pressure-bearing capacity of the pipes, hydrants and hoses, one or more relief valves shall be fitted; and
e. the suction connections at the ends of the ship shall be located as deep as practicable.

6. Fire pipes and hydrants

1. Where a mechanical fire pump is required, a fixed fire-line system shall be installed.
2. The arrangement of the main and branch fire pipes shall be such that a jet of water can be directed to all parts of the ship by means of a single hose not more than 15 metres in length.
3. All fire pipes shall be made of steel.
4. The ship shall have only one type of hydrant, and its minimum diameter shall be 40 millimetres.
5. Pipes and hydrants shall be so installed that fire hoses can easily be coupled to the hydrants, and if cargo is carried on deck, the hydrants shall be so installed that they are easily within reach.
6. Valves shall be fitted to all hydrants.
7. The internal diameter of the main fire-fighting pipe shall be at least 50 millimetres.

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8. The fire pipe system may also be used for washing the deck.

7. Fire hoses and nozzles

Where fire hoses are required, the following conditions shall apply, namely:

a. there shall be at least one hose for each hydrant and at least three hoses on the ship;
b. the hoses shall be kept in clearly marked hose boxes or on reels adjacent to the hydrants;
c. the hoses shall be made of approved material;
d. the hoses shall be at least 10 metres and not more than 15 metres in length;
e. the diameter of the hoses shall be at least 40 millimetres;
f. whatever the hose diameter, all hoses shall have standard couplings;
g. all hoses shall be provided with a nozzle of 12 millimetres diameter and the necessary couplings.

8. Portable fire extinguishers

1. Portable fire extinguishers are divided into the following classes, namely:

a. Class A fire extinguishers, designed for extinguishing fires of fibrous materials such as wood, paper and machinery;

b. Class B fire extinguishers, designed for extinguishing fires of flammable liquid substances;

c. Class E fire extinguishers, designed for extinguishing fires of electrical equipment;

d. in addition, various portable fire extinguishers may also be in combined form.

2. The weight of any portable fire extinguisher ready for use shall not exceed 20 kilograms.

3. Each portable fire extinguisher shall be tested every year.

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4. Portable fire extinguishers shall be of an approved type.

5. All portable fire extinguishers of the same class and capacity shall be of the same type.

6. The portable fire extinguishers of any ship or floating equipment shall not be of more than two trade marks.

7. At least one of the portable fire extinguishers designated for use in any space shall be kept at the entrance to that space.

9. Fire buckets

Fire buckets shall be made of non-combustible material, shall have a capacity of 9 litres, and a lanyard at least 4 metres in length shall be attached to each fire bucket.

10. Fixed CO2 system

Where a fixed CO2 system is required, the following conditions shall apply, namely:

a. the arrangement of piping shall be so laid out that it is effective for, and evenly distributed to, the cargo holds;

b. the alarm system shall be so arranged that an alarm sounds automatically as soon as the release cabinet is opened;

c. the CO2 bottles shall be kept in a locked locker;

d. the CO2 gas contained in the bottles shall be tested once a year, and the bottles shall be pressure-tested at least once every 12 years at a pressure 50% in excess;

e. the necessary warning signs shall be affixed in the proper places;

f. the quantity of gas released shall be at least equal to 30% of the total volume of the largest protected cargo space.

11. Approval

All classes of fire-fighting equipment shall be approved by the inland navigation safety administration or the Department of Shipping.

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Chapter 3 — Distribution of fire-fighting equipment

12. Application

1. The requirements described in this Chapter concerning fire pumps, fire pipe systems, hydrants, fire hoses and nozzles shall apply only to new ships.

2. On existing ships and floating equipment, the fire pumps, fire pipe systems, hydrants, fire hoses and nozzles already provided shall be properly maintained and kept in working condition.

13. Cargo ships

1. Every inland cargo ship shall carry the fire-fighting equipment described below, according to the scale mentioned below, namely:—

a. where the total engine power is from 12 to 40 kilowatts—

b. two Class BE fire extinguishers, each of 9 litres capacity, in the machinery space;

ii. two fire buckets in accessible places.

b. where the total engine power exceeds 40 kilowatts but does not exceed 180 kilowatts—

c. four Class BE fire extinguishers, each of 9 litres capacity, in the machinery space;

ii. six fire buckets in accessible places;

iii. one sand box containing at least 50 litres of sand, with a scoop, in the machinery space.

iv. where the total engine power exceeds 180 kilowatts but does not exceed 450 kilowatts—

v. one fire pump in accordance with rule 5, which may be driven by the main engine;

vi. a fire pipe system and hydrants in accordance with rule 6, and two hydrants in the engine room;

vii. fire hoses and nozzles in accordance with rule 7;

viii. two Class BE fire extinguishers, each of 9 litres capacity, and in addition thereto, in the machinery space p.249 two further Class BE fire extinguishers, each of 6 litres capacity;

ix. nine fire buckets in accessible places;

x. one sand box containing at least 50 litres of sand, with a scoop, in the machinery space;

xi. one fire axe.

xii. where the total engine power exceeds 450 kilowatts but does not exceed 800 kilowatts—

xiii. one fire pump of at least 15 cubic metres capacity, or a pump in accordance with rule 5 which shall be driven independently without the aid of the main engine;

xiv. a fire pipe system and hydrants in accordance with rule 6, and at least 2 hydrants in the engine room;

xv. fire hoses and nozzles in accordance with rule 7;

xvi. two Class BE fire extinguishers of 9 litres capacity, and four fire extinguishers of 9 litres capacity in the machinery space;

xvii. nine fire buckets in accessible places;

xviii. two sand boxes containing at least 50 litres of sand, with a scoop, in the machinery space;

xix. two fire axes;

xx. one crow bar.

2. In addition to the equipment required under sub-rule (1), all inland cargo ships shall have the following fire extinguishers, namely:

a. one Class AB fire extinguisher of 9 litres capacity on every accommodation deck;

b. one Class ABE fire extinguisher of 6 litres capacity in the wheel house;

c. one Class AB fire extinguisher of 6 litres capacity in the galley, and if there is electrical equipment there, the fire extinguisher shall be of Class ABE.

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14. Dumb barges

Each cargo ship other than those carrying dangerous cargo shall have the following fire-fighting equipment, namely:

a. four Class ABE fire extinguishers, each of 9 litres capacity, placed in accessible places;

b. if there are any persons on board the dumb barge, then in addition to the above fire extinguishers, at least one Class AB fire extinguisher of 9 litres capacity in the accommodation area and one Class AB fire extinguisher of 6 litres capacity in the galley shall be kept;

c. six fire buckets.

15. Inland ships or barges carrying dangerous cargo

Every inland ship or barge carrying dangerous cargo shall have the equipment mentioned in rules 13 and 14, and in addition thereto a fixed CO2 fire-extinguishing installation in accordance with rule 10 for the cargo-carrying compartment.

16. Tankers

1. Every inland tanker shall carry fire-fighting equipment in accordance with the provisions of rule 13, and in addition thereto each shall have two Class B fire extinguishers of 36 litres capacity.

2. In the case of new ships, all air pipes and ventilation passages leading to the cargo tanks shall be fitted with flame screens.

3. All electrical installations above the tank deck and inside the pump room shall be of explosion-proof type and intrinsically safe.

4. In the case of new ships, no door or other exit may open towards the tank deck.

17. Open passenger launches

Every inland open passenger launch shall carry the fire-fighting equipment described below according to the scale mentioned below, namely:

1. where the total engine power is up to 40 kilowatts, or a maximum of 100 passengers, or a maximum length (L) of 20 metres—

a. two Class BE fire extinguishers of 9 litres capacity in the machinery space; p.251

b. two fire buckets in accessible places;

c. two Class ABE fire extinguishers of 9 litres capacity on the accommodation deck;

d. two Class ABE fire extinguishers of 6 litres capacity in the wheel house;

e. two Class ABE fire extinguishers of 6 litres capacity in the galley, and if there is electrical equipment there, the fire extinguishers shall be of Class ABE.

2. where the total engine power exceeds 40 kilowatts but does not exceed 180 kilowatts, or the number of passengers exceeds 100 but does not exceed 200, or the length (L) exceeds 20 metres but does not exceed 30 metres—

a. four Class BE fire extinguishers, each of 9 litres capacity, in the machinery space;

b. six fire buckets in accessible places;

c. one sand box containing at least 50 litres of sand, with a scoop, in the machinery space;

d. four Class ABE fire extinguishers of 9 litres capacity on every accommodation deck;

e. one Class ABE fire extinguisher of 6 litres capacity in the wheel house;

f. one Class BE fire extinguisher of 6 litres capacity in the galley, and if there is electrical equipment there, the fire extinguisher shall be of Class ABE.

3. If the total engine power of an open launch exceeds 180 kilowatts, or the number of passengers exceeds 200, or the length (L) exceeds 30 metres, the fire-fighting equipment shall be the same as for ships under rule 18.

18. Passenger ferries

Every passenger ferry shall carry the fire-fighting equipment described below, namely:

1. where the total engine power exceeds 180 kilowatts, or the number of passengers is up to 200, or the maximum length (L) is 30 metres—

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a. one fire pump in accordance with rule 5, which may be driven by the main engine;

b. fire pipes and hydrants in accordance with rule 6, with at least two hydrants in the engine or engine room;

c. fire hoses and nozzles in accordance with rule 7;

d. four Class BE fire extinguishers, each of 9 litres capacity, in the machinery space;

e. six fire buckets with lanyards attached:

Provided that the lanyards must be at least two metres longer than the height from the place where the buckets are kept to the light water line, and at least 4 metres in length, and the buckets shall be kept in accessible places;

f. one sand box containing at least 50 litres of sand, with a scoop, in the machinery space;

g. one fire axe;

h. 6 Class BE fire extinguishers of 9 litres capacity on the accommodation deck;

i. one Class BE fire extinguisher of 9 litres capacity in the wheel house;

j. four Class AB fire extinguishers of 9 litres capacity in the galley and the ship’s cook house, and if there is electrical equipment there, the fire extinguishers shall be of Class ABE.

2. where the total engine power exceeds 180 kilowatts but does not exceed 450 kilowatts, or there are more than 200 but not more than 400 passengers, or the length (L) exceeds 30 metres but does not exceed 45 metres—

a. one fire pump of at least 15 cubic metres per hour capacity, or a pump in accordance with rule 5 which shall be driven independently without the aid of the main engine;

b. fire pipes and hydrants in accordance with rule 6:

Provided that the arrangement of the hydrants shall be such that two jets of water can be directed to all parts of the ship by one hose not more than 15 metres long and one double hose 15 + 15 metres long, and there shall be at least two hydrants in the engine room; p.253

c. fire hoses and nozzles in accordance with rule 7, and in addition thereto 50% extra hoses, which shall be fitted with couplings;

d. four Class BE fire extinguishers of 9 litres capacity, and in addition two further Class BE fire extinguishers of 9 litres capacity in the machinery space;

e. ten fire buckets, each of 9 litres capacity, with lanyards attached:

Provided that the lanyards must be at least two metres longer than the height from the place where the buckets are kept to the light water line, and at least 4 metres in length, and the buckets shall be kept in accessible places;

f. two sand boxes containing at least 50 litres of sand, with a scoop, in the machinery space;

g. two fire axes;

h. at least 8 Class AB fire extinguishers of 9 litres capacity on the accommodation deck;

i. one Class ABE fire extinguisher of 9 litres capacity in the wheel house;

j. one Class AB fire extinguisher of 9 litres capacity in the galley and the ship’s cook house, and if there is electrical equipment there, the fire extinguisher shall be of Class ABE.

3. where the total engine power exceeds 450 kilowatts but does not exceed 800 kilowatts, or the number of passengers exceeds 400 but does not exceed 800, or the length (L) exceeds 45 metres but does not exceed 60 metres—

a. one fire pump of at least 20 cubic metres per hour capacity, or a pump in accordance with rule 5 which shall be driven independently without the aid of the main engine;

b. one fire pump in accordance with rule 5 which may be driven by the main engine;

c. a fire pipe system and hydrants in accordance with rule 6:

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Provided that the arrangement of the hydrants shall be such that two water jets can be directed to all parts of the ship by one single hose not more than 15 metres long and one double hose 15 + 15 metres long, and there shall be at least two hydrants in the engine room;

d. fire hoses and nozzles in accordance with rule 7, and in addition thereto 50% extra hoses fitted with couplings;

e. eight Class BE fire extinguishers of 9 litres capacity, and in addition four further Class BE fire extinguishers, each of 9 litres capacity, in the machinery space;

f. twelve fire buckets, each of 9 litres capacity, with lanyards attached:

Provided that the lanyards must be at least two metres longer than the height from the place where the buckets are kept to the light water line, and at least 4 metres in length, and the buckets shall be kept in accessible places;

g. two sand boxes containing at least 50 litres of sand, with a scoop, in the machinery space;

h. four fire axes;

i. Class AB fire extinguishers of 9 litres capacity shall be placed in the places described below, namely:

j. one fire extinguisher in each cabin corridor, and if the corridor is more than 30 metres long, two fire extinguishers shall be kept;

ii. one fire extinguisher for every 100 square metres in other passenger spaces;

iii. one fire extinguisher in each crew compartment;

iv. at least 10 Class AB fire extinguishers of 9 litres capacity on every accommodation deck.

j. two Class ABE fire extinguishers of 6 litres capacity in the wheel house; p.255

k. two Class AB fire extinguishers of 9 litres capacity in the galley and in all alleyways, and if there are electrical installations there, the fire extinguishers shall be of Class ABE.

4. where the total engine power exceeds 800 kilowatts, or the number of passengers exceeds 800, or the length (L) exceeds 60 metres—

a. one fire pump of at least 20 cubic metres per hour capacity, or a pump in accordance with rule 5 which may be driven independently of the main engine;

b. a second fire pump in accordance with rule 6, which may be driven by the main engine;

c. an emergency fire pump with an independent drive and with the valves mentioned in rule 5 shall be installed in the forward part of the ship, which shall be connected to the fire main;

d. a fire pipe system and hydrants in accordance with rule 6:

Provided that the arrangement of the hydrants shall be such that two water jets can be directed to all parts of the ship by two single hoses not more than 15 metres long, and there shall be at least four hydrants in the engine room;

e. fire hoses and nozzles in accordance with rule 7, and in addition thereto 50% extra hoses fitted with couplings;

f. eight Class BE fire extinguishers, each of 9 litres capacity, in the machinery space, and in addition six further Class BE fire extinguishers, each of 9 litres capacity;

g. twelve fire buckets, each of 9 litres capacity, with lanyards attached:

Provided that the lanyards shall be at least two metres longer than the height from the place where the buckets are kept to the LIGHT WATER LINE, and at least 4 metres in length, and the buckets shall be kept in accessible places;

h. two sand boxes containing at least 50 litres of sand, with a scoop, in the machinery space;

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i. six fire axes;

j. two crow-bars;

k. Class AB fire extinguishers of 9 litres capacity shall be kept in the following places, namely:

l. one fire extinguisher in each cabin corridor, and if the length of the corridor exceeds 30 metres, two fire extinguishers;

ii. one fire extinguisher for every 100 square metres in other passenger spaces;

iii. one fire extinguisher in each crew compartment;

iv. at least 12 Class AB fire extinguishers of 9 litres capacity on every accommodation deck;

v. two Class ABE fire extinguishers of 6 litres capacity in the wheel house; and

n. two Class AB fire extinguishers of 9 litres capacity in the galley and in all galleys, and if there are electrical installations there, the fire extinguishers shall be of Class ABE.

19. Road ferries

Class BE fire extinguishers of 9 litres capacity shall be provided on road ferries as follows, namely:

a. where the total engine power is up to 40 kilowatts, two;

b. where the total engine power exceeds 40 kilowatts but does not exceed 180 kilowatts, four;

c. where the total engine power exceeds 180 kilowatts, six;

d. where the total engine power exceeds 360 kilowatts, eight;

e. where the total engine power exceeds 720 kilowatts, one fixed CO2 fire-extinguishing system;

f. 10 fire buckets.

20. Floating equipment

Portable fire extinguishers shall be kept on all floating equipment as follows, namely: p.257

a. in a compartment in which oil treatment or transfer equipment, or machinery of any kind of 40 kilowatts total power, is installed, one Class BE fire extinguisher of 9 litres capacity;

b. in a compartment in which machinery of any kind of total power up to 180 kilowatts is installed, two Class BE fire extinguishers of 9 litres capacity;

c. in a compartment in which machinery of any kind of total power up to 450 kilowatts is installed, four Class BE fire extinguishers of 9 litres capacity;

d. in a compartment in which machinery of any kind of power exceeding 450 kilowatts is installed, four Class BE fire extinguishers of 9 litres capacity and two Class AB fire extinguishers of 9 litres capacity;

e. on all floating equipment, two Class BE fire extinguishers and two fire extinguishers of 9 litres capacity.

21. Fishing vessels and service vessels

Fishing vessels and service vessels shall have fire extinguishers similar to those for cargo ships in accordance with rule 13.

Chapter 4 — Structural fire safety

22. Application

Unless expressly provided otherwise elsewhere in this Chapter, the provisions of this Chapter shall apply only to new ships, and none of them shall apply to open ships.

23. Location of machinery

1. All machinery spaces, including tanks and outer casings adjacent to accommodation, shall be enclosed by bulkhead walls of steel at least 5 millimetres thick.

2. The lining on the bulkhead walls shall be of non-combustible material.

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3. All doors of machinery spaces shall be made of steel.

24. Exhaust pipes

1. This rule shall apply to all inland ships.

2. Each exhaust pipe of an engine of power exceeding 40 kilowatts shall be lagged with non-combustible lagging material in such a manner that the surface of the pipe can never become heated above 60 degrees Celsius. This lagging shall be well protected (by galvanised steel).

25. Sight-glass

1. This rule shall apply to all inland ships.

2. All sight-glasses connected to fuel oil tanks shall be made of approved oil- and heat-resistant material.

3. All sight-glasses shall be provided with arrangements to protect them from mechanical damage.

4. A valve capable of self-closing shall be fitted at the lower part of the sight-glass.

26. Pump room

1. Pump rooms shall be kept safe in the same manner as machinery spaces.

2. There shall be no connection between a pump room and machinery spaces.

27. Stairways

1. If an internal stairway provides communication between two decks, there shall be a door made of non-

combustible material at the side of the deck.

2. If an internal stairway provides communication between more than two decks, it shall be protected by a watertight steel wall at least 4 millimetres thick, and a door shall be fitted on each deck.

3. The stairway door shall be arranged to open outwards and shall be made of non-combustible material.

28. Air inlets

1. All air intakes other than those of machinery spaces shall be provided with closing arrangements which can be closed from outside. p.259

2. If the closing arrangement is within a trunk or casing, the location of the closing arrangement shall be clearly indicated outside.

3. These closures or closers shall be well maintained and kept serviceable at all times.

Chapter 5 — Arrangements for protection from explosion and fire

29. Lamps and electrical installations

1. No open lamp of any kind may be kept in the tanks, hold, store or open deck of oil-carrying tankers.

2. All electrical wiring connection systems shall be insulated, and conduit pipes shall be used where necessary.

3. The electrical installations of pump rooms, paint, lamp and battery lockers shall be of explosion-proof type.

30. Muster list and drill

1. The master of the ship shall record in a muster list or general list the particulars of those among the crew of the ship or floating equipment who will work at the fire stations, and shall have it displayed in a conspicuous place.

2. In the event of a fire, the crew shall proceed to their stations of duty in accordance with the particulars of the muster list and perform their respective duties.

3. For this purpose, a fire-fighting exercise or fire drill shall be held at least once every month, and the particulars of this drill shall be recorded in the log book.

31. Combustible and readily flammable materials

1. No dangerous goods or cargo may be carried on a passenger ship.

2. Any combustible or readily flammable material weighing more than 20 kilograms or exceeding 1 cubic metre in volume shall be stored away from the spaces designated for passengers and kept covered with tarpaulin or other suitable material.

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3. If oil is spilled on the deck, in the bilges or in machinery spaces, it shall be cleaned up with cotton rags.

4. When handling combustible oil, fuel or cargo, the following signals or lights shall always be displayed, namely:

a. at night, display of a red light visible along the horizon for at least 1.5 kilometres;

b. in the daytime, hoisting of a red flag.

5. A laden oil or chemical tanker shall always display the above-described light or flag.

32. Restrictions for passengers

1. Passengers shall not be permitted free access to machinery and cargo spaces.

2. Passengers shall not be allowed to enter any dangerous area, and all dangerous places shall be kept restricted.

3. In areas where smoking is prohibited, the words “No Smoking” shall be written up.

33. Soldering or welding

No hot repair work, soldering work or gas-cutting work may be carried out in any part of a ship or floating equipment where there are readily flammable or combustible materials, without fire-prevention arrangements and suitable fire extinguishers having been put in place.

34. Welding on tanker ships

Where welding or any other hot work is to be carried out on a tanker ship, or hot work is to be carried out on a tank carrying combustible substances on other ships, a “gas free” certificate shall be obtained from the Explosives Department and the hot work permit shall be followed.

35. Ventilation

Adequate ventilation arrangements shall be provided in machinery, battery, pump rooms and cargo spaces.

Inland Ships Examination Syllabus (Inland Master, Driver etc.) Rules, 2001

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Chapter 1 — Introduction

1. Short title

These Rules shall be called the Inland Ships Examination Syllabus (Inland Master, Driver etc.) Rules, 2001.

2. Syllabus for the examination for the Third Class Inland Master’s certificate of competency

Every candidate for the Third Class Inland Master’s certificate of competency shall take part in an oral examination, and the oral examination shall be held on the following subjects, namely:—

a. general principles of navigation on rivers, including the inland ships navigation rules;

b. storm signals, inland and international;

c. matters relating to the ports of Bangladesh;

d. knowledge of rivers and pilotage;

e. fire-fighting and life-saving appliances, their use and maintenance;

f. management of inland ships;

g. emergency procedures;

h. the provisions of the Inland Shipping Ordinance, 1976 (Ordinance No. 72 of 1976);

i. boxing up the compass and determining the course;

j. stability and free board (elementary knowledge);

k. use of binoculars;

l. ship survey.

3. Syllabus for the examination for the Second Class Inland Master’s certificate of competency

1. A candidate for the Second Class Inland Master’s certificate shall take part in a written examination on the following subjects, namely:—

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a. work relating to cargo carried on ships (elementary knowledge);

b. tides and currents;

c. determination of speed, distance and ship’s passage time;

d. compass and binoculars (elementary knowledge).

2. If the candidate passes the written examination, he may take part in the oral examination on the following subjects, namely:

a. general principles of navigation on rivers, including the inland ships navigation rules;

b. storm signals, inland and international;

c. matters relating to the ports of Bangladesh;

d. knowledge of rivers and pilotage;

e. fire-fighting and life-saving appliances, their use and maintenance;

f. knowledge of the life-saving and fire-protection rules for inland ships;

g. management of inland ships;

h. emergency procedures and beaching;

i. the provisions of the Inland Shipping Ordinance, 1976 (Ordinance No. 72 of 1976);

j. boxing up the compass and determining the course;

k. practical knowledge of river charts;

l. stability and free board, and ship construction;

m. use of binoculars;

n. ship survey;

o. as optional subjects: carriage of dangerous goods, carriage of liquid substances in bulk in tanks, chemicals and gas.

4. Syllabus for the examination for the First Class Inland Master’s certificate of competency

1. A candidate for the First Class Inland Master’s certificate of competency shall take part in a written examination on the following subjects, namely:

a. stability, free board and ship construction (elementary knowledge);

b. tides and currents; p.267

c. chart work, including determination of speed, distance and ship’s passage time;

d. loading and lashing of cargo carried on ships (elementary knowledge).

2. If the candidate passes the written examination, he may take part in the oral examination on the following subjects, namely:—

a. general principles of navigation on rivers, including the inland ships navigation rules;

b. international regulations for safe navigation;

c. matters relating to the ports of Bangladesh;

d. storm signals, inland and international;

e. knowledge of rivers and pilotage;

f. fire-fighting and life-saving appliances, their use and maintenance;

g. life-saving and fire safety regulations for inland ships;

h. management of inland ships;

i. signals and communication;

j. the Inland Shipping Ordinance, 1976 (Ordinance No. 72 of 1976), the rules made under that Ordinance and the notifications issued thereunder;

k. survey and dry docking of inland ships;

l. reading the weather;

m. as optional subjects: carriage of dangerous goods, carriage of liquids and chemical substances in bulk in tanks, and carriage of gas.

5. Syllabus for the examination for the Third Class Engine Driver's certificate of competency

A candidate for the Third Class Engine Driver's certificate of competency shall take part in an oral examination, and the oral examination shall be held on the following subjects, namely:

a. various types of internal combustion engines (working knowledge as required);

b. names of the various parts of an internal combustion engine;

c. engine overhauling, measurement of wear of the various parts of the engine due to use, replacement of engine parts and keeping the engine in working order;

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d. various methods of supplying air and fuel oil to an internal combustion engine;

e. method of starting the engine;

f. various faults preventing the engine from working properly, and rectification of those faults;

g. precautionary measures to be taken to prevent breakdown of engine and machinery;

h. method of running the ship;

i. dry docking of the ship and method of repair during docking;

j. ship survey;

k. precautions for the prevention of fire;

l. maintenance and operation of fire-fighting equipment;

m. cost of maintenance of various pumps, valves, cocks and pipes;

n. maintenance of batteries, self-starters and other electrical gear commonly used on small vessels;

o. air compressors, air bottles, storage tanks and their maintenance;

p. damage control and emergency procedures;

q. life-saving appliances;

r. stability and free board (elementary knowledge).

6. Examination for the Second Class Engine Driver's certificate of competency

A candidate for the Second Class Engine Driver's certificate of competency shall take part in written and oral examinations on the following subjects, namely:

Written examination

a. Simple arithmetic, namely: addition, subtraction, multiplication and division, decimal numbers and square roots; oil and store materials; capacity of tanks, speed of the ship and similar matters;

b. Magnetism and electromotive force (e.m.f) including batteries, voltage, current, resistance and power, units of current and resistance; distribution of circuit current, resistance of conductors, Ohm's law, simple electric circuits, effects of electric current. p.269

Oral

c. Making, maintaining and charging secondary cells (acid and alkaline); moving iron (deflection type); working principle, construction and maintenance of motors and generators; safety fittings and switch boards; personal safety and precautions;

d. Matters relating to the working of various types of internal combustion engines, names of the various parts of the engine, their maintenance, repair and assembly; lubrication and cooling systems; method of starting the engine; gearing system, alignment of the crankshaft (CRANKSHAFT), replacement and repair of bearings; repair and replacement of liners, pistons, air start valves, fuel injectors, fuel pumps and engine governors etc.; fault finding and rectification relating to engine operation; repair and maintenance of the turbo-charger system; engine safety devices; emergency operation of the engine, precautions to be followed regarding the storage of lubricating and fuel oil; precautions against the occurrence of fire; fire-fighting; maintenance of fire-fighting equipment, keeping it in a usable condition and its use; use of life-saving appliances and equipment; various pumps (Pump); valves, cocks and pipes; anchor windlass etc.; power steering, dry docking and various kinds of repairs carried out in dock; ship surveying and various kinds of survey equipment; damage control and emergency procedures; secondary cells, charging, moving coil, moving iron (deflection type), magnetic field; lines of force, conductors, insulators, fields produced by current in conductors; motors and generators, maintenance of electrical equipment in operation, electrical switch boards and safety fittings, personal safety and safety precautions. Stability and free board, and ship construction.

7. Examination for the First Class Engine Driver’s certificate of competency

A candidate for the First Class Engine Driver’s certificate of competency shall take part in written and oral examinations on the following subjects, namely:

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Written

a. Mathematics: correct calculation, averages, percentages, ratios; square roots with the aid of statistical tables, exchange of units; rearrangement of formulae, use of brackets and factors, simplification of fractions; determination of the air pressure of an air bottle at a given measurement, determination of the loads on various parts of machinery and on shafts when the necessary data are given;

b. Magnetism and electricity: all electric circuits, Ohm’s law, series and parallel electric circuits, electromotive force, voltage, units of current, voltage, resistance and power, distribution of circuit current, change of dimensions (shape) of a body, variation of the resistance of a conductor due to the temperature coefficient of resistance.

Oral

c. Effects of electric current, such as chemical, magnetic and thermal effects; secondary cells, moving coil, moving iron instruments (deflection type), magnetic field, lines of force, field created by the flow of current through a straight conductor; working principle and construction of motors and generators, exchange, synchronising (Synchroniging), load sharing and parallel running, maintenance of electrical equipment, safety fittings on electrical switch boards, personal safety and precautions;

d. Working principle of internal combustion engines, difference between petrol and diesel engines, difference between four- and two-stroke cycles; construction, maintenance and operation of cylinder heads, liners, pistons, crank shaft bearings, turbo-chargers, fuel injectors, air starting valves, fuel pumps and governors, timing of fuel injectors and fuel pumps, air starting systems; maintenance, proper placement and operation of various types of pumps, such as centrifugal, gear, screw, reciprocating, mono pumps; principles of operation, maintenance and working condition of power steering (hydraulic, all-electric and mechanical); lubrication and cooling systems of marine diesel engines; maintenance of coolers, valves, cocks and piping systems; inlet, exhaust valves and p.271 setting the timing of fuel pumps; characteristics of various fuels and lubricants, storage and handling; operation of pressure gauges, thermometers, barometers, boiler water level indicators; boiler operation and maintenance of boiler mountings and boiler water; fire prevention, detection and extinguishing, fire-fighting appliances and equipment, refrigeration; maintenance of electric motors, generator switch-gear and safety devices; ship survey, life-saving appliances, structural arrangement of the ship, damage control and emergency procedures, power-driven windlasses and cranes; fault finding and permanent, temporary or emergency repair in the event of failure of the electrical power supply in essential areas such as navigation lights, steering gear, navigation equipment, engine starting systems etc.; restarting of engines, pumps, steering gear, compressors etc. after a black-out or dead-ship/inactive condition; use of life-saving appliances and equipment at open sea, fire-fighting at open sea. Stability, free board and construction knowledge of the ship.

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Inland Ships (Manning, Examination and Certification) Rules, 2001

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Chapter 1 — Preliminary

1. Short title

These Rules shall be called the Inland Ships (Manning, Examination and Certification) Rules, 2001.

2. Definitions

1. Unless there is anything repugnant in the subject or context, in these Rules—

a. “Ordinance” means The Inland Shipping Ordinance, 1976 (LXXII of 1976);

b. “inland water” means “inland water” as defined in Section 2 (f) of the Ordinance;

c. “ISSA” means the Inland Ship Safety Administration;

d. “chemical tanker” means a ship used for carrying large quantities of liquid chemical substances in tanks;

e. “gas carrier” means a ship used for carrying large quantities of gas in containers under pressure or in a refrigerated condition;

f. “ship” or “inland ship” means “inland ship” as defined in section 2 (e) of the Ordinance;

g. “Driver” means any person of the engine department holding a certificate of competency of the appropriate grade issued by the Principal Surveyor, who is responsible for the mechanical operation of the ship;

h. “Principal Surveyor” means the head of the ISSA;

i. “Director General” means the Director General of the Department of Shipping;

j. “Master” means any person of the deck department holding a certificate of competency of the appropriate grade issued by the Principal Surveyor, who, being in overall charge of the ship, is responsible for the navigation of the ship;

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k. “Sukani” means any deck seaman holding a certificate of competency for holding the helm (wheel) of a ship;

l. “recognised institute” means any institute recognised by the Director General.

3. Application

Unless otherwise provided, these Rules shall apply to all ships and floating equipment irrespective of size and type.

Chapter 2 — Manning of inland ships, etc.

4. Minimum safe manning certificate

1. All ships shall carry a minimum safe manning certificate issued by the Principal Surveyor.

2. Subject to the availability of the necessary life-saving appliances, the owner of a ship may employ, in addition to the number of certificated Masters, Inland Marine Engineers and Drivers mentioned in the minimum safe manning certificate, additional Masters and Drivers and ratings.

5. Restrictions on manning

1. No person who is not a citizen of Bangladesh shall be employed on any ship without the prior written approval of the Principal Surveyor.

2. Unless the competency to perform the relevant work is clearly mentioned in the requisite certificate of competency, no person shall be employed as Master, Inland Marine Engineer or Driver on any oil tanker, chemical tanker or gas carrier, nor shall be employed in charge of the loading, unloading and carriage of any dangerous goods.

3. No person who does not hold the requisite certificate of competency, or has not obtained a dispensation from the Principal Surveyor, shall be employed in charge of any post on a ship.

4. If a person does not hold a certificate of competency or, in lieu thereof, a dispensation, that person shall not be permitted to be given charge of a navigational watch or an engine-room watch.

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5. No person shall be employed in any post on any ship unless a medical officer approved by the ISSA has certified him to be medically fit.

6. Unless specially permitted by the Principal Surveyor, no person above the age of 58 years shall be employed on any ship.

Chapter 3 — Examination of personnel of inland ships

6. Examinations, etc.

1. The Principal Surveyor shall issue notifications to inform all concerned about the dates, places and time schedules of the various examinations.

2. The fees for the certificate of competency examinations to be held under these Rules shall be paid at the rates fixed by the Principal Surveyor.

3. A candidate shall have to obtain fifty per cent marks to pass the written examination in the relevant subject and sixty per cent marks to pass the oral examination.

7. Examination Board

1. For holding examinations in subjects other than dispensation, eyesight test and examinations in subjects prescribed for that purpose under these Rules, a Masters' Examination Board and a Drivers' Examination Board shall be constituted by the Principal Surveyor.

2. Each Board to be constituted under sub-rule (1) shall consist of a Chairman and one or more members.

3. The Chief Examiner of Masters, holding a First Class deck certificate of competency, shall be the Chairman of the Masters' Examination Board, and in his absence another examiner holding the same certificate of competency, nominated by the Principal Surveyor, shall act as Chairman. The other members of the Masters' Examination Board shall also be holders of First Class deck certificates of competency.

4. The Chief Examiner of Drivers, who shall be a Marine Engineer Officer (Motor) holding a First Class certificate of competency, shall be the Chairman of the Drivers' Examination Board, and in his absence an examiner holding the same certificate of competency, nominated by the Principal Surveyor, shall act as Chairman.

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The other members of the Drivers' Examination Board shall also be holders of First Class Marine Engineer Officer (Motor) certificates of competency.

5. Notwithstanding anything contained in this rule,—

a. all examiners of the Examination Board shall have practical experience of ships navigating on the rivers of Bangladesh and of inland water transport;

b. the examinations for the Karnaphuli River and Pashur River endorsements shall be conducted with the participation of representatives of the Chittagong Port Authority and the Mongla Port Authority respectively.

8. Examination fund

1. There shall be an examination fund constituted from the fees collected from the candidates for the holding of the examinations to be conducted under these Rules.

2. The Principal Surveyor shall be the drawing and disbursing officer of the examination fund mentioned in sub-rule (1), who shall maintain proper and audited accounts thereof.

3. Money may be spent from the examination fund in the following cases connected with the examinations to be held under these Rules, namely:

a. purchase of stationery, including question papers, answer scripts, pens, pencils, etc., required for holding the written examinations;

b. purchase of equipment such as eyesight test lanterns, signal lanterns, magnets, compasses, sextants, barometers, ship models, channel markings, marine engines, pumps, valves, machinery parts, etc.;

c. purchase of instruments such as parallel rulers, set squares, compasses, dividers, calculators, computers, etc.;

d. purchase of publications such as nautical tables, nautical almanacs, tide tables, meteorological codes, navigation charts, etc.;

e. purchase of technical books, periodicals and treatises necessary for giving guidance to examiners and for setting question papers;

f. procurement of such furniture as is necessary for holding examinations, including such articles as may be necessary for the storage and safekeeping of stationery, equipment, instruments and publications;

g. purchase of any other article necessary for holding examinations.

4. Honoraria for performing duties in connection with the conduct of examinations under these Rules may be paid in the following manner; namely:

a. if the examiner is an officer employed by the ISSA, forty per cent of the fees received for the examination in the relevant subject;

b. if the examiner is not an officer of the ISSA, in that case an amount sanctioned from the examination fund, including travelling allowance and daily allowance having regard to the status of the examiner, on the recommendation of the Principal Surveyor and subject to the approval of the Government;

c. in the case of the examinations, the examiners concerned shall receive honoraria for setting question papers at rates fixed by the Principal Surveyor.

9. Appointment of cadets

Unless exempted by the Principal Surveyor on grounds of non-availability, every ship of 450 kilowatts or greater power may employ one Deck Personnel Training Centre cadet and one Marine Technology cadet as deckhand and greaser respectively.

10. Service book

There shall be a service book, in the form prescribed by the Principal Surveyor, for recording information relating to the joining of and discharge from service of the Master, Driver, Coxswain and ratings of every ship, and every event or matter entered in the service book, from joining the ship until leaving the ship, shall have to be endorsed by the Engineer and Ship Surveyor of the nearest survey office or by an officer nominated by him.

11. Certificate of competency

1. The Principal Surveyor shall issue the certificates of competency of the Master, Driver, Coxswain and ratings of ships.

2. A certificate of competency issued under sub-rule (1) shall remain valid for service on vessels for 5 years from the date of its issue.

3. The Principal Surveyor shall, on the written application of any Master and Driver, revalidate their certificates of competency subject to the following conditions, namely:

a. during the period of validity of the certificate of competency, within the 5 years preceding the application for revalidation, the applicant has been employed in service on board a ship for a total of 12 months; or

b. within the three months preceding the date of the application for revalidation, he has been employed in service on a vessel at one grade below the competency of his certificate;

c. an approved medical practitioner has certified the applicant to be medically fit.

4. A copy of the certificate of competency issued under sub-rule (1) shall be duly preserved and recorded in the ISSA.

12. Provisional certificate of competency

1. Until the certificate of competency is issued under rule 11, a provisional certificate of competency shall be issued, for a period not exceeding six months, to the Master and Driver who have passed the examination of the relevant grade.

2. At the time of receiving the certificate of competency, the Master or, as the case may be, the Driver concerned shall return the provisional certificate of competency.

13. Issue of duplicate certificate of competency

1. If the certificate of competency of the Master or Driver of any ship is lost or torn, the Master or Driver concerned shall forthwith apply in writing to the Principal Surveyor for a duplicate of the certificate of competency, on payment of the prescribed fee.

2. On receipt of an application under sub-rule (1), the Principal Surveyor shall issue a duplicate of the relevant certificate of competency to the Master or Driver concerned.

14. Previous certificates

The certificates of competency of the various grades of Master, Inland Marine Engineer and Driver issued before the coming into force of these Rules shall be deemed to be certificates issued under these Rules.

15. Health certificate

1. All seamen of a ship shall have a health certificate valid for twelve months and shall keep it with them while on board the ship.

2. A medical practitioner approved by the ISSA shall issue the said certificate.

3. The medical practitioner shall issue the certificate after examining the general health, including eyesight and hearing.

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16. Requirement of watchkeeping

1. A Master, Inland Marine Engineer, Driver and Coxswain shall work only in the capacity for which he has obtained the ship’s certificate, or in a capacity below it.

2. No person shall, without holding the proper certificate under these Rules, perform any work for which a certificate of competency is required.

3. If any person dies during a voyage, or any person becomes incapable of performing his duties, and no person holding the same or a lower certificate, or no such person, is available on board the ship, the senior-most person concerned, even though not a certificate holder, may act in place of the deceased person until the next specified port or destination is reached.

4. Subject to the provisions of sub-rule (3), no person who is not duly qualified in accordance with these Rules shall be employed to perform the relevant work.

5. If any person contravenes any provision of these Rules, he shall be punishable under the relevant section of the Ordinance.

17. Exemptions

The Principal Surveyor may, subject to such conditions as he may prescribe, declare in any special case all or any of the provisions of these Rules to be inapplicable, and may also cancel or amend such declaration.

18. Syllabus of the examinations

The syllabus of the examinations under these Rules shall be prescribed by the Principal Surveyor.

Chapter 4 — Examination for Master’s certificate of competency

19. Application for examination

1. A candidate for the Master’s certificate of competency examination shall apply to the Chief Examiner, in the form prescribed for that purpose, together with his service book, testimonials and three copies of attested photographs, not less than 15 days before the fixed date of the examination.

2. The owner of the ship concerned shall certify as to the presence of any candidate on board the ship, and shall issue the testimonial mentioned in sub-rule (1) to that effect.

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20. Qualifications required for the Master’s certificate of competency examination

1. Any candidate having the following qualifications may appear in the examination for the Third Class Master’s certificate of competency of a ship, namely:

a. the age of the candidate shall not be less than twenty-one years, and if the Chief Examiner expresses doubt

about the age of the candidate, then a certificate shall have to be produced as to his age;

b. the candidate must be physically and mentally fit to perform duties while on board a ship, and must be free from stammering and loss of hearing;

c. before the examination, the candidate shall have to pass an eyesight test (without spectacles), a colour-distinguishing test and a lantern test, to be conducted by the Chief Examiner or an examiner nominated by him;

d. if any candidate fails in the eyesight test, he may appear again in the said test after a period of three months from the date of failure, and if any candidate fails in the lantern test, he may appear again in such test only three times, at intervals of three months between such examinations;

e. the candidate shall have to work for at least three years in the deck department of a power-driven ship, of which the last one year he shall have to work as Sukani;

f. the candidate must have passed Class 8;

g. candidates who have successfully completed a 1-year marine apprenticeship course at the Deck Personnel Training Centre or any other approved educational institution shall have to serve, before appearing in the examination, for at least two years on ships of 110 kilowatts and greater power, of which the last one year's service shall be as Sukani;

h. if any candidate fails to pass the written or oral examinations three times in succession, then he shall not be considered eligible to appear in the examination again unless he has served for at least three months of additional countable service as Sukani before filling in the papers for the next examination;

i. a candidate fulfilling the conditions of clauses (a), (b) and (c) of this rule shall not be considered eligible to appear in the examination unless he has served on board a ship for at least twelve months within the 2 years preceding the date of applying for the examination and filling in the papers.

2. To appear in the examination for the Second Class Master's certificate of competency, the candidate shall have to fulfil the following conditions:

a. the candidate must hold a Third Class Master's certificate of competency and have at least two years' service experience as in-charge, or two and a half years as Assistant Master, or three years as Sukani, on a ship at least 24 metres in length or with an engine power of 137 kilowatts;

b. subject to having twelve months' service experience on inland ships, at least ten years' active service experience as Leading Seaman, Petty Officer or Chief Petty Officer of the executive branch of the Bangladesh Navy;

c. three years' service experience as helmsman on a foreign-going ship and 1 year's service experience on inland ships;

d. completion of training at the Fisheries Academy and at least three years' service experience on fishing ships.

3. To appear in the examination for the First Class certificate of competency, the candidate shall have to fulfil the following conditions, namely:

a. the candidate must hold a Second Class Master's certificate of competency, and he shall have to have served for at least three years as a minimum Assistant Master on board a ship at least 36 metres in length or with an engine power of at least 185 kilowatts, or as Master on a ship 37 metres in length or with an engine power of at least 137 kilowatts;

b. subject to having at least six months' service experience on inland ships, three years' experience of watchkeeping work as a commissioned officer of the executive branch of the Bangladesh Navy;

c. subject to having at least six months' service experience on inland ships, experience as a Deck Officer of at least Fourth Class on a foreign-going ship.

4. A candidate having the qualifications mentioned in sub-rules (1), (2) and (3) shall have to fulfil the following conditions before appearing in the relevant examination, namely:

a. the candidate shall have to pass the eyesight test (without spectacles) conducted by a medical officer nominated by the Chief Examiner or by the ISSA;

b. if any candidate fails in the eyesight test, he may pass the said test again after the lapse of three months from the date of such failure;

c. if any candidate fails in the lantern test, he may pass the said test again after the lapse of three months from

the date of such failure.

5. If any candidate fails three times in succession in a certificate of competency examination under rule 6—

a. a candidate for a First Class certificate of competency shall have to serve for not less than six months in the proper post on a ship before appearing again in the next examination;

b. a candidate for a Second Class certificate of competency shall have to serve for not less than six months in the proper post on a ship before appearing again in the next examination;

c. a candidate for a Third Class certificate of competency shall have to serve for not less than three months in the proper post on a ship before appearing again in the next examination.

21. Rules

Notwithstanding anything contained in rule 20, before taking part in any Master's certificate of competency examination, a Master of every class shall have to successfully complete the following courses, namely:

a. Preparatory Course;

b. Basic Fire Prevention and Fire Fighting Technique Course;

c. Personal Safety and Social Responsibility Course;

d. Elementary First Aid Course;

e. Personal Survival Technique Course;

For service on vessels, the validity of the courses from (b) to (e) shall remain in force for the next 5 (five) years from the date of receipt of the certificate.

Chapter 5 — Examination for Drivers' certificate of competency

22. Application for examination

1. For the ship's Driver certificate of competency examination, the candidate concerned shall apply to the Chief Examiner, in the form prescribed for that purpose, together with his service book, testimonials and three copies of attested photographs, not less than 15 days before the fixed date of the examination.

2. The owner of the ship concerned shall certify as to the presence of any candidate on board the ship, and shall issue the testimonial mentioned in sub-rule (1) to that effect.

23. Qualifications required for the ship's Driver certificate of competency examination

1. Any candidate having the following qualifications may appear in the examination for the Third Class Driver's certificate of competency of a ship, namely:

a. the age of the candidate shall not be less than twenty-one years, and if any doubt arises in the mind of the Chief Examiner as to the age of the candidate, then the candidate shall have to produce a certificate as to his age;

b. the candidate must have served for at least three years in the engine department on board a ship of at least 137 kilowatts power;

c. the candidate must have passed Class 8;

d. the candidate must be fit in eyesight, in the ability to distinguish colours, in hearing, and physically and mentally;

e. candidates who have successfully completed a 1-year marine

apprenticeship course at the Bangladesh Institute of Marine Technology or any other similar educational institution shall have to serve, before taking part in the examination, for at least two years in the engine department on board a power-driven ship of at least 137 kilowatts power.

2. Any candidate having the following qualifications may appear in the examination for the Second Class Driver's certificate of competency, namely:

a. the candidate must hold a Third Class Engine Driver's certificate of competency and must have 24 months' post-certificate service experience as Driver-in-charge on board a motor-driven vessel of at least 137

kilowatts propulsion power, or as second-in-charge on an inland vessel of at least 186 kilowatts propulsion power;

b. an Engine Mechanic or Leading Engine Mechanic of the Bangladesh Navy, who must have at least ten years' active service experience, including at least five years' work on motor-driven ships of the Navy;

c. must have experience as an engine-room rating of a foreign-going vessel with at least three years' watchkeeping service experience;

d. a candidate holding a diploma or certificate in Marine Technology from any recognised institute may be permitted to take part in the examination before joining any vessel, but their certificates shall not be issued until they have performed at least 6 months' active service on vessels of at least 186 kilowatts power;

e. if any candidate fails three times in succession in the written or oral examination, he shall have to serve for not less than six months additionally on a ship before appearing in the next examination.

3. Any candidate having the following required qualifications may appear in the examination for the First Class Driver's certificate of competency, namely:

a. any candidate holding a Second Class Driver's certificate of competency of inland ships who has served for at least three years in the post of Assistant Driver on a ship with an engine power of not less than 450 kilowatts, or as Driver on a ship with an engine power of not less than 186 kilowatts

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power has service experience of not less than 24 months in the post of Driver;

b. an Engine Room Artificer (Engine Mechanic), Chief Petty Officer (Engine Mechanic) and Petty Officer of the Engineering branch of the Bangladesh Navy, who has a total of 12 years' active service experience, including not less than 5 years on board ships of the Bangladesh Navy;

c. an engine-room rating counted as part of the engine-room watch of a foreign-going ship, who has not less than six years' service experience;

d. a candidate who has passed the diploma course in Marine Technology from a recognised institute and who has 24 months' service experience, including presence on board a motor-driven ship with an engine power of not less than 186 kilowatts;

e. any candidate holding a certificate in Marine Technology from a recognised institute who has 24 months' practical service experience, including presence on board a motor-driven ship with an engine power of not less than 186 kilowatts;

f. any candidate holding a diploma in Mechanical or Power Engineering from a recognised polytechnic institute, who has 24 months' practical service experience, including presence on board a motor-driven ship with an engine power of not less than 186 kilowatts;

g. if any candidate fails three times in succession in the examination, he shall have to serve for not less than six months on a ship before appearing in the next examination;

h. for obtaining a "Boiler Operator Licence" or endorsement, a candidate shall have to complete 24 months' service experience in a boiler-attached installation and shall have to pass the written and oral examinations on that subject.

24. Rules

Notwithstanding anything contained in rule 23, before taking part in any Driver's certificate of competency examination, a Driver of every class shall have to successfully complete the following courses, namely:

a. Preparatory Course;

b. Basic Fire Prevention and Fire Fighting Technique Course;

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c. Personal Safety and Social Responsibility Course;

d. Elementary First Aid Course;

e. Personal Survival Technique Course;

For service on vessels, the validity of the courses from (b) to (e) shall remain in force for the next 5 (five) years from the date of receipt of the certificate.

Chapter 6 — Examination for Coxswain

25. Conditions required for the ship’s Coxswain certificate of competency examination

Any candidate having the following qualifications may appear in the certificate of competency examination as a ship’s Coxswain, namely:

a. the age of the candidate shall not be less than twenty-one years, and if any doubt arises in the mind of the Chief Examiner regarding the age of the candidate, then the candidate shall have to produce a certificate as to his age;

b. he must have at least two years’ service experience on board a power-driven country vessel or an inland ship;

c. he must be physically and mentally fit, including in eyesight, the ability to distinguish colours, the relevant vision and hearing;

d. before appearing in the examination, the candidate must pass the eyesight test (without spectacles) to be conducted by the Chief Examiner or an examiner nominated by him;

e. if any candidate cannot pass the eyesight test, he may appear again in a similar test after a period of three months;

f. if any candidate cannot pass the oral and practical examinations, then he shall not be considered eligible to take part in the examination again unless he has completed at least six months of additional countable service before submitting the application for the next examination.

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Chapter 7 — Examination of Ratings

26. Majhi (Sukani)

1. All Majhis (Sukanis) shall have to be certificated.

2. Any deckhand, having six months’ experience of holding the helm under the supervision of a certificated Majhi (Sukani) of an inland ship, may apply for the Majhi’s certificate after serving for twelve months on deck.

27. Pumpman

1. All pumpmen shall have to be certificated.

2. A lascar or greaser may apply for the pumpman’s certificate after serving for 18 months on tankers.

28. Operator (Driver)

1. The operator of any power-driven country vessel shall have to hold an operator’s licence.

2. After serving for 12 months as an apprentice on a power-driven country vessel, he may apply to the Principal Surveyor for the operator’s licence examination.

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Chapter 1 — Preliminary

1. Short title

These Rules shall be called the Inland Ships (Minimum Safe Manning) Rules, 2001.

2. Definitions

1. Unless there is anything repugnant in the subject or context, in these Rules—

a. “Ordinance” means The Inland Shipping Ordinance, 1976 (LXXII of 1976).

b. “inland water” means “inland ship” [sic] as defined in section 2(f) of the Ordinance;

c. “ISSA” means the Inland Ship Safety Administration;

d. “ship” or “inland ship” means “inland ship” as defined in section 2(e) of the Ordinance.

e. “Principal Surveyor” means the head of the ISSA; and

f. “Director General” means the Director General of the Department of Shipping.

3. Minimum safe manning certificate

1. The Principal Surveyor shall issue a minimum safe manning certificate for each inland ship.

2. Every ship shall compulsorily carry the certificate issued under sub-rule (1).

4. Physical fitness for employment on ships

1. No person shall be employed on a ship unless he has been physically examined and declared fit by a medical officer appointed for that purpose by the Principal Surveyor.

2. If any person above 58 years of age, having been examined by a civil surgeon designated by the Principal Surveyor, is declared physically fit for service on ships, his certificate of competency may be revalidated for employment on any ship:

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Provided that the said person shall have to appear before an Examination Board constituted for that purpose at intervals of every two years.

5. Employment of additional Masters, etc., subject to availability of life-saving appliances

Persons in addition to the Engineers, Masters, Drivers and persons of other designations prescribed by the minimum safe manning certificate may be employed on a ship on the condition that the life-saving appliances prescribed by the rules are available on the ship concerned.

6. Employment of foreign nationals

No foreign national shall be employed as a seaman or in any other post on a ship without the prior written approval of the Director General.

7. Seaman’s book

1. The owner of a ship shall maintain, for the Master, Driver and other seamen working on the ship, a seaman’s book approved by the Principal Surveyor, in which the particulars of each seaman, his joining of service, date of discharge from service, leave and other matters shall be recorded, and the same shall be certified by the survey office concerned.

2. The entry on each page of the seaman’s book shall have to be endorsed by an Engineer and Ship Surveyor designated by the Principal Surveyor or by any officer approved by him.

8. Inspection to ensure compliance with the conditions of the minimum safe manning certificate

1. For the purpose of ensuring compliance with all the conditions of the minimum safe manning certificate issued under these Rules, the Principal Surveyor shall arrange for the inspection of every ship by an Engineer and Ship Surveyor appointed or nominated by him.

2. If, during an inspection under sub-rule (1), the Engineer and Ship Surveyor is satisfied that any ship has violated the conditions of the minimum safe manning certificate, then the Engineer and Ship Surveyor may declare the said ship unfit for use in any work within the inland waters.

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9. Inspection of foreign-flag ships

1. The Engineer and Ship Surveyor charged with responsibility under rule 8 may inspect every foreign-flag ship staying within the inland waters as to whether it is complying with the regulations relating to minimum safe manning prescribed by the country concerned.

2. If, during an inspection under sub-rule (1), the Engineer and Ship Surveyor is satisfied that any foreign-flag ship, while staying within the inland waters, has violated any condition of the regulations relating to minimum safe manning of the country concerned, then he may order the detention of the said ship and shall notify the matter for the removal of the defect, inconsistency or deficiency on the ship, and the detention order shall remain in force until the said deficiency, etc. has been removed.

10. Restrictions on employment in certain posts on ships

1. No person shall be employed in any post for the performance of, or for carrying out, the duties of supervision, handling and direct responsibility for the loading of cargo and cargo in transit on an oil tanker, chemical tanker or gas carrier unless he holds a certificate issued by the Authority prescribed for that purpose, certifying him to be capable of performing those duties.

2. No person shall be employed as a navigational watch of a ship, or as in-charge in that post, unless he is qualified to that effect or has been granted an exemption (dispensation) by the Principal Surveyor.

3. On every inland ship of 186 kilowatts or greater power, at least one cadet trained at the Deck Personnel Training Centre and one other cadet trained at the Bangladesh Institute of Marine Technology shall have to be employed as lascar and greaser respectively:

Provided that if cadets of this standard are not available, the Principal Surveyor may exempt any ship from this obligation.

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11. Classification of ships, etc.

1. The minimum number of Masters/Engineers/Drivers/seamen to be employed on a ship, according to class, shall be determined in accordance with Appendix A.

2. The minimum number and standard of Inland Masters and seamen to be employed on any ship shall be determined in accordance with Appendix B.

3. The minimum number and standard in respect of Inland Marine Engineers and Engine Drivers and greasers shall be determined in accordance with Appendix C.

Appendix A

Class	Description
1st Class	All passenger ships of 20 metres load water line or greater length, navigating by day and night within the inland waters of Bangladesh.
2nd Class	Passenger ships navigating within the inland waters only in calm water and during the daytime (if such ships have adequate lighting facilities, including a generator, then permission may be given to navigate up to a maximum of 10 o'clock at night).
3rd Class	All cargo or towing ships of 20 metres load water line or greater length, navigating during the daytime within the inland waters of Bangladesh.
4th Class	All cargo or towing ships navigating within the inland waters of Bangladesh only in calm water and during the daytime.
5th Class	All non-commercial ships navigating during the daytime in river-port areas within the inland waters of Bangladesh; if they have adequate lighting facilities, including a generator, then permission may be given to navigate up to a maximum of 10 o'clock at night.

Appendix B

Column 1: Class of ship. Column 2: Kilowatt power. Column 3: Class of certification required.

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Column 1: Class of ship	Column 2: Kilowatt power	1st Class Master	2nd Class Master	3rd Class Master	Seamen
1st Class	447 kilowatts and above	1	1	–	8
1st Class	186 kilowatts and above but less than 447 kilowatts		1	1	7
1st Class	11 kilowatts and above but less than 186 kilowatts			2	4
*2nd Class	447 kilowatts and above	1	–	–	6
*2nd Class	186 kilowatts and above but less than 447 kilowatts		1		5
*2nd Class	11 kilowatts and above but less than 186 kilowatts			1	4
3rd Class	447 kilowatts and above	1	1		7
3rd Class	186 kilowatts and above but less than 447 kilowatts		1	1	5
3rd Class	11 kilowatts and above but less than 186 kilowatts			2	4
*4th Class	447 kilowatts and above	1			6
*4th Class	186 kilowatts and above but less than 447 kilowatts		1		4
*4th Class	11 kilowatts and above but less than 186 kilowatts			1	3

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Column 1: Class of ship	Column 2: Kilowatt power	1st Class Master	2nd Class Master	3rd Class Master	Seamen

*5th Class	447 kilowatts and above	1			4
*5th Class	186 kilowatts and above but less than 447 kilowatts		1		3
*5th Class	11 kilowatts and above but less than 186 kilowatts			1	3

* If ships of the 2nd, 4th and 5th Class are operated for more than eight hours, then there shall be at least two Inland Masters on such ships. The Chief Inland Master shall have to hold a certificate of competency corresponding to the registered kilowatt power of the ship. The junior Master/Masters shall have to hold a certificate of competency one grade below that certificate corresponding to the kilowatt power of the ship. Such ships shall have to carry at least one additional seaman over the number of seamen shown in Column 3.

Appendix C

Column 1: Class of ship. Column 2: Registered kilowatt power. Column 3: Class of certificate required.

Column 1: Class of ship	Column 2: Registered kilowatt power	Inland Marine Engineer	1st Class Driver	2nd Class Driver	3rd Class Driver	Greaser
1st Class	1120 kilowatts and above	1	1			3
1st Class	447 kilowatts and above but less than 1120 kilowatts		1	1		3

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Column 1: Class of ship	Column 2: Registered kilowatt power	Inland Marine Engineer	1st Class Driver	2nd Class Driver	3rd Class Driver	Greaser
1st Class	186 kilowatts and above but less than 447 kilowatts			1	1	3
1st Class	37 kilowatts and above but less than 186 kilowatts					
*2nd Class	1120 kilowatts and above	1				3
*2nd Class	447 kilowatts and above but less than 1120 kilowatts		1			3
*2nd Class	186 kilowatts and above but less than 447 kilowatts			1		2
*2nd Class	37 kilowatts and above but less than 186 kilowatts				1	1
3rd Class	1120 kilowatts and above	1	1			3
3rd Class	447 kilowatts and above but less than 1120 kilowatts		1			3
3rd Class	186 kilowatts and above but less than 447 kilowatts			1	1	3
3rd Class	37 kilowatts and above but less than 186 kilowatts					

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Column 1: Class of ship	Column 2: Registered kilowatt power	Inland Marine Engineer	1st Class Driver	2nd Class Driver	3rd Class Driver	Greaser
*4th Class	1120 kilowatts and above	1				3
*4th Class	447 kilowatts and above but less than 1120 kilowatts		1			3
*4th Class	186 kilowatts and above but less than 447 kilowatts			1		2
*4th Class	37 kilowatts and above but less than 186 kilowatts				1	1
*5th Class	1120 kilowatts and above	1				2
*5th Class	447 kilowatts and above but less than 1120 kilowatts		1			2
*5th Class	186 kilowatts and above but less than 447 kilowatts			1		2
*5th Class	37 kilowatts and above but less than 186 kilowatts				1	1

* If such ships are operated for more than eight hours, then at least two certificated Inland Marine Engineers/Drivers shall have to be kept on such ships. The Inland Marine Engineer and Engine Driver shall have to hold a certificate of competency corresponding to the registered kilowatt power of the ship. The junior Driver/Drivers shall have to hold a certificate of competency one grade below that certificate corresponding to the kilowatt power of the ship. On such ships the number of greasers shall not be less than two.

p.305 # Inland Ships (Navigation) Rules, 2001

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p.309 ## Chapter 1 — Preliminary

1. Short title

These Rules shall be called the Inland Ships (Navigation) Rules, 2001.

2. Definitions

Unless there is anything repugnant in the subject or context, in these Rules—

a. “Ordinance” means the Inland Shipping Ordinance, 1976 (LXXII of 1976);

b. “inland water” means “inland water” as defined in section 2(f) of the Ordinance;

c. “inland ship” and “ship” mean “inland ship” as defined in section 2(e) of the Ordinance;

d. “ISSA” means the Inland Ship Safety Administration;

e. “tanker” means a ship carrying a large quantity of liquid substance in tanks or containers;

f. “sailing ship” means any ship propelled by means of sails, and, if propelling machinery is fitted to it, that machinery is not used or is used only occasionally;

g. “BIWTA” means the Bangladesh Inland Water Transport Authority;

h. “BIWTC” means the Bangladesh Inland Water Transport Corporation;

i. “floating equipment” means miscellaneous equipment which is not self-propelled, such as dredgers, floating cranes, etc., used for performing various work in a floating condition;

j. “fishing vessel” means a ship used for the purpose of catching fish and built for that purpose;

k. “passenger” means any person among those on board an inland ship other than the master, officers and crew of the ship, but a child passenger under one year of age shall not be included;

p.310 (l) “passenger ship” means a ship fitted with a weather deck used for carrying more than 12 passengers, and a ship having more than one deck shall also be included;

m. “power-driven ship” means any inland ship propelled by machinery; and

n. “road ferry” means an open flush-deck ship engaged in ferrying 12 or more passengers or one or more vehicles across.

3. Application

Unless otherwise provided, these Rules shall apply to all ships and floating equipment.

4. Design and construction

All ships and floating equipment shall be designed (design, drawing, plan), constructed and equipped to the satisfaction of the Engineer and Ship Surveyor, in compliance with good naval-architectural practice, river-worthiness, safety and strength.

5. Responsibility

1. Nothing in these Rules shall exonerate any inland ship or the owner, master or crew of a ship from the penalty for the following acts, namely:

a. neglect to comply with these Rules;

b. neglect to observe any precaution which is ordinarily to be observed by seamen.

2. Due regard shall be had to all dangers of navigation and collision and to any special circumstances, including the limitations of the inland ships concerned, and where so required, departure from these Rules may be necessary.

Chapter 2 — Lights and Shapes

6. General

1. All rules contained in this Chapter concerning lights shall be complied with from sunset to sunrise in all weathers. In conditions of restricted visibility the lights shall also be exhibited from sunrise to sunset. While the lights are being exhibited, p.311 no light shall be exhibited which may be mistaken for the prescribed lights.

2. The lights shall be of a type approved by ISSA.

3. During the day, the provisions concerning shapes shall be complied with.

4. This Chapter shall apply to all ships, vessels and floating equipment, irrespective of their type, size or mode of propulsion.

7. Navigation lights

1. “Masthead light” means a bright white light so constructed as to show an unbroken light over an arc of the horizon of 225 degrees and so fixed on the ship as to show the light from right ahead to 112.5 degrees on either side of the ship, that is, from right ahead to 22.5 degrees abaft the beam on either side, and this light shall be placed over the centre line of the ship, vessel or floating equipment.

2. “Side light” means a green light on the starboard side and a red light on the port side, each so constructed as

to show an unbroken light over an arc of the horizon of 112.5 degrees and so fixed as to show the light from right ahead to 22.5 degrees abaft the beam on the starboard side and on the port side respectively.

3. “Stern light” means a bright white light so constructed as to show an unbroken light over an arc of the horizon of 135 degrees, so fitted as to show the light 67.5 degrees on either side from right aft of the ship, and placed as nearly as practicable at the centre line and stern of the ship.

4. “Towing light” means a yellow light having the same characteristics as the stern light.

8. Power-driven ships

A power-driven ship when under way shall carry the following lights, namely:

a. in the case of a ship or floating equipment of 24 metres or more in length:

b. a masthead light on the foremast at least 5 metres above the hull of the ship;

p.312 Provided that, if the ship has no foremast, then at least 2 metres above the covering roof of the fore part of the ship:

Provided further that the light shall be visible from a distance of at least 3 kilometres;

ii. abaft the masthead light and at least 2 metres below the masthead light, side lights visible from a distance of at least 1.5 kilometres, which shall be fitted with inboard screens in such a way that the screen is at least 1 metre away from the light and the lights cannot be seen from the bow:

Provided that the screens shall be painted black;

iii. a stern light visible from a distance of at least 1.5 kilometres;

b. in the case of a ship or floating equipment of more than 12 metres but less than 24 metres in length:

c. a masthead light on the foremast at least 3 metres above the hull of the ship:

Provided that, if the ship has no foremast, then at least 1 metre above the covering roof of the fore part of the ship:

Provided further that the light shall be visible from a distance of at least 2 kilometres;

ii. abaft the masthead light and 1 metre below it, side lights visible from a distance of at least 1 kilometre, which shall be fitted with inboard screens in such a way that the screen is at least 0.6 metre away from the light and the lights cannot be seen from the bow:

Provided that the screens shall be painted black;

iii. a stern light visible from a distance of at least 1 kilometre;

iv. in the case of a ship or floating equipment of more than 6.5 metres but less than 12 metres in length:

p.313 (i) a masthead light at least 1.5 metres above the hull, which shall be visible from a distance of at least 1 kilometre;

ii. side lights at least 0.5 metre below the masthead light, visible from a distance of at least 1 kilometre;

iii. a stern light visible from a distance of at least 1 kilometre;

iv. in the case of a ship, boat or floating equipment of less than 6.5 metres in length:

v. side lights visible from a distance of 1 kilometre;

vi. a white light visible over the 360-degree horizon and over a distance of 1 kilometre.

9. Lights and signals, etc. of towed and pushed vessels

Any inland ship engaged in towing shall exhibit a towing light vertically above the stern light. There shall be an additional masthead light placed vertically at least 1 metre above the first masthead light.

2. The lights shall be visible from the distances mentioned in rule 8.

3. During the day, a towing ship shall exhibit a black diamond shape 1 metre in height, placed where it can best be seen.

4. Where one or more inland ships or floating equipment are lashed together side by side, the whole group of ships shall be considered as one for the purpose of exhibiting the side lights.

5. When a power-driven ship is being pushed ahead, green and red side lights shall be exhibited on the foremast of the ship.

6. A ship or equipment being towed shall exhibit side lights or a stern light.

7. If the length of the ships, etc. being towed exceeds 200 metres, during the day 1 black diamond shape of 1 metre in height shall be exhibited, placed where it can best be seen.

8. A rowing ship or a sailing ship shall carry at the fore part of the ship a white light visible all round the horizon from a distance of at least 1 kilometre.

p.314 ### 10. Road ferries

A road ferry shall exhibit the following lights and shapes, namely:

a. at night, three red lights visible from a distance of at least 1 kilometre over the horizon, placed on an equilateral triangle set at right angles across the centre line, each of which shall be placed at a distance of at least one metre from the others;

b. during the day, two black balls of not less than 0.5 metre in diameter, placed horizontally at right angles across the centre line, at a distance of at least one metre from each other and four metres above the deck.

11. Fishing vessels and boats

A ship or boat engaged in fishing shall exhibit the following lights and shapes, namely:

a. in addition to the lights mentioned in rule 8, at night two lights visible from a distance of at least 1 kilometre, which shall be placed one metre apart from each other;

b. during the day, a shape consisting of two cones with their apexes together in a vertical line, one above the other, and if the length of the ship is less than 20 metres, a basket may be exhibited.

12. Ships at anchor

Every ship, while at anchor or made fast to a buoy, shall during the day exhibit at its fore part a black ball shape of at least 0.5 metre in diameter, and at night shall exhibit a white light visible all round the horizon from a distance of at least 1.5 kilometres, and in both cases it shall be exhibited at a height of at least 5 metres above the hull of the ship or 2 metres above the covering roof.

13. Ships aground on a shoal

A ship aground on a shoal shall, during the day, where they can best be seen:

a. carry 3 black ball shapes of not less than 0.5 metre in diameter placed in a vertical line, one above the other at a distance of at least one metre; and

p.315 (b) at night the ship shall carry the white light described in rule 12 and, in addition, shall show 2 red lights, one above the other at a distance of not less than 1 metre, thus vertically, visible all round the horizon from a distance of at least 1.5 kilometres, placed where they can best be seen.

14. Ships not under command

1. When a power-driven ship under way is not under command, it shall:

a. during the day, where they can best be seen, show two black balls of at least 0.5 metre in diameter, placed vertically one above the other at a distance of not less than 1 metre; and

b. at night, show 2 red lights visible all round the horizon from a distance of at least 1.5 kilometres.

2. If a ship not under command is at anchor, it shall show only the lights and shapes mentioned in rule 12.

15. Ships engaged in underwater operations

A ship engaged in dredging work, diving work or similar underwater work shall show the following lights or shapes, namely:

a. at night, two red lights placed vertically one above the other at a distance of not less than one metre, which shall be visible over the horizon from a distance of at least 1.5 kilometres on the side of the ship on which the work is being carried out;

b. at night, two green lights placed vertically one above the other at a distance of not less than one metre, which shall be visible over the horizon from a distance of at least 1.5 kilometres on the side of the ship on which the work is not being carried out;

c. during the day, two black balls placed vertically one above the other at a distance of not less than one metre, which shall be visible over the horizon on the side of the ship on which the work is being carried out; and

d. during the day, two double cones (cone) placed vertically one above the other at a distance of not less than 1 metre from each other, p.316 which shall be visible over the horizon on the side of the ship on which the work is not being carried out.

16. Working with inflammable oil

When a ship carries inflammable oil, fuel or goods, the following lights or shapes shall be shown, namely:

a. at night, a red light visible all round the horizon from a distance of at least 1.5 kilometres;

b. during the day, a red flag shall be hoisted.

2. A tanker full of oil or chemical substances shall at all times show, in addition to the navigation lights, the light or flag mentioned in sub-rule (1) or the anchor lights mentioned in rule 8.

17. Searchlights

All power-driven ships navigating at night shall have a searchlight, remotely controlled from the wheelhouse and capable of projecting a beam of light to a distance of at least 80 metres from the ship, and the said light shall be capable of being rotated on both sides of the ship over an arc of the horizon from ahead to 22.5 degrees abaft the beam.

2. A power-driven ship of less than 5.5 metres in length shall carry a torchlight and shall show a flash of light when other ships approach towards it.

Chapter 3 — Sound Signals

18. Sound signals in fog

In fog, mist, heavy rain and rainstorms, whether by day or by night, signals shall be used in the following manner, namely:

a. a power-driven ship making way shall sound a prolonged blast of the whistle at intervals of not more than 1 minute;

b. a power-driven ship making way, if stopped anywhere, shall sound two prolonged blasts of the whistle in rapid succession at intervals of not more than 1 minute;

p.317 (c) a power-driven ship at anchor shall sound the horn rapidly for about 5 seconds at intervals of not more than 1 minute.

d. a power-driven ship aground on a shoal or near a waterway shall sound the horn rapidly for about 5 seconds at intervals of not more than 1 minute and thereafter shall sound the horn three times distinctly;

e. a power-driven ship which has fallen into a condition of being not under command, or whose ability to manoeuvre has become restricted, shall sound the whistle at intervals of not more than 1 minute with one prolonged blast followed by short blasts in rapid succession;

f. when a power-driven ship is being towed alongside or astern by chain or rope, it shall give the sound signal in the manner mentioned in clause (a) or (b), whichever is applicable; and

g. when a power-driven ship tows one or more inland ships tied by chain or rope, it shall give the sound signal of clause (e), and if the number of ships being towed is more than one, the last ship shall, immediately after the sound signal given by the towing ship, ring the bell 3 times distinctly.

19. General warning signals

In the following cases one prolonged blast of the whistle shall be sounded to give warning, namely:

a. when a power-driven ship is proceeding towards its destination;

b. for the purpose of attracting the attention of other inland ships for a clear course; and

c. when proceeding towards any bend.

20. Sound signals for inland ships within sight of one another

Every power-driven ship making way shall indicate its course by means of its whistle or siren with the following signals, namely:

p.318 (a) one short blast of the whistle to indicate that its course is to starboard or to the right side of the ship;

b. two short blasts of the whistle to indicate that its course is to port or to the left side of the ship; and

c. three short blasts of the whistle to indicate that the engines are going astern.

21. Attracting the attention of another power-driven ship

If a power-driven ship needs to attract the attention of another power-driven ship, it shall sound one prolonged blast of the whistle immediately followed by 5 short blasts.

Chapter 4 — Navigation Regulations

22. Speed of ships in fog, etc.

1. In fog, mist, heavy rain and rainstorms, a power-driven ship shall proceed cautiously at a moderate speed, having regard to the existing circumstances, or shall anchor in a safe place with the engine stopped or running.

2. A power-driven ship, on hearing, apparently forward of its beam, the fog warning signal of another inland ship whose position cannot be ascertained with certainty, shall, so far as the circumstances require, stop its engine and shall navigate with caution until the danger of collision is over.

23. Ships in head-on situations

When two power-driven ships meet each other head-on or nearly head-on so as to involve risk of collision, each ship shall alter its course to starboard so that each ship may pass the other on its port side.

24. Crossing ships

When two power-driven ships are crossing each other so as to involve risk of collision, the inland ship which has the other ship on its starboard side shall keep out of the way of the other ship.

p.319 ### 25. Avoiding overtaking ships while proceeding

Every power-driven ship which is directed by these Rules to keep out of the way of another ship shall, if the circumstances of the case permit, avoid passing ahead of the other ship.

26. Overtaking another ship

Notwithstanding anything contained in these Rules, every power-driven ship proceeding ahead shall keep out of the way of the ship which it is overtaking.

27. Mutual responsibilities of ships

1. A power-driven ship shall keep out of the way of the following ships, namely:

a. a ship not under command;

b. a ship restricted in its ability to manoeuvre;

c. a ship engaged in fishing; and

d. a sailing ship.

2. A sailing ship under way shall keep out of the way of the following ships, namely:

a. a ship not under command;

b. a ship restricted in its ability to manoeuvre; and

c. a ship engaged in fishing.

3. A ship engaged in fishing, when under way, shall, so far as possible, keep out of the way of the following ships:

a. a ship not under command;

b. a ship restricted in its ability to manoeuvre.

28. Responsibility of ships proceeding on the same course

1. Where one of two ships is to keep out of the way, the other ship shall keep its course and speed.

2. If a ship proceeding on the same course finds that the ship which is required to give way is not taking appropriate action in accordance with these Rules to avoid collision, the ship (the ship proceeding on the same course), if power-driven, shall attract the attention of the other ship by sounding one prolonged blast of the whistle followed by five short and rapid blasts.

p.320 (3) When, for any reason, the ship required to keep its course and speed finds itself so close that collision cannot be avoided by the action of the give-way ship alone, it shall take such action as will best aid to avoid collision and shall indicate its own action by giving the appropriate sound signal.

29. Navigation in narrow channels

1. In a narrow channel, a power-driven ship shall, so long as it is safe, keep to that side of the waterway which lies on the starboard side of the said ship.
2. When two power-driven ships, with or without sidetow, meet in a narrow waterway or place where, owing to the presence of a third ship, passing each other becomes difficult, the ship which is going against the current shall reduce its speed and shall keep very close to the bank on its starboard side until the other ship has completely passed it.
3. When two power-driven ships meet each other at a bend of a narrow channel, the inland ship which is going against the current shall stop and shall remain at that place until the other ship has completely passed it.

30. Getting under way at the start of a voyage

A power-driven ship shall not get under way from its anchorage, river bank or jetty when another ship is seen approaching from upstream or downstream at such a distance that there is doubt as to whether the approaching inland ship can pass it safely.

31. Junctions of waterways

When two power-driven ships are about to meet at the junction of two waterways, the inland ship in the wider of the two waterways shall not attempt to enter until the ship in the narrower waterway has passed it.

p.321 ### 32. Power-driven ships, sailing ships and fishing ships

No sailing ship or ship engaged in fishing shall obstruct the safe passage of any power-driven ship in a navigable waterway.

33. Clear view (clearview) for navigation

Navigation shall be controlled from a position from which as clear a view as practicable may be had, and the line of sight shall be unobstructed from right ahead to 22.5 degrees abaft the beam on both sides.

34. Navigation of inland ships within the limits of the international sea ports of Bangladesh

1. Unless a duly qualified master is present on board the ship with his certificate of competency bearing an endorsement for the Karnaphuli river or, as the case may be, the Pashur river, the said ship shall not navigate within the limits of the international sea ports of Bangladesh.
2. While going from one destination to another or from one side of a navigable waterway to the other, no power-driven ship shall obstruct the safe passage of, or create a risk of collision with, any sea-going ship overtaking or passing alongside it, and if necessary the power-driven ship shall reduce its speed or stop or reverse its engine or wait at a safe distance until the sea-going ship has passed.
3. All power-driven ships navigating within the limits of an international sea port of Bangladesh shall comply with the rules, regulations, orders and directions of the port authority concerned.
4. When passing sea-going ships within the limits of any international sea port of Bangladesh, a power-driven ship shall follow the International Collision Regulations (International Collision Regulation).
5. No inland ship shall cross the coast without the prior approval of the appropriate authority.

p.322 ## Chapter 5 — Equipment

35. Navigation equipment

All inland power-driven ships shall have the following navigation equipment, namely:

a. a ship of 6.5 metres or more in length and all passenger ships shall have a compass, and in the case of navigation at night the compass shall be illuminated;

b. a ship under clause (a) above shall keep a pair of binoculars; and

c. passenger ships and launches shall have an instrument for measuring wind speed.

36. Signalling equipment

Inland ships, boats and floating equipment shall keep the following sound-based warning signal equipment,

namely:

a. all the necessary lights and shapes previously mentioned in these Rules;

b. an electric or air-operated horn audible from a distance of not less than 2 kilometres;

c. for passenger ships and tankers, an additional manual horn (manual horn); and

d. a brass bell of at least 20 centimetres in diameter kept at the fore part of the ship or floating equipment.

37. Internal communication equipment

All inland power-driven ships shall have the following internal communication systems, namely:

a. a system of communication or signal transmission between the wheelhouse or bridge and the engine room or engine;

b. in passenger ships having a capacity to carry more than 200 passengers, one loud-hailer system each; and

c. if there is any emergency steering position, there shall be a system of communication or signal transmission between the wheelhouse and the emergency steering position.

p.323 ### 38. External communication equipment

1. All inland ships registered under the I.S.O., 1976 and all power-driven and floating equipment shall have a radio receiver (Radio Receiver).

2. All passenger ships carrying more than 400 passengers, all tankers and ships carrying dangerous goods shall have a VHF radio telephone.

39. Mooring and anchoring equipment

1. The ship shall have a sufficient number of mooring ropes and wires of adequate length.

2. The ship shall have at least one anchor with a chain of adequate length and another spare anchor.

3. The ship shall have a hand-operated or mechanical anchor windlass (Anchor Windlass) and capstan.

40. Bilge pump

All inland ships, boats and floating equipment shall have a hand-operated or mechanical bilge pump.

41. Gangway

All passenger ships shall have a gangway of sufficient length, which shall be at least 600 millimetres wide and fitted with ropes.

42. Vessels navigating or stationed in inland waters

Every vessel, floating structure or any other equipment navigating or stationed within inland waters shall be registered by the Inland Ship Safety Administration.

Chapter 6 — Restrictions on Navigation

43. Unregistered ships or floating equipment

1. No unregistered ship, boat or floating equipment which has not been surveyed shall navigate on the main waterways suitable for the navigation of ships extending from Baghabari, Dhaka, Narayanganj and Bhairab to the open sea, and on the waterways up to Khulna, Mongla, Barisal and Chandpur, or on any other inland waterway, or p.324 which has previously navigated or may later navigate in inland waters but is at present confined or stationary, nor shall any structure which directly or indirectly uses the water of the inland water limits be navigated or operated.

2. No unregistered ship or boat shall take on any passenger from, or land any passenger at, any landing station belonging to BIWTA or BIWTC or any ghat nearer than 1 kilometre therefrom.

3. If any unregistered boat or floating equipment navigates after sunset, is engaged in fishing or is at anchor, it shall show lights in accordance with these Rules.

44. Open passenger launches

No open passenger launch of less than 20 metres in length shall navigate in rough waters for an effective voyage of more than 2 hours.

45. Navigation in the open sea

1. No inland passenger ship shall navigate outside inland waterways.

p.325 # Inland Ships (Dangerous Goods) Carriage Rules, 2001

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p.329 ## Chapter 1 — Preliminary

1. Short title

These Rules shall be called the Inland Ships (Dangerous Goods) Carriage Rules, 2001.

2. Definitions

Unless there is anything repugnant in the subject or context, in these Rules—

a. “Ordinance” means The Inland Shipping Ordinance, 1976 (LXXII of 1976);

b. “inland water” means “inland water” as defined in section 2(f) of the Ordinance;

c. “ship” or “inland ship” means “inland ship” as defined in section 2(e) of the Ordinance;

d. “Inspector” means an inspector appointed by the Government for the inspection of dangerous goods and explosives;

e. “dangerous goods” means “dangerous goods” as defined in section 2(d) of the Ordinance.

Chapter 2 — Classification of Dangerous Goods

3. Classification of dangerous goods

1. Explosives;

2. Gases compressed, liquefied or dissolved under pressure, and non-flammable;

3. Compressed, liquefied or dissolved under pressure, and poisonous;

4. Inflammable liquids;

5. Inflammable solids;
6. Inflammable solids or substances liable to spontaneous combustion;

p.330 (7) Inflammable solids and substances which, in contact with water, emit inflammable gases;

8. Oxidizing substances;
9. Organic peroxides;
10. Poisonous substances;
11. Infectious substances;
12. Radioactive substances;
13. Corrosive substances; and
14. Miscellaneous dangerous articles, that is: any other substance or article which experience shows, or may show, to be of such a dangerous character that this class may apply to it.

Chapter 3 — Carriage of Dangerous Goods in Inland Ships

4. Shipper’s declaration

1. No dangerous goods shall be loaded on any inland ship until the shipper of the goods has furnished the owner or master of the inland ship with a written declaration to the effect that the goods proposed for carriage have been properly marked and labelled, and have been packed in a manner fit to withstand the ordinary risks of loading, unloading and carriage when dispatched on the ship.
2. Where dangerous goods are enclosed in a container, the person responsible for closing the goods in the container shall furnish the master with a declaration to the effect that “the goods have been securely packed and properly secured and stowed in the container”.

5. Marking

Dangerous goods kept in packages or ordinary receptacles shall not be taken on board an inland ship unless they have been clearly marked by a separate label or stencil stating the correct technical name and class of the goods.

p.331 ### 6. Packaging

1. No dangerous goods shall be taken on board an inland ship unless they have been packed in a manner fit to withstand the ordinary risks of loading, unloading and carriage, and unless the declaration has been furnished to the master by the owner of the ship in accordance with rule 4.
2. Where the quantity of dangerous goods is not large, their packaging—

a. shall be good and in good condition;

b. shall be such that if the goods in the package come into contact with the interior surface it is not dangerously affected by the substance being carried.

3. If the dangerous goods are radioactive, there shall be arrangements with adequate safety to protect all persons on the inland ship.
4. Where any absorbent or cushioning material is used in the packaging of receptacles carrying liquids, that material—

a. shall be capable of keeping to a minimum the danger that the liquid may create;

b. shall be so placed that movement of the receptacle is prevented;

c. shall ensure that the receptacle remains surrounded;

d. shall, where reasonably practicable, be present in sufficient quantity so as to be capable of absorbing the liquid if the receptacle is broken.

5. Receptacles containing dangerous liquids shall, according to the filling temperature, have sufficient ullage or empty space so as to allow for expansion of the liquid due to the highest temperature likely to be encountered during ordinary carriage.
6. Cylinders or receptacles for compressed gases shall be properly constructed, tested, maintained and correctly filled.

7. Segregation

Dangerous goods shall be segregated according to the following table. (The top row or the left-hand column represents the different classes of dangerous goods)

p.332 | | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | |—|—|—|—|—|—|—|—|—|—|—|—|—|—| | 1 | d | b | d | d | d | d | d | d | 0 | b | b | | 2 | | 0 | 0 | b | a | b | a | b | d | 0 | b | a | | 3 | | | 0 | 0 | 0 | a | 0 | 0 | b | 0 | a | 0 | | 4 | | | | b | a | b | a | b | d | 0 | b | a | | 5 | | | | | a | b | b | a | c | 0 | b | a | | 6 | | | | | | a | a | a | b | 0 | b | a | | 7 | | | | | | | | a | b | b | 0 | b | a | | 8 | | | | | | | | | b | b | 0 | b | a | | 9 | | | | | | | | | | b | 0 | a | a | | 10 | | | | | | | | | | | 0 | b | a | | 11 | | | | | | | | | | | | 0 | 0 | | 12 | | | | | | | | | | | | | b |

Explanation of the segregation codes:

0 : In this case there is no special segregation provision, but it must be remembered that goods of class 6 shall be kept separate from living quarters and foodstuffs.

a : The goods shall be kept separated at a distance of at least 3 metres.

b : Shall be kept in separate cargo spaces when stowed, and at a distance of at least 12 metres on deck.

c : Shall be separated by one cargo space vertically or horizontally, and shall be kept at a distance of at least 3 metres on deck.

d : Shall be separated by one cargo space longitudinally, and the separation on deck shall be at least 3 metres.

8. Storage or stowage

1. Dangerous goods shall be stowed and kept separate in accordance with rule 7.

2. There shall be adequate ventilation arrangements in the place where dangerous goods are stowed.

3. Where inflammable goods are stowed, all electrical installations there shall be of explosion-proof type.

p.333 (4) No explosives shall be stowed where there is any electrical equipment or cable.

5. Dangerous goods of classes 1 to 4 shall be stowed at a distance of at least 3 metres from the vertical wall of the engine room. The vertical wall of the engine room shall be completely watertight.

9. Carriage of dangerous goods in inland passenger ships

Other than ammunition and fireworks of the safety class, no other kind of dangerous goods shall in any circumstances be carried in an inland passenger ship.

10. Precautionary measures to be taken while loading and unloading dangerous goods on ships

While loading and unloading dangerous goods, the following precautionary measures shall be taken, namely:

a. boards inscribed “No Smoking” shall be displayed at conspicuous places and at the gangways, and no one shall smoke on deck while the loading and unloading operations are in progress;

b. there shall be no naked light on deck;

c. no person present on deck shall have matches or lighters;

d. all fire-extinguishing equipment shall be kept ready for immediate use;

e. the galley fire or the fire of the ship’s kitchen shall be kept extinguished;

f. the master must be present on deck while the loading and unloading operations are in progress;

g. the driver shall be present in the engine room to start the pump in case of fire or to deal with any other emergency;

h. no unauthorised person shall be allowed to board the inland ship; and

i. during the day, a square red flag of 0.5 × 0.5 metre size shall be displayed at the place which can best be seen from all sides, p.334 and at night a red light visible all round the horizon from a distance of at least 1.5 kilometres shall be shown.

11. Special measures

1. All dangerous goods of classes (1) and (7) shall be pre-inspected by the Inspector before being taken on board the ship.

2. All dangerous goods of classes (1), (2), (4), (6) and (7) shall be inspected and supervised by the Inspector while being loaded and unloaded.

3. The work of pre-inspection, inspection and supervision shall be carried out by the Inspector.

p.335 *(Divider / title page)*

Inland Ships (Tonnage Measurement) Rules, 2001

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p.339 ## Chapter 1 — Preliminary

1. Short title:

These Rules shall be called the Inland Ships (Tonnage Measurement) Rules, 2001.

2. Definitions:

Unless there is anything repugnant in the subject or context, in these Rules—

a. “Department” means the Department of Shipping;

b. “Ordinance” means The Inland Shipping Ordinance, 1976 (LXXII of 1976);

c. “inland water” means “inland water” as defined in section 2(f) of the Ordinance;

d. “gross tonnage” means the size of a ship determined under rule 11;

e. “ship” or “inland ship” means “inland ship” as defined in section 2(c) of the Ordinance;

f. “length (L)” means the distance, on the load water line, from the fore end of the stem at the fore part of the ship to the aft end of the rudder post, or, if the ship has no rudder post, to the centre of the rudder stock;

g. “new ship” means all ships the keel of which is laid on or after the date on which these Rules come into force, and all ships which are at an equivalent stage of construction;

h. “net tonnage” means the measure of the useful carrying capacity of a ship determined under rule 12;

i. “Administration” means the Inland Ship Safety Administration (ISSA);

j. “Principal Officer” means the Principal Officer of the Inland Ship Safety Administration;

k. “existing ship” means a ship which is not a new ship;

l. “Director General” means the Director General of the Department of Shipping;

p.340 (m) “passenger” means any person carried on board a ship or inland ship other than the master, officers and crew of the ship, but a child-passenger under one year of age shall not be included therein;

n. “passenger ship” means a ship with a weather deck used for the carriage of more than 12 passengers, and a ship having more than one deck shall also be included therein.

3. Application:

These Rules shall apply in the following cases, namely:

a. new ships;

b. any existing ship which is considered by the Department to have undergone significant alteration and modification in respect of its existing gross tonnage;

c. existing ships in respect of which the owner has requested the application of these Rules;

d. all existing ships up to 100 GRT, two years after the date on which these Rules come into force;

e. all existing ships from 100 to 500 GRT, three years after the date on which these Rules come into force;

f. all existing ships above 500 GRT, four years after the date on which these Rules come into force;

g. the distance, on the load water line, from the fore end of the stem at the fore part of the ship to the aft end of the rudder post, or, if the ship has no rudder post, to the centre of the stock.

4. Exemption:

These Rules shall not apply to ships owned by the Defence Services of Bangladesh and to ships of less than 6 metres in length.

Chapter 2 — General

5. Tonnage:

1. The tonnage of a ship shall include gross tonnage or net tonnage.

2. Gross tonnage and net tonnage shall be determined in accordance with the provisions of these Rules.

p.341 (3) In the case of a novel type of ship, the constructional features of which are such that the application of the provisions of these Rules would be unreasonable or impracticable, the gross tonnage and net tonnage shall be determined by the Department.

4. The tonnage determined under these Rules and shown in the tonnage certificate issued under rule 7 shall indicate the volume of the enclosed spaces of the ship concerned, shall not indicate weight or cargo-carrying capacity, and shall be used for the purpose of determining fees in respect of registration, survey, docking, undocking, taxation etc. of the ship, and for the use of the facilities of inland ports and channels and other similar purposes.

6. Tonnage determining authority:

Gross tonnage and net tonnage shall be determined by the Department, and the Administration (ISSA) may delegate the responsibility for such determination to persons or organisations recognised by it:

Provided that the Department shall in every case accept full responsibility for the determination of gross tonnage and net tonnage.

7. Issue of tonnage certificate:

1. A tonnage certificate shall be issued to every ship the gross and net tonnage of which has been determined under these Rules.

2. Any person or organisation authorised for this purpose by the Administration or the Principal Officer shall issue the certificate under sub-rule (1).

3. The tonnage certificate shall be issued in accordance with Appendix-1.

8. Cancellation of certificate:

If any alteration is made in the arrangement, construction, capacity, use of spaces and the approved number of passengers, the load line of the ship or the approved draught, as indicated in the survey certificate of the ship, such as would necessitate an increase in the gross tonnage or net tonnage, and in any case where the ownership of the ship changes, then, unless otherwise provided in these Rules, the validity of the tonnage certificate shall cease and the certificate shall be deemed to be cancelled.

9. Inspection:

1. An Engineer and Ship Surveyor or Inspector of the Administration, or any person authorised by the Principal Officer, may at any time board an inland ship for inspection under these Rules,

p.342 and such inspection shall be carried out for the purpose of ascertaining the following, namely:

a. whether a valid tonnage certificate has been issued to the ship under these Rules; and

b. whether the principal characteristics of the ship correspond with the particulars given in the certificate.

2. If, on inspection, it is found that the characteristics of any ship differ from those included in the tonnage certificate in such a way as would necessitate an increase in the gross tonnage and net tonnage, then the ship shall be re-measured under these Rules, and the previously issued tonnage certificate shall be withdrawn and a new certificate issued for the ship.

Chapter 3 — Determination of gross and net tonnage of ships

10. Definitions:

For the purpose of determining the gross tonnage and net tonnage of ships, unless there is anything repugnant in the subject or context, the terms used in these Rules for the determination of the gross tonnage and net tonnage of ships shall have the following meanings, namely:

1. “Upper deck” means the uppermost complete deck exposed to the weather, which has permanent means of closing all openings in the weather part thereof, and below which all openings in the sides of the ship are fitted with permanent means of watertight closing; and the lowest line of the exposed deck and the continuation of that line parallel to the upper deck shall be included as the upper deck.

2. “Moulded depth”—

a. The moulded depth is the vertical distance at the side of the ship measured from the top of the keel to the underside of the upper deck. Where the form of the lower part of the midship section is of a hollow character, or where thick garboards are fitted, the distance is measured

p.343 from the point where the line of the flat of the bottom continued inwards cuts the side of the keel;

b. In ships having rounded gunwales, the moulded depth is measured to the point of intersection of the moulded lines of the deck and the shell plating, the lines extending as though the gunwales were of angular design;

c. Where the upper deck is stepped and the raised part of the deck extends over the point at which the moulded depth is to be determined, the moulded depth shall be measured to a line of reference extending from the lower part of the deck along a line parallel with the raised part.

3. “Breadth” means the maximum breadth of the ship, measured amidships to the outer surface of the frames in a ship with a metal shell.

4. “Enclosed spaces” means the maximum breadth of the ship with a shell, measured amidships to the outer surface of the frames.

5. Notwithstanding the provisions of clause (4) of this rule, the spaces referred to in this clause and in clauses (b) and (c) of sub-rule 2 shall be called excluded spaces, and “excluded spaces” shall not be included in enclosed spaces unless at least one of the following three conditions is fulfilled:

fitted with shelves or other means for securing cargo or stores;

fitted with any means of closing the open spaces;

in the process of construction, any possible means of closing such open spaces is provided.

a. i. Except for a curtain plate of a depth not exceeding 25 millimetres below the depth of the adjacent deck beams, a space within an erection opposite an end opening extending from deck to deck, where this provision shall apply for the exclusion of the enclosed space between the line of the opening of such open space and a line, or a line parallel to the face of the opening, at a distance equal to or more than 90 percent of the breadth of the deck in way of the opening (Figure 1, Appendix-2).

p.344 (ii) If, by reason of any arrangement other than the gradual narrowing of the outside plating, the breadth of a space becomes less than 90 percent of the breadth of the deck, then only the space between the line of the opening and a parallel line drawn through the point at which the athwartship breadth of the space becomes equal to or less than 90 percent of the breadth of the deck shall be excluded from the volume of enclosed spaces (Figures 2, 3 and 4, Appendix-2).

iii. Where the space between these two spaces is completely open except for bulwarks or open rails, the permission to exclude one or both spaces under sub-clauses (a) (i) and (ii) shall not apply if the separating space between the two spaces in its path is less than half of the minimum breadth of the deck (Figures 5 and 6, Appendix-2).

b. A space under an overhead deck, open to the sea at the sides of the ship except for the necessary stanchions, with open rails or upper bulwarks and a curtain plate, or with stanchions fitted for support, on condition that the distance between the top of the rails or upper bulwark and the curtain plate shall not exceed 0.75 metre or one-third of the height of the space, whichever is greater (Figure 7, Appendix-2).

c. Where any space, with an opening of 0.75 metre or one-third of the height of the space, whichever is greater, towards the open side on the opposite side, is constructed side by side, and in such construction the open side is kept on one side, then the space to be excluded from the volume of enclosed spaces shall be

p.345 limited to a maximum of half of the breadth of the deck in way of the opening (Figure 8, Appendix-2).

d. A space in an erection immediately below an uncovered opening in the deck overhead:

Provided that such open space shall be exposed to the weather, and the space to be excluded from enclosed spaces shall be limited to the area of the opening (Figure 9, Appendix-2).

e. A recess in the boundary bulkhead of an erection which is exposed to the weather and the opening of which extends from deck to deck without means of closing:

Provided that the interior breadth shall not be greater than the breadth of the entrance, and its extension into the erection shall not be more than twice the breadth of its entrance (Figure 10, Appendix-2).

6. The cargo spaces to be included in the computation of net tonnage are the enclosed spaces appropriated for the carriage of cargo, if they have been included in the computation of Gross Tonnage. Such cargo spaces (cargo compartment) shall be permanently marked with two letters, and shall be so placed as to be readily visible and not less than 100 millimetres in height.

7. “Weathertight” shall mean that in any sea condition water will not easily penetrate.

11. Gross Tonnage:

The gross tonnage of a ship shall be determined by the following formula:

`GT = K1 × V`

where: V = total volume of all enclosed spaces of the ship in cubic metres.

`K1 = 0.2 + 0.02 log10 V` (or as tabulated in Appendix 3).

p.346 ### 12. Determination of Net Tonnage:

1. The net tonnage of a ship shall be determined by the following formula:

`NT = K2 × Vc × (4d / 3D)^2 + K3 × (N1 + N2/10)`

in which formula:

a. the factor `(4d / 3D)^2` shall not be taken as greater than unity;

b. the term `K2 × Vc × (4d / 3D)^2` shall never be taken as less than 0.25 GT;

c. NT shall never be taken as less than 0.30 GT; and

where Vc = total volume of cargo spaces in cubic metres.

`K2 = 0.2 + 0.02 log10 Vc` (or as tabulated in Appendix 3).

`K3 = 1.25 × (GT + 10,000) / 10,000`

D = moulded depth amidships in metres, as defined in rule 10;

d = draught amidships in metres, in accordance with this rule;

N1 = number of passengers in cabins with not more than 8 berths;

N2 = number of other passengers;

N1 + N2 = total number of passengers the ship is permitted to carry as indicated in the ship's survey certificate; where N1 + N2 is less than 13, N1 and N2 shall be taken as zero;

GT = gross tonnage of the ship determined in accordance with the provisions of rule 11.

2. The moulded draught referred to in sub-rule (1) of this rule shall be one of the following draughts:

a. for ships to which the Inland Ships (Free-board) Rules, 2001 apply, the draught corresponding to the Smooth Water Load Line;

p.347 (b) for passenger ships, the draught corresponding to the deepest subdivision load line assigned in accordance with the Inland Ships (Free-board) Rules, 2001;

c. for other ships, 75 percent of the moulded draught amidships of the ship, as defined in rule 10(2).

13. Change of net tonnage:

1. When the characteristics of a ship, such as V, Vc, d, N1, N2 as defined in rules 11 and 12, are altered, and where such alteration results in an increase in the net tonnage determined under rule 12, the net tonnage of the ship corresponding to the new characteristics shall apply without delay.

2. A ship to which the load lines referred to in clauses 2(a) and 2(b) of rule 12 are concurrently applicable shall be assigned only one net tonnage determined in accordance with the provisions of rule 12, or that shall apply as the load line appropriate to the part of the inland waterways in which the ship is engaged.

3. When the characteristics of a ship, such as V, Vc, d, N1, N2 as defined in rules 11 and 12, are altered, or the load line referred to in sub-rule (2) of this rule is altered due to a change in the inland waterway in which the ship is engaged, and for that reason the net tonnage determined in accordance with the provisions of rule 12 is decreased, a new tonnage certificate shall not be issued until twelve months have elapsed from the date of issue of the existing certificate:

Provided that this sub-rule shall not apply where, in the opinion of the Principal Officer, a major alteration such as the removal of a superstructure of the ship has been made.

14. Calculation of volume:

1. The volumes included in the calculation of gross and net tonnage shall be measured, irrespective of the fitting of insulation or the like, to the inner side of the shell or structural boundary plating in ships constructed of metal, and to the outer surface of the shell or to the inner side of the structural boundary surfaces in ships constructed of any other material.

p.348 (2) Volumes of appendages shall also be included in the total volume.

3. Volumes of spaces open to the water may be excluded from the total volume.

15. Measurement and calculation:

1. All measurements used in the calculation of volumes shall be taken to the nearest centimetre.

2. The volumes shall be calculated by generally accepted methods for the spaces concerned and with an accuracy acceptable to the Department.

3. The calculation shall be sufficiently detailed to permit easy checking.

16. Use:

The gross tonnage and net tonnage determined under these Rules shall be taken as the basis of statistical data relating to the parameters of inland ships or the true size of ships or useful carrying capacity, if they are used in the Inland Ships (Safety) Rules; the Government, autonomous bodies and other agencies, who use gross registered tonnage and net registered tonnage as the basis of charges, shall, when using them, carefully consider which parameter is most suitable for use in the light of their existing practice.

p.349 **Government of the People's Republic of Bangladesh Inland Ship Tonnage Measurement Certificate (Office Seal) Certificate under the Inland Ships (Tonnage Measurement) Rules, 2001 — see rule 7 (3).**

Name of ship	Registration number	Port of registry	Date*

* The date on which the keel of the ship was laid or on which the existing ship was at a similar stage of construction, rule 2(f), or, in the case of an existing ship, the date on which the ship underwent a major alteration or modification, rule 2(j).

Main dimensions—

Length, rule 2 (k)	Breadth, rule 10 (3)	Moulded depth amidships to upper deck, rule 10 (2)

Tonnage of the ship:

Gross tonnage .....

Net tonnage .....

This is to certify that the tonnage of this ship has been determined in accordance with the Inland Ships (Tonnage Measurement) Rules, 2001.

.....	.....
Place of issue of certificate	Date of issue of certificate
	.....
	Signature and seal of the person issuing the certificate

The undersigned declares that he has been duly authorised by the Principal Officer, Inland Ship Safety Administration (ISSA), to issue this certificate.

p.350 ## Appendix-1

Spaces included in tonnage	
Gross tonnage	**Net tonnage**
Name of space measured — Location — Length	Name of space measured — Location — Length
	Number of passengers, rule 12 (1) Number of passengers in cabins with not more than 8 berths ..... Number of other passengers .....
Excluded spaces, rule 10 (5) An asterisk (*) shall be placed against the spaces listed above where both enclosed and excluded spaces are included.	**Moulded draught**, rule 12 (2)
Place and date of original measurement	
Place and date of last previous re-measurement	
Remarks	

p.351 ## Appendix-2 ### Figures referred to in rule 10 (5)

O - Excluded spaces

C - Enclosed spaces

I - Spaces to be considered as enclosed spaces

Hatch portions shall be included as enclosed spaces

B - Breadth of the deck in way of the opening

In ships with rounded gunwales the breadth shall be measured as shown in Figure 11.

Figures

(Six line drawings, plan views of deck erections with hatched enclosed spaces, shown in a 2×3 grid; labels transcribed below.)

- **Figure 1** — Rule 10 (5) (a) (i): erection open at one end; dimensions marked B, B/2, ≥ 0.9B; regions marked O (excluded) and C (enclosed).
- **Figure 2** — Rule 10 (5) (a) (ii): erection open at one end with a narrowing; dimensions B, B/2, ≥ 0.9B, < 0.9B; regions O and C.
- **Figure 3** — Rule 10 (5) (a) (ii): tapering (bow-shaped) erection; dimensions B, B/2, < 0.9B, ≥ 0.9B; regions O and C.
- **Figure 4** — Rule 10 (5) (a) (ii): tapering erection with recess; dimensions B, B/2, < 0.9B; regions O, C and C.
- **Figure 5** — Rule 10 (5) (a) (iii): two erections separated by an open space with “OPEN RAILS OR BULWARK”; dimensions B, B/2, ≥ B/2, ≥ 0.9B, < 0.9B; regions C, O, C.
- **Figure 6** — Rule 10 (5) (a) (iii): two erections separated by a space containing a “HATCH OR ERECTION” with “OPEN RAILS OR BULWARK”; dimensions B, B/2, ≤ B/2, ≥ 0.9B, < 0.9B; regions C, I, C.

p.352 ## Appendix-2 ### Figures

Rule 10 (5) (b) — Figure 7

(Perspective drawing of a deck erection open at the sides with stanchions; heights marked H and h.)

h = AT LEAST H/3 OR 0.75 m (2.5 FEET) WHICHEVER IS THE GREATER

Rule 10 (5) (c) — Figure 8

(Perspective drawing of an erection marked THWARTSHIP CLOSED at both ends, with side openings of height h within total height H and "LENGTH OF EXCLUDED SPACE" marked; below it two plan views.)

h AT LEAST H/3 OR 0.75 m (2.5 FEET) WHICHEVER IS THE GREATER

Left plan: OPPOSITE SIDE OPENINGS — region O between hatched regions. Right plan: OPENING ON ONE SIDE ONLY — dimensions B, B/2, B1, B1/2; region O.

Rule 10 (5) (d) — Figure 9

(Perspective drawing of a deck erection with a rectangular opening ABCD in the deck above a space ABCDEFGH.)

ABCD = OPENING IN THE DECK. SPACE ABCDEFGH SHALL BE EXCLUDED FROM ENCLOSED SPACE

p.353 ## Appendix-2 ### Figures

Rule 10 (5) (e) — Figure 10

(Plan view of an erection C with a recess O of width w2 and depth l2 on one side (l2 < 2w2), and a recess I of width w1 and depth l1 on the other side (l1 > 2w1).)

SHIPS WITH ROUNDED GUNWALES — Figure 11

(Section drawing of a ship with rounded gunwales: DECK, SHELL, breadth B measured between the points of RADIUS R on each side.)

p.354 ## Appendix-3 ### Coefficients K1 and K2 referred to in rules 11 and 12 (1) ### V or Vc = volume in cubic metres

V or Vc	K1 or K2	V or Vc	K1 or K2	V or Vc	K1 or K2	V or Vc	K1 or K2
10	0.2200	45000	0.2931	330000	0.3104	670000	0.3165
20	0.2260	50000	0.2940	340000	0.3106	680000	0.3166
30	0.2295	55000	0.2948	350000	0.3109	690000	0.3168
40	0.2320	60000	0.2956	360000	0.3111	700000	0.3169
50	0.2340	65000	0.2963	370000	0.3114	710000	0.3170
60	0.2356	70000	0.2969	380000	0.3116	720000	0.3171
70	0.2369	75000	0.2975	390000	0.3118	730000	0.3173
80	0.2381	80000	0.2981	400000	0.3120	740000	0.3174
90	0.2391	85000	0.2986	410000	0.3123	750000	0.3175
100	0.2400	90000	0.2991	420000	0.3125	760000	0.3176
200	0.2460	95000	0.2996	430000	0.3127	770000	0.3177
300	0.2495	100000	0.3000	440000	0.3129	780000	0.3178
400	0.2520	110000	0.3008	450000	0.3131	790000	0.3180
500	0.2540	120000	0.3016	460000	0.3133	800000	0.3181
600	0.2556	130000	0.3023	470000	0.3134	810000	0.3182
700	0.2569	140000	0.3029	480000	0.3136	820000	0.3183
800	0.2581	150000	0.3035	490000	0.3138	830000	0.3184
900	0.2591	160000	0.3041	500000	0.3140	840000	0.3185
1000	0.2600	170000	0.3046	510000	0.3142	850000	0.3186
2000	0.2660	180000	0.3051	520000	0.3143	860000	0.3187
3000	0.2695	190000	0.3056	530000	0.3145	870000	0.3188
4000	0.2720	200000	0.3060	540000	0.3146	880000	0.3189
5000	0.2740	210000	0.3064	550000	0.3148	890000	0.3190
6000	0.2756	220000	0.3068	560000	0.3150	900000	0.3191
7000	0.2769	230000	0.3072	570000	0.3151	910000	0.3192
8000	0.2781	240000	0.3076	580000	0.3153	920000	0.3193
9000	0.2791	250000	0.3080	590000	0.3154	930000	0.3193

p.355 | V or Vc | K1 or K2 | V or Vc | K1 or K2 | V or Vc | K1 or K2 | V or Vc | K1 or K2 | |—|—|—|—|—|—|—|—| | 10000 | 0.2800 | 260000 | 0.3083 | 600000 | 0.3156 | 940000 | 0.3194 | | 15000 | 0.2835 | 270000 | 0.3086 | 610000 | 0.3157 | 950000 | 0.3195 | | 20000 | 0.2860 | 280000 | 0.3089 | 620000 | 0.3158 | 960000 | 0.3196 | | 25000 | 0.2880 | 290000 | 0.3092 | 630000 | 0.3160 | 970000 | 0.3197 | | 30000 | 0.2895 | 300000 | 0.3095 | 640000 | 0.3161 | 980000 | 0.3198 | | 35000 | 0.2909 | 310000 | 0.3098 | 650000 | 0.3163 | 990000 | 0.3199 | | 40000 | 0.2920 | 320000 | 0.3101 | 660000 | 0.3164 | 1000000 | 0.3200 |

Coefficients K1 or K2 for intermediate values of V or Vc shall be obtained by linear interpolation.

The Inland Shipping Ordinance, 1976

The Inland Shipping Ordinance, 1976

Ordinance No. 72 of 1976

Incorporating all of the following amendments up to 15-03-05:

a. The Inland Shipping (Amendment) Ordinance, 1979 (Ordinance No. 14 of 1979)

b. The Inland Shipping (Amendment) Ordinance, 1983 (Ordinance No. 19 of 1983)

c. The Inland Shipping (Amendment) Ordinance, 1989 (Ordinance No. 15 of 1989)

d. The Inland Shipping (Amendment) Act, 2005 (Act No. 13 of 2005)

The Inland Shipping Ordinance, 1976 Ordinance No. 72 of 1976

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The Inland Shipping Ordinance, 1976 Ordinance No. 72 of 1976

Chapter I — Preliminary

1. Short title

1. This Ordinance may be called the Inland Shipping Ordinance, 1976.
2. It extends to the whole of Bangladesh.

2. Definitions

In this Ordinance, unless there is anything repugnant in the subject or context,—

a. “certificate of competency” means a certificate of competency granted under section 36;

b. “certificate of registry” means a certificate of registry granted under section 18;

(bb) “certificate of service” means a certificate granted under section 37 of this Ordinance;

c. “certificate of survey” means a certificate of survey granted under section 9;

d. “dangerous goods” means such goods as the Government may, by notification in the official Gazette, declare to be dangerous goods for the purposes of this Ordinance;

(dd) “Director General” means the Director General, Department of Shipping;

e. “inland vessel” means any vessel which ordinarily plies on inland waters of any description, and includes any

vessel propelled wholly or in part by steam, liquid fuel, electricity or any other mechanical power, sailing vessels, dumb barges or any other vessel which is not mechanically propelled but is

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towed or pushed by any other mechanically propelled vessel, and such vessels are also included among inland vessels;

f. “inland waters” means any canal, river, lake or any navigable waterway of Bangladesh and such portion of a tidal waterway as the Government may, by notification published in the official Gazette, declare to be inland waters for the purposes of this Ordinance;

g. “for commercial purposes”, in relation to an inland vessel, means the carriage of passengers and goods for hire, remuneration, reward or any other valuable consideration, or use for any other work;

(gg) “owner” means—

a. in the case of a registered inland vessel, the person whose name is entered as owner in any register book maintained under this Ordinance;

b. in the case of any other inland vessel, the owner of the vessel or his partner (part-owner);

c. “passenger” means any person carried in an inland vessel, but who is not the master, an officer or any other member of the crew of the vessel;

d. “prescribed” means prescribed by rules made under this Ordinance;

e. “Registrar” means a Registrar appointed under this Ordinance;

f. “survey” means the survey of inland vessels under this Ordinance;

g. “Surveyor” means a Surveyor appointed under this Ordinance;

h. “voyage”, in relation to an inland vessel, means plying on inland waters or for the purpose of reaching a destination.

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Chapter II — Registration and Survey

3. Inland vessels to be surveyed and registered

1. Every inland vessel, other than an inland vessel under the Ministry of Defence, which plies or is intended to ply on inland waters, or is used or is intended to be used for any purpose, shall be surveyed and registered under this Ordinance.

2. No inland vessel shall be surveyed and registered under this Ordinance unless the design and plan of that vessel have been approved under section 5A, a Certificate of Compliance has been obtained in accordance with sub-section (3) of section 5B, and its owner is—

a. a citizen of Bangladesh;

b. a company registered in Bangladesh; or

c. a company which is not registered in Bangladesh but which has an agent or branch in Bangladesh which jointly has an organisation sufficient to conduct operations independently in Bangladesh on behalf of the said company.

3. Sub-section (1) shall not apply—

a. if any inland vessel, for the purpose of registration, proceeds on its first voyage to the place of registration without carrying any passengers or goods, or is proceeding to the nearest place of survey for the purpose of renewing its certificate of survey from the date on which the certificate of survey expires; or

b. to any sea-going vessel which has the right or permission to ply at any port, place or tidal place in Bangladesh; or

c. to any country-type wooden mechanically propelled vessel, including those with shallow engines, which is propelled by an engine of a maximum of 16 B.H.P.

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4. Places of survey and registration

1. For the purposes of this Ordinance, the Government may, by notification in the official Gazette, declare such places in Bangladesh as it considers suitable to be places of survey and registration.

2. The Government shall appoint—

a. for every place of survey, one or more Surveyors as may be required; and

b. for every place of registration, a Registrar.

5. Powers of Surveyors and Registrars

1. For the purpose of survey or registration, any Surveyor or Registrar may at any reasonable time go on board any inland vessel and inspect its hull, boilers, engines, any part thereof together with other machinery, and all the equipment and articles of the vessel:

Provided that the Surveyor or Registrar shall not obstruct the loading or unloading of goods of any vessel, nor delay or detain any vessel in plying beyond the time necessary for the survey or inspection.

2. The owner, master and crew of every vessel to be surveyed and registered shall afford the Surveyor and Registrar all reasonable facilities for that purpose, and shall furnish all information which the Surveyor and Registrar may require regarding the machinery or any other part and equipment of the vessel which are on board.

5A. Approval of design and plan of inland vessels

1. For the approval of the design and plan of an inland vessel, the owner shall apply, in the prescribed form, to the authority empowered by the Government for that purpose, together with the design and plan of the vessel.

(1A) The Government may fix fees for the application and the approval of the design by the competent authority.

2. On receipt of an application under sub-section (1), the Authority—

a. shall, if after verification the design and plan are found to conform to the prescribed specifications or standards, grant a certificate of approval in the prescribed form within forty-five days of receipt of the application; or

b. shall, if it appears that the design and plan do not conform to the prescribed specifications or standards, return the application to the owner within the period specified in clause (a), stating in detail the reasons for which they do not conform:

Provided that nothing shall prevent the owner from submitting a fresh application for approval of the design and plan after altering and amending the application so returned in accordance with the prescribed specifications and standards, and such application shall be disposed of within 15 days of its receipt.

5B. Construction survey, etc.

1. For the purpose of the construction or conversion of an inland vessel, the owner shall, after obtaining a certificate of approval of the design under section 5A, inform the authority empowered by the Government for that purpose, in the form prescribed by the Government, when and where the vessel is to be constructed or converted.

2. On receipt of the information referred to in sub-section (1), the said Authority shall, during the construction or conversion of the inland vessel, carry out construction survey, construction supervision, stability determination and inclining test.

3. After the satisfactory completion of the construction or conversion of the inland vessel, if it is found that the construction or modification of the inland vessel has been carried out in accordance with the prescribed specifications and standards, the person empowered under sub-section (2) who supervised such construction or conversion shall issue a Certificate of Compliance in the form prescribed by the Government.

4. The Government may fix fees for the construction survey, construction supervision, stability determination, inclining test and other related suitability tests.

5C. Delegation of functions to a classification society, etc.

1. The Government may, by notification in the Gazette, delegate the functions referred to in sections 5A and 5B, and the functions of a Surveyor for the purpose of surveying inland vessels under this Ordinance, to a classification society empowered by the Government for that purpose.

2. The classification society shall be accountable and liable to the Government for the functions so delegated.

3. The constitution, functions, responsibilities and capabilities of the classification society shall be determined by order issued by the Government.

4. The classification society shall be entitled to receive fees, at the rates and in the manner prescribed by the Government, for performing the functions delegated under sub-section (1) of section 5C.

5. Explanation: “classification society” means an organisation or a society empowered by the Government for the purpose of fulfilling the objects of this Ordinance.

5D. Marking of ships

1. Before the grant of a certificate of registry under section 18 of this Ordinance, every inland vessel for which an application for registration has been made under section 17 shall be marked permanently and conspicuously in the prescribed manner and to the satisfaction of the Registrar.

2. No alteration or any other amendment shall be made to the marked places of the vessel required under this section without the prior written permission of the Registrar.

3. If any owner or master neglects to make the marking required under this section, or fails to mark, or conceals, obliterates, alters or adds to the marking, he shall be liable to a fine of up to twenty thousand Taka, and if the Surveyor reports that the inland vessel is in an insufficiently or improperly marked condition, the vessel shall remain detained until all such defects have been rectified.

6. Fees for survey, etc.

For the survey of an inland vessel, the owner of the vessel shall apply to the Surveyor in the prescribed form, and shall pay to the Government a fee for the survey at the rate and in the manner prescribed by the Government, and shall in addition pay an additional fee for the travelling expenses of the Surveyor.

7. Declaration of Surveyor

1. The Surveyor shall, within seven days of the completion of the survey of any inland vessel, prepare a declaration in the prescribed form and send a copy thereof to the owner/master of the vessel, if he is satisfied that—

a. the hull, boilers, engines and other machinery of the vessel are in good condition for the specified voyage or service;

b. the equipment of the vessel and the certificates of competency of the master, engineer or driver are sufficient in accordance with the requirements of this Ordinance; and

c. the “free board mark” has been made on the body of the vessel in the prescribed manner.

2. The declaration referred to in sub-section (1) shall contain—

a. a detailed statement of the matters described in clauses (a) and (b) of that sub-section;

b. the time, if less than one year, for which the boilers, engines and other machinery and equipment of the vessel will remain serviceable;

c. the time, if any, after the expiry of which the hull, boilers, engines and other machinery and equipment of the vessel will, in the judgment of the Surveyor, not be fit for plying;

d. the number of passengers, whatever it may be, which the vessel is capable of carrying, and how many of them shall be accommodated in the cabins, on the deck and in any part of the deck and cabins of the vessel, which shall, as necessary, be determined in the prescribed manner having regard to the season, the nature of the voyage, the quantity of goods carried and other incidental circumstances; and

e. the deadweight tonnage and such other particulars (if any) as may be prescribed.

3. A copy of the declaration under sub-section (1) shall be sent simultaneously—

a. if it is for the grant of a certificate of survey, to such officer as the Government may appoint for that purpose; and

b. if it is for the grant of a certificate of registry, to the Registrar.

8. Cancelled

9. Grant of certificate of survey

1. The officer to whom a declaration is sent under sub-section (1) of section 7 shall, if he is satisfied that the said vessel has complied with the provisions of this Ordinance, within fifteen days of receiving the said declaration prepare two copies of the certificate of survey and arrange for them to be delivered to the owner or master by post or by any other means.

2. The certificate of survey shall state, in the prescribed form, that all the provisions of this Ordinance relating to the survey of inland vessels and the declaration of survey have been duly complied with, and shall further state—

a. all the particulars regarding the vessel contained in the declaration of survey in accordance with clauses (a), (b), (c) and (d) of sub-section (2) of section 7; and

b. such other matters as may be prescribed.

3. The person who has prepared the declaration of survey under section 7 shall not grant the certificate of survey:

Provided that the Surveyor who has completed the survey may, until the certificate of survey is issued, grant a temporary permit to ply to the said inland vessel, which shall not be effective for more than forty-five days, and the permit shall specifically state this.

4. Any certificate of survey may be cancelled or suspended by written order of the Government or of any person appointed by the Government in that behalf, if the said person, upon inspection for that purpose or otherwise, finds that—

a. the certificate has in any way been fraudulently or by forgery altered; or

b. the said certificate has been granted on the basis of false or erroneous information; or

c. the said inland vessel to which the certificate of survey has been given is not complying with the provisions of the Ordinance relating to the grant of such certificate.

S.R.O. No. 394-S/76: In exercise of the powers conferred by section 9(4) of the Inland Shipping Ordinance, 1976, the Government has conferred upon the Director General, Department of Shipping, Government of the People's Republic of Bangladesh, the power to cancel or suspend any certificate of survey under the said section.

10. Certificate of survey to be hung up in a conspicuous open place on the vessel

The owner or master of every inland vessel shall, as soon as the certificate of survey of the vessel is granted, affix a copy thereof in some conspicuous place on the vessel in such a manner that all persons on board the vessel can easily see and read it, and it shall be kept hung up in such place for so long as the certificate remains in force and the vessel remains in service.

11. Duration of certificate of survey

1. The certificate of survey of any inland vessel shall remain in force—

a. in the case of dumb barges and other vessels which cannot ply by themselves but are towed or pushed by other power-driven vessels, for four years from the date of issue of the certificate, and in the case of other inland vessels, for one year from the date of issue of the certificate; or

b. until the expiry of the period of serviceability of the hull, boiler, engine or other machinery or any other equipment of the vessel stated in the said certificate; or

c. until it is suspended or cancelled under sub-section (4) of section 9.

2. The owner or master of any inland vessel whose certificate of survey is about to expire under clause (a) of sub-section (1) shall, thirty days before the expiry of the said certificate, notify the Surveyor of the date of such expiry.

3. An owner or master who fails to do what is required under sub-section (2) shall, in addition to the survey fee payable under section 6, pay a fine of Tk. 100/- for every day during which he fails to comply with the said provision.

12. Renewal of certificate of survey

A certificate of survey which has expired may be renewed only after the said inland vessel has been surveyed anew in accordance with the provisions of this Chapter.

13. Delivery of cancelled and expired certificates

Where the certificate of survey of any inland vessel has expired, or has been cancelled or suspended, or has otherwise become inapplicable, it shall be returned by the owner or master of the vessel to the person directed by the Government.

S.R.O. No. 387-L/76: In exercise of the powers conferred by section 13 of the Inland Shipping Ordinance, 1976, the Government has delegated to the Director General, Department of Shipping, the power to receive certificates cancelled, suspended or rendered inapplicable under the said section.

14. Power of the Government to have survey made by more than one Surveyor

The survey of an inland vessel shall ordinarily be carried out by one Surveyor. However, the Government may, by written order, appoint more than one Surveyor for the purpose of surveying at any place, generally in respect of all inland vessels, or in special cases in respect of any particular vessel or any particular class of vessels.

15. Power of the Government to order a second survey

1. If any Surveyor, after surveying any inland vessel, refuses to give a declaration under section 7, or gives such a declaration with which the owner or master of the said vessel is dissatisfied, the officer appointed under section 7 may, on the application of the owner or master of that vessel, and on receipt of a fee double the previous survey fee, order a second survey, and may direct more than one Surveyor to survey the said vessel.

2. The Surveyors directed under sub-section (1) shall, after surveying the said vessel as soon as possible, refuse to give a declaration or give such declaration as appears to them reasonable, and the said decision or declaration shall be deemed to be final.

16. Procedure where more than one Surveyor is appointed

When the survey of a vessel is carried out by more than one Surveyor under section 15, each Surveyor shall perform his own duties in accordance with this Ordinance or the rules made under this Ordinance.

17. Application for registration

For the registration of an inland vessel, the owner shall apply in such form as shall contain the prescribed particulars, and there shall further be attached thereto—

a. a copy of the certificate of survey of the said vessel which is in force; and

b. the challan (receipt) of payment of the prescribed fee for registration.

18. Grant of certificate of registry

1. When the Registrar, after making the necessary enquiry regarding any inland vessel which is to be registered, is satisfied that everything in respect of the vessel is in order in accordance with the provisions of this Ordinance and the rules, the Registrar shall issue to the applicant a certificate of registry for the vessel, which shall contain the prescribed particulars.

2. The Registrar may refuse to register any inland vessel if the said vessel is defective in respect of its machinery and structural framework, or if the applicant fails to produce satisfactory evidence in support of his claim of ownership of the said vessel and of the information stated in the application:

Provided that if the Registrar refuses to register any inland vessel, he shall inform the applicant in writing of the reasons for such refusal.

3. Cancelled.

19. Display of registration number in a conspicuous place

When a certificate of registry is granted to any inland vessel under section 18, the Registrar shall give the vessel a registration number, which shall be inscribed in the prescribed manner in a conspicuous open place on the vessel.

20. Maintenance of register book

The Registrar shall maintain a register book in the prescribed manner, in which the particulars contained in the certificates of registry of all inland vessels which have been registered shall be recorded in detail.

21. Certificate of registry to be kept on board the vessel

The certificate of registry of every inland vessel shall at all times be kept on board the said vessel by the owner or master, and shall be produced for inspection by any person empowered by the Government.

22. Loss of certificate of registry

When the certificate of registry of any inland vessel cannot be found or is lost by reason of being kept in any place, or is destroyed, the Registrar shall, on receipt of the fee in accordance with the prescribed rules, grant a new certificate in place of the said certificate.

23. Notice of loss, etc., of vessel

1. When any inland vessel registered under this Ordinance is lost, taken by the enemy, burnt or broken up, the owner of the said vessel shall forthwith give notice thereof to the Registrar, if such notice has not previously been given, and

the Registrar shall thereupon record the same in the register book.

2. When, in any event described in sub-section (1), the certificate of registry is not lost or destroyed, the owner or master of the inland vessel shall deliver the said certificate—

a. if the event occurs at the place of registry, forthwith; and if it occurs elsewhere, within ten days of the vessel's arrival at the place of registry, to the Registrar; and

b. if there is no likelihood of the vessel arriving at the place of registry, within ten days of the occurrence of the event, to the Registrar.

24. Transfer of ownership

1. When the ownership of any registered inland vessel is changed, the changed ownership shall be entered by the Registrar in the certificate of registry of the said vessel.

2. If the change of ownership of any inland vessel is effected at the place of its Registrar, the owner or master of the said vessel shall forthwith deliver the certificate of registry to the Registrar for the purpose of endorsement under sub-section (1), and if the transfer is effected while the vessel is located elsewhere, it shall be delivered immediately after the vessel first arrives at the place of registry.

25. Report of acquisition of ownership outside Bangladesh to be made to the Registrar

If any inland vessel becomes, outside Bangladesh, the property of a Bangladeshi or of any company as described in sub-section (2) of section 3, the owner of the vessel shall, as soon as it arrives at a place of registry in Bangladesh, furnish accurate information to the Registrar, which shall contain the following particulars—

a. the time of arrival of the vessel at the place of registry;

b. the name of the vessel, if it has one;

c. the time and place of its purchase, and the name of the purchaser or purchasers;

d. the name of its master; and

e. its tonnage capacity, date of construction and other available particulars in detail.

26. Transfer of registered inland vessels

1. No inland vessel registered under this Ordinance, or any share thereof, shall be transferred without the prior permission of the Government to any person other than a citizen of Bangladesh who is resident in and carrying on business in Bangladesh, and the said transfer shall not be complete unless its bill of sale has been executed and attested before a Magistrate of the first class.

2. When the ownership of any inland vessel registered under this Ordinance is transferred, the purchaser and seller shall, within thirty days of the sale, jointly report to the Registrar together with the bill of sale or an attested copy thereof and the necessary fee, and the Registrar shall record the same in the register book:

Provided that the Registrar may, if he so wishes, after being otherwise satisfied as to the correct identity of the transferor and the transferee of the ownership, act on the basis of the report of one party instead of a joint report.

27. Registration of vessels transferred to a citizen of Bangladesh

1. If any inland vessel which is not registered under this Ordinance is transferred to any citizen of Bangladesh or to any company described in sub-section (2) of section 3, it shall be registered under this Ordinance.

2. Before registering any inland vessel required to be registered under sub-section (1), the Registrar may, for the purpose of ascertaining the genuineness of the transfer and the identity of the purchaser and seller, require the bill of sale, the certificate of transfer and the certificate of registry to be produced before him.

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3. Where any inland vessel is registered under sub-section (1), the Registrar shall record in his register book all the particulars of the original certificate of registry of the said vessel.

28. Registration where the instrument of transfer is lost

When any bill of sale of an inland vessel is lost, becomes illegible, or becomes illegible by reason of being eaten by insects or mutilated, the Director General may, if he so wishes—

1. if the transfer authorised by the instrument has not taken effect, subject to proof of such loss, destruction, obliteration or mutilation; and

2. if the transfer has been completed, subject to proof of the other incidental matters executed;

direct the Registrar to enter the transfer in his register book, or to do what is required as if it had not been lost, destroyed, obliterated or mutilated.

29. Converted vessel to be registered anew

1. If any inland vessel registered under this Ordinance is so converted that the previous particulars relating to its tonnage have become inconsistent, or an inconsistency has appeared with the particulars contained in the register book, the vessel shall not be deemed to be registered until the Registrar receives a written application from him in that behalf and, on receiving from the Surveyor a certificate of survey containing detailed particulars relating to the conversion, enters the particulars relating to the conversion in the register, or, if he considers it proper, registers the said vessel anew.

2. For the purpose of entering the particulars relating to the conversion of an inland vessel in the register under sub-section (1), the Registrar may call for the certificate of registry currently in force to be brought before him, and may either grant a new certificate in place of the existing certificate,

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or amend it with the particulars of the conversion, or endorse and sign on the certificate in force a memorandum relating to the conversion of the vessel.

3. For the purpose of the fresh registration of any vessel under sub-section (1), the Registrar shall proceed as if the vessel were being registered for the first time, and in that case, on the certificate of registry currently in force being delivered to him and the other necessary formalities relating to registration being duly completed, including payment of the fee prescribed for first registration, or, in the case of a change of ownership, on his being satisfied as to such matters as he considers necessary to verify, he shall register the vessel and grant a new certificate of registry.

4. When any inland vessel is registered anew under this section, its previous registration shall be deemed to be closed, except as regards any certificate of sale contained therein which is neither satisfied nor expired; but the names of the persons included in the previous register who are interested in being owners of the vessel shall be included in the new register book, and the new registry shall in no case prejudice any right of those persons.

30. Duration of certificate of registry, etc.

1. Subject to the other provisions of this Ordinance, a certificate of registry granted in favour of any inland vessel under this Ordinance shall remain in force for 30 (thirty) years from the first registration, unless the ship is totally lost, broken up or otherwise transferred within the said period; provided that the period of this registration may be extended by five years if, after a special docking survey in the prescribed manner, the vessel is considered by the Registrar to be fit to ply in a satisfactory manner, and subject to the condition of special docking, it may again be extended for the same period.

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2. The registration of any inland vessel registered under this Ordinance, other than dumb barges and other vessels which cannot ply by themselves but are towed or pushed by other mechanically propelled vessels, which has not been surveyed for three consecutive years, being a registration granted under section 11(1) of this Ordinance, shall stand cancelled from the date of expiry of the certificate of survey.

3. When any inland vessel is laid up, or is broken up, or otherwise remains out of use, the owner shall in such case forthwith, within 30 days, inform the Registrar in writing.

31. Submission of returns relating to registration, etc.

Every year, between the 15th of January and the 15th of July, for the preceding six months, the Registrar shall submit, in the specified form, to the Government or to any officer appointed by the Government, a report containing the information entered in the register, including the names of the persons concerned, relating to registrations made under this Ordinance, cancellations of registration, and all other matters including transfers, together with such other information as the Government may direct to be furnished.

No. 1 S.R.O. 386-L/76: In exercise of the powers conferred by section 31 of the Inland Shipping Ordinance, 1976, the Government has empowered the Director General, Department of Shipping, Government of the People's Republic of Bangladesh, as the officer to whom reports under the said section are to be submitted.

32. Reciprocal obligation as to recognition of certificates of survey and registry

1. If the Government is satisfied that, under the laws and provisions of any foreign State, any inland vessel which has obtained a certificate of registry under this Ordinance—

a. by virtue of such certificate, enjoys any special exemption when plying in that country; or

b. is required to comply with special conditions, such as fresh registration or payment of fees, etc., for plying on the inland waterways of that country, then the Government may, by notification in the Gazette,

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issue an order to the effect that a similar exemption or conditions, or any exemption or condition, shall be approved or imposed on a reciprocal basis when vessels of that particular country ply in the inland waters of Bangladesh.

2. The Government may, by publication of a notification in the official Gazette, issue an order to the effect that the reciprocal obligation relating to registration under sub-section (1) shall, as required, also apply in the case of certificates of survey, certificates of competency and licences granted under this Ordinance.

33. Prohibition of voyage without a certificate of survey

No inland vessel shall proceed on any voyage or ply on inland waters unless it has a valid certificate of survey and certificate of registry granted under this Ordinance which are applicable to the service and the voyage.

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Chapter III — Manning, Examination and Certification

34. Definition

In this Chapter, "Director General" means the Director General of the Department of Shipping constituted under the Bangladesh Merchant Shipping Ordinance, 1983 (Ordinance No. 26 of 1983).

35. Classification of inland vessels

The Director General may, by notification in the official Gazette, classify inland vessels according to his judgment.

36. Certificate of competency

1. The Director General shall, after holding an examination or test in the prescribed manner, grant certificates of competency for inland engineers and for masters and drivers of various grades for inland vessels.

2. No inland engineer, master or driver shall be engaged or employed on any inland vessel unless he holds a certificate of competency under sub-section (1).

37. Manning of inland vessels

1. No inland vessel shall ply on inland waters or be used for any work unless crew are engaged in accordance with sub-section (2).

2. The Director General may, by notification in the official Gazette, issue an order to the effect that any inland vessel shall carry such number, and such grades, of inland engineers, masters, drivers and crew as are specified in the order of that notification.

3. The Director General may exempt any inland vessel or any class of vessels from any requirement of an order issued under sub-section (2).

4. An exemption granted under sub-section (3) may apply for a specified period or in respect of one or more particular voyages.

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5. Notwithstanding anything contained in this section, the Director General may relax this section in the case of any person, if—

a. the said person can satisfy the Director General that he is reasonably competent to perform the duties of the post for which he is a candidate for engagement;

b. the owner of the vessel satisfies the Director General that, despite proper efforts, it has not been possible to find a suitably certificated person for engagement.

37(A). Uniform and identity card of crew of inland vessels

1. The Director General may order that—

a. the owners of inland vessels shall supply uniforms to the crew working on the vessels, in accordance with the pattern and standard specified in the written order;

b. an identity card shall be granted to every crew member, which shall be referred to as the crew identity card under this section, and which shall contain all the particulars of the crew member holding the identity card, as shall be specifically stated in the order, and every crew member shall apply with a view to obtaining such an identity card;

c. the requirement for the holder of an identity card to produce the identity card shall be stated, in such circumstances and to such person as shall be specified in the order;

d. the requirement for the surrender of a crew member's identity card shall be stated, in such circumstances as shall be specified in the order.

2. If any person is required to wear the standard uniform or to produce an identity card, and he fails to do so, he shall be guilty of an offence punishable with a fine which may extend to two thousand Taka, or his certificate of competency, if any, may be suspended for a period not exceeding three months.

3. If any person makes a statement which appears to the Director General to be false, or recklessly makes a statement for the purpose of obtaining an identity card for himself or for another person which is proved to be false in material particulars, he shall be punishable with a fine not exceeding five thousand Taka, and in that case the crew member's identity card shall be cancelled.

37(B). Training, etc., of inland engineers, masters and engine drivers

1. The Director General shall arrange training programmes for the persons primarily responsible for the safety of inland vessels, the safety of passengers or other safety arrangements, and for ratings, greasers, sukanis (helmsmen), engine drivers, masters and inland engineers, in management of special passenger gatherings, fire-fighting, personal survival techniques, safety of oil-carrying vessels and other such training, to be given from time to time.

2. If—

a. any person is nominated for training under sub-section (1), he shall be bound to undergo the said training;

b. any person refuses to undergo the said training without reasonable cause, his certificate of competency or identity card shall remain suspended until the training is undergone.

38. Appointment of examiners, etc.

For the examination of the competency of persons desirous of obtaining certificates of competency under this Ordinance, the Government may appoint such examiners or boards of examiners as it thinks fit, and shall pay them honoraria in the prescribed manner.

39. Grant of certificate of competency

In respect of an applicant desirous of obtaining a certificate of competency, on the basis of a report by the examiners or the board of examiners that he has passed the competency examination, and after the submission of satisfactory evidence regarding his skill and his habitual satisfactory conduct on board a vessel, the Director General or any officer appointed in that behalf shall grant the applicant the certificate of competency applied for:

Provided that if, before granting a certificate of competency, the Director General or the officer appointed in that behalf has reason to believe, in respect of any applicant, that the report of the examiners or the board of examiners has been improperly given, the applicant shall in that case be required to be re-examined.

40. Certificate to be made in duplicate

Any certificate of competency granted under this Ordinance shall be made in duplicate in the prescribed form; one copy shall be delivered to the person entitled to it, and the other copy shall be preserved and recorded in the prescribed manner.

41. Loss of certificate

If any certificate of competency granted under this Ordinance is lost, destroyed, or so obliterated or eaten by insects as to be deemed useless while its validity is still in force, then, if the person entitled to it can satisfy the authority empowered or responsible for granting certificates that such loss, destruction, obliteration or insect damage was not due to his negligence, a copy of the certificate shall be delivered to him on the basis of the record preserved under section 40, and its effect shall be the same as that of the original certificate.

42. Suspension and cancellation of certificate, etc.

1. Any certificate of competency granted under this Ordinance may be suspended or cancelled by the Director General if the holder of the said certificate—

a. is convicted of any offence under this Ordinance or any other non-bailable offence; or

b. is on duty on any inland vessel which proceeds on a voyage or is used in any service in contravention of the provisions of this Ordinance; or

c. is declared by a medical practitioner appointed by the Director General to be physically unfit for work; or

d. is, in the opinion of the Director General, considered unfit to act in accordance with the conditions stated in the certificate or licence; or

e. is reported by any court, authority or officer trying, investigating or enquiring under this Ordinance to the effect that—

(1) the casualty or sinking of any inland vessel or the abandonment of a vessel, or the loss or destruction of any such vessel, or any loss of life or any damage to or destruction of goods, was caused by any

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wrongful act or default of the holder of the certificate; or

(2) such person has been found guilty of incompetency, or of gross negligence of duty, drunkenness or other misconduct:

Provided that no person's certificate shall be suspended or cancelled without giving him an opportunity to show cause, and in the case referred to in clause (e) he shall also be given a copy of the report of the court, authority or officer.

2. The order suspending or cancelling a certificate of competency suspended or cancelled under sub-section (1) shall be delivered to the person mentioned therein within the specified time.

3. The Government, or any officer empowered by the Government in that behalf, may, if it so wishes, at any time revoke an order of cancellation or suspension made under sub-section (1), and may grant him a new certificate of competency without examination, and the said certificate, unless it contains any condition to the contrary, shall have the same effect as the cancelled certificate.

43. Cancelled

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Chapter IV — Inland Vessel Casualties

44. Inland vessel casualties and report thereof

1. An inland vessel casualty shall be deemed to have occurred when—

a. any inland vessel is lost, sunk, abandoned or partially destroyed;

b. any loss of life or destruction of property occurs by reason of any casualty, or such occurs on any such vessel; or

c. any such inland vessel causes damage or injury to any other vessel or its property or to any person on board.

2. Immediately upon the occurrence of every inland vessel casualty, and if this is not possible, within twenty-four hours of the occurrence of the casualty, the master of the said vessel or vessels, or, in the absence or physical incapacity of the master, any crew member of the vessel or vessels, or any passenger of the vessel or

vessels, or any person having knowledge of the casualty, shall report the casualty to the officer-in-charge of the nearest police station.

3. After receiving a report of any inland vessel casualty under sub-section (2), or after receiving news through a reliable source, the officer-in-charge of the said police station shall, as soon as possible but in no case later than twelve hours after the casualty, report in writing on the facts and circumstances relating to the casualty to the Director General, to the District Magistrate within whose jurisdiction the casualty occurred, and to the Upazila Executive Officer.

45. Investigation of inland vessel casualties

1. On the basis of the report received under section 44 regarding an inland vessel casualty, the Upazila Executive Officer—

a. shall forthwith investigate, or cause to be investigated, the said inland vessel casualty; and

b. shall, within seven days, submit to the Government and the District Magistrate a report describing the facts and circumstances relating to the casualty, together with his opinion, if any, stating the cause of the casualty and the responsibility therefor.

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2. The person conducting an investigation under sub-section (1)—

a. may go on board any vessel concerned and inspect any part thereof, its machinery or other articles;

b. may enter and inspect any place where this is necessary in the interest of the investigation;

c. may, if he considers it necessary, summon and examine any person who is acquainted with the matters and circumstances relating to the casualty; and

d. may cause to be produced before him such books, papers and documents as he considers necessary in the interest of the investigation.

3. Notwithstanding the provisions of sub-section (1), the Government may, if it so wishes, on the basis of the report received under sub-section (3) of section 44, appoint any officer in that behalf to investigate the said casualty on behalf of the Government, and the said officer—

a. shall submit to the Government, in the form of a report, a description of the facts and circumstances relating to the said casualty and his opinion, if any, stating the cause of the casualty and the responsibility therefor;

b. shall be entitled to the powers and facilities referred to in sub-section (2).

4. When any investigation is held under sub-section (3), no investigation shall be held under sub-section (1).

S.R.O. 390-L/76: In exercise of the powers conferred by sub-section 3 of section 45 of the Inland Shipping Ordinance, 1976, the Government has empowered the Special Marine Safety Officer of the Department of Shipping, Government of the People's Republic of Bangladesh, to conduct investigations under this section.

46. Trial of offences

1. Where, on consideration of a report given under sub-section (1) or (3) of section 45, the Government is of the opinion that such casualty has occurred by reason of the contravention of any section of this Ordinance or of any other law, and that it ought to be tried, the Government shall direct the District Magistrate or Upazila

p.462 Executive Officer or the officer investigating under sub-section (3) of section 45 to submit the report for trial —

a. to the court of any Magistrate of the first class within whose jurisdiction the casualty occurred; or

b. to a Marine Court constituted under section 47.

2. The report prepared under sub-section (1) shall contain—

a. the names of the accused persons who are to be sent up for trial;

b. a formal charge in respect of the offence committed; and

c. a list of the witnesses who are to be produced in support of each charge:

Provided that, notwithstanding the filing of a list of witnesses under clause (c), the District Magistrate or the Upazila Executive Officer or the investigating officer appointed under sub-section (3) of section 44, or the prosecution, shall not be barred from filing a list of additional witnesses at a later stage of the trial.

47. Constitution of Marine Courts

1. For the purpose of trying offences punishable under this Ordinance, one or more Marine Courts shall be constituted, each consisting of one Magistrate of the first class.

2. If the number of Courts is more than one, the Government shall, by notification in the official Gazette, determine the jurisdiction of the Courts.

3. There shall be not less than two and not more than four Assessors, of whom one shall be a person conversant with nautical matters and the others shall have experience of inland ship operation or of commercial or engineering matters.

4. Every Assessor of the Marine Court shall, except when absent with the permission of the Magistrate of the Court concerned for sufficient cause, be present at every proceeding of the Court, and shall give his opinion in writing, which shall be kept on the record of the proceeding.

48. Marine Court to have the powers of a Magistrate of the first class

The Marine Court shall have the same powers as a court of a Magistrate of the first class under the Code of Criminal Procedure (Act V of 1898).

p.463 ### 49. Procedure of trial

The Court constituted for the purpose of trial under this Chapter, which here means both the Courts mentioned in sub-section (1) of section 46, shall, so far as possible, follow the procedure for summary trials under the Code of Criminal Procedure, 1898 (Act V of 1898) in conducting the trial.

50. Arrest of witnesses

1. Any Court engaged in trial under this Chapter may, if it considers the attendance of any person necessary for evidence, issue a warrant of arrest against him, and may authorise any police officer to enter any inland vessel or any part thereof for the purpose of executing the arrest.

2. A police officer authorised to enter an inland vessel under sub-section (1) may, if he so desires, for the purpose of giving effect to the right of entry, call upon any police officer or customs officer or any other person to assist him, and may, where reasonably necessary for effecting the arrest, detain and delay the vessel for a maximum of forty-eight hours.

51. Power to inquire into charges of incompetency, etc.

1. Any Court engaged in trial under this Chapter may, if it considers it necessary, during the course of the trial inquire into any charge brought to the effect that the casualty was caused by the incompetency, misconduct, wrongful act or default of any such master, engineer, engine driver or any person holding a certificate or licence under this Ordinance.

2. For the purpose of inquiring into a charge under sub-section (1), the Court may issue summons on any person and shall give him full opportunity of defending himself in person or by any other means.

52. Special powers of the Court

1. Notwithstanding the provisions of the Code of Criminal Procedure, 1898 (Act V of 1898), any Court conducting a trial under this Chapter

p.464 may impose any punishment provided in this Ordinance for any offence described in this Chapter.

2. If, in any marine casualty, any loss of life occurs or damage to property is caused or any inland vessel is lost, and the Court engaged in the trial is satisfied that the said casualty or loss, damage or destruction has occurred by reason of the negligence, incompetency, misconduct or offence of any other vessel or of the master or any other officer of the vessel, then in that case the Court may, if it considers it necessary, without prejudice to the provisions of sub-section (1) or to its other powers, order compensation not exceeding five lakh taka, and may direct that the money of the said compensation be paid within three months—

a. if loss of life has occurred, to the heirs of the deceased person;

b. if any person has been injured or damage to property has been caused, to the injured person and to the owner of the damaged property, as the case may be; and

c. if any vessel has been damaged, to the owner of the damaged vessel.

3. This order of compensation shall be executable in the same manner as a decree of a civil court of competent jurisdiction, and if payment of the money of compensation fails, the Government may, if necessary, realise the amount of compensation by sale or attachment of the vessel concerned.

53. Report to the Government by the Court

The Court engaged in trial under this Chapter shall, after the conclusion of the trial, send a report to the Government together with its judgment containing the evidence recorded, and if it is a Marine Court, the opinions of the Assessors shall also be included in the said report.

p.465 ## Chapter V — Safety of Vessels and Passengers

54. Voyage prohibited without route permit, time-table, fare list and printed tickets

No inland vessel which is engaged in the carriage of passengers shall be used on any voyage or for any work—

a. unless it has obtained a route permit granted by the Government or by any organisation empowered by the Government and an approved fare list;

b. except in accordance with the specified route allotted and the conditions set out in the said route permit; and

c. except on the issue, in the prescribed manner, of printed tickets or receipts stating the amount paid as fare for the passengers or goods carried.

54A. Plying in coastal waters without permission prohibited

1. No inland vessel shall proceed on a voyage or be used for any service in coastal waters without the written permission of the Registrar of the vessel for plying in coastal waters.

2. The Registrar shall not issue any written permission under sub-section (1) except on the fulfilment of such conditions as may be specified by the Director General for this purpose.

3. The permission for plying in coastal waters shall remain in force for the period of validity of the certificate of survey granted under section 11.

4. The Government may exempt any inland vessel from the conditions of this section for a period not exceeding three months.

54B. Voyage prohibited without telecommunication equipment

1. An inland vessel carrying—

a. one hundred or more than one hundred passengers; or

b. more than two hundred gross tonnes of oil, gas or chemicals; or

c. more than three hundred gross tonnes of goods,

shall not be used on any voyage or for any work unless it is equipped with such telecommunication equipment as may be specified by the Government.

p.466 (2) The Government may exempt any class of inland vessels from the conditions of this section if it appears that the nature of the voyage on which it is engaged is such that the installation of telecommunication equipment would be unreasonable or unnecessary.

55. Voyage prohibited while storm signal is in force

Except for the purpose of rendering assistance to any vessel, craft or person in distress, no inland vessel shall undertake any voyage or be used for commercial purposes when a storm warning signal is displayed or declared, or where there is an apprehension of a storm.

56. Safety measures against explosion, fire, etc.

No inland vessel shall be used for commercial purposes on any inland voyage or work until it has been equipped or prepared, in accordance with the rules, with life-saving appliances, fire-extinguishing equipment and other equipment for the prevention of fire, explosion or collision, etc.

56A. Compliance with regulations for avoiding collision, etc.

Every inland vessel, while under way, shall proceed in compliance with the prescribed regulations for avoiding collision or for steering and sailing.

57. Carriage of dangerous goods

1. No inland vessel shall carry dangerous goods except in compliance with the prescribed rules and on taking precautionary measures accordingly.

2. No person—

a. shall take with him any dangerous goods without the permission of the owner or master of any inland vessel;

or

b. shall bring or take any dangerous goods on board any vessel for carriage unless the description and nature of such goods is clearly written on the outside of the package or receptacle and without the permission of the competent authority.

p.467 (3) If the owner or master of any inland vessel suspects, or has reason to suspect, that any luggage or parcel has been taken on board or landed from the vessel which contains dangerous goods, he may, if he considers it necessary—

a. refuse to carry it on the vessel;

b. open and inspect the contents thereof, if necessary; or

c. if it has been accepted for carriage, withhold its despatch until he is satisfied as to the goods contained therein.

4. If any dangerous goods are taken or loaded on board a vessel in contravention of this section, the owner or master may, if he considers it reasonable, throw such goods, together with the package or receptacle, into the water or destroy them, and they shall not be liable in any civil or criminal court for such act.

57A. Creating obstruction in any navigable waterway prohibited

No person (a) shall create any obstruction to the plying of any inland vessel in a navigable waterway by setting fishing nets or by any other means;

b. shall in any way damage, destroy or remove any buoy, beacon or such equipment or mark placed in a navigational channel.

58. Passenger vessels not to carry goods on upper deck

If any inland vessel is used for the carriage of passengers for commercial purposes, it shall not carry—

a. any passenger on the open roof; or

b. any goods on the upper deck; or

c. passengers or goods in such a manner as to contravene the provisions relating to passenger safety and carriage; or

d. any dangerous goods; or

e. any passenger or goods in any unauthorised place.

58A. Voyage prohibited without insurance or Marine Casualty Trust Fund

No inland vessel shall undertake a voyage unless the passengers and crew are insured by an insurance company engaged in the insurance business in Bangladesh, or the owner of the ship is a member of a Marine Casualty Trust Fund constituted in the manner prescribed by the Government.

p.468 ### 59. Maximum and minimum fares for passengers and goods

The Government, or any authority empowered by the Government in this behalf, may, having regard to the distance between two places on an inland waterway and the class of the vessel, by notification in the official Gazette—

a. fix the maximum and minimum rates of fare per mile for passengers of different classes;

b. fix the maximum and minimum freight rates for the carriage of goods of any kind;

c. having regard to the distance between two places on any inland waterway, declare the places for which the maximum and minimum fares for the carriage of passengers and goods have been fixed.

60. Publication of fare list for carriage of passengers and goods

1. After obtaining direction and approval from the Government or from any authority empowered by the Government in this behalf, the owner of an inland vessel shall, for the convenience of the public, publish a list showing—

a. the time-table of the departure of the vessel from the different places;

b. the fares for the carriage of passengers of different classes to different places; and

c. the freight for the carriage of goods of different kinds to different places.

2. The owner and master of every inland vessel in respect of which a list has been published under sub-section (1) shall keep a copy thereof affixed in such a conspicuous place on the vessel that the public may read the contents of the said list while it remains in force and while the vessel remains in use.

3. Notwithstanding the provisions of sub-section (1), any authority empowered by the Government in this behalf may, if it so desires, publish, for sale to the public, a consolidated time-table and fare list in respect of any or all classes of inland vessels, and such a list shall contain the information mentioned in clauses (a), (b) and (c) of sub-section (1).

p.469 ## Chapter V-A — Provisions for Protection of Inland Waterways from Pollution

60A. Pollution of the environment of inland waterways prohibited

1. No vessel plying on or near an inland waterway, or any installation of a vessel, shall be used without registration and the prescribed sewage disposal arrangements, and the activities of an inland vessel shall not be conducted in such a manner that the environment of the inland waterway may be polluted.

2. On receipt of an application and the prescribed fee from the owner or master, all inland vessels shall be issued by the Surveyor with a Pollution Prevention Certificate renewable on an annual basis;

3. All inland vessels carrying 12 persons, including passengers, master, officers and crew, shall have arrangements for potable water and sewage approved by the Department of Shipping;

4. The discharge of oily wastes and sewage from an inland vessel into an inland river route is prohibited, except in the cases described below, when:

a. the sewage is discharged from the inland vessel after being disinfected in a manner approved by the Department of Shipping and in a separated manner;

b. an effective sewage treatment plant arrangement approved by the Department of Shipping, or a sewage holding tank of the necessary holding capacity, is in operation on the inland vessel;

c. the discharge of sewage, oil or oily substances is necessary for the purpose of saving the vessel and the lives on board;

d. oil, oily mixtures or sewage wastes are discharged as a result of damage sustained by the vessel or its equipment, even after all reasonable precautionary measures have been taken to keep the discharge of wastes to a minimum level after the damage occurred;

e. wastes are discharged with the approval of the Department of Shipping for the purpose of combating a specific pollution incident and in order to reduce the extent of damage caused by pollution.

p.470 Explanation:

a. “Pollution” means such contamination of water, soil or air, or such alteration of the natural, chemical or biological composition, as occurs when a change is brought about in the normal temperature, taste, colour, odour, density, etc., of soil, water or air, including sound within the prescribed limits, or in any other normal condition; or when the environment is polluted or faced with a threat; or when the discharge of any liquid, gaseous or solid substance, radioactive or non-degradable waste or sewage brings about such a change in or threat of pollution to the air, water, soil or environment that domestic animals, birds, wildlife, insects, fish, plants or any other kind of living creature is harmed or rendered injurious or unacceptable; so that people are deprived of the benefits thereof, and harm is caused to the public interest or public health; or so that there is deprivation of the usefulness or benefit for household, business, industrial, agricultural, recreational or any other kind of purpose—such a condition shall be understood.

b. “Sewage” means:

c. any kind of waste discharged from toilets, urinals or scuppers;

ii. waste discharged from medical facilities (dispensary, sick bay, etc.) through wash basins, wash tubs or scuppers;

iii. waste discharged from spaces where living animals are kept;

iv. any other waste mixed with the wastes mentioned above;

5. Vessel owners shall obtain a Pollution Prevention Certificate:

a. in the case of old vessels, within three years of coming into force;

b. in the case of newly built vessels, immediately after coming into force.

60B.

The Pollution Prevention Certificate mentioned in section 60A shall be carried on the vessel at all times; and it shall be open to inspection by a person nominated by the Government or the Department of Shipping.

p.471 ## Chapter VI — Penalties and Procedure

61. Penalty for contravention of sections 33, 43, 54 and 56

The owner and the master of an inland vessel may each be punished with imprisonment for a term which may extend to two years, or with fine which may extend to thirty thousand taka, or with both, if—

a. Omitted

b. any inland vessel is used on any voyage or work on an inland waterway without the certificate of survey and certificate of registry required under section 33, or an attested copy thereof;

c. any inland vessel, in contravention of section 43, undertakes any voyage without a master, engineer or engine driver;

d. any vessel, in contravention of the law under section 54, plies on any route other than the prescribed route without obtaining a valid route permit and time-table;

e. Omitted

f. any vessel is not equipped with equipment and appliances for the prevention of fire, explosion or collision, etc., in accordance with section 56.

61A. Penalty for contravention of sections 55 and 56A

The master of an inland vessel shall be punished with imprisonment for a term which may extend to three years, or with fine which may extend to thirty thousand taka, or with both, if the said inland vessel contravenes the conditions of section 55 or 56A.

62. Penalty for contravention of sections 10, 19 and 60

The owner and the master of an inland vessel shall both be punished with imprisonment for a term which may extend to three months, or with fine which may extend to five thousand taka, or with both, if—

a. for the purposes of section 10, the certificate of survey of the vessel is not kept hung up;

b. for the purposes of section 19, the registration number is not displayed on the ship;

c. the list published under section 60 is not kept hung up on the vessel in accordance with section 60(2).

p.472 ### 63. Penalty for contravention of sections 13, 23 and 24

The owner or the master of an inland vessel, or both, shall be punished with imprisonment for a term which may extend to three months, or with fine which may extend to five thousand taka, or with both, if he fails, without reasonable cause, to deliver up—

a. a certificate of survey cancelled under section 13;

b. the certificate of registry or an attested copy thereof under section 23(2), when the vessel has been missing, has been taken by the enemy, has been burnt or has been broken up; or

c. for the purposes of section 24(2), the certificate of registry or an attested copy thereof for the endorsement of transfer of ownership.

64. Penalty for transfer of ship in contravention of section 26

1. Whoever transfers or attempts to transfer any inland vessel in contravention of sub-section (1) of section 26 shall be punished with rigorous imprisonment for a term of three years, or with fine which may extend to thirty thousand taka, or with both.

2. Both the transferor and the transferee of the ownership of an inland vessel may be punished with imprisonment for a term which may extend to three months, or with fine which may extend to twenty thousand taka, or with both, if he fails to report the change of ownership under sub-section (2) of section 26.

65. Penalty for contravention of section 42

If any person fails to deliver up any suspended or cancelled certificate of competency or licence under sub-section (2) of section 42, he shall be punished with imprisonment for a term which may extend to three months, or with fine which may extend to fifteen thousand taka, or with both.

66. Penalty for navigating a vessel without certificate, etc.

Any person may be punished with imprisonment for a term which may extend to three years, or with fine which may extend to fifty thousand taka, or with both, if he—

a. employs as master, engineer or engine driver of an inland vessel any person of whom he is not satisfied as to his having a proper and valid certificate of competency or licence and as to his competency;

b. without being qualified for a proper valid certificate of competency or licence, or without having obtained a certificate of competency or licence, acts as master, engineer or engine driver on any voyage of such a vessel.

67. Penalty for carrying excess passengers

If any inland vessel, on any voyage on a commercial basis, carries on the said vessel or in any part thereof passengers in excess of the number specified in the certificate of survey, then—

p.473 (a) Omitted.

b. the owner or his representative or the master who is present at the time of the voyage or at the time of loading or unloading of goods may be fined three hundred taka for each passenger, subject to a maximum of one lakh taka.

67A. Penalty for irregular loading of goods on an inland vessel

If the owner or representative of an inland vessel is present on the vessel or at the terminal at the time of loading or unloading of goods, or the master of the vessel, or both, shall be punished with imprisonment which may extend to two years, or with fine which may extend to fifty thousand taka, or with both, if they load goods on the inland vessel in such a manner that the lives and property on board the said vessel may be endangered.

67B. Penalty for loading excess goods (solid, liquid, gaseous or semi-solid)

If the owner or representative of an inland vessel, who is present on the vessel or at the terminal at the time of the voyage or at the time of loading or unloading of goods, or the master, carries goods in excess of the load line mark mentioned in the certificate of registry of any inland vessel, the owner or master shall be punished with imprisonment which may extend to two years, or with fine which may extend to—

1. ten thousand taka for an inland vessel with a capacity of up to one hundred and fifty gross tons; and

2. fifty thousand taka in the case of an inland vessel with a capacity exceeding one hundred and fifty gross tons, or with both.

68. Penalty for contravention of sections 57 and 58

If any inland vessel carries any dangerous goods in contravention of sub-section 1 of section 57, or any passengers or goods, or dangerous goods, in contravention of section 58, then—

a. the owner of the vessel shall be punished with fine which may extend to three thousand taka; and

b. the master shall be punished with imprisonment for a term which may extend to three months, or with fine which may extend to three thousand taka, or with both.

p.474 ### 68A. Penalty for contravention of section 57A

Any person contravening section 57A may be punished with imprisonment for a term which may extend to one year, or with fine which may extend to twenty-five thousand taka, or with both, and in such case the trial Court may direct the Government to confiscate the articles, etc., creating the obstruction to navigation.

68B. Penalty for contravention of sections 54A, 54B and 58A

The owner and the master of an inland vessel shall each be punished with imprisonment which may extend to two years, or with fine which may extend to thirty thousand taka, or with both, if any of them contravenes section 54A or 54B or 58A.

69. Penalty for carrying dangerous goods, etc.

Any person shall be punished with imprisonment for a term which may extend to one year, or with fine which may extend to twenty-five thousand taka, or with both, if he, in contravention of sub-section 2 of section 57, takes with him, delivers or consents to carry any dangerous goods on any inland vessel, and the Government shall confiscate goods of this kind so carried or supplied.

70. Penalty for endangering a ship by reason of misconduct, etc.

1. Any person appointed to or on duty in any post on an inland vessel shall be punished with imprisonment for a term which may extend to five years, or with fine which may extend to ten thousand taka, or with both, if he wilfully breaches discipline or neglects his duty—

a. by doing any such act by reason of which the vessel sinks, is destroyed or any part thereof is damaged, or there is apprehension of loss of life or limb of any person on board that vessel or any other vessel, or any property is destroyed or partly damaged; or

b. by refusing or failing to perform any such duty required of him, the performance of which could have saved any vessel from sinking, destruction or damage, or any person from loss of life or limb.

2. Where, as a result of a marine casualty caused by any inland vessel, any loss of life occurs or any person is injured or any property of the vessel or of any other vessel is destroyed, and such casualty has occurred by reason of the fault of any vessel, or the incompetency or misconduct of, or the contravention of any law by, the owner, master or any officer or crew member of the inland vessel, then the owner, master or any other officer or crew

p.475 member of that vessel, or each of them, may be punished with imprisonment for a term which may extend to five years, or with fine which may extend to one lakh taka but not less than ten thousand taka, or with both.

71. Penalty for failure to report a casualty

Any person may be punished with imprisonment for a term which may extend to six months, or with fine which may extend to ten thousand taka, or with both, if he, without proper cause, fails to report any casualty to the officer-in-charge of the nearest police station under sub-section (2) of section 44.

71A. Penalty for contravention of section 60A

1. If any person contravenes section 60A, he shall be punished with imprisonment which may extend to one year, or with fine which may extend to one lakh taka, or with both, and compensation may be realised as damages for any kind of actual loss, which shall include the cost of clean-up, or of the disposal arrangements, or of taking the necessary measures subject to satisfaction.

2. The extent of the environmental damage and other related costs shall be determined by a committee constituted by the Government for this purpose.

72. Penalty for contravention of other sections

If any person wilfully contravenes any section of this Ordinance for the contravention of which no penalty is mentioned elsewhere in this Ordinance, he shall be punished with fine which may extend to a maximum of ten thousand taka.

72A. Offences committed by companies, etc.

1. If any offence under this Ordinance is committed by a company, then every person who, at the time of the commission of the offence, was in charge of or responsible for the conduct of the business of that company shall also be deemed to be guilty, like the company, and shall be sent up for trial and punished:

Provided that such person may be relieved of the punishment imposed in this sub-section if he can prove that the offence was committed without his knowledge or that he exercised all due diligence to prevent the commission of such offence.

2. Notwithstanding anything contained in sub-section (1), if any offence under this Ordinance has been committed by a company and it

p.476 is proved that the said offence was committed with the consent or connivance of, or by reason of the negligence of, any director, manager, secretary or other officer of the company, then such director, manager, secretary or other officer shall be deemed to be guilty and shall be sent up for trial and punished.

Explanation — For the purposes of this section,

a. “company” means any body corporate and a firm or association,

b. “director”, in relation to a firm, means a partner of the said firm.

73. Trial of offences

1. Unless otherwise provided in this Ordinance, no offence punishable under this Ordinance, except offences punishable under the rules made under this Ordinance, shall be tried except by a Magistrate of the first class or a Marine Court.

2. Unless the Government, by notification in the Gazette, specifies any other place in this behalf, any offence punishable under this Ordinance may, subject to the condition of sub-section (1), be tried by the local court within whose jurisdiction the offence was committed and the offender is found.

73A. Trial by a Magistrate or by the Magistrate of the Marine Court

Unless contrary to any law described in this Ordinance or any law for the time being in force, the Magistrate of the first class of the Marine Court constituted under section 47 may, as a Magistrate of the first class, also try any offence punishable under this Ordinance.

74. Realisation of fine by attachment of property

Where the owner or master of any inland vessel has been ordered to pay a fine for an offence punishable under this Ordinance, but the persons punished are not paying the amount of the fine, in that case the Court imposing the punishment may, if necessary, realise the amount of the said fine by attaching the property and selling the vessel, or by selling the necessary machinery, equipment or furniture, or whatever is required thereon.

p.477 ## Chapter VII — Miscellaneous

75. Omitted

76. Power to modify the application of the Ordinance

1. The Government may, by notification published in the official Gazette, declare that the sections of Chapters II and III, or any of those sections, shall not apply to any inland vessel mentioned in the notification, or shall apply with such modifications as are mentioned in the notification.

2. The Government may exempt any foreign ship not registered under this Ordinance from the conditions of Chapters II and III, and the Government may also specify the conditions on which it may ply on the inland waterways of Bangladesh.

77. Certificated master to be deemed a pilot under Act 15 of 1908

Every master of an inland vessel who holds a valid certificate of competency granted under this Ordinance shall, for the purposes of section 30 of the Ports Act, 1908 (Act 15 of 1908), be deemed to be the pilot of that vessel in that port.

78. Surveyors and Registrars to be public servants

Every Surveyor, Registrar and other officer discharging duties under this Ordinance shall be deemed to be a public servant within the meaning of the Penal Code (Act 45 of 1860).

79. Power to regulate the movement of goods, etc.

1. If the Government considers that the regulation of the movement of passengers and goods on any waterway and the fixing of the routes of inland vessels has become indispensable, in that case the Government, or any authority empowered in this behalf, may, fixing the conditions necessary for the carriage of passengers and goods, by written direction direct in writing the owner or master of an inland vessel to the effect that—

p.478 (a) the said vessel shall ply on the specified route or routes mentioned in the said order; and

b. the vessel shall be used for the carriage work mentioned in the said order.

S.R.O. No. 391-L/76: In exercise of the power under section 79 of the Inland Shipping Ordinance, the Government has delegated power to the Bangladesh Inland Water Transport Authority for the purpose of fulfilling the objects of this section.

2. Whoever fails to comply with the rule in sub-section 1, or to follow a requisition or obey a direction, may be punished with imprisonment for a term which may extend to two years, or with fine which may extend to ten thousand taka, or with both.

80. Facilities for loading and unloading of goods, etc.

1. The owner or master of an inland vessel—

a. shall afford reasonable facilities for the receipt, despatch and unloading of all goods which are to be carried on his vessel, and, where there is a contract for transhipment with another vessel, for the work of shipment; and

b. shall not, at the time of such receipt, despatch, unloading or shipment, discriminate between particular persons, commercial establishments or classes of goods.

2. For contravention of any condition of sub-section (1), the owner and the master of an inland vessel shall each be punished with imprisonment for a term which may extend to two years, or with fine of ten thousand taka, or with both.

81. Inspection and detention to ensure compliance with the provisions of the Ordinance

1. The Government may, for the purpose of seeing whether the sections of this Ordinance and the provisions

made under this Ordinance are being duly complied with, appoint such officers as it considers necessary, and the officers so appointed—

a. may at all reasonable times go on board any inland vessel and inspect any part thereof, any machinery, equipment, goods and passengers;

b. may inspect, or compel the production of, the certificate of registry, certificate of competency, route permit, fare list, freight list,

p.479 time-table and such other documents of the said vessel as in his judgment are necessary;

c. may examine the owner or master of the vessel, or any other person on duty there, and take a written statement from him;

d. Omitted

e. may inspect and examine whether an inland ship is carrying excess passengers or is loaded in a proper manner.

2. If any officer, after an inspection under sub-section (1) of this section or section 81A or 81B, considers that any offence has been committed, in that case he may file a complaint in writing before any court having jurisdiction to try such an offence, and the said court may take cognizance of the offence for trial on the basis of such complaint.

81A. Temporary detention, etc., of an inland vessel plying in a hazardous condition

Where, after inspecting and examining any inland vessel under sub-section 1 of section 81, any officer gives his opinion that it contravenes any section of this Ordinance or any rule made thereunder, by reason of which the voyage and the lives and property thereon are at risk, he may then temporarily detain the ship, and may issue in writing such documents as he considers necessary, and such a written direction to the effect that the ship shall remain detained until it is considered fit for safe navigation to the satisfaction of the said officer.

81B. Power to detain a vessel plying without a certificate of registry or a temporary plying permit from the Surveyor

1. Where any officer appointed under section 81 has reason to believe that any inland vessel is without a certificate of survey as required under sub-section (1) of section 18 or sub-section (1) of section 9, or without a permit as required under sub-section 3 of section 9, in that case he may seize and detain the inland vessel.

2. Notwithstanding anything contained in sub-section (1), where such officer or person has reason to believe that any inland vessel is plying without a certificate of survey as required under sub-section 3 of section 9, he may seize the inland vessel

p.480 and, instead of detaining it, may confiscate the certificate of registry and the certificates of competency of the master and driver, and shall issue an acknowledgement in this regard.

3. If the owner of any detained inland vessel cannot be found, in that case, after reasonable investigation, the vessel may be publicly auctioned in the manner prescribed by the Government, or may be taken under the control of the Government for the purpose of stopping unlawful plying.

81C. Obtaining assistance from law-enforcing agencies or other authorities

The Director General, or any person appointed by him, or any officer appointed under the provisions of this Ordinance may, while performing any function or exercising any power under this Ordinance, request any law-enforcing agency or any other Government or statutory authority to render the necessary assistance, and on receipt of such request the said agency or authority shall render assistance.

81D. Delegation of powers

The Government may, by notification published in the Gazette, delegate the provisions of this Ordinance to the Director General or any other competent authority, to be exercised in such manner as the conditions specifically mentioned in the said notification provide.

81E. Special provision regarding fines

Notwithstanding anything contained in section 32 of the Code of Criminal Procedure, 1898 (Act V of 1898), any Magistrate of the first class may impose a fine exceeding ten thousand taka under this Ordinance.

82. Power to make rules

1. The Government may, by notification in the Gazette, make rules for the purpose of carrying out the objects of this Ordinance.

2. In particular, and without prejudice to the generality of the foregoing power, such rules may provide for all of the following matters:

3. when, where and in what manner survey work shall be carried out;

4. the duties of Surveyors and, if more than one Surveyor is appointed, the duties of each Surveyor;

p.481 (3) the method of determining the position of the free-board deck of any inland vessel and of marking the position of that deck on both sides of the vessel by “free-board lines”;

4. the forms for preparing the Surveyor’s declaration, the certificate of survey and the certificate of registry, etc., and the nature of the particulars to be stated therein;

5. the rates, varying with the place of survey and registration, at which survey and registration fees shall be payable;

6. the cases in which, and the extent to which, survey or registration may be dispensed with;

(6A) the classification of masters and drivers;

(6AA) the syllabus and examination curriculum and tests for certificates of competency;

7. the time, place and manner of holding examinations for persons desiring to obtain certificates of competency, and the fees for the posts for all such examinations;

8. the qualifications required of persons desiring to obtain certificates of competency;

9. the regulation of the grant of certificates of service and the fees payable therefor;

10. the regulation of the grant of licences under section 38 and the fees payable therefor;

11. the forms in which, and the authorities by which, the aforesaid certificates and licences shall be granted, and the manner in which copies thereof shall be preserved;

12. the manner in which, and the conditions subject to which, dangerous goods may be carried on any inland vessel;

13. the steps and precautions to be taken for the prevention of explosion and fire on an inland vessel;

14. the fire-extinguishing appliances and other equipment to be kept on an inland vessel;

15. the carrying and exhibition of lights, and the making of sound and visual signals, on inland vessels;

16. the carrying and exhibition of lights or other signals by any inland vessel or craft plying on an inland waterway;

p.482 (17) the regulations for the prevention of collision, steering and sailing to be observed by inland vessels;

18. the method of towing (Towing) or pushing (Pushing) of any vessel by any inland vessel;

19. the speed limit of any inland vessel plying in specified places;

20. all the equipment, machinery and other articles which it is necessary to keep and carry for the purpose of the safety of the vessel, life and property;

21. the regulation of navigation with a view to preventing a threat to the safety of any other vessel and small craft, or to avoiding damage to the signals and marks placed for the convenience of navigation, or to the banks of rivers, navigable canals or the banks of rivers or any property of navigable creeks;

22. the regulation of the carriage of passengers and matters to be observed by passengers on any inland vessel;

23. the causes and circumstances in which the refusal of admission of passengers to, and the removal of passengers from, any inland vessel is necessary;

24. the arrangements for the supply of drinking water free of charge for the use of passengers;

25. the provision of pollution prevention and of other necessary amenities, including urinals and latrines, with a view to ensuring amenities and cleanliness for the passengers of inland waterways and vessels;

26. the arrangements for the accommodation of separate spaces for women and children;

27. the arrangements for the issue of tickets and the display of receipts for the fares paid for the carriage of passengers and goods;

28. the arrangements for the printing of, or otherwise fixing the price on, tickets issued to passengers;

29. the removal or arrest of any person contravening the law or committing contempt on an inland vessel;

30. the conditions and rules relating to the appointment of Assessors (ASSESSOR);

31. the preparation of complaints, the production of certificates or other documents, and the inspection and detention of inland vessels under section 81; and

p.483 (32) such other matters, for the purpose of carrying out the objects of this Ordinance, as are considered necessary or expedient.

3. The rules made under this Ordinance may provide that any person contravening or disobeying such rules may be punished with imprisonment for a term which may extend to a maximum of six months and with fine of five thousand taka, or with both.

4. For the purpose of arresting any person under the rules made under this Ordinance, the procedure for arrest by private persons under section 59 of the Code of Criminal Procedure, 1898 (Act V of 1898) shall apply so far as possible.

83. Repeal and savings

1. The Inland Mechanically Propelled Vessels Act, 1917 (Act No. 1 of 1917), hereinafter referred to as the said Act, is hereby repealed.

2. Without prejudice to section 24 of the General Clauses Act, 1897 (Act No. 10 of 1897), and unless the Government otherwise directs—

a. any rule, notification, order or exemption prepared, issued or granted under the Act in force immediately before the coming into force of this Ordinance shall be deemed to have been made or issued under the corresponding sections of this Ordinance;

b. any officer or authority appointed or empowered under the said Act shall be deemed to have been appointed or empowered under this Ordinance;

c. any certificate, licence or other document issued under the said Act before the coming into force of this Ordinance shall be deemed to have been granted or made in accordance with the provisions of this Ordinance.

p.484 # Charter of Duties of Supervisors

Old inland ships/vessels

1. At the very commencement of the work, the Supervisor shall examine all technical information and documents of the inland ship/vessel, and shall inspect the ship/vessel at the specified place within the specified time. The Supervisor shall make a recommendation together with a precise description of the stability of the inland ship/vessel and shall bear all responsibility therefor. To this end the Supervisor shall execute the following documents.

1. He shall check the correctness and suitability of the lines plan, and if he finds any difference from the shape of the hull, he shall note it and correct the lines plan. If the difference is very great, the Supervisor shall take point-to-point measurements of the hull and prepare the plan afresh, and shall draw the as-built general arrangement and midship section drawings.

2. Using the correct lines plan, he shall prepare a stability booklet, in which the following information shall be included for at least 3 waterlines, including the load waterline:

a. Area of the waterplane and its centroid.

b. Volume of displaced water and its centroid.

c. Transverse moment of inertia and the centre of the floating surface, and

d. Position of the centre of buoyancy.

3. He shall determine the position of the centre of gravity by conducting an inclining test. The inclining test shall be carried out in the presence of the Government Inland Vessel Surveyor, which the Supervisor must ensure. The results of the test shall include the following information:

a. Date and place of the test;

b. Name and address of the owner;

c. Depth of water and position of the mooring lines with reference to the prevailing surrounding weather.

d. Forward, aft and mean draught and volume of displaced water.

e. Weights used in the inclining test.

f. Position of the weights used, length of the pendulum and deflection, to be shown on a general GA plan.

g. Light weight of the inland ship/vessel and the position of the corresponding centre of gravity.

p.485 (4) The position of the centre of gravity shall be determined in different loading conditions, e.g. 50%, 100% and 150% load. In the case of passenger-carrying inland ships/vessels, the realistic positions of the loads must be worked out and shown on a GA plan.

5. GZ curves shall be prepared for the different loading conditions, and the corresponding KB, KG, KM, BM and GM shall be specifically stated.

6. He shall determine the recommended load, the corresponding draught, free-board and loading plan, in respect of which the Supervisor shall bear all responsibility relating to the stability of the inland ship/vessel.

7. He shall draw the draught and load line marks on the hull in the MKS system. The load line mark shall be simple but serviceable, like the example below.

[Figure: Load line mark. A rectangular Deck Line of 300 mm length is placed above; the vertical distance from the Deck Line down to the Minimum Freeboard line is marked “Required Free Board”. A ring (annulus) of 25 mm band width is centred on the Minimum Freeboard line, which is a horizontal bar 450 mm long and 25 mm high passing through the centre of the ring. A sectional sketch at the right shows the Deck Line at the deck edge on the side shell.]

The marks shall be of white or yellow colour on a dark background, or of black colour on a light background.

8. a. He shall prepare a report and send it to the Director General of the Department of Shipping for approval, and shall give a signed copy to the vessel owner. The following enclosures must accompany the report.

p.486 (1) General arrangement plan and midship section according to the as-built ship;

2. Correct lines plan of the hull;

3. Inclining test data;

4. Stability booklet;

5. Detailed load line mark; and

6. Loading plan.

b. The seal and signature of the Supervisor must be on every drawing and on every page of the report.

New inland ships/vessels

2. a. In the case of a new inland ship/vessel, the Supervisor may proceed in two stages:

1. Stage 1: designing the ship.

2. Stage 2: supervising during the construction of the ship.

b. In the case of a particular ship/vessel, it is desirable that the same Supervisor be involved in both of the above stages.

3. In the case of Stage 1:

1. The Supervisor shall perform the following functions.

a. He shall prepare the design of the inland ship/vessel in accordance with the Inland Steel Ships Construction Rules, 2001, and shall submit it for the approval of the Inland Shipping Safety Administration (ISA) of the Department of Shipping relating thereto.

b. He shall prepare the stability booklet and a detailed description of all equipment and machinery; and

c. He shall prepare all structural and workshop drawings.

2. He shall send one set of all the above information to the Director General of the Department of Shipping for approval and preservation. He shall submit a further copy to the Registrar of Inland Vessels for preservation. The Registrar shall use the information of this set as a reference in future for the supervision of the construction of the vessel, the inclining test and the examination of the final stability booklet.

p.487 (3) The Supervisor in charge during Stage 1 shall remain fully responsible for the expected stability and strength of the inland vessel after its construction.

4. While preparing the design of the inland ship/vessel, the Supervisor shall comply with the Inland Steel Ships Construction Rules, 2001 of Bangladesh, but he may take the assistance of any rules for the purpose of

making a safe design.

4. In the case of Stage 2:

1. The Supervisor shall perform the following functions.

a. He shall examine all information and drawings received and, if he finds any deficiency, shall bring it to the notice of the owner and the Director General of the Department of Shipping. Such an observation shall not relieve the Supervisor in charge of Stage 1 of his responsibility.

b. He shall carry out the lofting check of the inland ship/vessel and prepare the necessary correct lines plan.

c. He shall supervise all stages of construction until the final stability booklet is prepared after carrying out the inclining test.

d. He shall supply the results obtained and the recommendations to the owner, the building shipyard/dockyard and the Director General of the Department of Shipping at the following stages during construction.

1. After completion of lofting.

2. After laying of the keel.

3. After erection of the frames.

4. After construction of the hull.

5. After installation of the engine, propulsion system and rudder.

6. After launching.

7. After construction of the superstructure.

8. After installation of the system fittings.

e. He shall carry out the inclining test in the presence of the Government Inland Vessel Surveyor and prepare the final stability booklet.

f. He shall draw the draught and load line marks on the ship/vessel in accordance with paragraph 1(7).

2. For the purpose of observing and checking each stage during the construction of the vessel in accordance with paragraph (d) above, the Supervisor must ensure the presence of the Government Inland Vessel Surveyor at least

p.488 once.

5. The MKS system shall be used at all times.

6. “Code of Conduct” for Supervisory Panel members. (Rules of conduct)

A Supervisory Panel member:

1. shall work in the light of honesty towards the ship owner, with fidelity towards the client, loyalty to the country, and strengthened by personal dignity and high ideals.

2. shall, by professional competence, uphold their supremacy by providing for the highest safety, environment, health and welfare of the public.

3. shall refrain from involving themselves, or allowing their name and signature to be used, merely in exchange for money.

4. shall refrain from using misleading information in advertisements.

5. shall inform their client or employer of any kind of business connection, interest or relationship which influences their judgment or impairs the quality of the work.

6. shall refrain from using any kind of improper or questionable method, and from taking work in exchange for bribes or commission.

7. may, with the full knowledge and consent of all interested parties, accept cash or any other form of remuneration for a particular job from only one source.

8. shall build their business reputation by competence or work and shall not compete unfairly with others.

9. shall co-operate in increasing the efficiency of the work by exchanging general information and experience with colleagues and other Supervisory Panel members, and by contributing to journals, conferences and learned professional societies.

10. shall, during the period of membership of the Supervisory Panel, in addition to increasing their own professional competence, provide opportunities for increasing the competence of other professionals and of the professionals subordinate to them.

p.489 7. Accountability and penalty

If, for any kind of casualty or accident, it is found that there has been any deficiency in the work of a Supervisory Panel member, or if it is found that a Supervisory Panel member has done anything which has breached the “Code of Conduct” (rules of conduct), then he shall be immediately and permanently expelled from the Supervisory Panel. In addition, legal action shall be taken against him in the light of the existing law in accordance with the recommendation of the following committee.

The Director General of the Department of Shipping shall constitute a committee including at least 3 members together with a convener. In the committee, 1 or more members shall be from the Department of Shipping, and at least 1 member (not below Associate Professor) shall be from the Department of Naval Architecture and Marine Engineering of the Bangladesh University of Engineering and Technology or from among the surveyors of any classification society. Any person in Bangladesh possessing the necessary qualifications and extensive experience for the determination of errors and deficiencies may also be included in this committee. No one from among the Supervisory Panel of the ship concerned or those involved in the registration process may be a member of this committee. The Supervisory Panel member of the ship concerned shall certainly have the right of self-defence before the above committee.

8. The Supervisor shall follow the following rules and advise the ship owner accordingly.

1. The Inland Ships (Passenger) Rules, 2001
2. The Inland Ships (Manning, Examination and Certification) Rules, 2001
3. The Inland Ships (Minimum Safe Manning) Rules, 2001
4. The Inland Ships (Fire Safety) Rules, 2001
5. The Inland Ships (Dangerous Goods) Rules, 2001
6. The Inland Ships (Stability) Rules, 2001
7. The Inland Ships (Free-board) Rules, 2001

p.490 (8) The Inland Ships (Navigation) Rules, 2001

9. The Inland Ships (Life Saving) Rules, 2001
10. The Inland Ships (Tonnage Measurement) Rules, 2001
11. The Inland Ships Steel Ships Construction Rules, 2001 and
12. Notifications issued from time to time by the Department of Shipping regarding ship construction, space measurement and other relevant requirements.

9. Supervisor’s fee:

Old inland ships/vessels

	Fee
(1) Passenger ship/vessel, maximum length up to 19.51 metres (64 feet)	Taka 7,500/=
(2) Passenger ship/vessel, maximum length up to 30.48 metres (100 feet)	Taka 10,000/=
(3) Passenger ship/vessel, maximum length more than 30.48 metres (100 feet)	Taka 10,000/=
(4) Cargo ship/vessel, maximum length up to 30.48 metres (100 feet)	Taka 7,000/=
(5) Cargo ship/vessel, maximum length more than 30.48 metres (100 feet)	Taka 10,000/=

New inland ships/vessels

Stage 1:

	Fee
(1) Passenger ship/vessel, maximum length up to 19.51 metres (64 feet)	Taka 12,000/=
(2) Passenger ship/vessel, maximum length up to 30.48 metres (100 feet)	Taka 18,000/=

(3) Passenger ship/vessel, maximum length more than 30.48 metres (100 feet)	Taka 25,000/=
(4) Cargo ship/vessel, maximum length up to 30.48 metres (100 feet)	Taka 8,000/=
(5) Cargo ship/vessel, maximum length more than 30.48 metres (100 feet)	Taka 15,000/=

p.491 Stage 2:

	Fee
(1) Passenger ship/vessel, maximum length up to 19.51 metres (64 feet)	Taka 15,000/=
(2) Passenger ship/vessel, maximum length up to 30.48 metres (100 feet)	Taka 20,000/=
(3) Passenger ship/vessel, maximum length more than 30.48 metres (100 feet)	Taka 30,000/=
(4) Cargo ship/vessel, maximum length up to 30.48 metres (100 feet)	Taka 10,000/=
(5) Cargo ship/vessel, maximum length more than 30.48 metres (100 feet)	Taka 20,000/=

The above fees may be re-fixed at any time in the light of discussion between the ship owners and the Panel members.

10. At the commencement of the work, the Supervisor shall obtain an entry number for the work concerned from the office of the Chief Engineer/Chief Surveyor (ISA). No Panel member may supervise more than 12 ships in a year. In the case of any member, if this quota of 12 inland ships is filled up within one year, written permission shall have to be obtained from the Department of Shipping for the supervision of additional ships.

11. The list of Supervisory Panel members shall be rearranged/reconstituted every two years, or at any time if necessary.

12. The Director General of the Department of Shipping reserves the right to drop any member from, or to include any new member in, the Supervisory Panel list in the light of the work performed.

(Signed) (Golam Rabbani) Commodore Director General Department of Shipping

p.492 # Storm Warning Signals for Sea Ports

Plant more trees — storms and tempests will lessen. To save life and property, obey the weather signals.

Signal number	**Meaning of the signals**
No. 1 Distant Cautionary Signal — Day: one black cylinder (rectangle) hoisted; Night: two white lights, one above the other	After leaving port, ships may encounter disturbed weather. There is a region of squally weather in a distant area where the wind speed is 61 km per hour, which may turn into a cyclonic storm.
No. 2 Distant Warning Signal — Day: one black cylinder (vertical); Night: two red lights, one above the other	A storm has formed in the distant deep sea. There the sustained wind speed is 62–88 km per hour. The port is not yet affected by the storm, but ships leaving port may fall into danger on the way.
No. 3 Local Cautionary Signal — Day: one black inverted cone (apex down); Night: white light above, red light below	The port and the ships anchored in the port are apprehended to be affected by disturbed weather. Squally wind may blow over the port, and the sustained speed of the gusty wind may be 40–50 km per hour.
No. 4 Warning Signal — Day: one black cone (apex up); Night: red light above, white light below	The port is threatened by a cyclonic storm. The probable wind speed is 51–61 km per hour, but the time has not yet come to be so dangerous as to require final preparations for the cyclone.
No. 5 Danger Signal — Day: one inverted cone (apex down) above one black diamond (square); Night: white, white, red lights from top to bottom	The port is in the grip of a storm of small or moderate intensity accompanied by squalls. The maximum sustained wind speed in the storm may be 62–88 km per hour. The storm may cross the coast keeping the port to its left.
No. 6 Danger Signal — Day: one cone (apex up) above one black diamond (square); Night: red, white, white lights from top to bottom	The port is in the grip of a storm of small or moderate intensity accompanied by squalls. The maximum sustained wind speed in the storm may be 62–88 km per hour. The storm may cross the coast keeping the port to its right.

p.493 | Signal number | Meaning of the signals | |—|—| | **No. 7 Danger Signal** — Day: two cones apex to apex (hourglass) above one black diamond (square); Night: white, red, white lights from top to bottom | The port is in the grip of a storm of small or moderate intensity accompanied by squalls. The maximum sustained wind speed in the storm may be 62–88 km per hour. The storm may cross the coast over or near the port. | | **No. 8 Great Danger Signal** — Day: one inverted cone (apex down) above one black square; Night: white, red, red lights from top to bottom | The port may fall in the grip of a severe cyclonic storm of violent or maximum intensity accompanied by squalls. The maximum sustained wind speed in the storm may be 89 km per hour or more. The violent storm will

cross the coast keeping the port to its left. | | **No. 9 Great Danger Signal** — Day: one cone (apex up) above one black square; Night: red, red, white lights from top to bottom | The port is in the grip of a severe cyclonic storm of violent or maximum intensity accompanied by squalls. The maximum sustained wind speed in the storm may be 89 km per hour or more. The violent storm will cross the coast keeping the port to its right. | | **No. 10 Great Danger Signal** — Day: two cones apex to apex (hourglass) above one black square; Night: red, white, red lights from top to bottom | The port is in the grip of a severe cyclonic storm of violent or maximum intensity accompanied by squalls. The maximum sustained wind speed in the storm may be 89 km per hour or more. The violent storm will cross the coast over or near the port. | | **No. 11 Communication Failure Signal** — Day: two cones apex to apex (hourglass); Night: one red light | All communication with the weather warning centre has broken down, and the local officers consider the weather to be extremely disturbed. |

○ — White light

● — Red light

p.494 # Warning Signals for the Inland River Ports of Bangladesh

Signal number	Meaning of the signals
No. 1 Inland Cautionary Signal — Day: one black diamond; Night: white light above, red light below	There is a possibility of the port area being affected by transient squally weather. This signal is also displayed in the case of a nor'wester (kalbaishakhi) with a maximum wind speed of 60 km per hour. This signal also urges keeping a careful watch on the current state of the weather.
No. 2 Inland Warning Signal — Day: one black ball; Night: one red light	The port area may be affected by a storm of intensity equivalent to a depression, with a wind speed not exceeding 61 km per hour, or by a nor'wester (kalbaishakhi) storm with a wind speed of 61 km per hour or more. There is a possibility of vessels being caught by either of these. Vessels of 65 feet in length or less shall quickly proceed to safe shelter.
No. 3 Inland Danger Signal — Day: one black cone (apex up); Night: two red lights, one above the other	The port area is affected by a storm. A cyclonic storm with a maximum sustained wind speed of 62–88 km per hour may strike the port area shortly. All kinds of vessels shall take safe shelter immediately.
No. 4 Inland Great Danger Signal — Day: one black cylinder (rectangle); Night: three red lights, one above the other	The port area is affected by a cyclonic storm of violent or maximum intensity, and it will strike the port area shortly. The maximum sustained wind speed in the storm is 89 km per hour or more. All kinds of vessels shall remain in safe shelter.

Published by the Meteorological Officers' Council of Bangladesh Meteorological Complex, Agargaon, Dhaka Phone: 9111068, 9118574, 813071, 9114388

Bangladesh Meteorological Department in the service of the public

Notice regarding proper photographs for the issue of certificates

Government of the People’s Republic of Bangladesh Department of Shipping 141–143 Motijheel C/A (8th floor) Dhaka-1000

No. DGSL-18/Certificate Misc./05/2351 — Date: 13/03/2007

It is hereby notified for the information of all concerned that, for the purpose of obtaining a certificate from this Department, the applicant shall submit photographs of the following description for all kinds of certificates (where applicable):

The photographs must—

- be recent (taken within 6 months) and in colour;
- be taken from the front, with both ears equally visible, and the face, with the head and shoulders, shown in full;
- have a white/light background;
- be taken wearing clothing of a light colour.

The measurements of the photographs must be—

- Stamp size: width 24 mm, length 28 mm.
- Passport size: width 35 mm, length 45 mm.
- From the chin to the top of the head: 25–35 mm (for passport-size photographs).

The following photographs are not acceptable—

[Photograph, crossed out] Not acceptable because the background of the photograph is of a dark colour.

[Photograph, crossed out] Not acceptable because the photograph is computer-printed.

[Photograph, crossed out] Not acceptable because the photograph is taken wearing a cap.

[Photograph, crossed out] Not taken from the front; not acceptable because one ear is not visible.

[Photograph, crossed out] Not acceptable because it is black-and-white [unclear].

2. On the other hand, passport-size and stamp-size photographs of the following specimens shall be acceptable —

[Photograph: Passport size] Acceptable because it is taken from the front, with a light-coloured background and the face shown in full. Measurements: width 35 mm, length 45 mm.

[Photograph: Stamp size] Acceptable because it is taken from the front, with a light-coloured background and the face shown in full. Measurements: width 24 mm, length 28 mm.

Any photograph other than the above specimens shall in no case be acceptable.

(Signed) (Captain A. K. M. Shafiqullah) Director General (Current Charge)

Appendix: Translator’s notes on doubtful readings

Items the translation could not resolve with certainty from the scan, by section.

Construction Rules (pp. 2–31)

- p.6: the page-number column is faint on the scan; values 52–60 and 61–64 were read after image enhancement and are believed correct but a few (e.g. 54 for sections 89–92, 62 for sections 103–108) could be off by one.
- p.24 (rule 11(6)): the two lines “t + 2 মিলিমিটার এবং / t + 2 মিলিমিটার অন্যান্য ক্ষেত্রে” are printed identically in the original; the first probably should carry a different qualifier (e.g. “inside the angle bar frame”) but the printed text was reproduced as is.
- p.30 (rule 30(6)): formula printed as “১০০ (N−১) (N+১) শতকরা”; reproduced literally as 100 (N−1) (N+1) per cent (the intended division/sign is not shown in the print).

Construction Rules (pp. 32–61)

- p.39, rule 53(3): “১ মিলিমিটার” (1 millimetre) as the maximum spacing between cleats — printed as such, but probably a misprint (likely 1 metre); translated as printed and marked [sic].
- p.41, rule 60(2): formula printed as “O.lx lm²” — read as 0.1 × l m² (l = continuous bulwark length in metres).
- p.47, rule 79(1): “তফসিলের সারণি-৪৫” — table number read as 45 (print slightly faint).
- p.55, rule 93(1): sentence is garbled in the original (“1.1 গুণ ছেদক ক্ষেত্র … ১.৩৬ গুণের সমপরিমাণ”) — translated literally; exact intended meaning uncertain.
- p.55, rule 94(1): “স্টিয়ার আপস বা স্প্যাংসমূহ (Spanges)” — rendered as “steer-ups or spangles (Spanges)”; the printed English term is unclear.
- p.58, rule 98(6): “স্টিচ ওয়েবসমূহ” — rendered as “stitch webs” (probably “stitch welds”).
- p.61, rule 101(1), heading of Chapter 3: the chapter has no printed title on this page beyond “অধ্যায়-৩”.

Construction Rules (pp. 62–91)

- p.76, rule 142(e)(1): the second bilge-pump capacity formula is printed “Q = ৫.৭৫/০০০০d² (৩০ মিটার পর্যন্ত দীর্ঘ নৌযানের জন্য)” — the divisor is missing a leading digit (apparently 10000, as in the first formula) and the length condition is printed identically to the first formula (“up to 30 metres”), where “over 30 metres” is evidently intended. Translated as printed with the divisor read as 10000.
- p.85, rule 166(2) and p.91, rule 186(1): the word printed as “পানিবাহী” (“water-carrying”) in both places appears to be a misprint for “পরিবাহী” (conductor); translated as “conductor”.
- p.89, rule 179(1): the light type printed as “UNC” (in Latin letters) could not be identified; reproduced verbatim.
- p.89, rule 177(3): the phrase “নিবিড়ভাবে প্রস্ফুটিত (ইনট্রিনসিক্যালি সেইফ)” is obscure; translated as “closely spaced (intrinsically safe)” following the bracketed English.
- p.77: the printed page carries handwritten marginal notes in English (“filling pipe & overflow pipe dia = as per rule = air vent pipe dia × 0.8”; “air vent pipe dia = air vent pipe dia × 0.8” / “overflow pipe dia = air vent pipe dia”), also at the top of pp.78–79; these are reader annotations, not part of the Rules, and were not translated.

Construction Rules (pp. 92–121)

- p.92: “বিনুনীকৃত ধাতব আর্মর” — read as “braided metal armour”; the word is slightly indistinct in the scan.
- p.93 sub-rule (8): “০.১৫ ডিগ্রি সেলসিয়াস” for silicone rubber — figure reproduced as printed (0.15); the printed value appears to be a misprint (likely 85 or similar) but cannot be verified from the scan.
- p.95 sub-rule (1) of rule 193: “৫০ কিলোওয়াট হার নির্ধারিত” — read as “rated at 50 kilowatts”; the qualifier (e.g. “below”/“not exceeding”) is not printed.
- p.102 rule 204(3): “অনুপাত অনধিক ৬৫” — reproduced as printed (65); may be a misprint for 6.5 [or 15].
- p.106 rule 210(1): “খোলা প্লেটিং” — read as “open plating” (possibly a misprint of “খোলের প্লেটিং”, hull plating).
- p.106 rule 212: “০.৫ জুড়িয়া” — read as 0.5 L (the “L” is not printed).
- p.109 rule 217: two consecutive sub-rules numbered (2) as printed.
- p.111 rule 223(4): “অতিক্রম করিতে পারিবে” — “may exceed” as printed (likely intended “shall not exceed”).
- p.113 rule 228(2)(d): “হসার প্লেটিং” — the word before “প্লেটিং” is unclear; left as [unclear].
- p.118 rule 245(4): “(closur)” — printed English term reproduced as printed (presumably “closure”).
- p.120 rule 250(2)(q): “নোঙরকরণ ও বুকিং সরঞ্জাম” — “বুকিং” rendered as “[mooring]” (likely a misprint of “মুরিং”).
- p.121 rule 252(1)(f): “গর্তসমূহ” (holes) — rendered “holes [bearings]”; the intended term is unclear.

Construction Rules (pp. 122–151)

- p.140, Table 5, longitudinal framing column, L = 20: printed “৮.৫” (8.5), which is out of sequence with the following values (5.5, 5.5, 7.0 …); reproduced as printed.
- p.140, Table 6: values printed as “৮.০/৮.৫” for L = 30, 40 and then “৫.০/৫.৫/৬.০” for L = 50–70; reproduced as printed (printing may be inconsistent).
- p.142, Table 11: column headings printed “প্রস্থ” (breadth) then “পুরুত্ব” (thickness); a handwritten annotation

swaps them. Reproduced as printed.
- p.143, Table 13, draught 1, frame spacing 500, B = 4: “২৫” — faint, possibly “৩৫”.
- p.143, Table 13: the third draught block is printed “২.৫” (2.5) although the sequence (1, 1.5, ?, 3, 3.5, 4) suggests 2; reproduced as printed.
- p.143, Table 13, draught 3.5, frame spacing 550, B = 4: “১১৫” — possibly “১৩৫”.
- p.143, Table 13, draught 4, frame spacing 500, B = 12: “১১২৯” — possibly “১২২৯”.
- p.145, Table 18: unit for web framing spacing printed “মিলিমিটারে” (millimetres) although values 3.0–5.0 appear to be metres.
- p.147, Table 22, P (kN) header row: printed “১০ ২০ ৩০ ৪০ ৬০ ৮০ ১০ ১২ ১৫ ১৮ ২১ ১৪ ২৮ ৩২”; the later entries are presumably 100, 120, 150, 180, 210, 240, 280, 320 (the “১৪” is probably a misprint for 24).
- p.148, Table 23, diameter 76.1 / thickness 2.9, first column: “৩৭” — probably “৯৭”.
- p.148, Table 23, diameter 88.9 / thickness 3.2, third column: “৯৫” — faint.
- p.149, Table 24: both value columns bear the same printed heading; column (2) and (3) are presumably minimum and maximum distances.
- p.149, Table 25, title: “সংঘর্ষ এবং পশ্চাৎ শীর্ষ পানিরোধী দেয়ালসমূহের পুরুত্ব” — translated as collision and after peak bulkheads.

Construction Rules (pp. 152–180)

- p.152, Table 29: the fourth frame-spacing column header is printed “২.৪” but partly obscured by a handwritten annotation; read as 2.4.
- p.153, Table 33: row b=4, column I=7 printed “২৫০” (250), which breaks the increasing sequence (likely 350); reproduced as printed.
- p.154, Table 34 note: both I and S are printed as “অনালম্বিত বিস্তার” (unsupported span); S is probably spacing; reproduced as printed with [sic].
- p.154, handwritten marginal notes (not part of the printed text) ignored.
- p.156, Table 37 header reads “ডেক গার্ডারসমূহের পারস্পরিক ব্যবধান, মিলিমিটারে” (millimetres) though the values 2.0–3.2 appear to be metres; reproduced as printed.
- p.159, Table 44: third column header printed as “dt” and in the note as “dI”; read as d1 (nominal chain diameter). Row 70 SWL printed “৬.৩০” (6.30); read as 630 in line with the next row.
- p.159, Table 44 note: “নামমাত্র () ব্যাস” — the bracket is empty in print.
- p.161, Table 46 grade headers “গেড (ক১)/(ক২)” rendered as Grade (a1)/(a2).
- p.165, Table 52: row 64 angle “৮০০×৬৫×১০” (800×65×10) and row 66 bulb “১৪০×৬৮” (140×68), row 60 bulb “১২০×৮০” (120×80) appear to be misprints; reproduced as printed.
- p.166, Table 52: modulus “১৩০” appears between 96 and 105 (probably 100); “২১×৭.৫”, “২২×৫০×৬.৫”, “১৩০×৭৫×২২”, “২০০×১.৫”, “১৮০×১১৮”, “২৩০×১৮” appear to be misprints; reproduced as printed.
- p.167, Table 52: row 230 bracket “১৬০×৬” (160×6) probably 260×6; row 370 bulb and flat bar both “২৪০×১১”; row 320 flat bar “২২০×৯”; reproduced as printed.
- p.168, Table 52: row 420 “৩২০×৫৫×৮৫”; row 580 “৩৬×৬০×৯.৫”; row 520 values placed in bulb column in print; reproduced as printed.
- p.169, Table 52: row 780 “৪০×৭০×১০.৫”; row 860 “৪১১×১৩”; rows 1080 and 1120 “৪৪০×৮০৮×১১” / “৪৫০×৮০৮×১১.৫”; reproduced as printed.
- p.170, Table 53: row 180 col 4 “৭.১” (7.1), row 400 col 5 “৩৭৭” (377), row 600 col 4 “১১.১” (11.1) appear to be misprints; reproduced as printed.
- p.171, Table 53: row 1450 col 2 “৪৬৫” (465, probably 565); row 1600 col 2 “৫৭৫” repeated; row printed “২০০” (should be 2000); reproduced as printed.
- p.176, Table 61: row 2.5, 3–4 core “২৯” (29, probably 19); row 70, 2-core “১৬৬” (166); reproduced as printed.
- p.177, Table 63: row 16, 2-core “৯১” (91, probably 81); reproduced as printed.
- p.178, Table 64: Rubber/PVC at 55° C shows only a dash/dots in print; rendered as “–”.
- p.179, Table 66: item “পার্শ্ব খোলা অনুদৈর্ঘ্যসমূহ” translated literally as “side open longitudinals” (possibly “side shell longitudinals”).
- p.179, Tables 66–67: unit printed as KN/mm² (probably N/mm² intended); reproduced as printed.

Passenger / Free-board Rules (pp. 181–205)

- p.199, rule 2(b) and 2(e): the printed definitions appear to be transposed/garbled in the original (“অভ্যন্তরীণ জলপথ” defined as “inland ship” under section 2(e), and “জাহাজ বা অভ্যন্তরীণ জলপথ” defined as “inland water” under section 2(e)); translated as printed and marked [sic].
- p.203, rule 13(2): “International Load Line Convention, 1996” printed thus (the Convention is 1966); reproduced as printed.
- p.204, rule 13(2)(d): formula for 55 ≤ L ≤ 60 m printed as “F = 8 × (L−24) + 300”, apparently a misprint (likely intended 10 ×); reproduced as printed.
- p.204, rule 17(1): the text says “মালবাহী জাহাজসমূহে” (cargo ships) within the tanker rule; translated as printed and marked [sic].
- p.205, rule 18(1): printed “বর্ষব্যাপী অশান্ত জলসীমায় ও মৌসুমী শান্ত জলসীমায় শান্ত ঋতুতে” (year-round rough / seasonal calm), apparently inverted compared with rules 16–17; reproduced as printed.

Stability / Life Saving Rules (pp. 206–238)

- p.216, formula for k where cotθ < b/h: the printed formula reads `k = (Cosθ/8){1+tanθ/(b/h)}−`

(cosθ/{12(b/h)²} (1+cot²θ/2); bracketing is ambiguous in the print and has been reproduced as printed.
- p.216, value of coefficient k for b/h = 10 at 5°: printed as ০.০১ (0.01); this appears to be a print error for 0.06 (IMO table) but has been reproduced as printed.
- p.219, item (d): “P = ০.০৩২২ (টন/মিটার)” reproduced as printed (tonne/metre; the usual unit is t/m²).
- p.224, sub-rule (4): the English in brackets is printed as “hydrostatic data corresponding to the inclining letter” (evidently “condition”); reproduced verbatim with [sic]. Also “জাহাজের পশ্চাদভাগ (stem)” — the bracketed term for the after part is printed as “stem” (presumably “stern”).
- p.234, item (h): “৫ কিলো-নিউটন (KN)” — printed as a weight (“ওজনবিশিষ্ট”) of 5 kilo-newtons; reproduced as printed with [sic].
- p.236, rule 11(1): “৩টি অনুমোদিত প্যারাশুট” — the word “rocket” is not printed after “parachute”; supplied in square brackets.

Fire Safety / Examination Syllabus Rules (pp. 239–271)

- p.249, rule 13(1)(c)(iv): the original says “two Class BE fire extinguishers of 9 litres capacity and, in addition thereto, in the machinery space two further Class BE fire extinguishers of 6 litres capacity” — translated as printed (the location of the first two extinguishers is not stated in the original).
- p.251, rule 17(1)(e): the original reads “two Class ABE fire extinguishers … and if there is electrical equipment there, the fire extinguishers shall be of Class ABE” (apparent printing error, ABE repeated; reproduced as printed).
- p.254, rule 18(3)(i): the list contains both an item lettered (i) and sub-items (i)–(iv) (original uses ঝ and অ/আ/ই/ঈ); the fourth sub-item (ঈ) is printed as a sub-item in the original and is reproduced as such.
- p.257, rule 20(e): “two Class BE fire extinguishers and two fire extinguishers of 9 litres capacity” — the class of the second pair is not printed in the original.
- p.269, rule 6(d): “নোঙরের কপিকল” translated as “anchor windlass”; “পরিবারের কারেটজনিত ক্ষেত্রে” appears to be a misprint of “পরিবাহীর কারেন্টজনিত ক্ষেত্র” and was translated as “fields produced by current in conductors”.
- p.270, rule 7(a): “বিনিময় ক্রয়” (literally “exchange purchase”) translated as “exchange of units” — exact intended term uncertain.
- p.270, rule 7(c): “বিনিময় সীনক্রোনাইজিং (Synchroniging)” — English spelling reproduced as printed; “বিনিময়” rendered as “exchange”.

Manning / Minimum Safe Manning Rules (pp. 272–304)

- p.297, rule 2(1)(b): the printed text defines “অভ্যন্তরীণ জলপথ” (inland water) as “inland ship” as defined in section 2(f) — apparently a misprint for “inland water”; reproduced as printed with [sic].
- p.301, Appendix B: some cells in the Master columns are blank in the original while others show “–”; blanks and dashes reproduced as printed (both meaning nil).
- p.303, Appendix C: the rows “37 kilowatts and above but less than 186 kilowatts” for 1st Class and 3rd Class are entirely blank in the original (no figures printed); reproduced as blank.

Tonnage Measurement Rules (pp. 335–355)

- p.339, rule 2(f) sub-clause reference and p.340 rule 3(g): the text of 3(g) duplicates the definition of length and appears to be a printing anomaly in the original; translated as printed.
- p.343, rule 10(4): the printed Bengali defines “আবদ্ধ স্থানসমূহ” (enclosed spaces) with wording identical to the definition of breadth (apparent misprint in the original); translated as printed.
- p.343, rule 10(5): the three conditions following “unless at least one of the following three conditions is fulfilled” are printed without (i)/(ii)/(iii) labels; reproduced as printed.
- p.349, certificate heading: “সনদ বিধি ৭ (৩) দ্রষ্টব্য” and rule references “বিধি ২ (ট)” for length and “২(ঞ)” for date — references given as printed, though they do not match the lettering of rule 2 on p.339.
- p.354, row “৭০ / ০২৩৬৯”: decimal point missing in the print; read as 0.2369.

Inland Shipping Ordinance 1976 (pp. 431–461)

- p.439, section 3(3)(c): “শ্যালো ইঞ্জিনসহ” — rendered as “including those with shallow engines”; the phrase is slightly garbled in the original.
- p.442, section 5C: the Explanation is numbered “(১)” in the original, apparently a misprint; reproduced as printed.
- p.445, S.R.O. number read as “৩৯৪-এস/৭৬” (394-S/76); the letter after the hyphen is small and could be “এল” (L).
- p.452, section 29(4): “সন্তোষজনকও নয় এবং মেয়াদ উত্তীর্ণও নয় এমন বিক্রয় সংক্রান্ত সার্টিফিকেট” — translated as “certificate of sale which is neither satisfied nor expired”; the original Bengali is awkward.
- p.453, section 30(2): the sentence structure in the original is garbled (reference to section 11(1) inserted mid-sentence); translated as faithfully as possible.
- p.457, section 37(B)(1): “নাবিক বিশেষ যাত্রী সমাগম ব্যবস্থাপনা” — rendered as “management of special passenger gatherings”; the word “নাবিক” appears to be a stray word in the original.

Inland Shipping Ordinance 1976 (pp. 462–494)

- p.469, section 60A(3): “যাত্রী, মাস্টার, কর্মকর্তা এবং ক্রুসহ ১২ জন ব্যক্তি বহনকারী” — translated literally as “carrying 12 persons, including passengers, master, officers and crew”; the original does not say “more than” 12.

- p.470, section 60B: the section is printed without a heading; the number “৬০খ” is followed directly by the text.
- p.478, S.R.O. number read as “৩৯১-এল/৭৬” (391-L/76); the letter after the hyphen is small.
- p.479, section 81B(1): the original reads “ধারা ১৮ এর উপধারা (১) অথবা ধারা ৯ এর উপধারা (১)”; “section 18” is reproduced as printed although the context suggests a cross-reference to the certificate of survey/registry provisions.
- p.484, item (2)(c): “আড়াআড়ি জড়তার ভ্রামক এবং ভাসমান তলের কেন্দ্রবিন্দু” — rendered as “transverse moment of inertia and the centre of the floating surface (i.e. centre of flotation)”.
- p.484, item (3)(c): “পানির গভীরতা এবং বিদ্যমান পারিপার্শ্বিক আবহাওয়ার সাপেক্ষে মুরিং লাইন্স সমূহের অবস্থান” — translated as “depth of water and position of the mooring lines with reference to the prevailing surrounding weather”; the original is awkward.
- p.485, figure: dimensions read as 300 mm (deck line length), 450 mm (freeboard bar length), 25 mm (ring band width and bar height); the sectional sketch at the right is described, not reproduced.
- p.492–494, storm-signal tables: the day/night signal shapes are described in words from the pictograms (cylinder/rectangle, cone, diamond, ball, white/red lights); the “black cylinder” for Signal No. 1 and No. 2 is drawn as a rectangle.
- Loose insert (s218): the file number is read as “ডিজিসেল-১৮/সনদ বিবিধ/০৫/২৩৫১” (“DGSL-18/Certificate Misc./05/2351”); the prefix letters are small and blurred. The caption of the fifth rejected photograph (“সাদা-কালো বিধায় গ্রহণযোগ্য নয়”, black-and-white) and the captions of the two acceptable specimens are faded and partly reconstructed from the legible fragments. The parenthetical after the Director General’s name is read as “(চঃদাঃ)” = current charge.